

Table of Content

[Introduction to ONE-TRAX](#)

[Opening ONE-TRAX](#)

[Create a new well from a previous well](#)

[Start a new well without a previous well](#)

Well File Setup

[Engineering Units](#)

[Loss Data Categories](#)

[Currency Symbol and Tax Rate](#)

[First fluid check type / First well file date](#)

[Well Header Information](#)

[Well Information](#)

[Marketing Codes](#)

[Well Casing Intervals \(Cost\)](#)

[Pit Information](#)

[General Setup](#)

[Loss Setup](#)

[Active Products/Equipment/Screens](#)

[Products Setup](#)

[Equipment Rental Setup](#)

[Screens Setup](#)

[Benchmark Setup](#)

[Equipment Properties Setup](#)

Drilling Fluids and Equipment module

[Drilling Fluids Extra Properties](#)

[1- General](#)

[2- Pumps Bits](#)

[3- Mud Properties](#)

[4- Well Geometry](#)

[Well Casing Interval \(Cost\)](#)

[Daily Casing Volume](#)

[5- Comments](#)

[6- Solids Equipment](#)

[Shaker Screen Inventory](#)

[Shaker Screen Transactions](#)

[Equipment Solids Analysis](#)

[Cuttings Retention](#)

[Equipment Details and Usage](#)

[7- Time Distribution](#)

[8- Inventory/Hydraulic/Concentration](#)

[Daily Loss Print Selection](#)

[Benchmark Evaluation](#)

[Product Concentration](#)

[Hydraulic Results](#)

[Vol. Accounting and Product Inventory](#)

Table of Content

Reservoir Drill-in Fluids module

Additional Setup

Daily Data Entry

1- General

2- Pumps Bits

3- Fluid Properties

4- Well Geometry

Daily Casing Volume

5- Well Information

6- Displacement

7- Time Distribution

8- Solids Equipment

9- Inventory and Volume Accounting

Completion Fluids module

Additional Setup

Daily Data Entry

1- General

2- Pumps Bits

3- Fluid Properties

4- Well Geometry

5- Well Information

6- Displacement

NTU Data / Pump Rate vs Pressure

Spacer Details

7- Time Distribution

8- Inventory and Volume Accounting

9- Filtration

10- Equipment

Table of Content

Waste Treatment and Disposal module

Additional Setup

Daily Input Choices

Input Source and Recovery Type

Disposal Category and Waste Properties

Tank Type

Water Treatment Input Choices

Injection Well List

Transportation and Disposal Price List

Daily Data Entry

1- General

2- Transactions

3- Dewatering, Cuttings Disposal & Fixation, Water Treatment

4- Solids Control, Haul-off, Cuttings Injection

5- Inventory

Unscheduled Events

Reports : Daily Report, Recap Report

Special Cases: Side Track, Pilot Hole, Solids Analysis data when using CaCO₃.



ONE-TRAX[®]
Wellsite

Introduction to ONE-TRAX

- What is ONE-TRAX?
 - Company-wide Standard Database (Software).
 - This database holds technical and financial data of our rig operations.
- The ONE-TRAX data is used to:
 - Create 'Daily' & 'Recap' reports of our rig services.
 - Create financial reports and invoices (if used with Oracle).
 - Analyze standard parameters across the organization.
 - Document lessons learned and best practices.
 - Keep performance historical data.
 - Prove our expertise to our clients (through hard data).
 - Track our performance.
 - Compare actual vs. planned data.
 - Document products & equipment performance.

Introduction to ONE-TRAX

- What are the benefits of using ONE-TRAX?
 - Easier and faster reporting process.
 - Faster invoicing and payment.
 - Global data mining.
 - Quicker and reliable access to operational data.
 - Better practice to preserve data.

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ONE-TRAX[®] Wellsite

Illustrated instructions for using
ONE-TRAX Wellsite

Opening ONE-TRAX[®]

ONE-TRAX[®] opens under the ECCCP security program.

Single-Click on the ONE-TRAX to activate.



Opening ONE-TRAX[®]

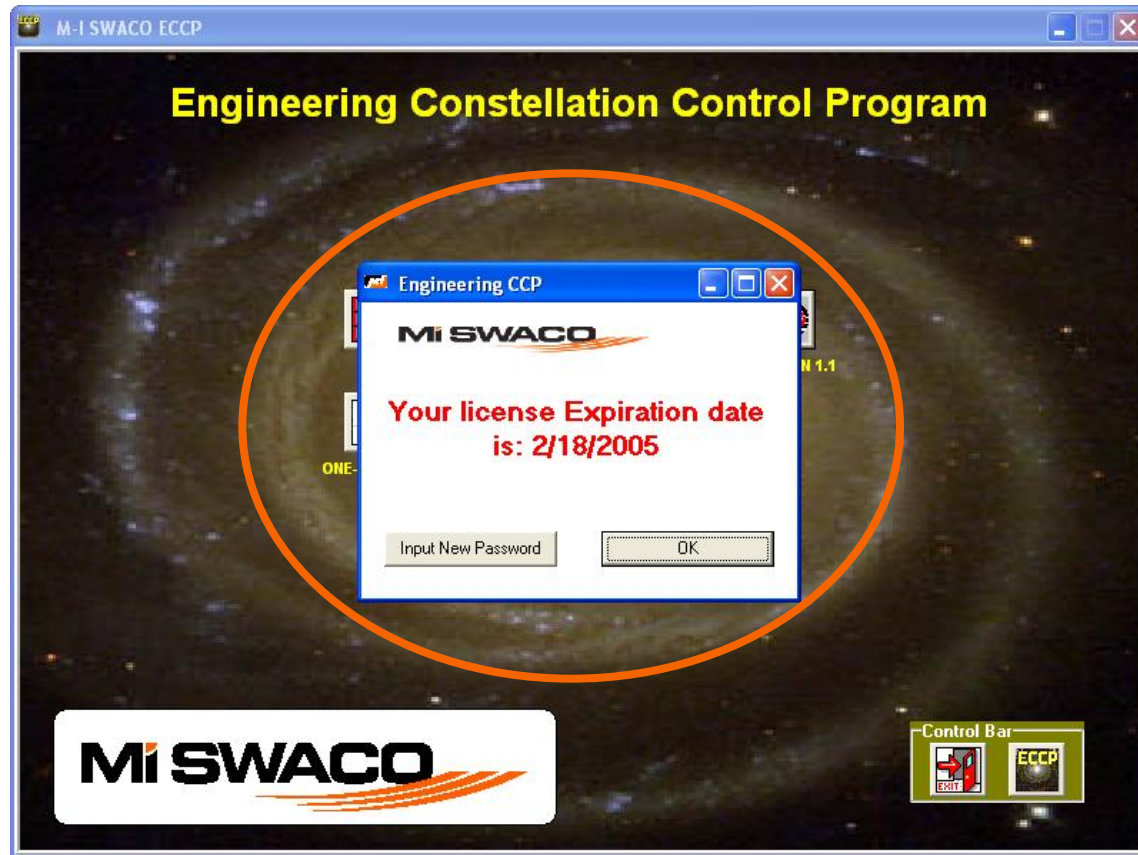
If this is your first time use, you will be prompted to fill out the License Request Form and submit to Houston to obtain a password.



This password will be good for one year.

Opening ONE-TRAX[®]

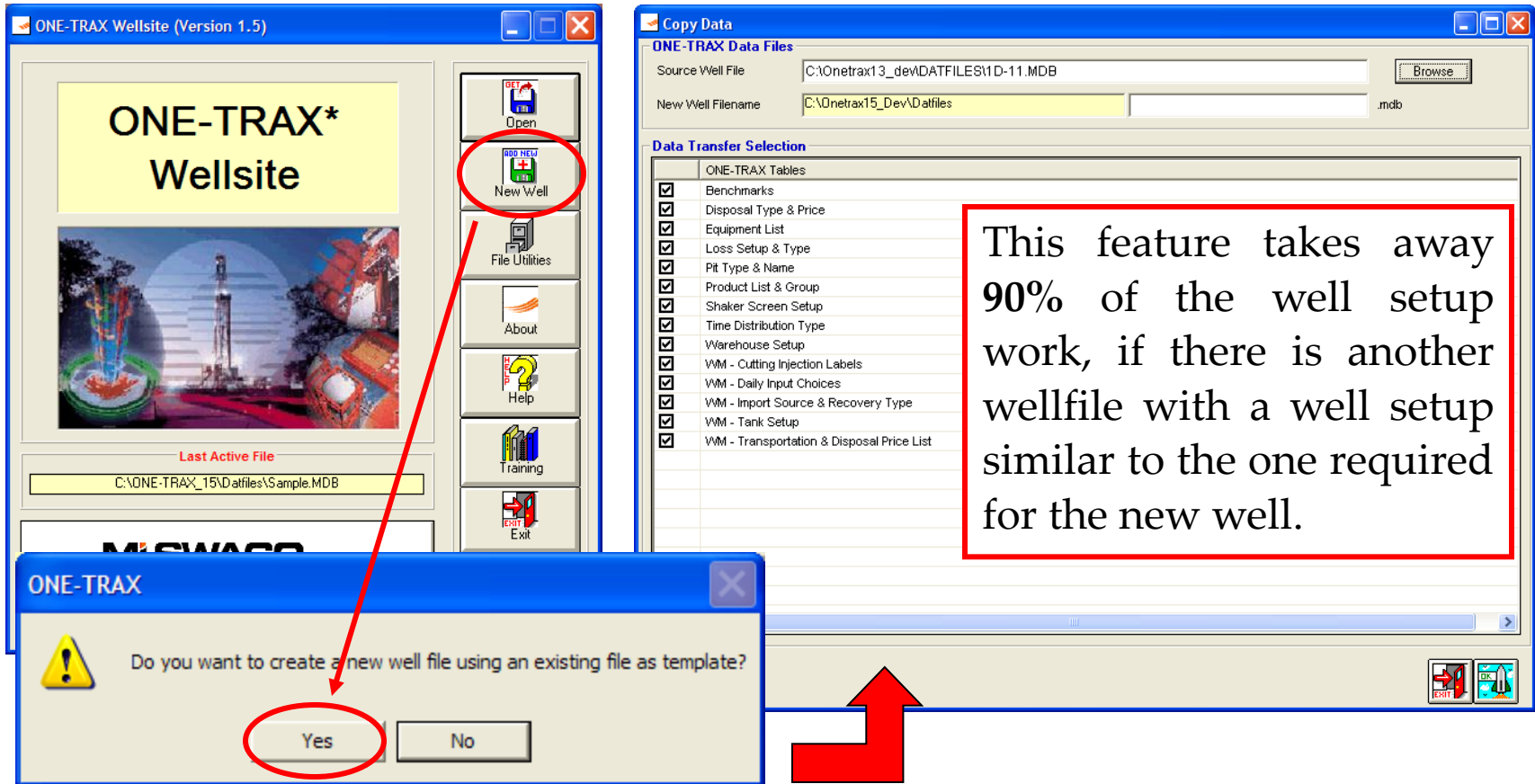
During the last month or so of activation, this warning will appear to remind you to apply for a new password.



ONE-TRAX 1.5 – New Feature

Create New Well from a previous Well

ONE-TRAX 1.5 has a built-in feature that allows the user to 'create a well from a previous well', just like using i-OT (Interact OT).



Starting a New Well (without using a reference well setup)



Click on the
“**New Well**”
button to begin

Starting a New Well – Setting up your units

This is the units 'system' used for the well. Click "Change" to select a different set of units.

The units used in the wellfile can only be selected when the new well file is created. This selection **can not** be changed once you click the 'OK' button on this screen.

New Well Information

Mi SWACO

Unit Selection for New Well

Standard Oilfield Change

Loss Data Categories

Custom Change

Financial Settings

Currency Decimal Places 2

Currency Symbol \$

Tax Rate (%) 0

Solids Analysis Equations

Water-Based Mud M-I

Oil-Based/Synthetic-Based Mud M-I

New Well File Name

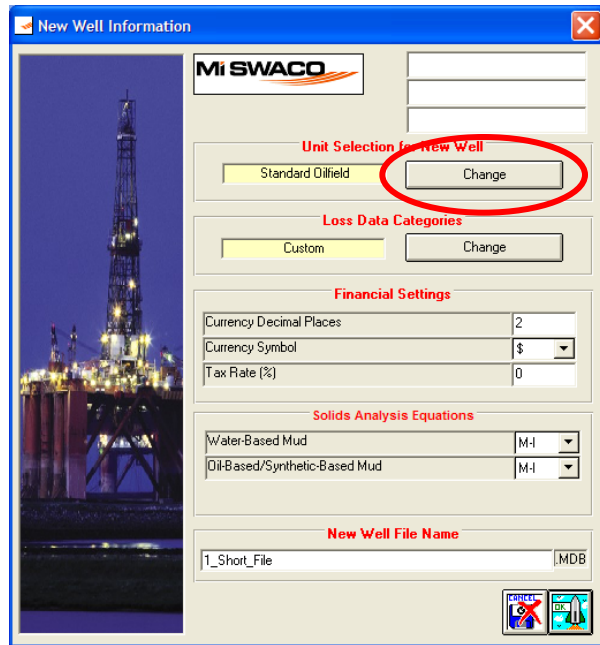
1_Short_File .MDB

Cancel OK

Starting a New Well – Setting up your units (cont.)

Ten pre-defined unit sets are available.

Select your units system from this list



The 'New Well Information' dialog box contains several sections. The 'Unit Selection for New Well' section is highlighted with a red circle around the 'Change' button. Other sections include 'Loss Data Categories', 'Financial Settings', 'Solids Analysis Equations', and 'New Well File Name'.

New Well Information

Unit Selection for New Well

Standard Oilfield **Change**

Loss Data Categories

Custom **Change**

Financial Settings

Currency Decimal Places: 2

Currency Symbol: \$

Tax Rate (%): 0

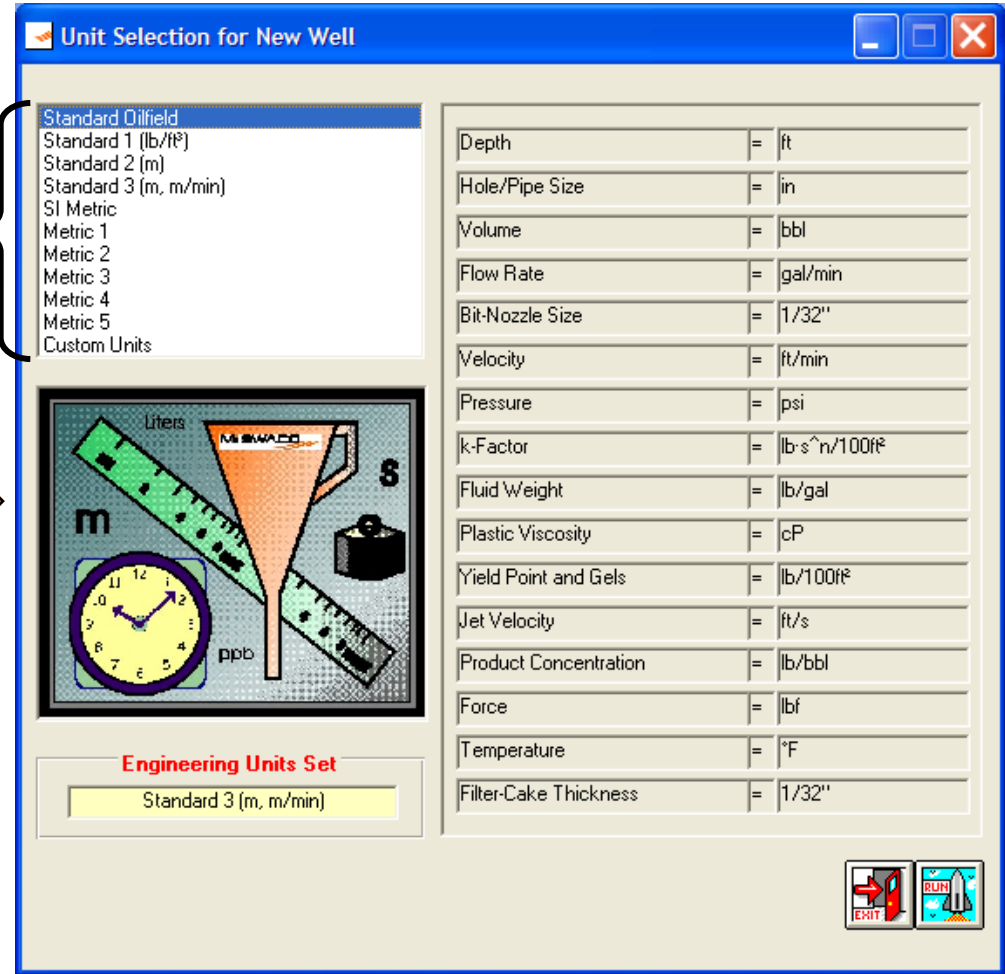
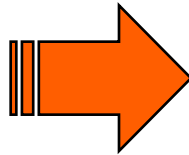
Solids Analysis Equations

Water-Based Mud: M-I

Oil-Based/Synthetic-Based Mud: M-I

New Well File Name

1_Short_File .MDB



The 'Unit Selection for New Well' dialog box shows a list of unit sets on the left and a table of unit definitions on the right. The 'Engineering Units Set' is selected at the bottom.

Unit Selection for New Well

Standard Oilfield
Standard 1 (lb/ft³)
Standard 2 (m)
Standard 3 (m, m/min)
SI Metric
Metric 1
Metric 2
Metric 3
Metric 4
Metric 5
Custom Units

Depth	=	ft
Hole/Pipe Size	=	in
Volume	=	bbl
Flow Rate	=	gal/min
Bit-Nozzle Size	=	1/32"
Velocity	=	ft/min
Pressure	=	psi
k-Factor	=	lb·s ⁿ /100ft ²
Fluid Weight	=	lb/gal
Plastic Viscosity	=	cP
Yield Point and Gels	=	lb/100ft ²
Jet Velocity	=	ft/s
Product Concentration	=	lb/bbl
Force	=	lbf
Temperature	=	°F
Filter-Cake Thickness	=	1/32"

Engineering Units Set

Standard 3 (m, m/min)

EXIT **RUN**

Starting a New Well – Setting up your units (cont.)

Choosing “Custom Units” allows you to pick your units using the drop down menu for each property listed on this screen.



Starting a New Well – Setting up your units (cont.)

Make sure you click the “OK” button before you leave this screen, to update the unit selection made (shown in the yellow box).



Unit Selection for New Well

Standard Oilfield
Standard 1 (lb/ft³)
Standard 2 (m)
Standard 3 (m, m/min)
SI Metric
Metric 1
Metric 2
Metric 3
Metric 4
Metric 5
Custom Units

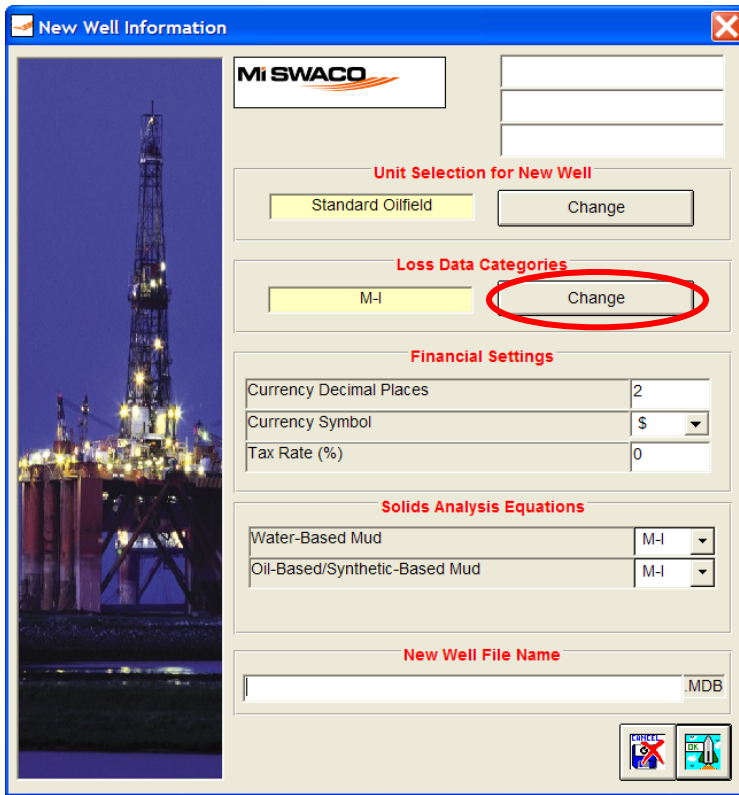
Engineering Units Set
Standard Oilfield

Depth	=	ft
Hole/Pipe Size	=	in
Volume	=	bbl
Flow Rate	=	gal/min
Bit-Nozzle Size	=	1/32"
Velocity	=	ft/min
Pressure	=	psi
k-Factor	=	lb s ⁿ /100ft²
Fluid Weight	=	lb/gal
Plastic Viscosity	=	cP
Yield Point and Gels	=	lb/100ft²
Jet Velocity	=	ft/s
Product Concentration	=	lb/bbl
Force	=	lbf
Temperature	=	°F
Filter-Cake Thickness	=	1/32"

EXIT OK

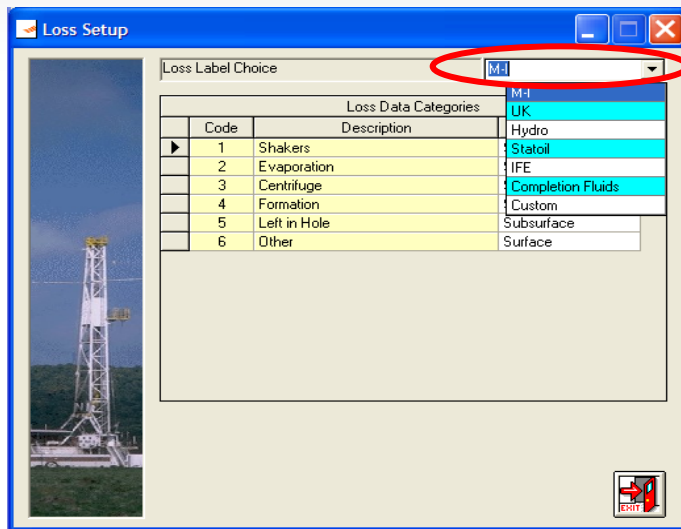
Starting a New Well – Loss Data Categories

The 'Loss Data Categories' section is used to indicate where can you lose fluid during your project. This setup is used for your volume accounting.

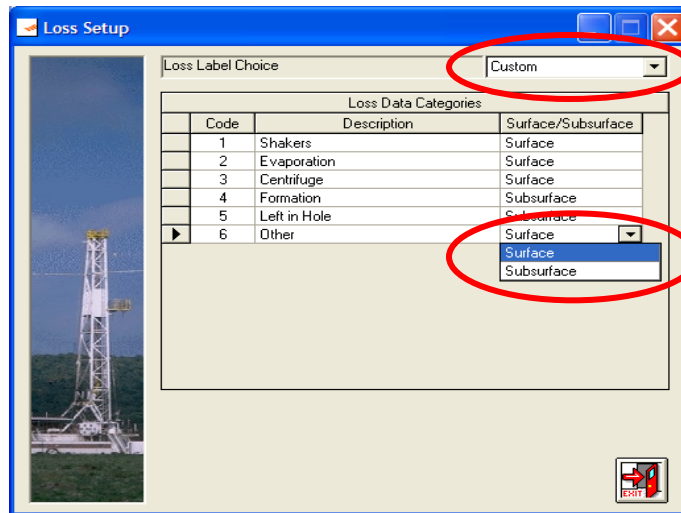


Click the “**Change**” button to indicate the categories required in your project.

Starting a New Well – Loss Data Categories (cont.)



Use the drop down menu to select any pre-defined list of “Loss Categories”. If you need to add ‘categories’ to the list selected you can do that later in the well setup.



Choosing “Custom” from the list gives you the ability of editing all the category labels.

Select either ‘Sub-surface’ or ‘surface’ for each ‘loss category’, depending whether the fluid is lost *down-hole* or *on surface*, respectively.

Starting a New Well – Currency Symbol & Tax Rate

New Well Information

Mi SWACO

Unit Selection for New Well

Standard Oilfield Change

Loss Data Categories

Custom Change

Financial Settings

Currency Decimal Places 2

Currency Symbol USD

Tax Rate (%) 0

Solids Analysis Equations

Water-Based Mud M-H

Oil-Based/Synthetic-Based Mud M-H

New Well File Name

NewWellFile .MDB

OK

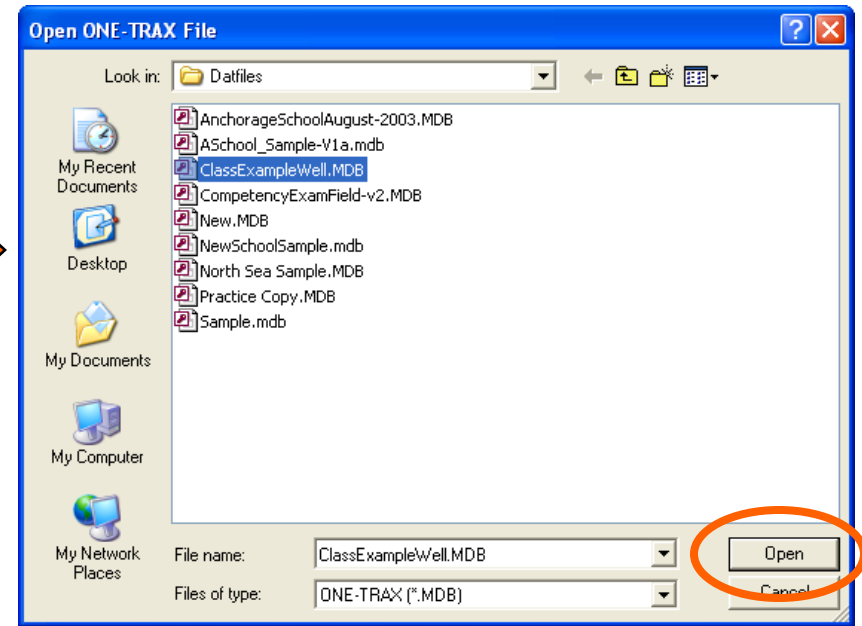
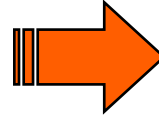
Properly select your currency symbol. You won't be able to change this selection once you click the 'OK' button of this screen.

If applicable, enter the Tax Rate. It will be applied to products that you mark as taxable. You can change this entry later on, if needed.

After completing all of the setup items on the New Well Information screen you are ready to create the new well file. Enter a new well file name and then click the OK button.

Starting a New Well – Completing the Setup process

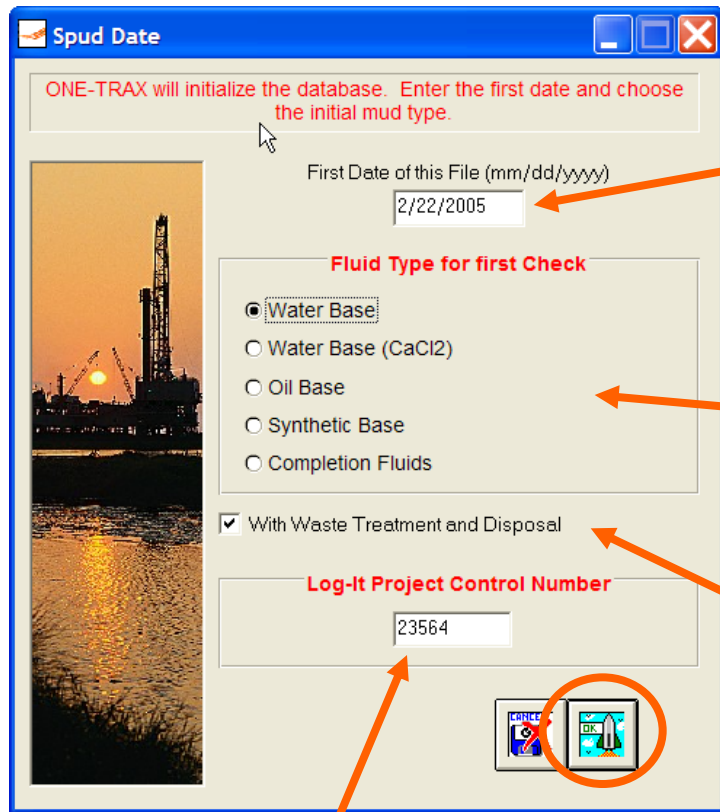
Click the “**OPEN**” button



Choose the Well File you created in the previous steps and click ‘Open’.

Starting a New Well – One-time data screen

*This form appears **only** the first time a well is opened*



The screenshot shows a software window titled 'Spud Date'. At the top, a red text box states: 'ONE-TRAX will initialize the database. Enter the first date and choose the initial mud type.' Below this is a text field for 'First Date of this File (mm/dd/yyyy)' containing '2/22/2005'. An orange arrow points to this field. Underneath is a section titled 'Fluid Type for first Check' with five radio button options: 'Water Base' (selected), 'Water Base (CaCl2)', 'Oil Base', 'Synthetic Base', and 'Completion Fluids'. An orange arrow points to the 'Water Base' option. Below this is a checked checkbox labeled 'With Waste Treatment and Disposal'. An orange arrow points to this checkbox. At the bottom is a text field for 'Log-It Project Control Number' containing '23564'. An orange arrow points to this field. In the bottom right corner, there are two icons: one with a red 'X' and one with a green checkmark. The checkmark icon is circled in orange.

Enter here the first date that you need to enter data (this date may precede the spud date). No dates may be added prior to this date. **This entry can not be changed after you click the 'OK' button on this screen.**

Properly select the type of fluid tested on your first day. **This entry can not be changed after you click the 'OK' button on this screen.**

Click the check-box if there are 'Waste Treatment and Disposal' services for this project. This option can also be selected at a later time.

If available at this time, enter here the 'Log-It Project Control Number' assigned to this well.

ONE-TRAX Main Sections

The screenshot displays the ONE-TRAX software interface. At the top, a blue title bar shows 'File Name: SAMPLE.MDB'. Below this, a form contains three fields: 'Operator' with the value 'ABC Oil Company', 'Well Name' with 'Gash #12', and 'Unit System' with 'Standard Oilfield'. The main area is divided into two sections. The upper section, titled 'Well Information' in red, is enclosed in an orange border and contains ten buttons arranged in two columns: 'Well Header Information', 'Loss Setup', 'Well Casing Intervals (Cost)', 'Active Products/Equipment/Screens', 'Well Survey', 'Benchmark Setup', 'Pit Information', 'Equipment Properties Setup', 'General Setup', 'Unscheduled Events', and 'Cost Overview'. The lower section, titled 'ONE-TRAX Modules' in red, is enclosed in a blue border and contains four buttons: 'Drilling Fluids and Equipment', 'RDF Module', 'Completion Fluids', and 'Waste Treatment and Disposal'. To the left of the 'ONE-TRAX Modules' section is a small image of three workers in orange safety gear. At the bottom left is a 'Recap Reports' button, and at the bottom center is an 'Activate GRAFX' button with a small icon. A red 'EXIT' button is located at the bottom right.

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
Cost Overview	

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

Activate GRAFX

EXIT

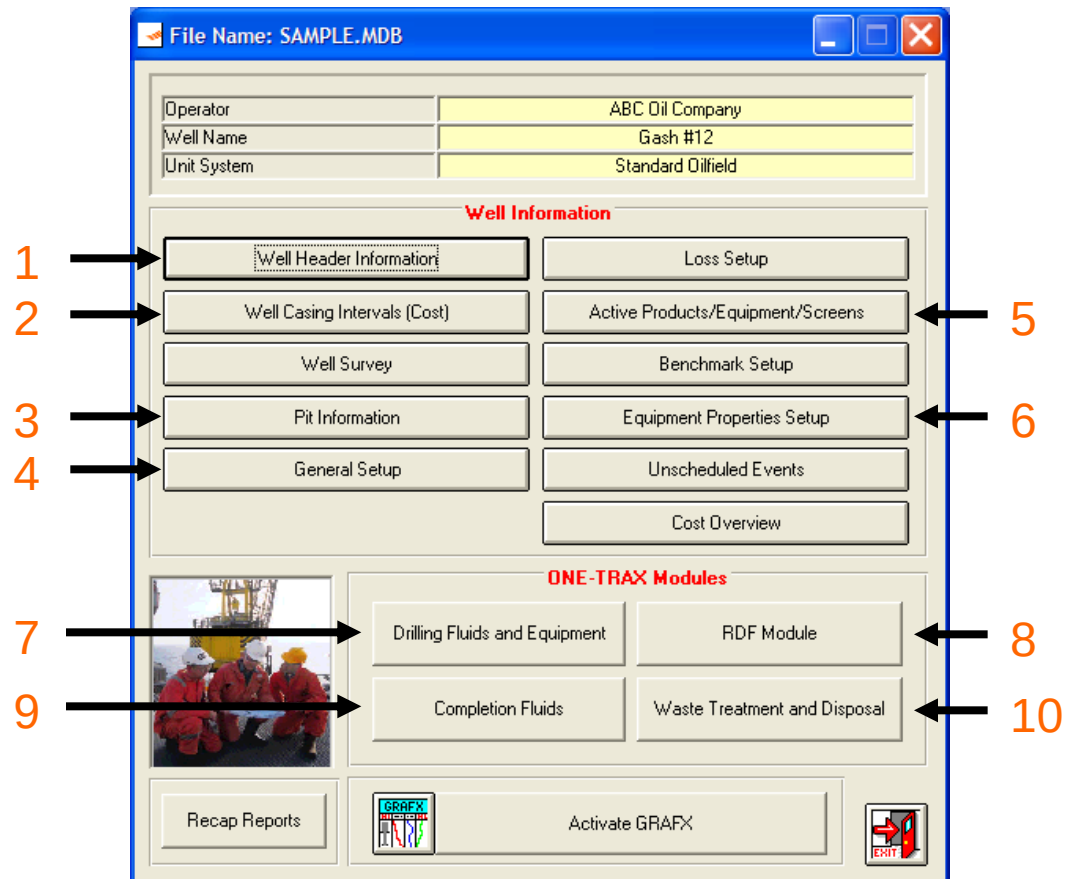
Project Setup and
Common Data

Daily Data Entry
Modules

ONE-TRAX 1.5 – New Features

The biggest change of this new version is the addition of the “**RDF Module**” (item number 8), however there are actual changes in all the sections highlighted here.

The second most significant change to the software was adding the option of using the ‘API 5th edition’ method to calculate hydraulic parameters.



Setup Process - Project Main Screen

Before starting entering the daily data, specific project information needs to be entered, such as: *who's the client?*, *what products & equipment will be used?*, *is this an offshore project?*, *where is this well been drilled?*, etc., etc., etc.

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
Cost Overview	

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

GRAFX

Activate GRAFX

EXIT

Here is where you enter the general information of your project.

Setup Process - Project Main Screen

The screenshot shows the 'Project Main Screen' for a file named 'SAMPLE.MDB'. At the top, there are three input fields: 'Operator' (ABC Oil Company), 'Well Name' (Gash #12), and 'Unit System' (Standard Oilfield). Below these is the 'Well Information' section, which contains a grid of buttons. Arrows 1 through 8 point to the following buttons: 1 points to 'Well Header Information', 2 to 'Well Casing Intervals (Cost)', 3 to 'Well Survey', 4 to 'Pit Information', 5 to 'Loss Setup', 6 to 'Active Products/Equipment/Screens', 7 to 'Benchmark Setup', and 8 to 'Equipment Properties Setup'. The 'Unscheduled Events' and 'Cost Overview' buttons are highlighted with an orange border. Below the 'Well Information' section is the 'ONE-TRAX Modules' section, which contains four buttons: 'Drilling Fluids and Equipment', 'RDF Module', 'Completion Fluids', and 'Waste Treatment and Disposal'. At the bottom, there is a 'Recap Reports' button, a 'GRAFX' button with a small icon, and an 'Activate GRAFX' button. An 'EXIT' button is located in the bottom right corner.

Well Information	
Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

ONE-TRAX Modules	
Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Visit each section (starting with "1") and provide all the required information.

There is no set-up done in these two buttons.

These sections will be addressed later in this presentation.

Setup Process - Well Header Information

Clicking the “**Well Header Information**” button will open a new screen that gathers “Well General Data”.

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information (highlighted)

Loss Setup

Well Casing Intervals (Cost)

Active Products/Equipment/Screens

Well Survey

Benchmark Setup

Pit Information

Equipment Properties Setup

General Setup

Unscheduled Events

Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment

RDF Module

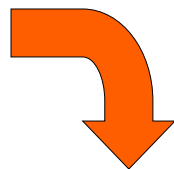
Completion Fluids

Waste Treatment and Disposal

Recap Reports

Activate GRAFX

EXIT



File Name: C:\ONE-TRAX_15\DATAFILES\SAMPLE.MDB

1- Well Information

2- Marketing Codes

Offshore

Air Gap (ft)	84
Water Depth (ft)	220
Sea Floor Temp (°F)	40

Well Data

Operator	ABC Oil Company	M-I SWACO Engineer 1	Ian Scott
Well Name	Gash #12	M-I SWACO Engineer 2	David Jones
Field/Area	Offshore Northcours	Spud Date (mm/dd/yyyy)	12/7/1999
Description	Exploration	TD Date (mm/dd/yyyy)	12/21/1999
Location	Out Back	TD Days	10
Warehouse	North Course	Re-Entry Depth (ft)	100
Contractor	Drillem Enterprises	Latitude NS Indicator	
Rig Name	Ocean	Longitude EW Indicator	
Project Engineer	John Smith	Surface temp (°F)	70
		Temp Gradient (°F/100ft)	2

End of Well

Total Depth (ft)	9002	Total Days	13	Maximum Temperature (°F)	180
TVD (ft)	7584	End Date (mm/dd/yyyy)	7/8/2002		
Horiz. Displacement (ft)	100	Total Cost (\$)	404183.78		

Comments

Sample file

Log-It Project Control Number

Edit

EXIT

Setup Process - Well Header Information

“Surface Temperature”, “Temperature Gradient” and “Sea Floor Temperature” are three new inputs required by the ‘API 5th edition’ to improve the accuracy of the hydraulics’ calculations.

File Name: C:\ONE-TRAX_15\DATAFILES\SAMPLE.MDB

1- Well Information

2- Marketing Codes

Offshore

Air Gap (ft)	84
Water Depth (ft)	220
Sea Floor Temp (°F)	40

Well Data

Operator	ABC Oil Company	M-I SWACO Engineer 1	Ian Scott
Well Name	Gash #12	M-I SWACO Engineer 2	David Jones
Field/Area	Offshore Northcours	Spud Date (mm/dd/yyyy)	12/7/1999
Description	Exploration	TD Date (mm/dd/yyyy)	12/21/1999
Location	Out Back	TD Days	10
Warehouse	North Course	Re-Entry Depth (ft)	100
Contractor	Drillem Enterprises	Latitude NS Indicator	
Rig Name	Ocean	Longitude EW Indicator	
Project Engineer	John Smith	Surface temp (°F)	70
		Temp Gradient (°F/100ft)	2

End of Well

Total Depth (ft)	8002	Total Days	13	Maximum Temperature (°F)	180
TVD (ft)	7584	End Date (mm/dd/yyyy)	7/8/2002		
Horiz. Displacement (ft)	100	Total Cost (\$)	404183.78		

Comments

Sample file

Log-It Project Control Number

Edit

Sea Floor Temp (°F) 40

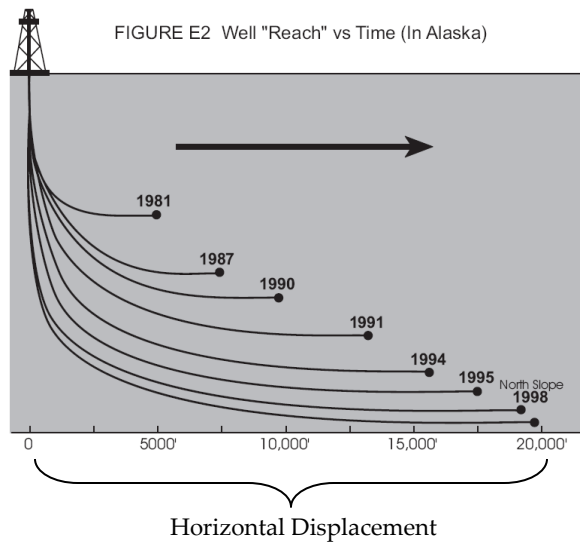
Note: using the “API 5th edition” rules is optional in ONE-TRAX 1.5, but if you decide to follow this new regulation, then you must remember to enter these data for proper analysis.

Surface temp (°F) 70 Temp Gradient (°F/100ft) 2

Setup Process - Well Header Information (cont)

“Horizontal Displacement” was added to the ‘End of Well’ section of the ‘Well Information’ tab.

This data will be used to identify Extended Reach wells.



File Name: C:\ONE-TRAX_15\DATFILES\SAMPLE.MDB

1- Well Information		2- Marketing Codes	
		Offshore	
		Air Gap (ft)	84
		Water Depth (ft)	220
		Sea Floor Temp (°F)	40
Well Data			
Operator	ABC Oil Company	M-I SWACO Engineer 1	Ian Scott
Well Name	Gash #12	M-I SWACO Engineer 2	David Jones
Field/Area	Offshore Northcours	Spud Date (mm/dd/yyyy)	12/7/1999
Description	Exploration	TD Date (mm/dd/yyyy)	12/21/1999
Location	Out Back	TD Days	10
Warehouse	North Course	Re-Entry Depth (ft)	100
Contractor	Drillem Enterprises	Latitude NS Indicator	
Rig Name	Ocean	Longitude EW Indicator	
Project Engineer	John Smith	Surface temp (°F)	70
		Temp Gradient (°F/100ft)	2
End of Well			
Total Depth (ft)	8002	Total Days	13
TVD (ft)	7584	End Date (mm/dd/yyyy)	7/8/2002
Horiz. Displacement (ft)	1213	Total Cost (\$)	404183.78
Sample file		Comments	
		Log-It Project Control Number	
		Edit	

Horiz. Displacement (ft) 1213

Setup Process - Well Header Information (cont.)

File Name: C:\ONE-TRAX 15\DATFILES\SAMPLE.MDB

1- Well Information

Offshore

Air Gap (ft)	84
Water Depth (ft)	220
Sea Floor Temp (°F)	40

Well Data

Operator	ABC Oil Company	M-I SWACO Engineer 1	Ian Scott
Well Name	Gash #12	M-I SWACO Engineer 2	David Jones
Field/Area	Offshore Northcours	Spud Date (mm/dd/yyyy)	12/7/1999
Description	Exploration	TD Date (mm/dd/yyyy)	12/21/1999
Location	Out Back	TD Days	10
Warehouse	North Course	Re-Entry Depth (ft)	100
Contractor	Drillem Enterprises	Latitude NS Indicator	
Rig Name	Ocean	Longitude EW Indicator	
Project Engineer	John Smith	Surface temp (°F)	70
		Temp Gradient (°F/100ft)	2

End of Well

Total Depth (ft)	8002	Total Days	13	Maximum Temperature (°F)	180
TVD (ft)	7584	End Date (mm/dd/yyyy)	7/8/2002		
Horiz. Displacement (ft)	100	Total Cost (\$)	404183.78		

Comments

Sample file

Log-It Project Control Number

Edit

Offshore Projects **must** fill in this data. Some of these values will affect your volume calculations.

This is to be filled in at the end of the well

After completing click "Save", then go to Tab #2: "Marketing Codes"

Setup Process - Well Header Information

Marketing Code		
Primary Mud Type	2045	VERSADRIL
Well Type	0040	Extended-Reach
Contract Type	0003	Integrated Services
Completion Fluid Type	0021	Potassium Bromide

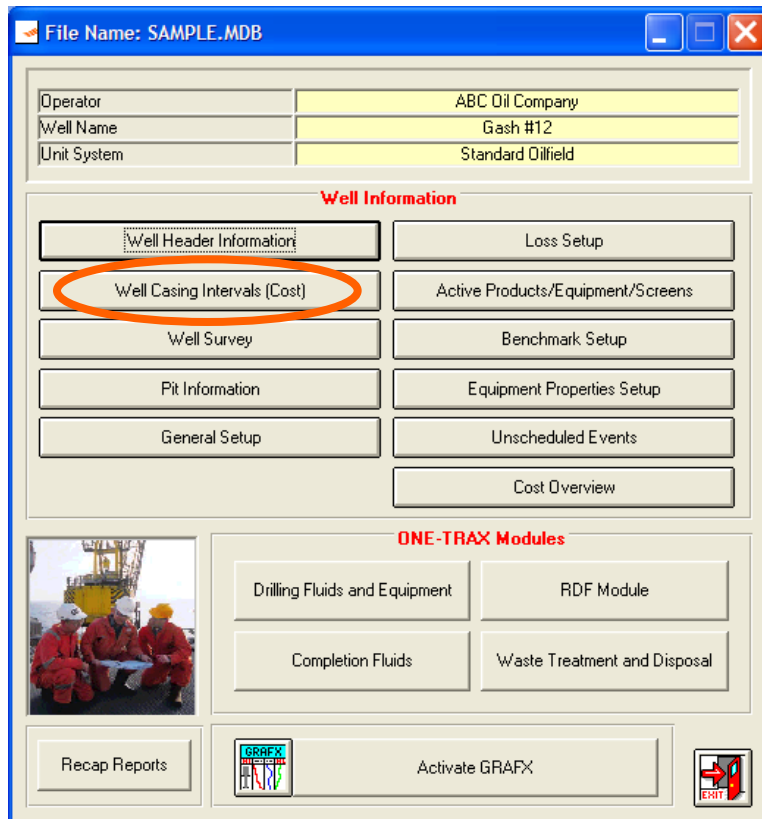
These data is **widely** used in ONE-TRAX Central queries, for tender responses and marketing.

This section **must** be filled out prior to uploading the well file to ONE-TRAX Central database.

If any of these data is not available at the beginning of the well, it can be entered later at any time.

Setup Process - Well Casing Intervals (Cost)

Click the “**Well Casing Intervals (Cost)**” button to open the following screen. Then, just click “New” and follow the instructions on the next slide.



File Name: SAMPLE.MDB

Operator: ABC Oil Company
Well Name: Gash #12
Unit System: Standard Oilfield

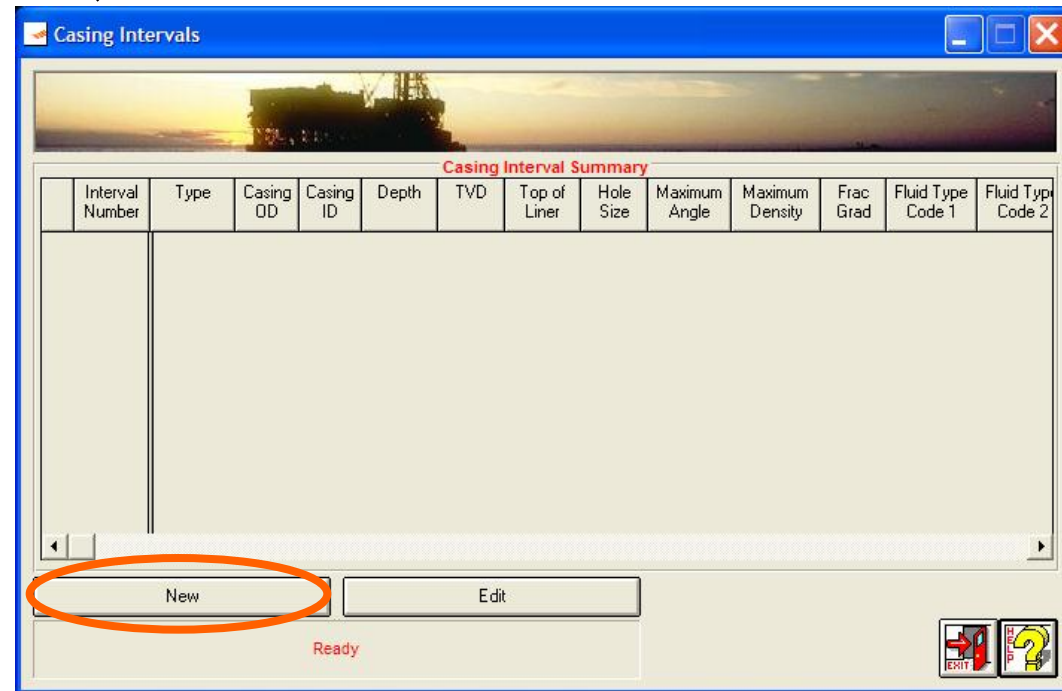
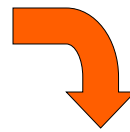
Well Information

Well Header Information | Loss Setup
Well Casing Intervals (Cost) | Active Products/Equipment/Screens
Well Survey | Benchmark Setup
Pit Information | Equipment Properties Setup
General Setup | Unscheduled Events
Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment | RDF Module
Completion Fluids | Waste Treatment and Disposal

Recap Reports | Activate GRAFX



Casing Intervals

Casing Interval Summary

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top of Liner	Hole Size	Maximum Angle	Maximum Density	Frac Grad	Fluid Type Code 1	Fluid Type Code 2

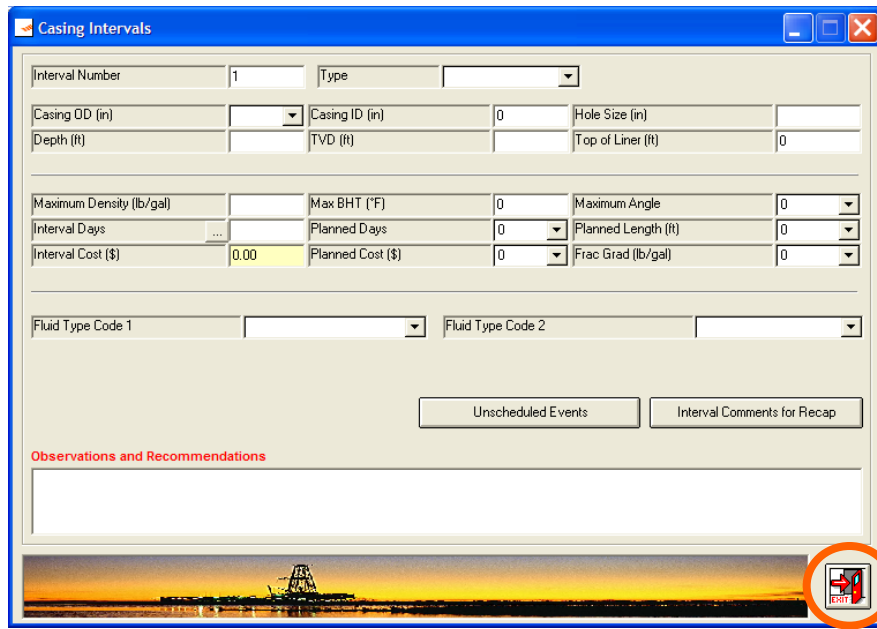
New | Edit

Ready

Setup Process - Well Casing Intervals (Cost)

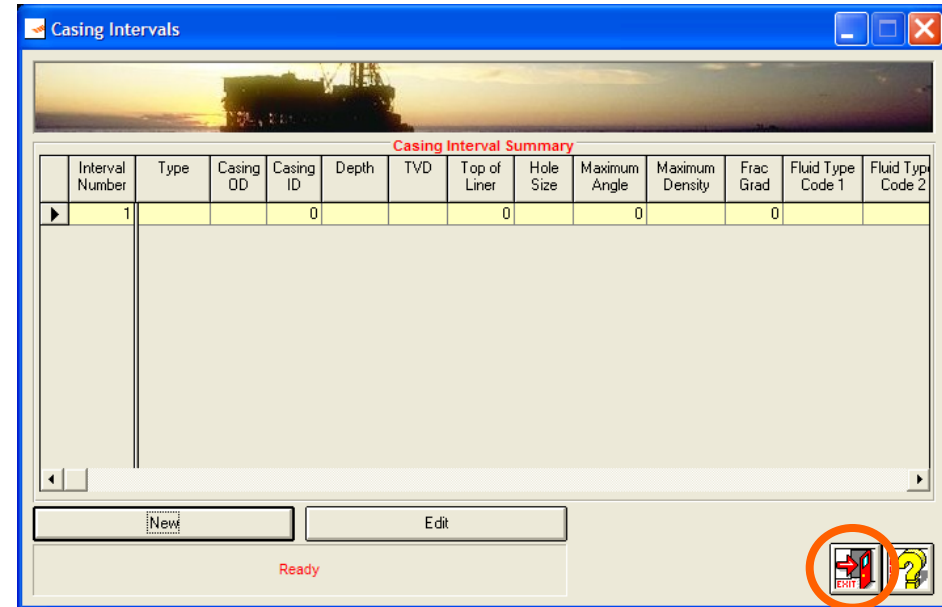
At this point you don't need to enter more information about the 1st interval of your well, so just click "Exit" to come back to the "Casings Intervals Summary" table.

You should now see a yellowed line indicating that your 1st 'Interval Cost' has been created. Now click "Exit" to continue your well setup.



The 'Casing Intervals' window displays various input fields for configuring a casing interval. The 'Interval Number' is set to 1. The 'Interval Cost (\$)' field is highlighted in yellow and contains the value 0.00. Other fields include Casing OD, Casing ID, Depth, TVD, Top of Liner, Hole Size, Maximum Density, Maximum Angle, Interval Days, Planned Length, Planned Cost, Frac Grad, Fluid Type Code 1, and Fluid Type Code 2. There are also buttons for 'Unscheduled Events' and 'Interval Comments for Recap'.

Interval Number	Type	Casing OD (in)	Casing ID (in)	Depth (ft)	TVD (ft)	Top of Liner (ft)	Hole Size (in)	Maximum Angle	Maximum Density	Frac Grad	Fluid Type Code 1	Fluid Type Code 2
1			0				0	0	0	0		



The 'Casing Intervals' window displays the 'Casing Interval Summary' table. The table has 13 columns: Interval Number, Type, Casing OD, Casing ID, Depth, TVD, Top of Liner, Hole Size, Maximum Angle, Maximum Density, Frac Grad, Fluid Type Code 1, and Fluid Type Code 2. The first row is highlighted in yellow, indicating the 1st interval cost has been created. The 'Exit' button is circled in red.

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top of Liner	Hole Size	Maximum Angle	Maximum Density	Frac Grad	Fluid Type Code 1	Fluid Type Code 2
1			0			0		0		0		

Setup Process - Well Casing Intervals (Cost)

This tag was changed from "Max Temperature" to "Max BHT"

Casing Intervals

Interval Number	1	Type	Casing		
Casing OD (in)	20	Casing ID (in)	19	Hole Size (in)	
Depth (ft)	800	TVD (ft)	800	Top of Liner	
Maximum Density (lb/gal)	8.6	Max BHT (°F)	75	Maximum Annulus Pressure (psi)	
Interval Days	4	Planned Days	0	Planned Length (ft)	
Interval Cost (\$)	8938.50	Planned Cost (\$)	N/A	Frac Grad (lb/gal)	

Fluid Type Code 1: Spud Mud Fluid Type Code 2: CALDRIL

Buttons: **Interval Comments for Recap**

Observations and Recommendations

Interval Comments for Recap

Doyon rig 141 moved from MPS-05 and rigged up over slot MPS-03 on July 27th, 2002. The rig was accepted on July 27th at 19:30 hrs. The diverter and annular preventer were rigged up prior to the rig move. The rig picked up the 4" drill pipe.

712 bbls of new mud was built at the rig prior to spudding on July 28th, 2002 at 13:00 hours. Soda Ash was added to precipitate any hardness before adding the MI Gel.

Drilling commenced with the well being drilled to a depth of 3,522' MD. A bit run was made as planned. MI Gel and Gelex were used for viscosity exclusively during the interval. After the trip bottoms up was dumped due to excessive mud weight.

Near the end of the interval PolyPAC Supreme UL was added to reduce fluid loss to 4 - 5 cc API. Lubricants were added at 1% each of Flo-Lube and Lubetex to aid the sliding and also to help getting casing to bottom. The last 1,500' MD was drilled with approximately 60% sliding in order to get the programmed 90-degree turn and build in the well.

The well TD of 7,320' MD was reached and the rig circulated 3 bottoms up prior to pumping out of the hole. An attempt was made to pull the pipe out of the hole without pumping at 3,000' MD but the well started swabbing. After cutting and slipping the drill line the rig ran back in the hole encountering bridges at 1,400 & 2,200' MD. The rig continued tripping in to 4,275' MD with the hole being noted as sticky. It was decided to trip out to run a hole opener. It was picked up and run to 300' MD before reaming the upper part of the hole to 2,900' MD. The bit was then run to 4,200' MD before reaming to bottom. All reaming was done at an elevated RPM and flow rate, 150 RPM & 550 gpm. The hole was circulated three bottoms up and the system was treated with 4 sacks of Tannathin and 2 sacks of Caustic Soda in preparation for running casing. A slug was pumped before tripping out with no problems.

The 7-5/8" casing was run to bottom at 7,309' MD without problems. The fluid was circulated and treated for the cement job with Spersene CF, SAPP and water until the funnel viscosity was 45 sec/qt. with a 9.0 ppg weight going in and coming out.

Haliburton pumped the cement job as follows: 10 bbls water, released the bottom plug, pumped 40 bbls water, 50 bbls Alpha spacer at 10.2 ppg, 855 bbls of lead cement at 10.7 ppg, 61 bbls tail cement at 15.8 ppg, 30 bbls water, released the top plug and pumped it to bottom with 309 bbls of mud. Mud returns were lost after 713 bbls of lead cement were pumped. Cement was not returned to surface so a top job was performed. The BOP was nipped down and 1" pipe was run to 729'. The mud was flushed out of the annulus with 150 bbls of water followed by 150 bbls of cement. A total of 22 bbls of cement was returned to surface. Total cost for this interval was \$53,263.29. This total includes engineering service charges of \$7,800.00

Added a new section: "Interval Comments for Recap" (that accepts up to 10,000 characters), to improve the recapping functionalities in ONE-TRAX

Setup Process – Pit Information

The next step in the well setup process is the “**Pit Information**”.

File Name: SAMPLE.MDB

Operator: ABC Oil Company
Well Name: Gash #12
Unit System: Standard Oilfield

Well Information

Well Header Information | Loss Setup

Well Casing Intervals (Cost) | Active Products/Equipment/Screens

Well Survey | Benchmark Setup

Pit Information | Equipment Properties Setup

General Setup | Unscheduled Events

Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment | RDF Module

Completion Fluids | Waste Treatment and Disposal

Recap Reports | Activate GRAFX

EXIT

Pits and Tanks Information

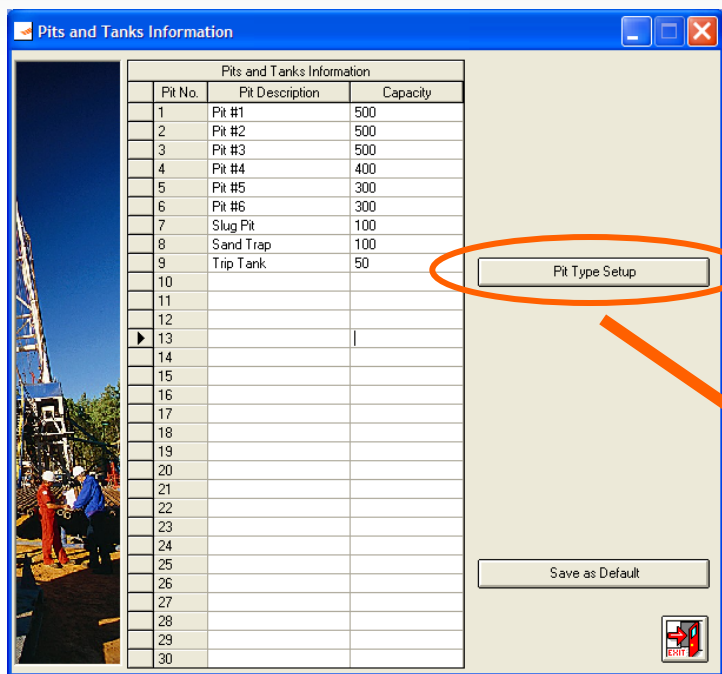
Pit No.	Pit Description	Capacity
1	Pit #1	500
2	Pit #2	500
3	Pit #3	500
4	Pit #4	400
5	Pit #5	300
6	Pit #6	300
7	Slug Pit	100
8	Sand Trap	100
9	Trip Tank	50
10		
11		
12		
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Pit Type Setup

Save as Default

EXIT

Setup Process – Pit Information (continuation)



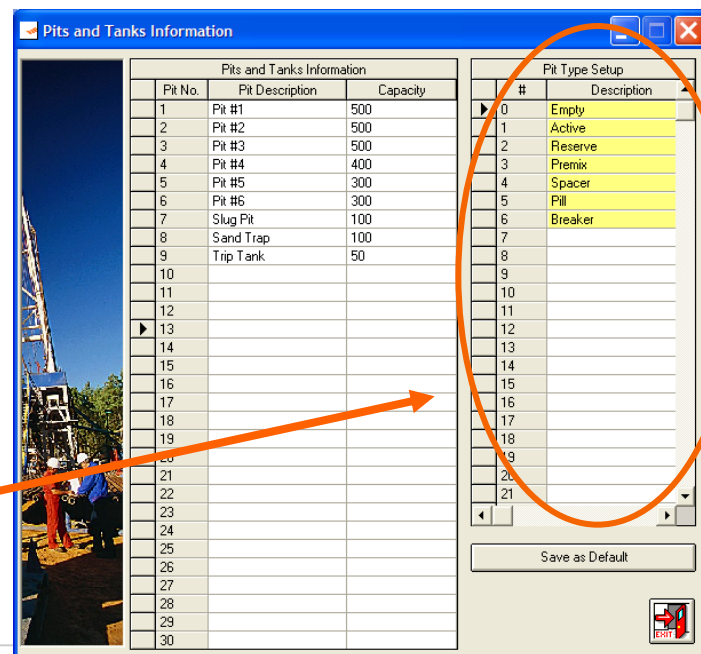
Pit No.	Pit Description	Capacity
1	Pit #1	500
2	Pit #2	500
3	Pit #3	500
4	Pit #4	400
5	Pit #5	300
6	Pit #6	300
7	Slug Pit	100
8	Sand Trap	100
9	Trip Tank	50
10		
11		
12		
13		
14		
15		
16		
17		
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30		

Pit Type Setup

Save as Default

Assign here a 'name' and capacity to all the pits in your mud system. Pit names are descriptive only (so their use is determined solely by the 'Pit Type' selected on a daily basis in the "Volume Accounting" tab).

Suggestion: **do not** use the words "Active", "Reserve" or "Premix" as part of the pit name, because it could mislead the data entry during the course of the well.



Pit No.	Pit Description	Capacity
1	Pit #1	500
2	Pit #2	500
3	Pit #3	500
4	Pit #4	400
5	Pit #5	300
6	Pit #6	300
7	Slug Pit	100
8	Sand Trap	100
9	Trip Tank	50
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		
25		
26		
27		
28		
29		
30		

#	Description
0	Empty
1	Active
2	Reserve
3	Premix
4	Spacer
5	Pill
6	Breaker
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	

Save as Default

The 'Pit Type Setup' list displays all the different uses that a pit may have at any given day, during the project. There are 6 standard types of pits: empty, active, reserve, premix, spacer, pill and breaker. Additional 'pit types' may also be defined, if required.

Setup Process – Pit Information (continuation)

Pits and Tanks Information		
Pit No.	Pit Description	Capacity
1	Active-1	600
2	Active-2	600
3	Active-3	600
4	Reserve-1	400
5	Reserve-2	400
6	Reserve-3	400
7	Premix-1	50
8	Premix-2	100
9	Premix-3	100
10	BaseOil-1	500
11	Base #2	1000
12	Boat	4000
13	SW Leg	3000
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		
25		
26		
27		
28		
29		
30		

Pit Type Setup	
#	Description
0	Empty
1	Active
2	Reserve
3	Premix
4	Spacer
5	Pill
6	Breaker
7	Base Oil
8	Brine
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	

Save as Default

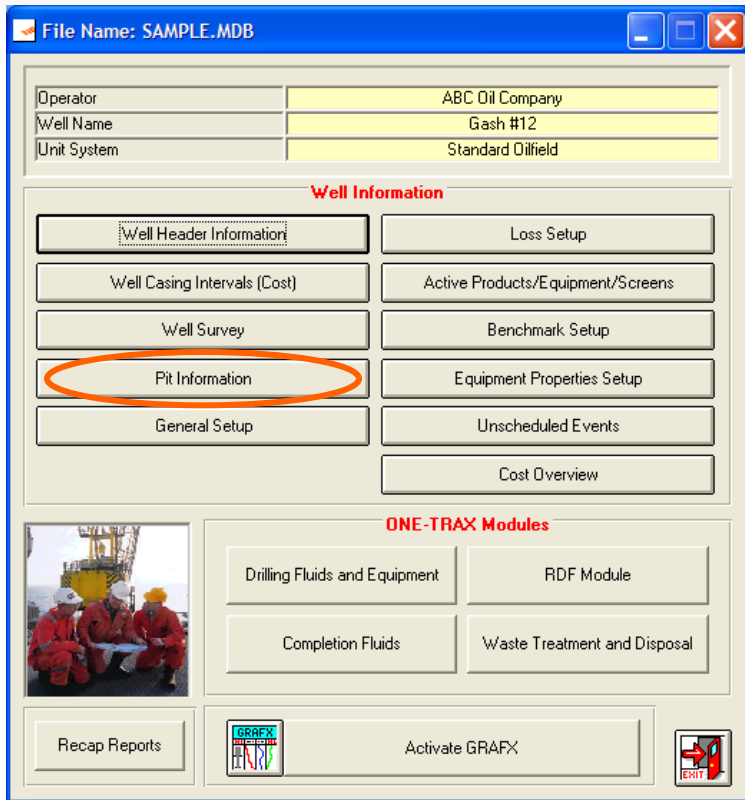
EXIT

Two new “Pit Descriptions” were added to the default list of ‘Pit Type’:

- Pill
- Breaker

These ‘descriptions’ will address some requirements on the new “RDF Module” but will also be available on the “Completion Fluids” module.

Setup Process – General Setup



File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
Cost Overview	

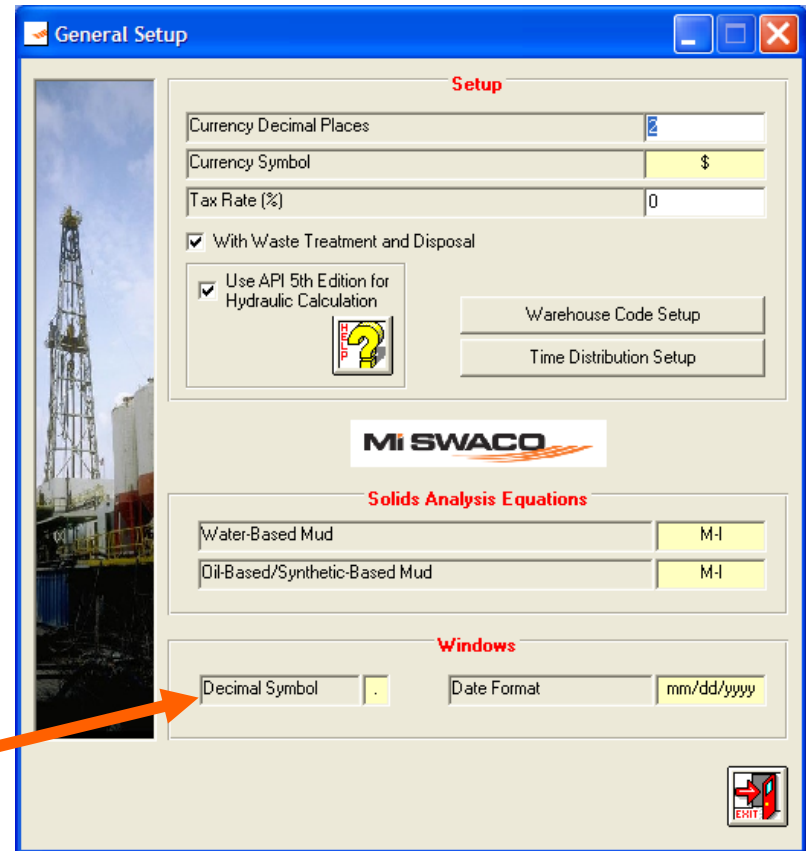
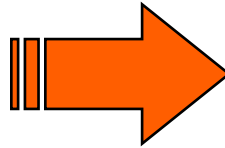
ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

Activate GRAFX

The 'General Setup' screen summarizes key file settings and allows you to change "Currency Decimal Places" and "Tax Rate %".



General Setup

Setup

Currency Decimal Places	2
Currency Symbol	\$
Tax Rate (%)	0

☒ With Waste Treatment and Disposal

☒ Use API 5th Edition for Hydraulic Calculation

Warehouse Code Setup

Time Distribution Setup

MI SWACO

Solids Analysis Equations

Water-Based Mud	M-I
Oil-Based/Synthetic-Based Mud	M-I

Windows

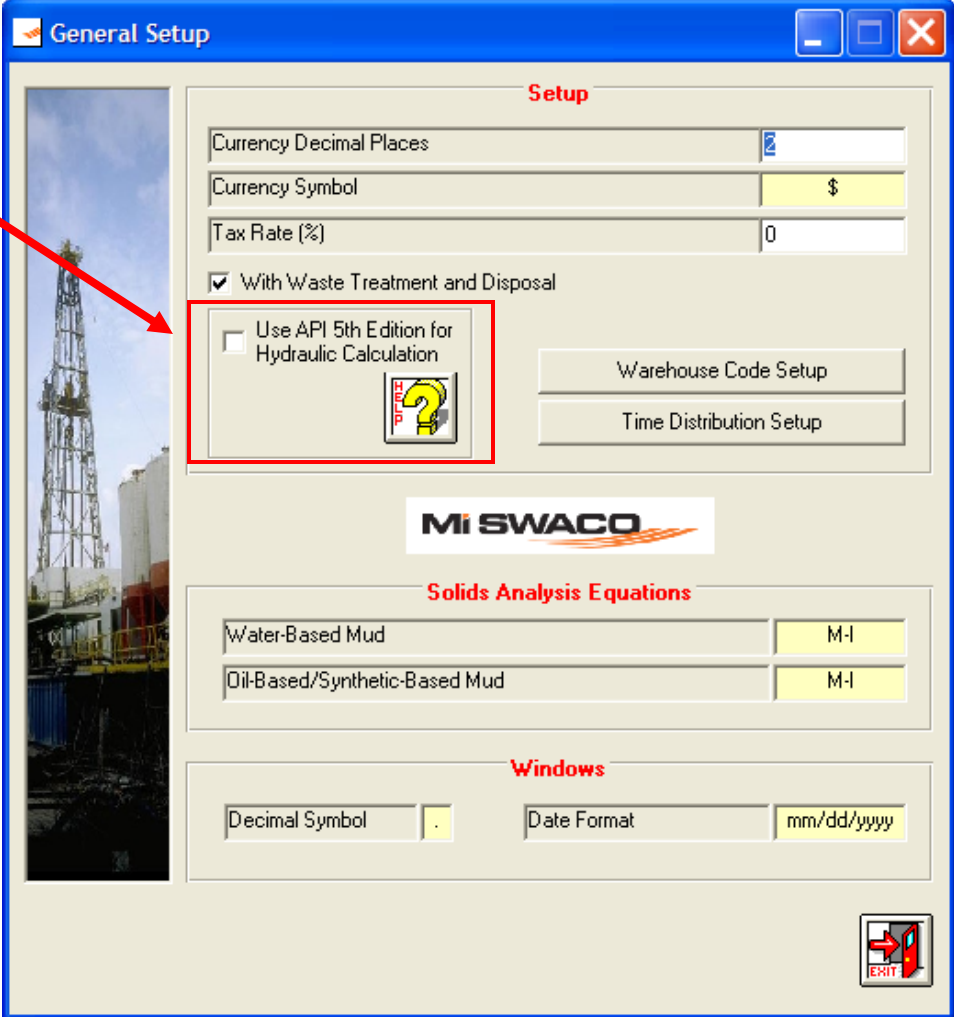
Decimal Symbol	.
Date Format	mm/dd/yyyy

The decimal symbol and the date format are controlled by MS Windows settings. These can be changed in the "control panel - regional settings" option.

Setup Process – General Setup (continuation)

Here is where you select whether you want to use the new “API 5th edition” method or keep using the 4th edition.

A more detailed explanation of the new data requirements (if using the API 5th edition) is displayed when clicking the “Help” button on this screen.



General Setup

Setup

Currency Decimal Places: 2

Currency Symbol: \$

Tax Rate (%): 0

☒ With Waste Treatment and Disposal

☐ Use API 5th Edition for Hydraulic Calculation 

Warehouse Code Setup

Time Distribution Setup

MI SWACO

Solids Analysis Equations

Water-Based Mud: M-I

Oil-Based/Synthetic-Based Mud: M-I

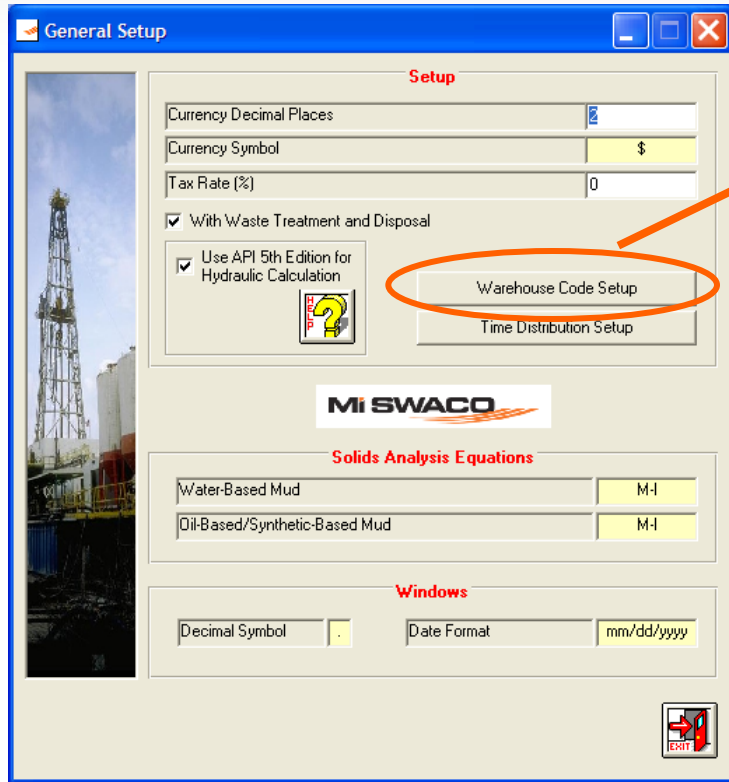
Windows

Decimal Symbol: .

Date Format: mm/dd/yyyy



Setup Process – General Setup (continuation)



The General Setup window contains several sections. The 'Setup' section includes fields for 'Currency Decimal Places' (2), 'Currency Symbol' (\$), and 'Tax Rate (%)' (0). It also has checkboxes for 'With Waste Treatment and Disposal' and 'Use API 5th Edition for Hydraulic Calculation'. A 'Warehouse Code Setup' button is circled in orange. The 'Solids Analysis Equations' section has dropdowns for 'Water-Based Mud' and 'Oil-Based/Synthetic-Based Mud', both set to 'M-I'. The 'Windows' section has dropdowns for 'Decimal Symbol' and 'Date Format' (mm/dd/yyyy). The Mi SWACO logo is at the bottom.

Setup

Currency Decimal Places: 2
Currency Symbol: \$
Tax Rate (%): 0

☒ With Waste Treatment and Disposal
☒ Use API 5th Edition for Hydraulic Calculation

Warehouse Code Setup
Time Distribution Setup

Solids Analysis Equations

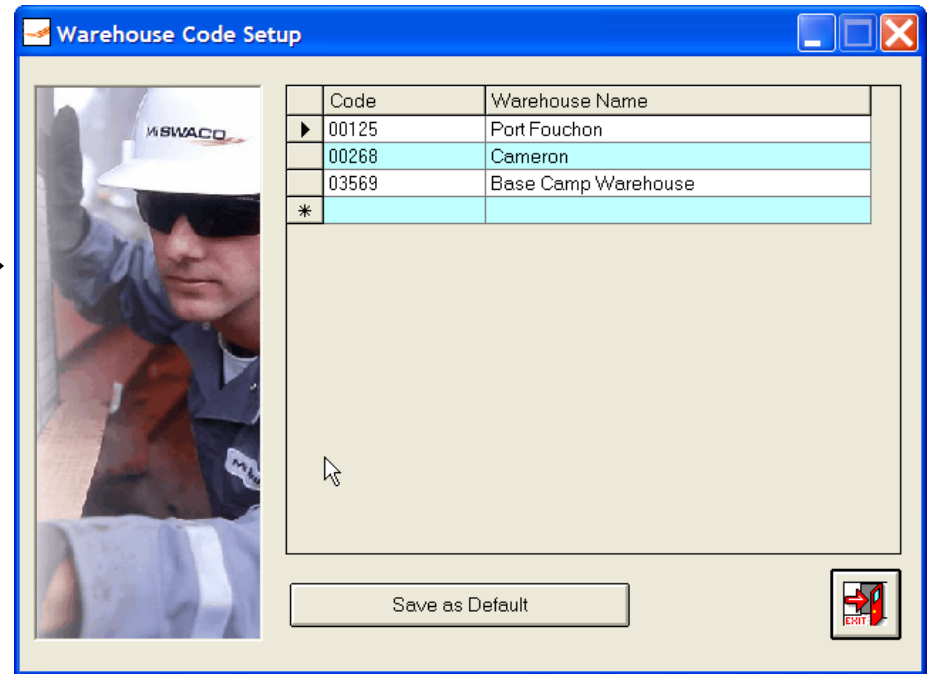
Water-Based Mud: M-I
Oil-Based/Synthetic-Based Mud: M-I

Windows

Decimal Symbol: .
Date Format: mm/dd/yyyy

Mi SWACO

Click on the “**Warehouse Code Setup**” to list the Warehouses used in the project. New Warehouses can be added at any time.



The Warehouse Code Setup window displays a table with columns 'Code' and 'Warehouse Name'. The table contains three entries: 00125 Port Fouchon, 00268 Cameron, and 03569 Base Camp Warehouse. A new entry is indicated by an asterisk in the 'Code' column. The window also features a 'Save as Default' button and the Mi SWACO logo.

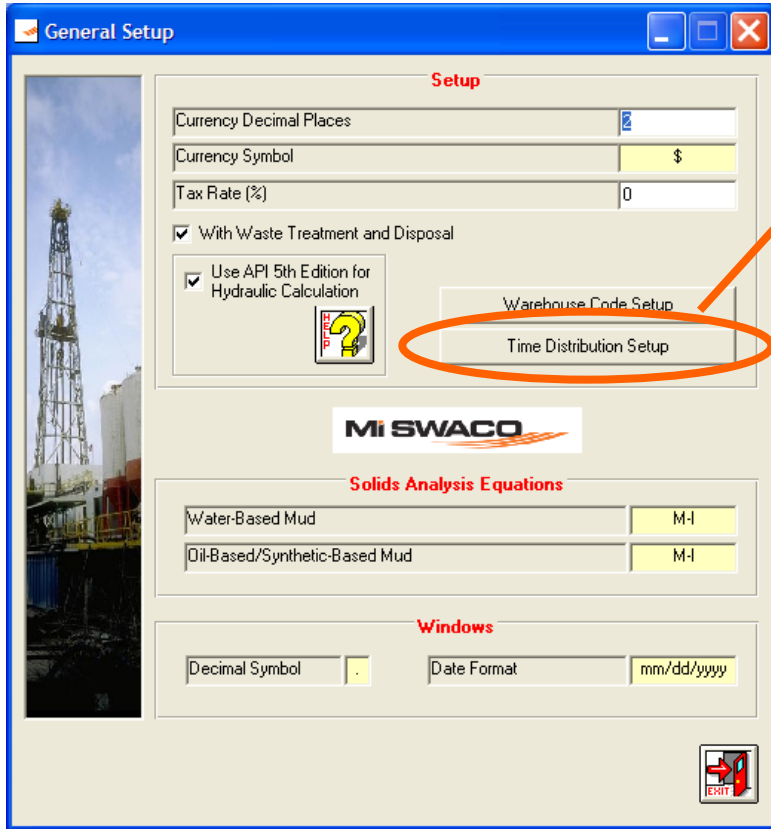
Warehouse Code Setup

Code	Warehouse Name
▶ 00125	Port Fouchon
00268	Cameron
03569	Base Camp Warehouse
*	

Save as Default

Mi SWACO

Setup Process – General Setup (continuation)



The General Setup window contains several sections: Setup, Solids Analysis Equations, and Windows. The Setup section includes fields for Currency Decimal Places, Currency Symbol, Tax Rate (%), and checkboxes for With Waste Treatment and Disposal, and Use API 5th Edition for Hydraulic Calculation. The Solids Analysis Equations section includes fields for Water-Based Mud and Oil-Based/Synthetic-Based Mud. The Windows section includes fields for Decimal Symbol and Date Format. The Mi SWACO logo is displayed in the center. The Time Distribution Setup button is circled in orange, and an orange arrow points from it to the Time Distribution Type Setup window.

General Setup

Setup

Currency Decimal Places: 2

Currency Symbol: \$

Tax Rate (%): 0

☒ With Waste Treatment and Disposal

☒ Use API 5th Edition for Hydraulic Calculation

Warehouse Code Setup

Time Distribution Setup

Mi SWACO

Solids Analysis Equations

Water-Based Mud: M-I

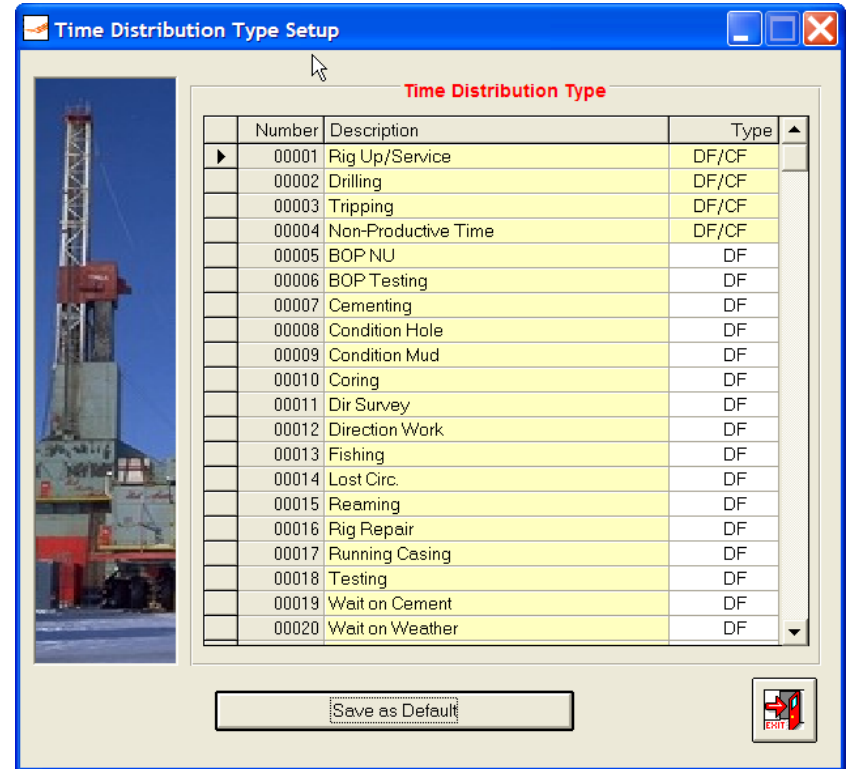
Oil-Based/Synthetic-Based Mud: M-I

Windows

Decimal Symbol: .

Date Format: mm/dd/yyyy

Click on **“Time Distribution Setup”** to see/edit the list of categories that describe the rig activity, during a 24h period.



The Time Distribution Type Setup window displays a table of categories that describe rig activity during a 24h period. The table has columns for Number, Description, and Type. The categories are listed in a table with a scroll bar on the right. The Mi SWACO logo is displayed on the left. The Save as Default button is at the bottom.

Time Distribution Type Setup

Time Distribution Type

Number	Description	Type
00001	Rig Up/Service	DF/CF
00002	Drilling	DF/CF
00003	Tripping	DF/CF
00004	Non-Productive Time	DF/CF
00005	BOP NU	DF
00006	BOP Testing	DF
00007	Cementing	DF
00008	Condition Hole	DF
00009	Condition Mud	DF
00010	Coring	DF
00011	Dir Survey	DF
00012	Direction Work	DF
00013	Fishing	DF
00014	Lost Circ.	DF
00015	Reaming	DF
00016	Rig Repair	DF
00017	Running Casing	DF
00018	Testing	DF
00019	Wait on Cement	DF
00020	Wait on Weather	DF

Save as Default

Setup Process – Loss Setup

Click the “**Loss Setup**” button to add additional ‘fluid loss categories’ or make modifications to the existing labels.

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

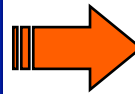
ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

Activate GRAFX

EXIT



Loss Setup

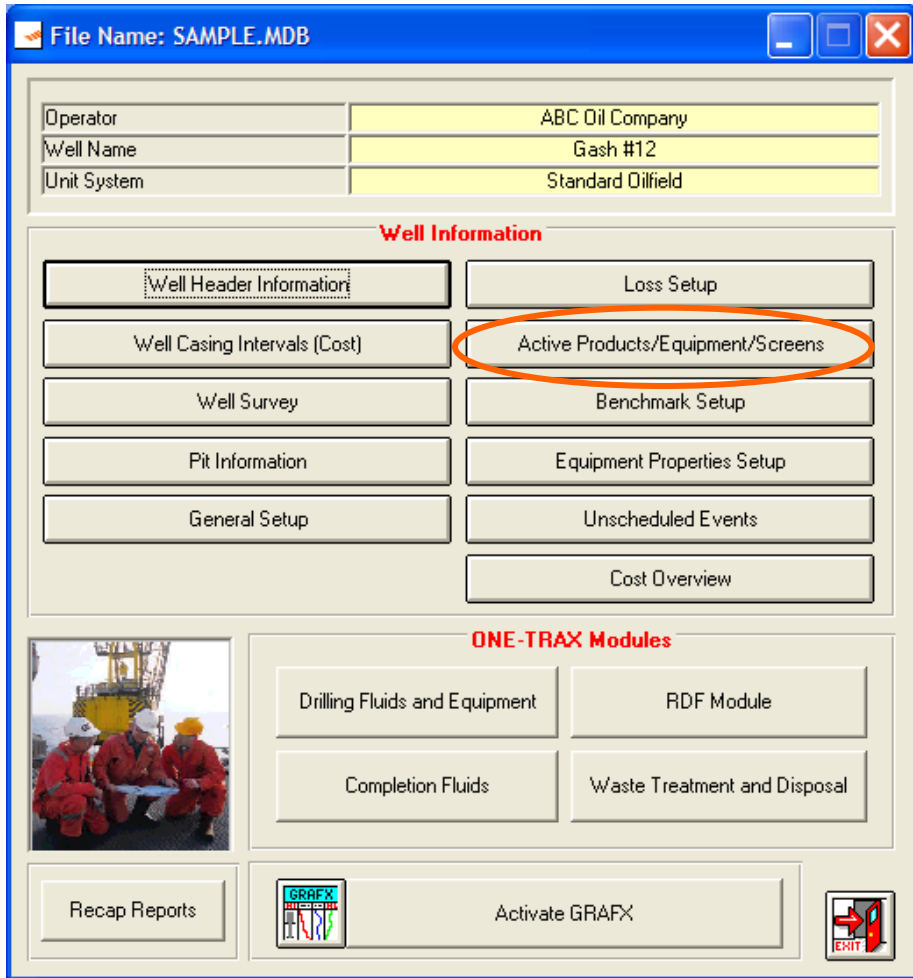
Loss Data Categories			
	Code	Description	Surface/Subsurface
▶	1	Shakers	Surface
	2	Other/Solids	Surface
	3	Centrifuge	Surface
	4	Tripping	Surface
	5	Evaporation	Surface
	6	Returned	Surface
	7	Behind Csg/In Hole	Subsurface
	8	Lost to Formation	Subsurface
	9	Cuttings Boxes	Surface
	10	Sweeps (Sea Floor)	Subsurface
	11	Discharged	Surface
	12	Mud Cleaner	Surface
	13	Surface Lines	Surface
	14	Interface	Surface
	15	Filtration Unit	Surface

Add Delete

Save as Default

EXIT

Setup Process – Active Products/Equipment/Screens



Click the “**Active Products / Equipment / Screens**” button to specify the list of items that will be used in this project, including Engineering charges.

New products, equipment and/or screens can be added at any time.

Here is where 80% of the setup mistakes come from, so extra care must be taken during this process.

Setup Process – Active Products/Equipment/Screens

Master List:

Full list of
M-I SWACO
products

Select your product from
the 'Master List' and use
the "Add Product" button to
add it to your project

Active List:

Products selected
to be used on
this project

Each product to be used on the project must
be listed in here. The product price, cost
category, grouping for pie charts, s.g., unit
size and several other parameters are
specified in this list.

Active Product List

1- Product Setup 2- Equipment Rental Setup 3- Shaker Screen Setup

Master Product List

Product Code	Product Name	Unit Size	Unit	Packing
M0009232	CENTRALIZER 9.9 IN OD (11 3/4 IN)	1		EA
	6.4 LB.	1		EA
		50 LB		BG
		50 LB		BG
	T	1		EA
	N HOLE FOR	1		EA
		1 KG		

Choice

- ☐ Product List1
- ☐ Product List2
- ☐ Product List3
- ☐ Product List4
- ☐ Product List5
- ☒ Master List

Name Code Name Code Product Name

Active Product List

No.	Product Code	Product Name	Abbreviation	Unit Size	Unit
-----	--------------	--------------	--------------	-----------	------

Add Product

Edit Product

Remove Product

Edit Group

Import Active List

Export Active List

Edit Short List

Recalc Cum Cost

Start Quantity

Edit Price/Packing

Full List Waste Treatment and Disposal List Completion Fluids

Setup Process – Active Products/Equipment/Screens

Active Product List

1- Product Setup | 2- Equipment Rental Setup | 3- Shaker Screen Setup

Master Product List

Product Code	Product Name	Unit Size	Unit	Packing
M0009232	CENTRALIZER 9.9 IN OD (11 3/4 IN)	1		EA
M0009300	LTTP GAGE RING, LOWER 20 - 26.4 LB.	1		EA
M0008046	#12 SAND	50 LB		BG
M0004956	#8 SAND	50 LB		BG
M0009461	1 IN JETTING HOUSING FOR MAJT	1		EA
M0009460	1 IN NPT JETTING PORT W/ 302 IN HOLE FOR	1		EA
W0009144	1,9-DECADIENE	1	KG	

Choice

☐ Product List1
☐ Product List2
☐ Product List3
☐ Product List4
☐ Product List5
☒ Master List

Sort By: ☐ Name ☐ Code Search By: ☐ Name ☐ Code Product Name

Enter Product Code

No.	Product Code	Product Name	Abbreviation	Unit Size	Unit
1	M0000003	M-1 BAR BULK	Bar-blk	1	TN
2	M0000321	M-1 GEL BULK	Gel-blk	1	TN
3	M0001993	DF-1	DF-1 Oil	1	BL
4	M0003212	CALCIUM CHLORIDE (94-97%) PELLETS	CaCl2	100	LB
5	M0004005	CALCIUM CHLORIDE BRINE (11.6 ppg)	CaClBrn	1	BL
6	M0004024	CAUSTIC SODA	Caustic	50	LB
7	M0002978	DUO-VIS	Duo-vis	50	LB
8	M0005469	GUAR GUM	GuarGum	50	LB
9	M0004111	LIME	Lime	50	LB
10	M0002786	VERSAMUL	V-Mul	200	LT

Save
Cancel
Clear
Edit Group
Import Active List
Export Active List
Edit Price/Packing

Full List | Waste Treatment and Disposal List | Completion Fluids

Enter the required information for each product. Scroll to your right to displayed all the fields that need to be filled in.

Be sure to correctly select the 'Daily Cost Code' for each product; this code is used on cost allocation and to make the product available in certain transactions (i.e. Base Oil, Whole Mud, etc.)

No.	Product Name	Unit Size	Unit	Pack	Price	S.G.	Calc Conc?	Unit Size	Unit	Pack	Price	S.G.	Calc Conc?	M-1 Prod?	Prod Group	Daily Cost Code	Calc W/Mgt Conc?	W/Mgt Cost Category	CF Cost Category
1	M-1 BAR BULK	1	TN	8K	105	4.2	Y	1	TN	8K	105	4.2	Y	Y	1	2	Y	1	1
2	M-1 GEL BULK	1	TN	8K	154	2.6	Y	1	TN	8K	154	2.6	Y	Y	2	2	Y	1	1
3	DF-1	1	BL	8BL	85	0.75	N	1	BL	8BL	85	0.75	N	Y	11	7		1	10
4	CALCIUM CHLORIDE (94-97%) PELLETS	100	LB	8G	25.5	2.2	Y	100	LB	8G	25.5	2.2	Y	Y	4	2	Y	2	4
5	CALCIUM CHLORIDE BRINE (11.6 ppg)	1	BL	8K	22.6	1.39	N	1	BL	8K	22.6	1.39	N	Y	12	4	Y	2	2
6	CAUSTIC SODA	50	LB	8G	25.5	2.13	Y	50	LB	8G	25.5	2.13	Y	Y	4	2	Y	2	4
7	DUO-VIS	50	LB	8G	320	1.5	Y	50	LB	8G	320	1.5	Y	Y	2	2	Y	2	4
8	GUAR GUM	50	LB	8G	75	1.5	Y	50	LB	8G	75	1.5	Y	Y	2	2	Y	2	4
9	LIME	50	LB	8G	6.5	2.34	Y	50	LB	8G	6.5	2.34	Y	Y	4	2	Y	2	4
10	VERSAMUL	200	LT	DM	485	0.9	Y	200	LT	DM	485	0.9	Y	Y	9	2	Y	2	4
11	VERSACLEAN MUD (11.5 ppg)	1	BL	8K	82.5	1.38	N	1	BL	8K	82.5	1.38	N	Y	13	4	Y	2	2
12	VERSACOAT	200	LT	DM	510	0.9	Y	200	LT	DM	510	0.9	Y	Y	10	2	Y	2	4

Setup Process – Active Products/Equipment/Screens

Enter Short Product Label Used for Reports						
Product Code	Product Name	Abbreviation	Unit Size	Unit	Packing	Price
M0001828	ALCOMER 72		25	KG	BG	

When adding an item to the active product list notice the fields:

- Product Code
- Product Name
- Unit Size
- Unit
- Packaging

These fields are locked to assure proper information for each product code and cannot be changed when locked in Yellow. These are called “**Approved Items**”

Active Product List

1- Product Setup | 2- Equipment Rental Setup | 3- Shaker Screen Setup

Master Product List

Product Code	Product Name	Unit Size	Unit	Packing
M0002993	ALCOMER 60	55	LB	BG
M0007966	ALCOMER 60	25	KG	BG
M0001783	ALCOMER 60 RD	25	KG	BG
M0007720	ALCOMER 630	25	KG	BG
M0001828	ALCOMER 72	25	KG	BG
M0005890	ALCOMER 72	50	LB	BG
M0000775	ALCOMER 72L	25	LT	CN

Sort By: ☒ Name ☐ Code Search By: ☒ Name ☐ Code Product Name:

Enter Short Product Label Used for Reports

No.	Product Code	Product Name	Abbreviation	Unit Size	Unit	Packing	Price
	M0001828	ALCOMER 72		25	KG	BG	
2	M0005108	CALCIUM CHLORIDE (94%)	CaCl2	50	LB		
3	M0001210	Calcium Chloride (11.6 ppg)	CaBrine	1	BL		
4	M0006659	CAUSTIC SODA	Caustic	50	LB		
5	M0001715	DF-1 Base Oil	DF-1	1	BL		
6	M0007915	DUO-VIS	Duovis	20	KG		
7	M0006520	GUAR GUM	Guar	50	LB		
8	M0004111	LIME	Lime	50	LB		
9	M0006011	M-I BAR BULK	Barite	1	TN		
10	M0000321	M-I GEL BULK	Gel Bk	1	TN		
11	M0002103	VERSAMUL	VMul	200	LT		

Save Cancel Clear Edit Group Import Active List Export Active List Edit Price/Packing

Full List Waste Treatment and Disposal List Completion Fluids

Setup Process – Active Products/Equipment/Screens

Enter Short Product Label Used for Reports						
Product Code	Product Name	Abbreviation	Unit Size	Unit	Packing	Price
R0088720	ENGINEERING SERVICE	ENG	1		EA	425

If for some reason you need to add non-approved items 1.5 version still allows this as the example above

If you edit the item in which the product number, description, unit size, unit and packaging line up as an approved item the product will automatically change to an approved item. As the example below shows the product code was changed to the proper code and saved in which the item changed to an approved item.

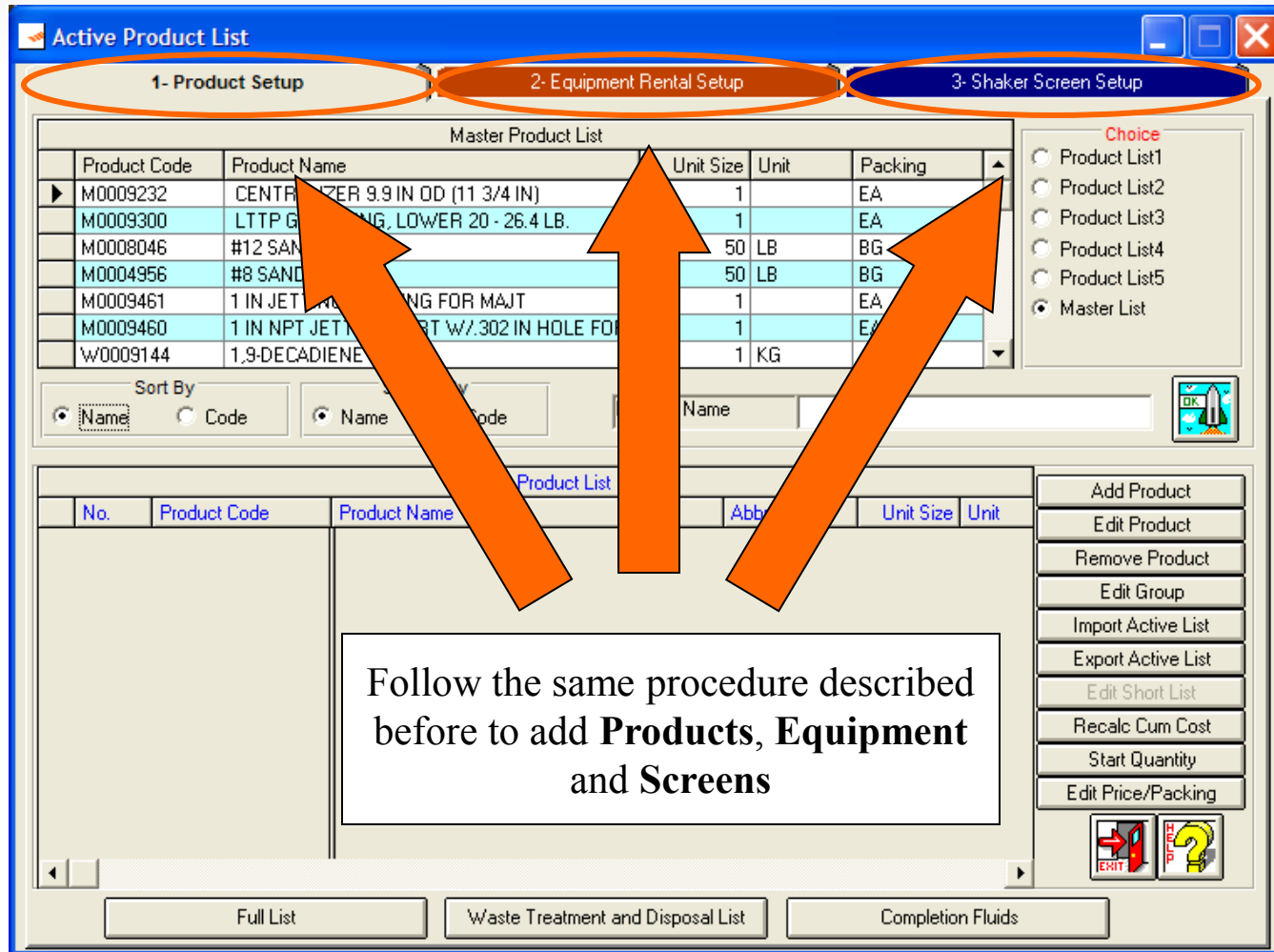
Enter Product Code						
Product Code	Product Name	Abbreviation	Unit Size	Unit	Packing	Price
R0088110	ENGINEERING SERVICE	ENG	1		EA	425

Setup Process – Active Products/Equipment/Screens

Active Product List						
	No.	Product Code	Product Name	Abbreviation	Unit Size	Unit
▶	24	P0012257	2-Micron ABS Filter Cartridge	Cartrdg	1	
	39	M0001828	ALCOMER 72	Alc 1	25	KG
	40	M0001828	ALCOMER 72	Alc 2	25	KG
	3	M0001210	Calcium Chloride (11.6 ppg)	Calcline	1	BL
	2	M0005108	CALCIUM CHLORIDE (94%)	CaCl2	50	LB
	33	M0001210a	Calcium Chloride Spike	CaCl_br	1	BL
	25	P00122562	Cartridge Filter Unit	Filters	1	
	4	M0006659	CAUSTIC SODA	Caustic	50	LB
	35	R00881103	Completion Fluids Engineer	CF-Eng	1	
	5	M0001715	DF-1 Base Oil	DF-1	1	BL
	31	M0007844	DIATOMACEOUS EARTH	D.E.	50	LB
	38	M0007915	DUO-VIS	hjkh	20	KG
	3	M0007915	DUO-VIS	hjkh	20	KG

In the 1.5 Version of ONE-TRAX you will be able to add multiple items with the same product code as the example shows above:

Setup Process – Active Products/Equipment/Screens



Setup Process – Active Products/Equipment/Screens

“Equip. Detail” was added to the tab #2 of the “Active Products/Equipment/Screens”. This new field is used to keep track of the correct M-I SWACO name of the equipment, while the “Description” name can still be modified to meet specific reporting needs.

Active Product List

1- Product Setup 2- Equipment Rental Setup 3- Shaker Screen Setup

Master Equipment List

Equipment Code	Description
NC-014	518 FSV Centrifuge

Choice

☐ Equip List1
☐ Equip List2
☐ Equip List3
☐ Equip List4
☐ Equip List5
☒ Master List

Sort By: ☒ Name ☐ Code Search By: ☒ Name ☐ Code Equipment Name: fsv

Product Editor (Input Data)

Serial No.	Equipment Code	Description	Equip. Detail
50138	NC-014	518 Centrifuge (Verti-G dedicated)	518 FSV Centrifuge
T2005	NC-016	Mongoose Shaker #1	Mongoose Shaker
12025	NC-016	Mongoose Shaker #2	Mongoose Shaker
12173	NC-016	Mongoose Shaker #3	Mongoose Shaker
21056	62-8442	Mud Cleaner 8T4	Mud Cleaner 8T4 w/o pump
50258	NC-014	518 Centrifuge - Drilling Fluid	518 FSV Centrifuge
50138	NC-014	518 Centrifuge (Verti-G dedicated)	518 FSV Centrifuge
V120	NC-018	Vaccum Unit	75 HP Vaccum Unit
A12345	NA002	Auger	Auger
C00000	NC-021	Rig Vaccum Tank	Rig Vaccum Tank (RVT)
2222	NC-010	Verti-G Dryer	Verti-G Dryer

Save
Cancel
Clear
Copy Equip
Edit Short List
Import Active List
Export Active List

Full List Waste Treatment and Disposal List

Keeping unchanged the entry in the 'Equip. Detail' field will improve the data consistency in **ONE-TRAX Central**, when querying for specific equipment models.

Setup Process – Active Products/Equipment/Screens

Active Product List

1- Product Setup **2- Equipment Rental Setup** 3- Shaker Screen Setup

Master Equipment List

Equipment Code	Description
R0089051	10-3/4" 'PUP' Brush Tool
R0089050	10-3/4" 'PUP' Scraper
R0089052	10-3/4" 'PUP' Scraper/Brush
NC-009	1850 Centrifuge
62-8409	202 Microclone w/o pump
R0089032	21" Riser Brush Rental
NC-007	3N1 212 / 8T4 w/o pump
NC-006	3N1 212 /6T4 w/o pump

Sort By: ☒ Name ☐ Code Search By: ☒ Name ☐ Code Equipment Name:

Choice:

- ☐ Equip List1
- ☐ Equip List2
- ☐ Equip List3
- ☐ Equip List4
- ☐ Equip List5
- ☒ Master List

Active Equipment List

Serial No.	Equipment Code	Description	Rental Price	Stand-By Price
12005	NC-001	ALS Shaker #1	0	0
12025	NC-001	ALS Shaker #2	0	0
12173	NC-001	ALS Shaker #3	0	0
21056	62-8442	Mud Cleaner 8T4	0	0
50258	62-8411	Swaco 518 Centrifuge #1	350	230
50138	62-8411	Swaco 518 Centrifuge #2	350	230
V120	62-9070	Vacuum Unit	600	250
A12345	62-3745	1 ft Auger	50	50
C00000	62-3754	Vacumn Container	20	20
F10234	62-0785	Cartridge Unit	140	70
56312	C0007	Casing Scrapper	260	30
21553	C0017	Cartridge Filter Unit	160	80
25866	NC-002	BEM 600 Shaker #4	0	0

Buttons: Add Equipment, Edit Current Equip, Remove Equip, Copy Equip, Edit Short List, Import Active List, Export Active List, Recalc Cum Cost

Buttons: Full List, Waste Treatment and Disposal List

It is important to add all equipment that will be tracked, even if it is not M-I SWACO rental equipment. Example - to report information regarding rig shakers on the mud report, they must be set up in the active list.

Setup Process – Active Products/Equipment/Screens

Active Product List

1- Product Setup | 2- Equipment Rental Setup | **3- Shaker Screen Setup**

Master Screen List

Code	Description
JBM6MK010H	SCREEN, BEM600 10# SS frmd pt
JBM6MK020H	SCREEN, BEM600 20# SS frmd pt
JBM6MK040H	SCREEN, BEM600 40# SS frmd pt
JBM6MK060H	SCREEN, BEM600 60# SS frmd pt
JBM6XR105P	SCREEN, BEM600 XR105 SS frmd pt
JBM6XR120P	SCREEN, BEM600 XR120 SS frmd pt
JBM6XR145P	SCREEN, BEM600 XR145 SS frmd pt
JBM6XR165P	SCREEN, BEM600 XR165 SS frmd pt

Sort By: ☒ Name ☐ Code Search By: ☒ Name ☐ Code Screen Name:

Active Screen List

Code	Description	Mesh Size	Price	Discount %
XSJUE084L	84 Mesh Magnum XR	84	325	0
XSJUE110L	110 Mesh Magnum XR	100	325	0
XSJUE140L	140 Mesh Magnum XR	140	325	0
XSJUE175L	175 Mesh Magnum XR	175	325	0
XSJUE210L	210 Mesh Magnum XR	210	325	0
XSJUE250L	250 Mesh Magnum XR	250	325	0

Choice

☐ Screen List1
☐ Screen List2
☐ Screen List3
☐ Screen List4
☐ Screen List5
☒ Master List

Add Screen
Edit Screen
Remove Screen
Edit Short List
Import Active List
Export Active List
Recalc Cum Cost
Start Quantity

EXIT ?

It is important to add all screens that will be tracked even if they are not M-I SWACO screens.

Setup Process – Benchmarks

- The **Benchmark Setup** section allows you to track key parameters against a desired value (target).
- This is a two step process:
 - Selecting the parameter of interest.
 - Indicating its target value (or acceptable range).

The screenshot displays the SWACO ONE-TRAX software interface. At the top, a blue title bar shows 'File Name: SAMPLE.MDB'. Below this, a table contains the following information:

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Below the table, the 'Well Information' section is visible. It contains a grid of buttons for various setup options. The 'Benchmark Setup' button is highlighted with a red oval.

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

Below the 'Well Information' section, there is a 'ONE-TRAX Modules' section. It contains a grid of buttons for different modules:

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

At the bottom of the interface, there is a 'Recap Reports' button, a 'GRAFX' logo, an 'Activate GRAFX' button, and an 'EXIT' button.

Setup Process – Benchmarks (step 1)

File Name: SAMPLE.MDB

Operator: ABC Oil Company
Well Name: Gash #12
Unit System: Standard Oilfield

Well Information

Well Header Information | Loss Setup
Well Casing Intervals (Cost) | Active Products/Equipment/Screens
Well Survey | **Benchmark Setup**
Pit Information | Equipment Properties Setup
General Setup | Unscheduled Events
Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment | RDF Module
Completion Fluids | Waste Treatment and Disposal

Recap Reports | Activate GRAFX

Select all the parameters to benchmark from the list and then click “Add”.

1

Benchmarks

Benchmark Definitions

Mud Properties

Depth ft
TVD ft
Mud Weight lb/gal
MW Temp °F
Funnel Viscosity s/qt
R600
R300
R200
R100
R6
R3
PV cP
YP lb/100ft²
10s Gel lb/100ft²
10m Gel lb/100ft²
30m Gel lb/100ft²
API Fluid Loss cc/30 min
HTHP Fluid Loss cc/30 min
HTHP FL Temp °F
Cake API 1/32"
Cake HTHP 1/32"
Solids %Vol
Oil %Vol
Water %Vol

Water-Base

Add >> << Remove

New Benchmark Target Data Type: Numeric
Reset benchmarks to: Drilling_Fluid

Description	Unit	ONE-TRAX Data	Fluid Type	Is
Mud Weight(WBM)	lb/gal	Prop5	WBM	N
PV(WBM)	cP	Prop15	WBM	N
YP(WBM)	lb/100ft²	Prop16	WBM	N
API Fluid Loss(WBM)	cc/30 min	Prop20	WBM	N
Mud Weight(OBM)	lb/gal	Prop5	OBM	N
PV(OBM)	cP	Prop15	OBM	N
YP(OBM)	lb/100ft²	Prop16	OBM	N
HTHP Fluid Loss(OBM)	cc/30 min	Prop21	OBM	N
Mud Interval Cost/Unit Vol	\$/bbl	Calculated Value		Y
Mud Engineer Cost	\$	CumMudEngrCost		Y

Use the drop-down menus to display more parameters.

Setup Process – Benchmarks (step 2)

Go to the “Target Entry” tab and enter the value (or range) targeted for each parameter, on each interval section.

File Name: SAMPLE.MDB

Operator: ABC Oil Company
Well Name: Gash #12
Unit System: Standard Oilfield

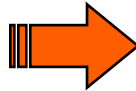
Well Information

Well Header Information | Loss Setup
Well Casing Intervals (Cost) | Active Products/Equipment/Screens
Well Survey | **Benchmark Setup**
Pit Information | Equipment Properties Setup
General Setup | Unscheduled Events
Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment | RDF Module
Completion Fluids | Waste Treatment and Disposal

Recap Reports | Activate GRAFX



Benchmarks

Benchmark Definitions | **Target Entry**

Description	Whole Well	Interval1	Interval2	Interval3	Interval4	Interval5
Top	1					
Bottom	2					
Mud Weight(WBM) - min (lb/gal)		8.9	9.8			
Mud Weight(WBM) - max (lb/gal)		9.1	10.0			
PV(WBM) - min (cP)		20	15			
PV(WBM) - max (cP)		25	20			
YP(WBM) - min (lb/100ft²)		15	20			
YP(WBM) - max (lb/100ft²)		20	25			
API Fluid Loss(WBM) - min (cc/30 min)			4.0			
API Fluid Loss(WBM) - max (cc/30 min)			6.0			
Mud Weight(OBM) - min (lb/gal)				11.5		
Mud Weight(OBM) - max (lb/gal)				12.0		
PV(OBM) - min (cP)				25		
PV(OBM) - max (cP)				30		
YP(OBM) - min (lb/100ft²)				20		
YP(OBM) - max (lb/100ft²)				25		
HTHP FL Temp(OBM) - min (°F)				3.0		
HTHP FL Temp(OBM) - max (°F)				5.0		

Data Type: ☐ Numeric ☐ Min/Max ☐ Text ☒ All

Setup Process – Equipment Properties Setup

Click the “Equipment Properties Setup” button to specify equipment performance parameters that will be tracked during your project.

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

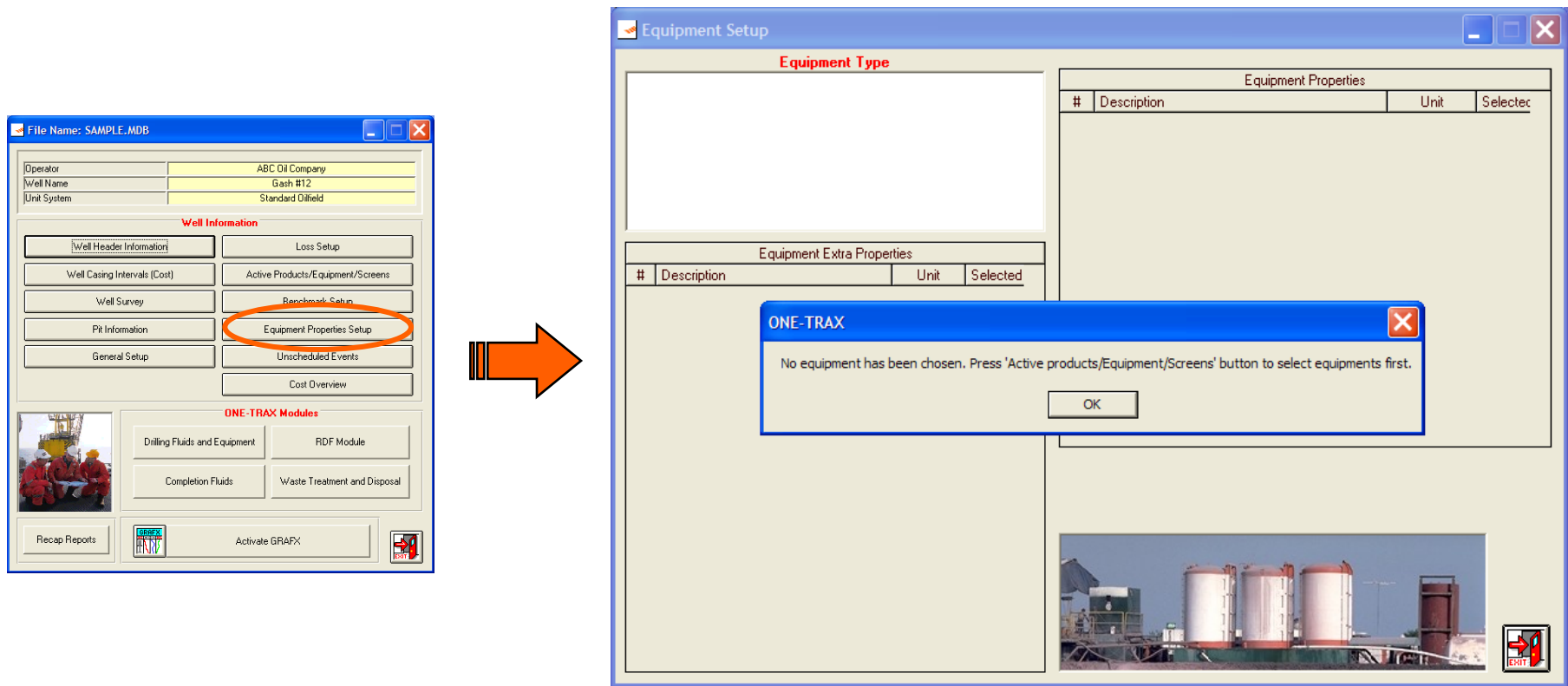
Recap Reports

Activate GRAFX

EXIT

Setup Process – Equipment Properties Setup

You will get this message only if you press the “Equipment Properties Setup” button before adding your equipment to your “Active Equipment List”; so you need to go to the “Active Products / Equipment / Screens” section **before** clicking the “Equipment Properties Setup” button.



Setup Process – Equipment Properties Setup

List of the different equipment types defined in your Active Equipment List

The screenshot shows the 'Equipment Setup' window. On the left, under 'Equipment Type', a list includes: 1 - Centrifuge, 4 - Mud Cleaner, 5 - Shale Shaker, 6 - Others, 11 - Vacuum System, and 13 - Cuttings Container. Below this is the 'Equipment Extra Properties' table with columns for #, Description, Unit, and Selected. A red box highlights rows 7 through 13 with the text 'Type in here any additional property required...'. To the right is the 'Equipment Properties' table with columns for #, Description, Unit, and Selected. It lists 17 properties, most of which are pre-selected. At the bottom is the 'Centrifuge Flexible Units Selection' section with dropdowns for 'Flow Rate' (gal/min) and 'Mass' (Ton), and a 'Change' button. An 'EXIT' button with a red arrow is in the bottom right corner.

#	Description	Unit	Selected
1	Flow Rate	gal/min	☒
2	WT In	lb/gal	☒
3	WT Out	lb/gal	☒
4	Discard WT	lb/gal	☒
5	Flow Rate Out	gal/min	☒
6	Discard Flow Rate	gal/min	☒
7	Discard Mass Rate	Ton/Hour	☒
8	Mass Discharged	Ton	☒
9	Cum Mass Discharged	Ton	☒
10	Bowl Speed	rpm	☒
11	G-Force		☒
12	Main Drive Torque	amp	☒
13	Differential Speed		☒
14	Back Drive Torque	amp	☒
15	Feed Pump Torque	amp	☐
16	Vibration		☐
17	Weir Setting		☐

#	Description	Unit	Selected
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐
9			☐
10			☐
11			☐
12			☐
13			☐
14			☐
15			☐
16			☐
17			☐
18			☐
19			☐
20			☐

Centrifuge Flexible Units Selection

Flow Rate: gal/min
Mass: Ton
Change

EXIT

Use the check boxes to select the pre-defined properties you want to track

Extra units setup only for the Centrifuge

Setup Process – Equipment Properties Setup

Do the same for each
“Equipment Type”

The screenshot shows the 'Equipment Setup' window. It has a blue title bar and a menu bar. The main area is divided into three sections. Section 1 is the 'Equipment Type' list on the left, containing: 1 - Centrifuge, 4 - Mud Cleaner, 5 - Shale Shaker, 6 - Others, 11 - Vacuum System, and 13 - Cuttings Container. Section 2 is the 'Equipment Properties' table in the top right, which lists pre-defined properties with their units and selection checkboxes. Section 3 is the 'Equipment Extra Properties' table at the bottom left, which allows for custom properties. A red-bordered box highlights the first few rows of this table with the text 'Type in here any additional property required...'. At the bottom right, there is a small image of industrial equipment and an 'EXIT' button. Orange arrows indicate the flow from the equipment type selection to the properties setup and then to the extra properties section.

Equipment Type

- 1 - Centrifuge
- 4 - Mud Cleaner
- 5 - Shale Shaker
- 6 - Others
- 11 - Vacuum System
- 13 - Cuttings Container

Equipment Properties

#	Description	Unit	Selected
1	Pressure	psi	☒
2	Deck Angle	°	☒
3	G-Force	G's	☒
4	Vibration Angle	°	☒
5	Type of Motion		☒
6	Flow over shaker	%	☒
7	Size of Cones	in	☒
8	Cones Used		☒
9	Input Head	ft	☒

Equipment Extra Properties

#	Description	Unit	Selected
1			☐
2			☐
3			☐
4			☐
5			☐
6			☐
7			☐
8			☐
9			☐
10			☐
11			☐
12			☐
13			☐
14			☐
15			☐
16			☐
17			☐
18			☐
19			☐
20			☐

Type in here any additional property required...

EXIT

Use the check
boxes to select the
pre-defined
properties you
want to track

Setup Process

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

GRAFX
Activate GRAFX

EXIT

Now the setup process
if finished.

ONE-TRAX[®] Wellsite

Daily Input Procedures

Daily Data Input Modules

The screenshot displays the ONE-TRAX software interface. At the top, a blue title bar shows 'File Name: SAMPLE.MDB'. Below this, a 'Well Information' section contains a table with the following data:

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Below the table, a grid of buttons is organized under the heading 'Well Information'. The buttons include: 'Well Header Information', 'Loss Setup', 'Well Casing Intervals (Cost)', 'Active Products/Equipment/Screens', 'Well Survey', 'Benchmark Setup', 'Pit Information', 'Equipment Properties Setup', 'General Setup', 'Unscheduled Events', and 'Cost Overview'. To the left of this grid is a small image of three workers in red safety gear. Below the grid, a section titled 'ONE-TRAX Modules' is highlighted with an orange border and contains four buttons: 'Drilling Fluids and Equipment', 'RDF Module', 'Completion Fluids', and 'Waste Treatment and Disposal'. At the bottom left is a 'Recap Reports' button, and at the bottom right is an 'EXIT' button with a red arrow icon. A 'GRAFX' logo and an 'Activate GRAFX' button are also visible.

ONE-TRAX Modules

- Drilling Fluids and Equipment
 - Complete Drilling Fluids Reporting.
 - Solids Control Equipment Usage and Cost Tracking.
 - Screen Usage / Inventory Reporting.
- Waste Treatment and Disposal
 - Waste Treatment Operations Reporting: Dewatering, Water Treatment, Thermal Desorption & Fixation.
 - Waste Disposal Operations Reporting: Cutting Injection, Haul-off & Discharge.
- Reservoir Drill-in Fluid
 - Full RDF Reporting.
- Completion Fluids
 - Similar Capabilities to DF.
 - Displacement Reporting.

Daily Data Input – Drilling Fluids and Equipment Module

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

GRAFX
INTERSECTION

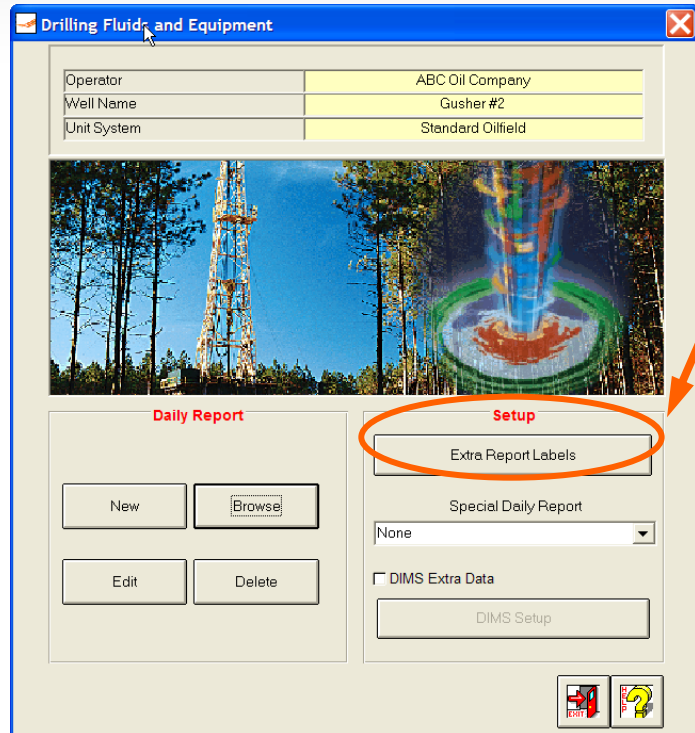
Activate GRAFX

EXIT

Drilling Fluids and Equipment module

Daily Data Input – Drilling Fluids and Equipment Module

Drilling Fluid Extra Properties



Drilling Fluids and Equipment

Operator: ABC Oil Company
Well Name: Gusher #2
Unit System: Standard Oilfield

Image: A photograph of an oil rig in a forest.

Buttons: New, Browse, Edit, Delete

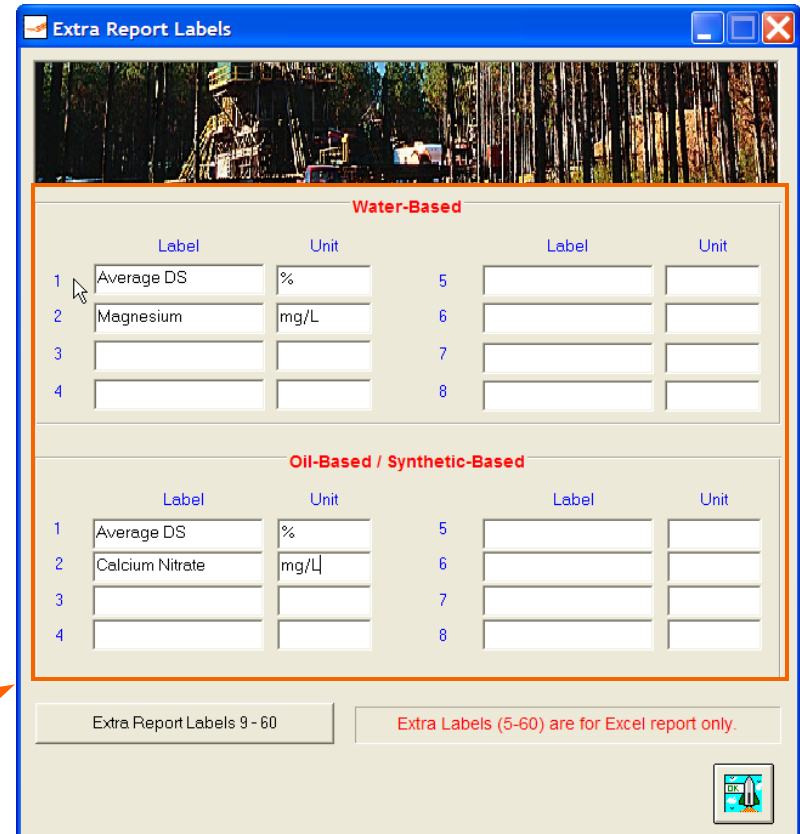
Special Daily Report: None

☐ DIMS Extra Data

DIMS Setup

Extra Report Labels (circled in red)

If special fluid tests are required in the daily report, click on the “**Extra Report Labels**” button to add them to the ‘mud check’ parameters’ list.



Extra Report Labels

Image: A photograph of an oil rig in a forest.

Water-Based

	Label	Unit		Label	Unit
1	Average DS	%	5		
2	Magnesium	mg/L	6		
3			7		
4			8		

Oil-Based / Synthetic-Based

	Label	Unit		Label	Unit
1	Average DS	%	5		
2	Calcium Nitrate	mg/L	6		
3			7		
4			8		

Extra Report Labels 9 - 60

Extra Labels (5-60) are for Excel report only.

Up to eight additional labels may be added for each of Water-Based or Oil / Synthetic-Based reports.

Daily Data Input – Drilling Fluids and Equipment Module

Drilling Fluid Extra Properties

Extra Report Labels

Water-Based

	Label	Unit		Label	Unit
1	Average DS	%	5		
2	Magnesium	mg/L	6		
3			7		
4			8		

Oil-Based / Synthetic-Based

	Label	Unit		Label	Unit
1	Average DS	%	5		
2	Calcium Nitrate	mg/L	6		
3			7		
4			8		

Extra Report Labels 9 - 60

Extra Labels (5-60) are for Excel report only.

If you need more than eight extra properties, click the button highlighted here and up to 60 more properties can be entered.

Extra Properties Setup

#	Water-Based Mud		Oil-Based/Synthetic-Based Mud	
	Description	Unit	Description	Unit
9	Specification number		Specification number	
10	Temperature In		Temperature In	
11	H2S		H2S	
12	Excess gypsum		Excess gypsum	
13	Excess lime		WPS as chlorides	
14	Volume glycol		Volume glycol	
15	Volume lubricant		Volume lubricant	
16	Silicate		Silicate	
17	30rpm		30rpm	
18	60rpm		60rpm	
19	PPT fluid loss		PPT fluid loss	
20	PPT cake thickness		PPT cake thickness	
21	PPT test temperature		PPT test temperature	
22	PPT test pressure		PPT test pressure	
23	PPT disc permeability		PPT disc permeability	
24	Static sag test duration		Static sag test duration	
25	Static sag test temperature		Static sag test temperature	
26	Static sag test delta SG		Static sag test delta SG	
27	Dynamic sag test duration		Dynamic sag test duration	
28	Dynamic sag test temperature		Dynamic sag test temperature	

Predefined Setting: Statoil

Import Custom Setting Export Custom Setting

You can choose a pre-defined set of properties for Statoil, or a custom settings may be used.

Daily Data Input – Drilling Fluids and Equipment Module

Open New Daily Report, Edit, Browse or Delete.

Use the following buttons to either :

- Create a **NEW** day
- **BROWSE** through an existing day
- **EDIT** a previous day, or
- **DELETE** an entire day

Drilling Fluids and Equipment

Operator	ABC Oil Company
Well Name	Gusher #2
Unit System	Standard Oilfield

Image: Drilling rig and wellbore visualization

Daily Report

New Browse Edit Delete

Setup

Extra Report Labels

Special Daily Report

None

☐ DIMS Extra Data

DIMS Setup

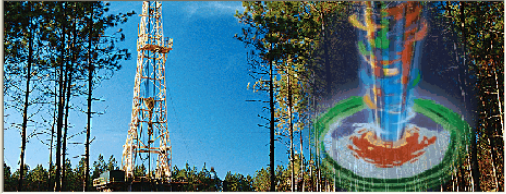
EXIT HELP

Daily Data Input – Drilling Fluids and Equipment Module

Browse through an existing day (also used for Editing)

Drilling Fluids and Equipment

Operator	ABC Oil Company
Well Name	Gusher #2
Unit System	Standard Oilfield



Daily Report

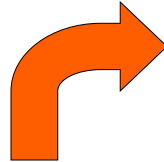
New **Browse** Edit Delete

Setup

Extra Report Labels

None

☐ DIMS



Drilling Fluids and Equipment

Operator	
Well Name	
Unit System	Standard Oilfield



Browse Daily Checks

Date	Depth	Activity	Fluid Type	Rpt Fo
8/14/2005				

Mud Checks Full List

Double click on the date of interest to open it

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc

1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Date	8/14/2005	Depth (ft)		Fluid Type	
Spud Date		TVD (ft)		Lithology	
Report		Bit Depth (ft)			
Present Activity					

Lithology Setup

Operator Representative

Contractor Representative

M-H SWACO Representative 1

M-H SWACO Representative 2

Telephones

Pagers/FAX Numbers

Rig Telephone Number

Warehouse Telephone



	DF Chem /Personnel	IFE/SC Engineer	Drilling Total	DF Equip. /Screens	Other Cost (DWM/CF)	Total
Daily Cost	0.00	0.00	0.00	0.00	0.00	0.00
Cumulative Cost	0.00	0.00	0.00	0.00	0.00	0.00

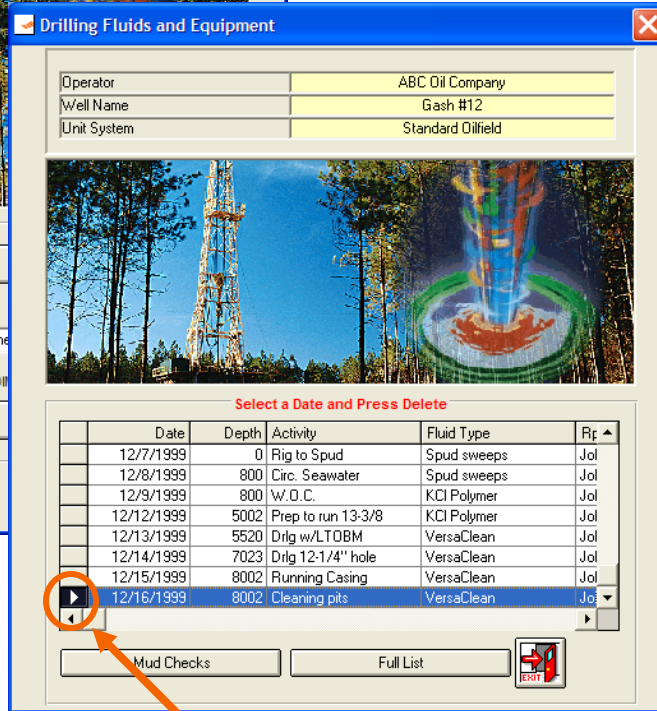
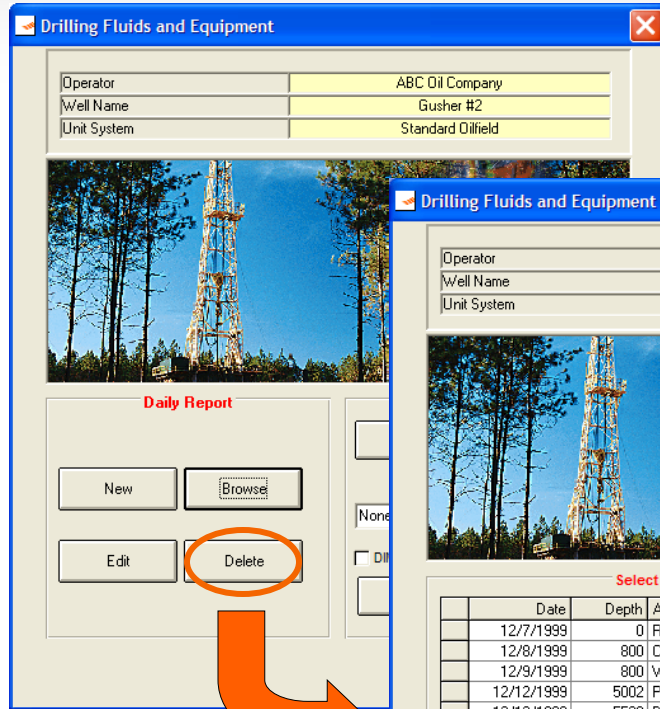
8/14/2005

EXIT ? F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

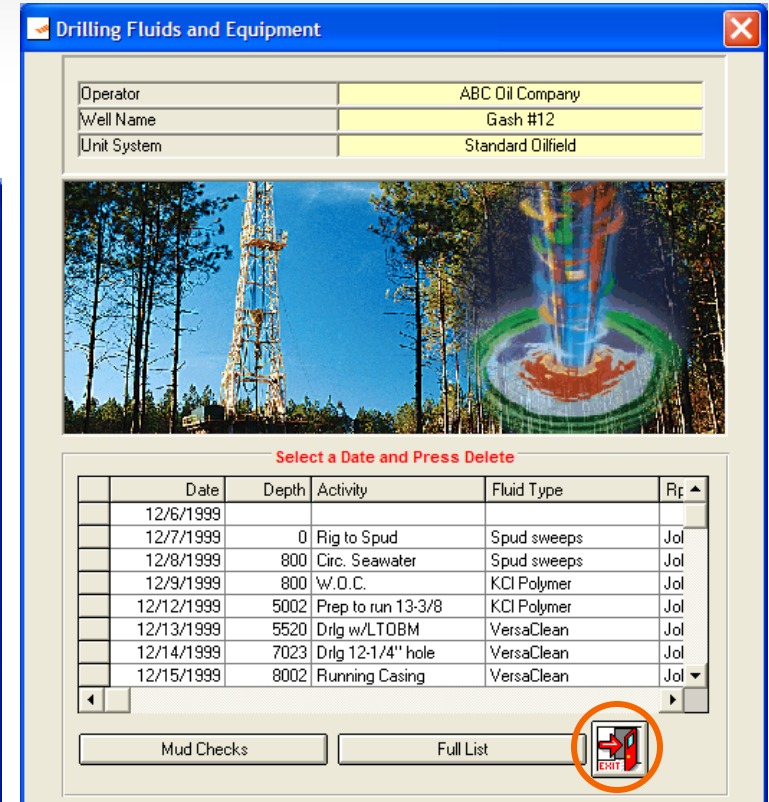
The "browsing bar" is only available if you use the "Browse" option to open the day.

Daily Data Input – Drilling Fluids and Equipment Module

Delete an entire day



Click at the very left of the row to highlight it and use the "Delete" key in your keyboard to delete the entire day of data.



Then just click "Exit"

Daily Data Input – Drilling Fluids and Equipment Module

Open New Daily Report

Drilling Fluids and Equipment

Operator	ABC Oil Company
Well Name	Gusher #2
Unit System	Standard Oilfield

ONE-TRAX

Copy Data from Previous Date?

Yes **No**

New Daily Check

Date for New Check (mm/dd/yyyy) 12/21/2004

Mud Check Type

☒ Water Base
☐ Water Base (CaCl₂)
☐ Oil Base
☐ Synthetic Base

When you click “**New**” the default date will be the next consecutive one, but this may be changed, if required. The **Mud Check Type** defaults to the last used, and must be changed if the fluid system changes. Click “OK”.

Click “Yes” to bring forward standard information on the well (Header, Geometry, etc.).

Daily Data Input – Drilling Fluids and Equipment Module

Daily Data Entry page - General Description

The daily data entry screen is divided into eight tabs to organize the information relevant to the project. Each tab holds information used in the Daily Report and/or the Recap Report. Some tabs have sub-sections, for additional data.

	DF Chem /Personnel	IFE/SC Engineer	Drilling Total	DF Equip. /Screens	Other Cost (D/W/M/CF)	Total
Daily Cost	48430.00	1450.00	49880.00	1774.00	7565.06	59219.06
Cumulative Cost	345735.50	5825.00	351560.50	6622.00	10464.29	368646.79

Each project dictates the information that's required for reporting & analysis, however this presentation will remark specific **key data** used to perform calculations that affect other parts of the program.

Daily Data Input – Drilling Fluids and Equipment Module

Daily Data Entry page - General Description

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Date	12/14/1999	Depth (ft)	7023	Fluid Type	VersaClean
Spud Date	12/7/1999	TVD (ft)	6486	Lithology	
Report	7	Bit Depth (ft)	7023		
Present Activity	Drig 12-1/4" hole			Lithology Setup	Well Survey

Operator Representative		John Able	
Contractor Representative		Peter Smart	
M-I SWACO Representative 1	Ian Scott	Rig Telephone Number	567-890-1234
M-I SWACO Representative 2	David Jones	Warehouse Telephone	678-901-2345
Telephones	123-456-7890		
	345-678-9012		
Pagers/FAX Numbers	234-567-8901		
	456-789-0123		

	DF Chem /Personnel	IFE/SC Engineer	Drilling Total	DF Equip. /Screens	Other Cost (DWM/CF)	Total
Daily Cost	48430.00	1450.00	49880.00	1774.00	7565.06	59219.06
Cumulative Cost	345735.50	5825.00	351560.50	6622.00	10464.29	368646.79

12/14/1999

PRINT ? [Icons] SAVE [Icon]

This section is common to all the tabs. These buttons are used for common tasks (described below).

Displays a help file section.

Printing a pre-defined daily report

Displays Excel based daily reports

Displays a Cost Overview screen

Daily Data Input – Drilling Fluids and Equipment Module

Tab #1 - General

Tab #1 is for general information and current depth data.

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

5- Comments
1- General

6- Solids Equipment
2- Pumps / Bits

7- Time Distribution
3- Mud Properties

8- Inv/Hyd/Conc
4- Well Geometry

Date	12/14/1999	Depth (ft)	7023	Fluid Type	VersaClean
Spud Date	12/7/1999	TVD (ft)	6486	Lithology	
Report	7	Bit Depth (ft)	7023		
Present Activity	Drig 12-1/4" hole			Lithology Setup	Well Survey

Operator Representative	John Able				
Contractor Representative	Peter Smart				
M-I SWACO Representative 1	Ian Scott	Rig Telephone Number	567-890-1234		
M-I SWACO Representative 2	David Jones	Warehouse Telephone	678-901-2345		
Telephones	123-456-7890 345-678-9012				
Pagers/FAX Numbers	234-567-8901 456-789-0123				



	DF Chem /Personnel	IFE/SC Engineer	Drilling Total	DF Equip. /Screens	Other Cost (DWM/CF)	Total
Daily Cost	48430.00	1450.00	49880.00	1774.00	7565.06	59219.06
Cumulative Cost	345735.50	5825.00	351560.50	6622.00	10464.29	368646.79

12/14/1999

EXIT ? ? ? ? ? ? ? ? ? ?

DAILY KEY DATA

1. **Depth:** determines your well geometry, affecting: Downhole Vol., Tot. Vol., Depth Drilled, Cuttings Drilled & Cuttings Discharged.
2. **Bit Depth:** sets your drill string total depth, affecting Fluid Volumes, Bottom-up & Drill Pipe length.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #1 - General

You can also identify the formation lithology (thru **Lithology Setup**) and the well directional profile (using **Well Survey**)

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

1- General 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Date: 12/14/1999 Depth (ft): 7023 Fluid Type: VersaClean
Spud Date: 12/7/1999 TVD (ft): 6486 Lithology:
Report: 7 Bit Depth (ft): 7023
Present Activity: Drlg 12-1/4" hole

Lithology Setup **Well Survey**

Operator Representative: John Able
Contractor Representative: Peter Smart
M-I SWACO Representative 1: Ian Scott Rig Telephone Number: 567-890-1234
M-I SWACO Representative 2: David Jones Warehouse Telephone: 678-901-2345
Telephones: 123-456-7890
345-678-9012
Pagers/FAX Numbers: 234-567-8901
456-789-0123

	DF Chem /Personnel	IFE/SC Engineer	Drilling Total	DF Equip. /Screens	Other Cost (D/W/M/CF)	Total
Daily Cost	48430.00	1450.00	49880.00	1774.00	7565.06	59219.06
Cumulative Cost	345735.50	5825.00	351560.50	6622.00	10464.29	368646.79

Lithology

Formation Tops and Lithology			
Depth (ft)	Formation Top	Lithology	% Sand
0	Zephir 3	Breccia	1
1000	Promateris	Chalk	2
5500	Alpha 24	Claystone	10
*			

Well Survey Data

Depth	Inclination	Azimuth (°)
0	0	0
2000	1	18
5000	2	18
5500	5	17
6000	10	50
6500	20	60
7000	25	60
10000	45	80
*		

File Input
Comma Delimited Text

Daily Data Input – Drilling Fluids and Equipment Module

Tab #2 – Pumps / Bits

The rig pump data (liner diameter, stroke length, % efficiency & pump rate) is used to calculate the Pump Flow Rate, but this figure may be over-written if needed.

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Pump Data

Make and Model	Liner Diameter	Stroke Length	Rod Diam Duplex Only	Eff%	Pump Rate	Pump On Report	Riser Pump
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6	12		97		☐	☐
						☐	☐

Nozzles

Size	Quantity	TFA
14	x 2	0.69
13	x 3	
	x	

Extra ECD Input

Surface Code (1-5)	
Surface Pressure (psi)	
Ref. Flow Rate (gal/min)	
On/Off Bottom (psi)	

Bit and Drilling Data

Bit Description	Hycalog X-175	Bit Information
Bit Number	3	Bit Size/Hole Diameter 12.25
% Washout	5	Washout Hole Size 12.553
Rotary rpm	150	Pump Flow Rate 803
Rotating Hours	0	Pump Pressure 3000
Weight on Bit	24000	dP MWD
ROP	30	dP Motor
Riser Pump Flow Rate		Motor RPM

12/14/1999

DAILY KEY DATA

- Pump Rate:** this is used to calculate the total Pump Flow Rate, which affects the Total Circulating Time, Bottom-up and some indexes (i.e. pressure losses, jet velocity, annular velocity, annular critical velocity, etc.)
- Nozzles:** this info is utilized to calculate the Total Flow Area (TFA), used in several hydraulic calculations.
- Pressures:** used to calculate pressure losses, ECD, etc.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #2 – Pumps / Bits

The rig pump data (liner diameter, stroke length, % efficiency & pump rate) is used to calculate the **Pump Flow Rate**, but this figure may be over-written if needed.

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Pump Data

Make and Model	Liner Diameter	Stroke Length	Rod Diam Duplex Only	Eff%	Pump Rate	Pump On Report	Riser Pump
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6	12		97		☐	☐
						☐	☐

Nozzles

Size	Quantity	TFA
14	x 2	0.69
13	x 3	
	x	

Extra ECD Input

Surface Code (1-5)	
Surface Pressure (psi)	
Ref. Flow Rate (gal/min)	
On/Off Bottom (psi)	

Bit and Drilling Data

Bit Description	Hycalog X-175	Bit Information
Bit Number	3	Bit Size/Hole Diameter 12.25
% Washout	5	Washout Hole Size 12.553
Rotary rpm	150	Pump Flow Rate 803
Rotating Hours	0	Pump Pressure 3000
Weight on Bit	24000	dP MWD
ROP	30	dP Motor
Riser Pump Flow Rate		Motor RPM

12/14/1999

EXIT ? HELP PRINT SAVE

DAILY KEY DATA

4. **% Washout**: is the % increase in volume of the open-hole section.

Note: IT IS NOT THE SAME FIGURE GIVEN BY THE LOGGIN COMPANY.

5. **Bit Size / Hole Diameter**: it determines the open hole geometry (together with the **Depth**), affecting: Downhole Vol., Tot. Vol., Depth Drilled, Cuttings Drilled & Cuttings Discharged.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #2 – Extra ECD Input

These extra parameters are used to improve the accuracy on the hydraulics calculations. They are also present in Virtual Hydraulics

Extra ECD Input	
Surface Code (1-5)	
Surface Pressure (psi)	
Ref. Flow Rate (gal/min)	
On/Off Bottom (psi)	

Mud Report - ABC Oil Company Gash #12 - 12/13/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Pump Data

Make and Model	Liner Diameter	Stroke Length	Rod Diam Duplex Only	Eff%	Pump Rate	Pump On Report	Riser Pump
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6.5	12		97	80	☐	☐
EMSCO F-1000	6	12		97		☒	☐

Bit and Drilling Data

Nozzles		TFA	
Size	Quantity		
14	x 2		0.69
13	x 3		
	x		

Bit Description		Bit Information	
Bit Number	3	Bit Size/Hole Diameter	12.25
% Washout	5	Washout Hole Size	12.553

Rotary rpm	150	Pump Flow Rate	803
Rotating Hours	12	Pump Pressure	2700
Weight on Bit	22000	dP MWD	
RDP	29	dP Motor	
Riser Pump Flow Rate		Motor RPM	

12/13/1999

Extra ECD Data

- **Surface Code:** this is to estimate your surface equipment pressure loss. Select "1" to manually enter the "Surface Pressure" loss, or select the code that best describes your surface equipment from the table shown here.
- **Surface Pressure:** enter here the pressure lost on surface equipment if using "Surface Code" equal to "1".
- **Ref. Flow Rate:** flow rate at which the pressure losses through the MWD and Mud Motor were measured.
- **On/Off Bottom:** additional pressure drop observed when the drill string is on bottom.

Type	Standpipe, L x ID	Hose, L x ID	Swivel, L x ID
1	-----	-----	-----
2	40' x 3"	45' x 2"	4' x 2"
3	45' x 3.5"	55' x 2.5"	5' x 2.25"
4	45' x 4"	55' x 3"	5' x 2.25"
5	45' x 4"	55' x 3"	6' x 3"

Daily Data Input – Drilling Fluids and Equipment Module

Tab #3 – Mud Properties

Fill in the properties from each mud check in reverse order, where the “#1 -Primary” check is the last one done today.

Mud Report - ABC Oil Company Gash #12 - 12/12/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits **3- Mud Properties** 4- Well Geometry

Mud Properties		#1 - Primary	#2	#3	#4
▶ Sample From		In			
Time Taken		09:00			
Flow Line Temp	°F	90			
Depth	ft	5002			
TVD	ft	4750			
Mud Weight	lb/gal	10.5			
MW Temp	°F	86			
Funnel Viscosity	s/qt	45			
Rheology Temp	°F	120			
R600		50			
R300		38			
R200		26			
R100		22			
R6		17			
R3		14			
PV	cP	12			
YP	lb/100ft²	26			
10s Gel	lb/100ft²	14			
10m Gel	lb/100ft²	16			
30m Gel	lb/100ft²	18			

Current Solids Analysis Equations: ☒ M-I ☒ Weighted ☐ Unweighted Extra Properties

12/12/1999

EXIT ? HELP PRINT COPY PASTE SAVE

Be sure to fill in s.g.'s for base oils, weight materials and drill solids.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #3 – Mud Properties

Fill in the properties from each mud check in reverse order, where the “#1 -Primary” check is the last one done today.

Mud Report - ABC Oil Company Gash #12 - 12/12/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits **3- Mud Properties** 4- Well Geometry

Mud Properties		#1 - Primary	#2	#3	#4
▶ Sample From		In			
Time Taken		09:00			
Flow Line Temp	°F	90			
Depth	ft	5002			
TVD	ft	4750			
Mud Weight	lb/gal	10.5			
MW Temp	°F	86			
Funnel Viscosity	s/qt	45			
Rheology Temp	°F	120			
R600		50			
R300		38			
R200		26			
R100		22			
R6		17			
R3		14			
PV	cP	12			
YP	lb/100ft²	26			
10s Gel	lb/100ft²	14			
10m Gel	lb/100ft²	16			
30m Gel	lb/100ft²	18			

Current Solids Analysis Equations: ☒ M-I ☒ Weighted ☐ Unweighted Extra Properties

12/12/1999

To make sure you enter **ALL** the required data, you need to scroll down until you see the “Solids Analysis” section (in yellow).

This selection is used to correctly apply the “solids analysis” formulas to the data entered. This is a manual entry and **it will not automatically** when you enter a higher mud weight in the mud check.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #3 – Mud Properties

Solids Analysis on WATER BASED MUD

Mud Report - ABC Oil Company Gash #12 - 12/12/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Mud Properties		#1 - Primary	#2	#3	#4
Retort Mud Wt	lb/gal	10.5			
Retort Mud Temp.	°F	75			
K+ (From KCl)	mg/l	0			
Wt Additive SG		4.2			
Oil SG		.7			
Frac Bent	about 0.1111	.1111			
Chem Conc	lb/bbl	0			
Drill Solids SG		2.6			
NaCl	%	0.5			
NaCl	lb/bbl	6.3			
KCl	%	0.			
KCl	lb/bbl	0.			
LGS	%	3.4			
LGS	lb/bbl	31.3			
Bentonite	%	0.2			
Bentonite	lb/bbl	1.7			
Drill Solids	%	3.3			
Drill Solids	lb/bbl	29.6			
HGS	%	6.			

Current Solids Analysis Equations: M-I Weighted Unweighted Extra Properties

12/12/1999

Solids Analysis results

DAILY KEY DATA

- Retort Mud Weight:** this is the density of the fluid that is poured into the retort cell, at the indicated temperature.
- Frac. Bentonite:** is the ratio between the activity of the formation clay vs. the commercial bentonite (M-I GEL). **This value is not always "0.1111" so further tests may be required to calculate the correct value for your area.**
- Chem. Conc.:** is the amount of chemical in your fluid (in ppb, Kg/m3, etc) without including weighting material, bentonite, salt or liquid products. **Note: if you leave this value as "0", you'll get higher "% Drilled Solids" than the actual amount present in the fluid.**

Daily Data Input – Drilling Fluids and Equipment Module

Tab #3 – Mud Properties

Solids Analysis on OIL/SYNTHETIC BASED FLUID

Mud Report - ABC Oil Company Gash #12 - 12/13/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Mud Properties		#1 - Primary	#2	#3	#4
Retort Mud W/t	lb/gal	12			
Retort Mud Temp.	°F	76			
Base Fluid SG		.8			
Wt Matl SG		4.2			
Drill Solids SG		2.6			
Salt	%wt	15.66	20.84		
Salt	lb/bbl	12.88			
Adjusted Solids	%Vol	17.56	-2.3		
Oil/Water Ratio		76/24	15/44		
Avg SG Solids		4.1	-14.5		
LGS	%	1.6	-32.6		
LGS	lb/bbl	14.78	-296.96		
HGS	%	15.9	29.8		
HGS	lb/bbl	234.11	438.57		

Current Solids Analysis Equations: M-I

☒ CaCl2 ☐ NaCl Extra Properties

12/13/1999

EXIT ? F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

Solids Analysis results

Select the salt of your internal phase here...

DAILY KEY DATA

1. **Retort Mud Weight**: this is the density of the fluid that is poured into the retort cell, at the indicated temperature.

This parameter makes a **significant** impact on the solids analysis results, and **OBM/SBF Retort Mud Weight** is often unadjusted for temperature expansion/contraction. Make sure you understand how to correctly measure/estimate this value before you print your daily report.

This is the single cause of most solids analysis errors in ONE-TRAX reports when using OBM/SBF. **Note: leaving this value unadjusted (if required) produces higher "% Drill Solids" than the actual amount you have.**

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry

List drill string components from surface to TD.

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	T.J. Len (in)
DP	6609	5	4.276	6.375	3.75	21
DC	268	8.5	3			
DC	146	9.5	3			

Interval (Cost)

Number: 3

Print Order:

Pilot Hole

Hole Size: 0

Depth: 0

DAILY KEY DATA

- Interval (Cost) Number:** is used to indicate where will today's cost be allocated. For example: if the number in the box is equal to "3", it means that ONE-TRAX will consider all the cost generated by today's activity (rental, personnel, mud mixing, etc.) as part of the cost that belongs to the 3rd interval.

This number is critical to get each 'project interval cost' correctly calculated in the Recap report.

Pilot Hole data entry details

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

To assign your today's costs to a specific interval you need to do this...

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	T.J. Len (in)
	0.					

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number 0

Print Order

Pilot Hole

Hole Size 0

Depth 0

8/14/2005

Used the drop-down button to select the interval number where today's cost should be allocated to.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)	Intv #	Casing Type	Casing OD	Casing ID	Depth	TVD	Hole Size
	0.						1	0		0			

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number 0

Pilot Hole

Hole Size 0

Depth 0

8/14/2005

Click here and select from the list of available intervals.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
	0					

Interval (Cost)

Daily Casing / Volume
Well Casing Intervals (Cost)

Number
Print Order

Pilot Hole

Hole Size
Depth

8/14/2005

EXIT ? ? ? ? ? ? ? ?

Once selected the interval number will show up in the "Number" box.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
	0					

Interval (Cost)

Number 1

Intv #	Casing Type	Casing OD	Casing ID	Depth	TVD	Hole Size
1	0		0			

Pilot Hole

Hole Size 0
Depth 0

8/14/2005

EXIT ? ? ? ? ? ? ? ? ? ?

If the desired interval is not available in the list displayed (in this example “Interval #2” is not yet available), then follow the procedure explained next.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
	0					

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number 1
Print Order

Pilot Hole

Hole Size 0
Depth 0

8/14/2005

EXIT ? ? ? ? ? ? ? ?

Click here

[Back to: Table of Content](#)

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Casing Intervals



Casing Interval Summary

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top of Liner	Hole Size	Maximum Angle	Maximum Density	Frac Grad	Fluid Type Code 1	Fluid Type Code 2
▶ 1	0		0			0	12.25	0		0		

Click the “Edit” button to complete the information of ‘Interval #1’, before you can generate ‘Interval #2’.

New Edit

Ready

EXIT ?

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Fill in all the cells that have information in this screen shot...

The screenshot shows a software window titled "Casing Intervals" with a blue title bar and standard Windows window controls. The window contains several data entry fields organized into sections. The first section includes "Interval Number" (1) and "Type" (Casing). The second section includes "Casing OD (in)" (20), "Casing ID (in)" (19.124), "Hole Size (in)" (26), "Depth (ft)" (800), "TVD (ft)" (800), and "Top of Liner (ft)" (0). The third section includes "Maximum Density (lb/gal)" (8.6), "Maximum Temperature (°F)" (80), "Maximum Angle" (0.25), "Interval Days" (3), "Planned Days" (4), "Planned Length (ft)" (2), "Interval Cost (\$)" (10458.50), "Planned Cost (\$)" (10000), and "Frac Grad (lb/gal)" (10.5). The fourth section includes "Fluid Type Code 1" (Spud Mud) and "Fluid Type Code 2" (empty). Below these fields is a button labeled "Unscheduled Events". At the bottom, there is a section titled "Observations and Recommendations" with a text area containing the text "This interval was drilled without problems." The window also features a decorative banner image of an oil rig at sunset and a red "EXIT" button in the bottom right corner.

Interval Number	1	Type	Casing		
Casing OD (in)	20	Casing ID (in)	19.124	Hole Size (in)	26
Depth (ft)	800	TVD (ft)	800	Top of Liner (ft)	0
Maximum Density (lb/gal)	8.6	Maximum Temperature (°F)	80	Maximum Angle	0.25
Interval Days	3	Planned Days	4	Planned Length (ft)	2
Interval Cost (\$)	10458.50	Planned Cost (\$)	10000	Frac Grad (lb/gal)	10.5
Fluid Type Code 1	Spud Mud	Fluid Type Code 2			

Unscheduled Events

Observations and Recommendations

This interval was drilled without problems.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Casing Intervals

Casing Interval Summary

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top of Liner	Hole Size	Maximum Angle	Maximum Density	Frac Grad	Fluid Type Code 1	Fluid Type Code 2
1	F	20	19.124	800	800	0	26	0.25	8.6	10.5	1071	

Finally you can create a new interval cost...

To do that just click “New” as indicated here.

New Edit

Ready

EXIT ?

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Casing Intervals

Interval Number: 2 Type: [Dropdown]

Casing OD (in): [Dropdown] Casing ID (in): 0 Hole Size (in): [Dropdown]

Depth (ft): [Dropdown] TVD (ft): [Dropdown] Top of Liner (ft): 0

Maximum Density (lb/gal): [Dropdown] Maximum Temperature (°F): 0 Maximum Angle: 0 [Dropdown]

Interval Days: ... Planned Days: 0 [Dropdown] Planned Length (ft): 0 [Dropdown]

Interval Cost (\$): 0.00 Planned Cost (\$): 0 [Dropdown] Frac Grad (lb/gal): 0 [Dropdown]

Fluid Type Code 1: [Dropdown] Fluid Type Code 2: [Dropdown]

You don't need to enter any more information right now, so just exit.

Observations and Recommendations

[Text Area]

[Image of drilling rig at sunset]

[EXIT Button]

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

Casing Intervals



Casing Interval Summary

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top of Liner	Hole Size	Maximum Angle	Maximum Density	Frac Grad	Fluid Type Code 1	Fluid Type Code 2
1	F	20	19.124	800	800	0	26	0.25	8.6	10.5	1071	
2			0			0		0		0		

You will see your new “interval cost number” here...

Ready

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Well Casing Intervals (Cost)

[illegible]

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Daily Casing Volume

Describing your full well geometry

Daily Report [ABC Oil Company Gusher #2 12/14/2002]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
DP	6609	5	4.276	6.375	3.75	21
DC	268	8.5	3			
DC	146	9.5	3			

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number 3
Print Order

Pilot Hole

Hole Size 0
Depth 0

12/14/2002

EXIT ? F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

Click here to describe the current well geometry. This will affect your volumes calculations.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Daily Casing Volume

The screenshot shows the 'Well Geometry (12/14/2002)' window. On the left is a well diagram with depth markers at 304, 5002, and 7023. The main area contains a 'Daily Intervals' table, a 'Side-Track Setup Help' button, and a 'Hole Volume Summary' table. The 'Daily Intervals' table is highlighted with an orange box. The 'Side-Track Setup Help' button is highlighted with a blue box. The 'Hole Volume Summary' table is highlighted with a yellow background. The 'Hole Volume Detail' button is circled in orange.

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size
1	F	20	19	800	800	0	
2	F	13.375	12.515	5002	4747	0	1
3	O			7023	6188		12

Item	Volume
Drill String	119
Annulus	939
Below Bit	
Total Volume	1058
Drilled Hole	230

Parameter	Value
Bit Size/Hole Diameter	12.25
Washout Hole Size	12.553
Depth	7023
TVD	6188
Bit Depth	7023
Pilot Hole Depth	0
Pilot Hole Size	0
Air Gap	84
Water Depth	220
Riser ID	19.5
Riser OD	20.5
Riser Length	304
Riserless Drilling	☐

DAILY KEY DATA

1. **Daily Intervals info:**
Describe all your different geometry sections (from top to bottom) in here.

Use these buttons to add/delete intervals (as required) to accurately describe all your "geometry intervals".

If all is correct, press the "**Hole Volume Detail**" to obtain the section volume information.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #4 – Well Geometry / Daily Casing Volume

Hole Volume Results (12/14/2002)

Hole Volume Results

	Sec	Length	Hole Dia	Pipe OD	Pipe ID	ToolJ OD	Description	DS Vol	Ann Vol
▶	1	304	19.500	5	4.276	6.375	DP/Riser	5.33	104.65
	2	4698	12.515	5	4.276	6.375	DP/Casing	82.36	596.68
	3	1607	12.550	5	4.276	6.375	DP/Open Hole	28.17	205.47
	4	268	12.550	8.5	3		DC/Open Hole	2.34	22.20
	5	146	12.550	9.5	3		DC/Open Hole	1.28	9.54

Total DS, ft: 7023
Drill String, bbl: 119.48
Annulus, bbl: 938.54
Below Bit, bbl: .
Total Volume, bbl: 1058.02

Bottom Up Stks: 7854
Bottom Up Time, min: 49.1
Tubing Displacement bbl: 78.3

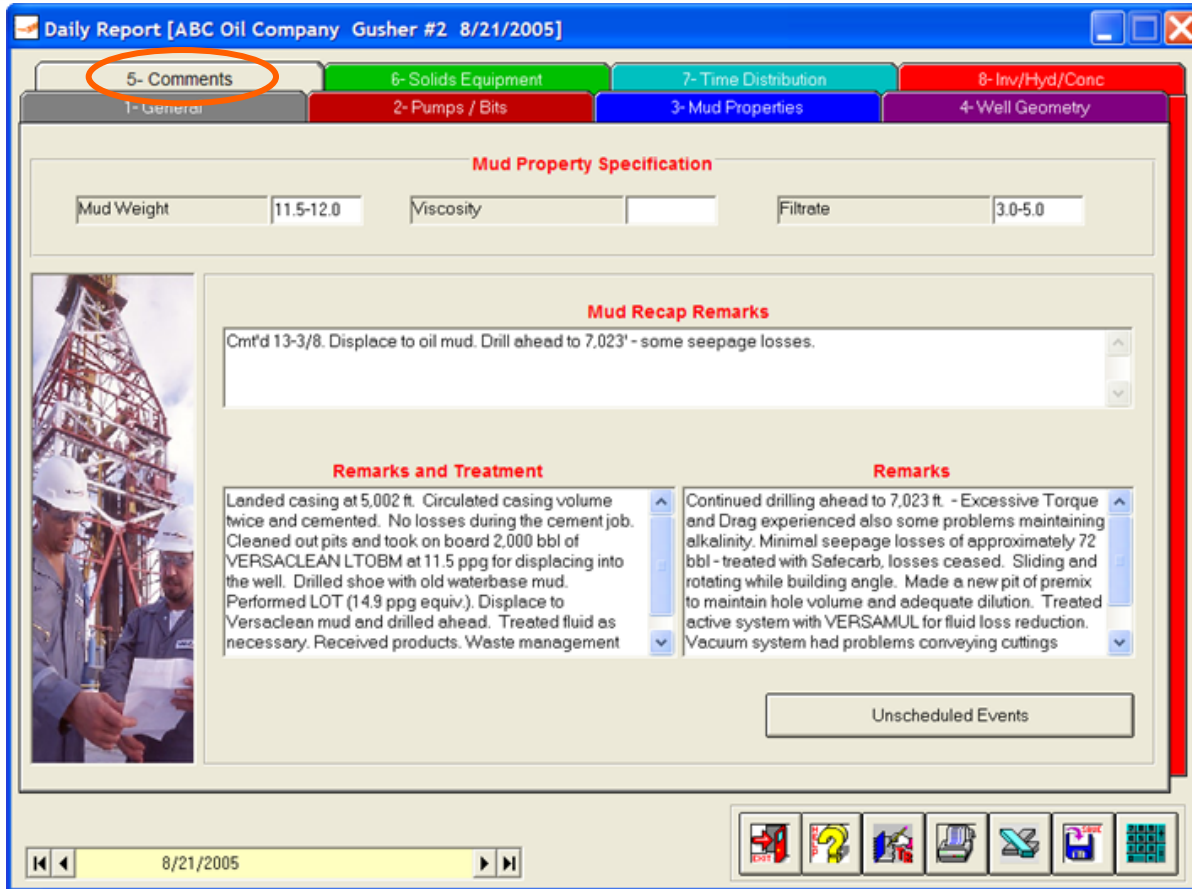
EXIT

This screen gives a complete breakdown of volumes by section.

Information is for review only, and cannot be changed in this screen.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #5 – Comments



Daily Report [ABC Oil Company Gusher #2 8/21/2005]

5- Comments (selected)

6- Solids Equipment

7- Time Distribution

8- Inv/Hyd/Conc

1- General

2- Pumps / Bits

3- Mud Properties

4- Well Geometry

Mud Property Specification

Mud Weight: 11.5-12.0

Viscosity:

Filtrate: 3.0-5.0

Mud Recap Remarks

Cmt'd 13-3/8. Displace to oil mud. Drill ahead to 7,023' - some seepage losses.

Remarks and Treatment

Landed casing at 5,002 ft. Circulated casing volume twice and cemented. No losses during the cement job. Cleaned out pits and took on board 2,000 bbl of VERSACLEAN LTOBM at 11.5 ppg for displacing into the well. Drilled shoe with old waterbase mud. Performed LOT (14.9 ppg equiv.). Displace to Versaclean mud and drilled ahead. Treated fluid as necessary. Received products. Waste management

Remarks

Continued drilling ahead to 7,023 ft. - Excessive Torque and Drag experienced also some problems maintaining alkalinity. Minimal seepage losses of approximately 72 bbl - treated with Safecarb, losses ceased. Sliding and rotating while building angle. Made a new pit of premix to maintain hole volume and adequate dilution. Treated active system with VERSAMUL for fluid loss reduction. Vacuum system had problems conveying cuttings

Unscheduled Events

8/21/2005

Insert on **Mud Recap Remarks** the observations you would like to display (as a one-liner) on the Well Recap.

Remarks refer to general rig operations (normally coming from the IADC sheet).

Remarks and Treatments refer to comments about any M-I SWACO activity or service provided during the day.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment

Daily Report [8/14/2005]

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc

1- General 2- Pumps / B's 3- Mud Properties 4- Well Geometry

Avg. Daily Drilled Solids for SP Calculation .0

Drilled Solids Removal System Performance(SP) * 0.

Mud Added / Drilled Hole Volume 0.

SRSP Recalculate

Shaker Screen Transactions

Shaker Screen Inventory

Equipment Details and Usage

SUS Compliance Testing

Cuttings Retention

Equipment Solids Analysis

IFE Remarks

Drilling Fluids Summary

Drilling Fluids Forward Plan

Solids-Control Summary

Solids-Control Forward Plan

D'WM Extra Properties

8/14/2005

Recommendation

If you follow this order, it will make the data entry process more simple.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Inventory

Click here to receive or transfer NEW and USED shaker screens

Daily Report [8/14/2005]

5- Comments | 6- Solids Equipment | 7- Time Distribution | 8- Inv/Hyd/Conc

1- General | 2- Pumps / Bits | 3- Mud Properties | 4- Well Geometry

Avg. Daily Drilled Solids for SP Calculation: .0
Drilled Solids Removal System Performance(SP) *: .0
Mud Added / Drilled Hole Volume: .0

SRSP Recalculate

Shaker Screen Transactions
Shaker Screen Inventory
Equipment Details and Usage
SUS Compliance Testing
Cuttings Retention
Equipment Solids Analysis

* - Based on API Recommended Practice 13C.

IFE Remarks

Drilling Fluids Summary | Drilling Fluids Forward Plan
Solids-Control Summary | Solids-Control Forward Plan
D'WM Extra Properties

8/14/2005

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Inventory

Shaker Screen Inventory

Daily Inventory										
Code	Description	Start	New				Costs			
			Received	Returned	Used	Final	Gross Price	Daily	Cumulative	
▶ XBM6MK010H	84 Mesh Magnum XR	2				2	253.00	0.00	506.00	
X100TF110C	110 Mesh Magnum XR	0				0	325.00	0.00	0.00	
XBXLK140H		10				10	325.00	0.00	650.00	
XBXLK175H									1625.00	
XBXLK210H									325.00	
XBXLK250H									1625.00	
W100TB084Y									0.00	
JBMEXR200P									0.00	
W100TB270C									0.00	

Use this button to receive or transfer shakers screens from your inventory

Daily Cost (\$) 1300
Cumulative Cost (\$) 4731

Delivery/Return Tickets Recalculate Cumulative Cost

EXIT

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Inventory

Transfer Ticket Transactions

Transfer Tickets Summary

Type	Ticket Number	Order By	Receive By	Warehouse Name	Warehouse

Click the **NEW** button, enter the information in the 'Transfer Ticket Summary' section and fill in the 'Transfer Ticket Detail' table.

Transfer Ticket Detail -

Product Description	Mesh Size	Price	Used			
			Amount from Ticket	Actual Amount	Amount from Ticket	Actual Amount
84 Mesh Magnum XR	84	253				
110 Mesh Magnum XR	110	325				
140 Mesh Magnum XR	140	325				
175 Mesh Magnum XR	175	325				
210 Mesh Magnum XR	210	325				
250 Mesh Magnum XR	250	325				
84 Mesh HC Magnum	84	253				
200 Mesh HC Magnum	200	325				
270 Mesh HC Magnum	270	325				

Buttons: ADD NEW, SAVE, DELETE, EXIT

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Inventory

Transfer Ticket Transactions

Transfer Tickets Summary

Type	Ticket Number	Order By	Receive By	Warehouse Name	Warehouse
Receive from Warehouse	G9962002	D. Jones	D. Jones	Aberdeen	102

When you finish entering the data, click the **SAVE** button

Transfer

Product Description	Mesh Size	Price	New		Used	
			Amount from Ticket	Actual Amount	Amount from Ticket	Actual Amount
84 Mesh Magnum XR	84	253	4	4		
110 Mesh Magnum XR	110	325				
140 Mesh Magnum XR	140	325	10	10		
175 Mesh Magnum XR	175	325				
210 Mesh Magnum XR	210	325				
250 Mesh Magnum XR	250	325				
84 Mesh HC Magnum	84	253				
200 Mesh HC Magnum	200	325				
270 Mesh HC Magnum	270	325				



Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

Click here to install, uninstall or dispose screens from your shale shakers

Daily Report [8/14/2005]

5- Comments
1- General

6- Solids Equipment
2- Pumps / Bits

7- Time Distribution
3- Mud Properties

8- Inv/Hyd/Conc
4- Well Geometry

Avg. Daily Drilled Solids for SP Calculation .0

Drilled Solids Removal System Performance(SP) * 0.

Mud Added / Drilled Hole Volume 0.

SRSP Recalculate

Shaker Screen Transactions

Shaker Screen Inventory

Equipment Details and Usage

SUS Compliance Testing

Cuttings Retention

Equipment Solids Analysis

* - Based on API Recommended Practice 13C.

IFE Remarks

Drilling Fluids Summary

Drilling Fluids Forward Plan

Solids-Control Summary

Solids-Control Forward Plan

D'WM Extra Properties

8/14/2005

EXIT ? ? ? ? ? ? ? ? ? ?

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

Shaker Screen Transactions

Transactions							
Trans No.	Action	Screen Description	Screen Code	Shaker Serial No.	Shaker Description	Position	
1	Transfer to Storage	175 Mesh Magnum XR	XBXMLK175	12005	ALS II Shaker	1	
2	Install Used Screen	210 Mesh Magnum XR	XBXMLK210	12005	ALS II Shaker	1	
3	Dispose from	140 Mesh Magnum XR	XBXMLK140	12005	ALS II Shaker	2	
4	Install New Screen	250 Mesh Magnum XR	XBXMLK250	12005	ALS II Shaker	2	
5	Dispose from	175 Mesh Magnum XR	XBXMLK175	12025	ALS II Shaker	2	
6	Install New Screen	250 Mesh Magnum XR	XBXMLK250	12025	ALS II Shaker	2	
7	Transfer to Storage	175 Mesh Magnum XR	XBXMLK175	12173	ALS II Shaker	2	
8	Install New Screen	250 Mesh Magnum XR	XBXMLK250	12173	ALS II Shaker	2	
9	Install New Screen	175 Mesh Magnum XR	XBXMLK175	12173	ALS II Shaker	3	
10	Dispose from	175 Mesh Magnum XR	XBXMLK175	12173	ALS II Shaker	3	

Screen Usage **Undo Last Transaction** **Reset** **Print Preview** **Exit**

Recalculate **Inventory**

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

Shaker Screen Transactions

1  2  3  4  5 

ALS II Shaker ALS II Shaker ALS II Shaker Mud Cleaner 8T4 w/o BEM 600 Shaker

6 N/A 7 N/A 8 N/A 9 N/A 10 N/A

11 N/A 12 N/A 13 N/A 14 N/A

Shaker Screens

1 Description ALS II Shaker Serial No. 12005

1 210 Mesh Magnum XR (Mesh: 210) 7

2 250 Mesh Magnum XR (Mesh: 250) 8

3 9

4 10

5 11

6 12

New Screen Dispose from Equipment Dispose from Storage

☐ Used Screen ☒ New Screen Install New Screen Transfer to Storage Exit

Click to select this position

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

Shaker Screen Transactions

1	2	3	4	5
ALS II Shaker	ALS II Shaker	ALS II Shaker	Mud Cleaner 8T4 w/o	BEM 600 Shaker

6	7	8	9	10
N/A	N/A	N/A	N/A	N/A

11	12	13	14
N/A	N/A	N/A	N/A

Shaker Screens

1	Description	Serial No.
1	ALS II Shaker	12005

Click to select this position

Select your shaker here

Here you have up to 14 different shakers, depending on how many you defined in the “Active Products/Equipment/Screens” section.

Select the shaker by clicking on to of its picture and then move to ‘Section 2’.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

Shaker Screen Transactions

In this section you select the position of the screen that you will work with.

You have up to 12 positions per shaker.

Shaker Screens

1	Description	Serial No.
1	210 Mesh Magnum XR (Mesh: 210)	7
2	250 Mesh Magnum XR (Mesh: 250)	8
3		9
4		10
5		11
6		12

New Screen

☐ Used Screen ☒ New Screen

Click to select this position

Section 2

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

BEM 3



3 screens = positions 1,2,3

BEM 600

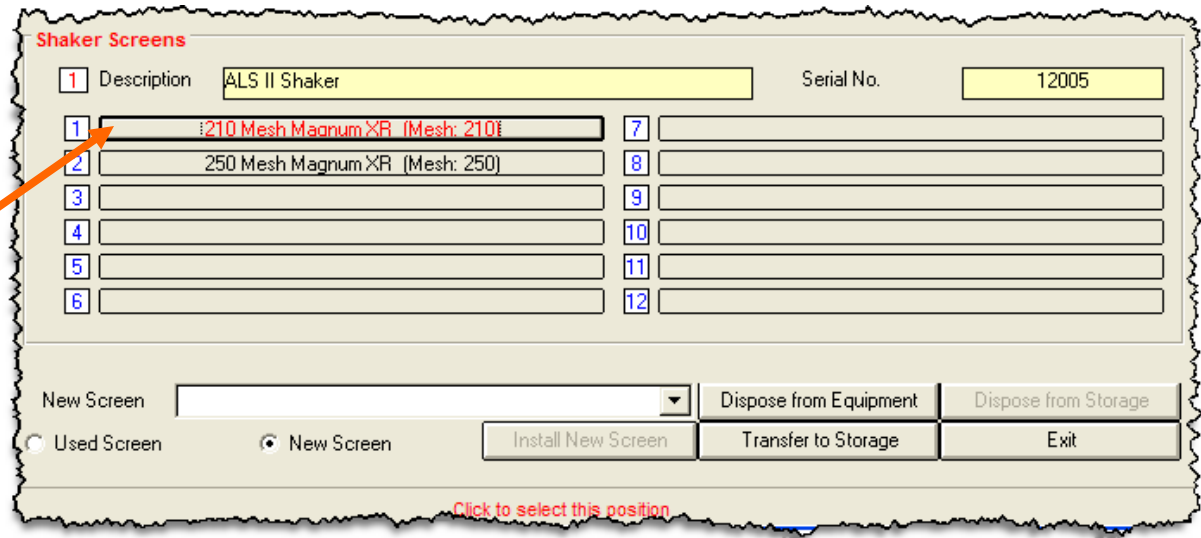


5 screens = positions 1,2,3,4,5

“Position #1” is always the screen closest to the flow line. Then use Position 2,3,4... all the way back to the discharge.

Click on the position that you want to work with.

In this case I will work with position #1.



1	Description	Serial No.
1	ALS II Shaker	12005
2	210 Mesh Magnum XR (Mesh: 210)	
3	250 Mesh Magnum XR (Mesh: 250)	
4		
5		
6		
7		
8		
9		
10		
11		
12		

New Screen

☐ Used Screen ☒ New Screen

Click to select this position

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

AFTER you have selected your screen position, follow the steps below...

Shaker Screens

1 Description ALS II Shaker Serial No. 12005

1 210 Mesh Magnum XR (Mesh: 210) 7

2 250 Mesh Magnum XR (Mesh: 250) 8

3 9

4 10

5 11

6 12

New Screen [Dropdown]

☐ Used Screen ☒ New Screen

Install New Screen Transfer to Storage Dispose from Equipment Dispose from Storage

Exit

Click to select this position

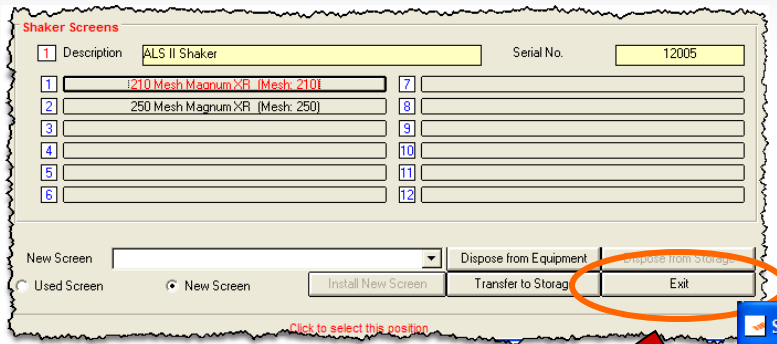
1.- If you are installing a screen you need to say if it is a NEW screen or a USED screen.

2.- Use the drop down list to select the correct screen from your inventory

3.- Use the these buttons to select the action that you need to do.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions



Shaker Screens

1 Description: ALS II Shaker Serial No. 12005

1	210 Mesh Magnum XR (Mesh: 210)	7	
2	250 Mesh Magnum XR (Mesh: 250)	8	
3		9	
4		10	
5		11	
6		12	

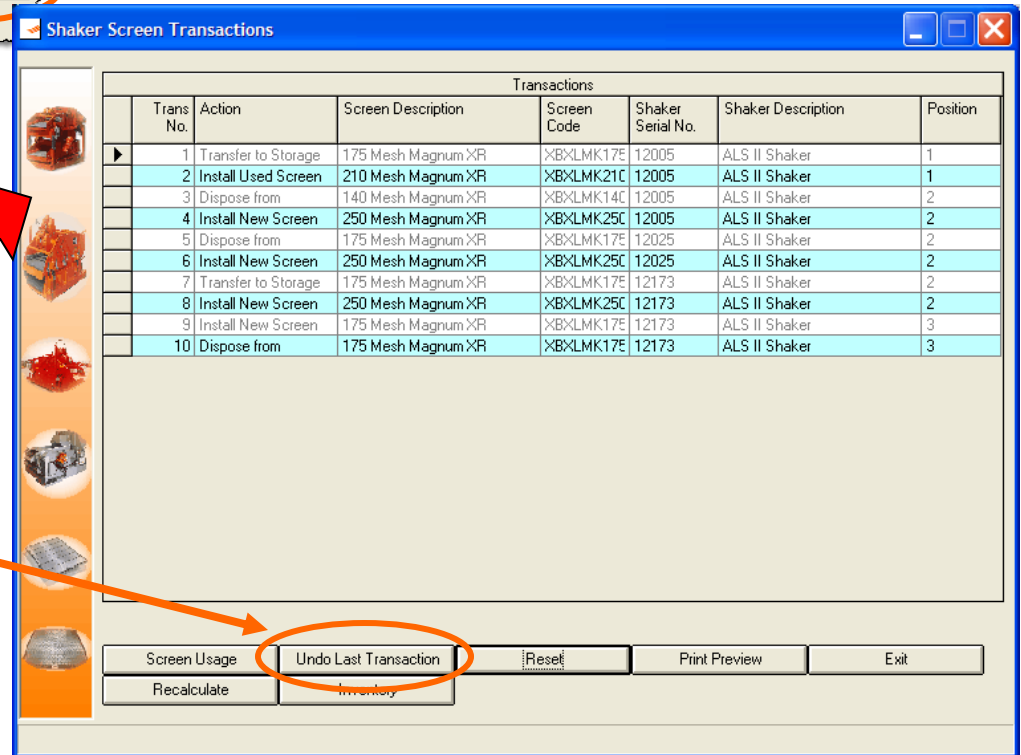
New Screen: [Dropdown] Disposed from Equipment [Dropdown] Disposed from Storage [Dropdown]

Used Screen [Radio] New Screen [Radio]

Install New Screen [Button] Transfer to Storage [Button] **Exit [Button]**

Click to select this position

When you click the EXIT button, this screen will appear...



Shaker Screen Transactions

Trans No.	Action	Screen Description	Screen Code	Shaker Serial No.	Shaker Description	Position
1	Transfer to Storage	175 Mesh Magnum XR	XBXL MK17E	12005	ALS II Shaker	1
2	Install Used Screen	210 Mesh Magnum XR	XBXL MK21C	12005	ALS II Shaker	1
3	Dispose from	140 Mesh Magnum XR	XBXL MK14C	12005	ALS II Shaker	2
4	Install New Screen	250 Mesh Magnum XR	XBXL MK25C	12005	ALS II Shaker	2
5	Dispose from	175 Mesh Magnum XR	XBXL MK17E	12025	ALS II Shaker	2
6	Install New Screen	250 Mesh Magnum XR	XBXL MK25C	12025	ALS II Shaker	2
7	Transfer to Storage	175 Mesh Magnum XR	XBXL MK17E	12173	ALS II Shaker	2
8	Install New Screen	250 Mesh Magnum XR	XBXL MK25C	12173	ALS II Shaker	2
9	Install New Screen	175 Mesh Magnum XR	XBXL MK17E	12173	ALS II Shaker	3
10	Dispose from	175 Mesh Magnum XR	XBXL MK17E	12173	ALS II Shaker	3

Screen Usage [Button] **Undo Last Transaction [Button]** [Reset] [Print Preview] [Exit]

Recalculate [Button] Inventory [Button]

If you make a mistake in a screen transaction, come to this screen and use this button. It will erase your last transaction.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Shaker Screen Transactions

Shaker Screen Transactions

Trans No.	Action	Screen Description	Screen Code	Shaker Serial No.	Shaker Description	Position
1	Transfer to Storage	175 Mesh Magnum XR	XB\LMK175	12005	ALS II Shaker	1
2	Install Used Screen	210 Mesh Magnum XR	XB\LMK210	12005	ALS II Shaker	1
3	Dispose from	140 Mesh Magnum XR	XB\LMK140	12005	ALS II Shaker	2
4	Install New Screen	250 Mesh Magnum XR	XB\LMK250	12005	ALS II Shaker	2
5	Dispose from	175 Mesh Magnum XR	XB\LMK175	12025	ALS II Shaker	2
6	Install New Screen	250 Mesh Magnum XR	XB\LMK250	12025	ALS II Shaker	2
7	Transfer to Storage	175 Mesh Magnum XR	XB\LMK175	12173	ALS II Shaker	2
8	Install New Screen	250 Mesh Magnum XR	XB\LMK250	12173	ALS II Shaker	2
9	Install New Screen	175 Mesh Magnum XR	XB\LMK175	12173	ALS II Shaker	3
10	Dispose from	175 Mesh Magnum XR	XB\LMK175	12173	ALS II Shaker	3

Screen Usage Recalculate Undo Last Transaction Inventory Reset Print Preview Exit

After exiting the 'screens transaction' section you can go to the

8/14/2005]

6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Solids for SP Calculation .0
Removal System Performance (SP) * .0
Filled Hole Volume .0

Shaker Screen Transactions
Shaker Screen Inventory
Equipment Details and Usage
SUS Compliance Testing
Cuttings Retention
Equipment Solids Analysis

SRSP Recalculate

* - Based on API Recommended Practice 13C.

IFE Remarks

Drilling Fluids Summary Drilling Fluids Forward Plan
Solids-Control Summary Solids-Control Forward Plan
D'WM Extra Properties

8/14/2005

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Equipment Solids Analysis

The first time you open this screen it will look like this...

Click the 'ADD NEW' button to start entering the data.

Serial Number	Description	Start Time	End Time	Print Order
---------------	-------------	------------	----------	-------------

Mud Type:

Sample Type:

Description:

Measured Depth:

Depth Drilled/Represented by Sample:

WMB Solids Analysis

Sample Density (lb/gal)		Wt Additive SG	
Water %Volume		Oil SG	
Oil %Volume		Frac Bent	
Solids %Volume		Chem Conc (lb/bbl)	
K+ (From KCl)		MBT	
Drill Solids SG		Chlorides (mg/l)	

☐ Weighted ☒ Unweighted

Date:

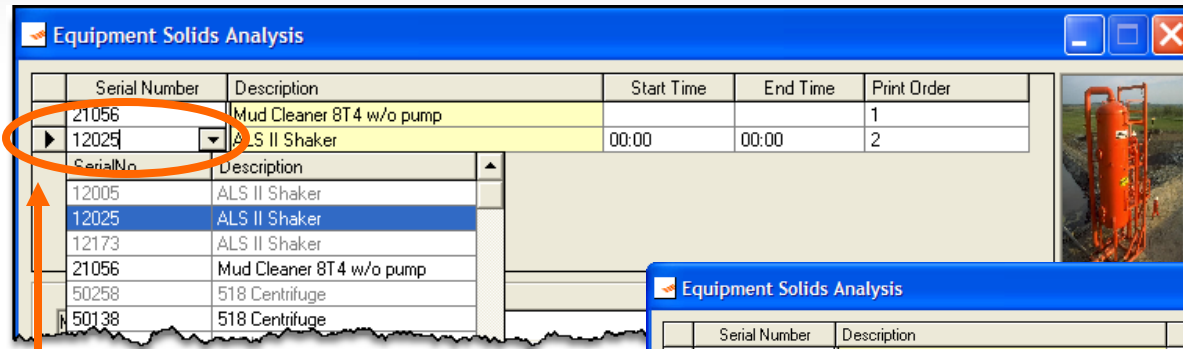
Calculated Results

NaCl%	
NaCl (lb/bbl)	
KCl%	
KCl (lb/bbl)	
LGS%	
LGS (lb/bbl)	
Bentonite%	
Bentonite (lb/bbl)	
Drill Solids%	
Drill Solids (lb/bbl)	
HGS%	
HGS (lb/bbl)	
Inert/React	
Average SG	

ADD NEW DELETE SAVE PRINT

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Equipment Solids Analysis



1- Click on the 'Serial Number' cell to select your equipment from the drop down list.

2.- Fill in the information highlighted in yellow and then click the 'RUN' button.

Equipment Solids Analysis

Serial Number	Description	Start Time	End Time	Print Order
21056	Mud Cleaner 8T4 w/o pump			1
12025	ALS II Shaker	00:00	00:00	2

Mud Type: Oil Base
Sample Type: Discharge
Description: Mud Cleaner Discard
Measured Depth: 7000
Depth Drilled Represented by Sample: 6900

Solids Analysis - Oil Base

Sample Density (lb/gal)		Base Fluid SG	
Water %Volume	19	HGS SG	
Oil %Volume	43	LGS SG	
Solids %Volume	52	Chlorides (mg/l)	

☒ CaCl2 ☐ NaCl

Calculated Results - Oil Base

Salt%wt	32.3
Salt (lb/bbl)	31.78
Adjusted Solids%	35.6
Oil/Water Ratio%	69/31
LGS%	27.5
LGS (lb/bbl)	250.06
HGS%	8.1
HGS (lb/bbl)	118.48
Average SG	2.96

Date: 8/31/2005

RUN

3- The results will show up on this side

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Cuttings Retention

This is used to calculate “Mud on Cuttings” & “Oil on Cuttings” on a discharge

Daily Report [8/14/2005]

5- Comments
1- General

6- Solids Equipment
2- Pumps / Bits

7- Time Distribution
3- Mud Properties

8- Inv/Hyd/Conc
4- Well Geometry

Avg. Daily Drilled Solids for SP Calculation .0

Drilled Solids Removal System Performance(SP) * 0.

Mud Added / Drilled Hole Volume 0.

SRSP Recalculate

Shaker Screen Transactions

Shaker Screen Inventory

Equipment Details and Usage

SUS Compliance Testing

Cuttings Retention

Equipment Solids Analysis

* - Based on API Recommended Practice 13C.

IFE Remarks

Drilling Fluids Summary

Drilling Fluids Forward Plan

Solids-Control Summary

Solids-Control Forward Plan

D'WM Extra Properties

8/14/2005

EXIT ? [Icons] SAVE [Icons]

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Cuttings Retention

The same process described before applies to the **Cuttings Retention** screen.

Cuttings Retention

Serial Number	Description	Start Time	End Time	Sample Mass	Sample Volume	Bit diameter	Print Order
12005	ALS II Shaker	03:33	23:00	110	50	12.25	1
21056	Mud Cleaner 8T4 w/o pump	21:00	22:00	121	50	12.25	2
50258	518 Centrifuge	23:00	23:59	90	50	12.25	3
50138	518 Centrifuge	21:00	22:00	94	50	12.25	4

Comments

Composite sample from all 3 shakers.

Measured Depth

7023

Depth Drilled Represented By Sample

6161

Cutting Retention for ALS II Shaker 12005

Fluid Weight (lb/gal)	11.6
% Base Fluid From Retort of Mud	56
Base Fluid SG	0.75
Drill Solids SG	2.6
Retort Cell Weight (gm)	342.6
Cell +Wet Cuttings Weight (gm)	373.3
Cell +Dry Cuttings Weight (gm)	363.3
Graduated Cylinder Weight (gm)	48.5
Water Volume in Cylinder (cc)	4.6
Wt Cylinder+Water+Base Fluid (gm)	58.3

Calculated Results

Mass of Wet Cuttings (gm)	30.7
Mass of Dry Cuttings (gm)	20.7
Mass of Base Fluid (gm)	5.2
Mass Balance Factor	0.993
Mass Base Fluid/kg Wet Cutting (gm/kg)	169.38
Mass Base Fluid/kg Dry Cutting (gm/kg)	251.21
Mud On Cuttings (bbl/bbl)	2.392
Wt% Base Fluid - Wet Cuttings	16.94
Wt% Base Fluid - Dry Cuttings	25.12

Date

8/21/2005

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Cuttings Retention

Cuttings Retention

Serial Number	Description	Start Time	End Time	Sample Mass	Sample Volume	Bit diameter	Print Order
12005	ALS II Shaker	03:33	23:00	110	50	12.25	1
21056	Mud Cleaner 8T4 w/o pump	21:00	22:00	121	50	12.25	2
50258	518 Centrifuge	23:00	23:59	90	50	12.25	3
50138	518 Centrifuge	21:00	22:00	94	50	12.25	4

SerialNo Description

12005 ALS II Shaker

12025 ALS II Shaker

12173 ALS II Shaker

21056 Mud Cleaner 8T4 w/o pump

50258 518 Centrifuge

50138 518 Centrifuge

V120 100 HP Vacuum Unit

HVD338 Horizontal Vacuum D-GASSER

Fluid Weight (lb/gal) 11.6

Mass of Wet Cuttings (gm) 25.10

1- Click on the 'Serial Number' cell to select your equipment from the drop down list.

2.- Fill in the information highlighted in yellow.

3- The results will show up on this side

Cuttings Retention

Serial Number	Description	Start Time	End Time	Sample Mass	Sample Volume	Bit diameter	Print Order
12005	ALS II Shaker	03:33	23:00	110	50	12.25	1
21056	Mud Cleaner 8T4 w/o pump	21:00	22:00	121	50	12.25	2
50258	518 Centrifuge	23:00	23:59	90	50	12.25	3
50138	518 Centrifuge	21:00	22:00	94	50	12.25	4

Comments Solids discharge

Measured Depth 7000

Depth Drilled Represented By Sample

Cutting Retention for 518 Centrifuge 50258

Fluid Weight (lb/gal)	11.6
% Base Fluid From Retort of Mud	56
Base Fluid SG	0.75
Drill Solids SG	2.6
Retort Cell Weight (gm)	342.6
Cell +Wet Cuttings Weight (gm)	367.7
Cell +Dry Cuttings Weight (gm)	359.5
Graduated Cylinder Weight (gm)	48.5
Water Volume in Cylinder (cc)	4
Wt Cylinder+Water+Base Fluid (gm)	56.5

Calculated Results

Mass of Wet Cuttings (gm)	25.10
Mass of Dry Cuttings (gm)	16.90
Mass of Base Fluid (gm)	4.00
Mass Balance Factor	0.992
Mass Base Fluid/kg Wet Cutting (gm/kg)	159.36
Mass Base Fluid/kg Dry Cutting (gm/kg)	236.69
Mud On Cuttings (bbl/bbl)	2.092
Wt% Base Fluid - Wet Cuttings	15.94
Wt% Base Fluid - Dry Cuttings	23.67

Date 8/21/2005

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Equipment Details and Usage

This is one of the most important screens for ES Operations

Daily Report [8/14/2005]

5- Comments
1- General

6- Solids Equipment
2- Pumps / Bits

7- Time Distribution
3- Mud Properties

8- Inv/Hyd/Conc
4- Well Geometry

Avg. Daily Drilled Solids for SP Calculation .0

Drilled Solids Removal System Performance(SP) * 0.

Mud Added / Drilled Hole Volume 0.

SRSP Recalculate

Shaker Screen Transactions

Shaker Screen Inventory

Equipment Details and Usage

SUS Compliance Testing

Cuttings Retention

Equipment Solids Analysis

* - Based on API Recommended Practice 13C.

IFE Remarks

Drilling Fluids Summary

Drilling Fluids Forward Plan

Solids-Control Summary

Solids-Control Forward Plan

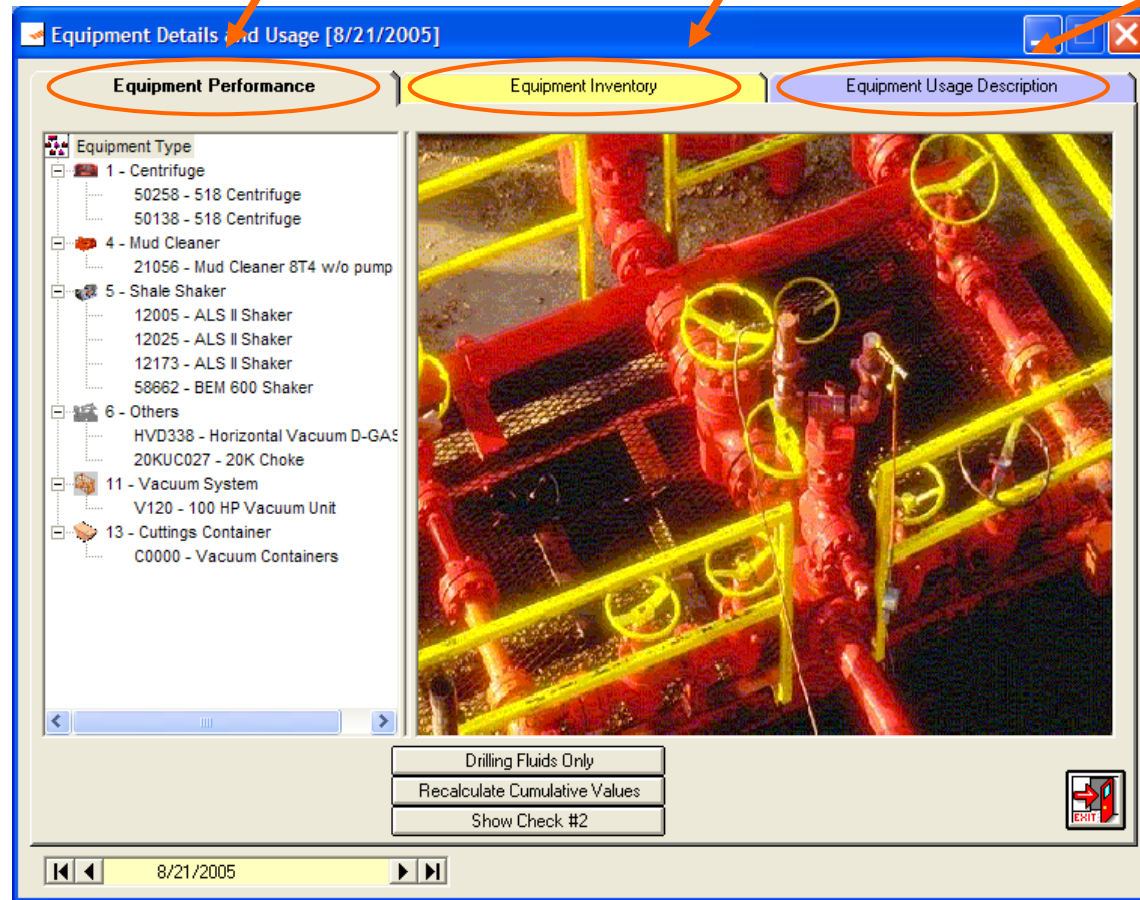
D'WM Extra Properties

8/14/2005

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Equipment Details and Usage

Enter here the **Equipment Performance Data**, **Equipment Costs** and **Downtime**



Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / ... / Performance Data

Equipment Details and Usage [9/14/2006]

The “Mud on Cuttings” value may come from the ‘Cuttings Retention Test’ section of ONE-TRAX (see previous slide), or it can be a manual entry value based on historical offset data.

SHAKERS

Equipment Type	Equipment Properties	Unit	#1 - Primary
1 - Centrifuge	Size / Specifications		30/38/38/38
Cent-VG - 518 Centrifuge	Hours turned on		22
Cent-#1 - 518 Centrifuge	Cum. Hrs turned on		760.5
Cent-#2 - 518 Centrifuge	Mud On Cuttings	m³/m³	1.463
4 - Mud Cleaner	Loss Type		Shakers
MC#1 - Mud Cleaner 8T4 Mongoose	Vol Discharged	m³	1
5 - Shale Shaker	Cum. Vol Discharged	m³	153.6
Shaker#1 - Mongoose Shaker	Vol Mud On Solids Discharged	m³	0.6
Shaker#2 - Mongoose Shaker	Cum. Vol Mud On Solids Discharged	m³	94.8
Shaker#3 - Derrick Shaker-1	% of Cuttings		18
Shaker#4 - Derrick Shaker-2	Deck Angle	°	
Shaker#5 - Derrick Shaker-3	G-Force	G's	
6 - Others	Vibration Angle		
VG-002 - Dry Cuttings Conveyor with			
VG-003 - Dryer Process Tank			
Cent-005 - OOC report laboratory			
VS-001 - Vacuum System Package			
7 - Cuttings Dryer			

This is an estimate of how much cuttings (in percentage) are discharged by this piece of equipment, where 100% is equivalent to the total volume drilled that day.

Example: if I estimate that my four (4) shakers discharge 80% of the cuttings drilled today, then each one discharges 20%; so, in this case “20%” would be the number entered as ‘% of Cuttings’.

DAILY KEY DATA

1. **Mud on Cuttings:** used to calculate the volume of fluid lost on cuttings.
2. **% of Cuttings:** used to calculate the volume of cuttings discharged by each shaker.

Back to: [Table of Content](#)

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / ... / Performance Data

The “Mud on Cuttings” value may come from the ‘Cuttings Retention Test’ section of ONE-TRAX, or it can be a manual entry value based on historical offset data.

CENTRIFUGES

Equipment Type			
1 - Centrifuge			
Cent-VG - 518 Centrifuge			
Cent-#1 - 518 Centrifuge			
Cent-#2 - 518 Centrifuge			
4 - Mud Cleaner			
MC#1 - Mud Cleaner 8T4 Mongoose			
5 - Shale Shaker			
6 - Others			
VG-002 - Dry Cuttings Conveyor with			
VG-003 - Dryer Process Tank			
Cent-005 - OOC report laboratory			
VS-001 - Vacuum System Package			
7 - Cuttings Dryer			
11 - Vacuum System			
VS-008 - Nemesis Rig Vac System			

Cent-VG - 518 Centrifuge (Drilling Fluids Only)			
Equipment Properties		Unit	#1 - Primary
Size / Specifications			518
Centrifuging Hrs			22
Cum. Centrifuging Hrs			715.5
Mud On Cuttings		m³/m³	0.855
Loss Type			Centrifuge
Vol Discharged		m³	9.2
Cum. Vol Discharged		m³	1414.9
Vol Mud On Solids Discharged		m³	4.2
Cum. Vol Mud On Solids Discharged		m³	671.1
Flow Rate		gal/min	18.9
WT In		sp.gr.	1.35
WT Out		sp.gr.	1.28
Discard WT		sp.gr.	2
Flow Rate Out		gal/min	17.1

DAILY KEY DATA

1. **Centrifuging Hrs**: used to calculate the total volume centrifuged that day.
2. **Mud on Cuttings**: used to calculate the volume of fluid lost on cuttings.
3. **Flow rate, Weight In, Weight Out and Discard Weight**: used to calculate the total volume discharged and returned, as well as the total mass processed, discharged and returned.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / ... / Performance Data

The “Mud on Cuttings” value may come from the ‘Cuttings Retention Test’ section of ONE-TRAX, or it can be a manual entry value based on historical offset data.

DRYERS

Equipment Type	VG-001 - VERTI-G Dryer (Drilling Fluids Only)		
1 - Centrifuge	Equipment Properties	Unit	#1 - Primary
4 - Mud Cleaner	Size / Specifications		15#
MC#1 - Mud Cleaner 8T4 Mongoose	Hours		22
5 - Shale Shaker	Cum. Hours		742.5
6 - Others	Mud On Cuttings	m³/m³	0.214
VG-002 - Dry Cuttings Conveyor with	Loss Type		Verti-G
VG-003 - Dryer Process Tank	Vol Discharged	m³	2.2
Cent-005 - OOC report laboratory	Cum.Vol Discharged	m³	330.4
VS-001 - Vacuum System Package	Vol Mud On Solids Discharged	m³	0.4
7 - Cuttings Dryer	Cum.Vol Mud On Solids Discharged	m³	61
VG-001 - VERTI-G Dryer	% of Cuttings		80
11 - Vacuum System	Main Drive G Force		
VS-008 - Nemesis Rig Vac System	Back Drive Differential		

This is an estimate of how much cuttings (in percentage) are discharged by this piece of equipment. 100% is equivalent to the total volume drilled that day.

Example: if I estimate that my four (4) shakers are discharging 80% of the volume drilled today, then each one of them discharges 20%. The number that should be entered in ‘% of Cuttings’ for the Verti-G should be lower than 80% (because the shaker system is feeding the Dryer), so the maximum the dryer can discharge is 80%.

Back to: [Table of Content](#) is between 55 and 70%.

DAILY KEY DATA

- Mud on Cuttings:** used to calculate the volume of fluid lost on cuttings.
- % of Cuttings:** used to calculate the volume of cuttings discharged by each shaker.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #6 – Solids Equipment / Equipment Details and Usage

Charge your equipment on this tab. Click on the “Charge Code” column to select ‘Full Charge’, ‘Stand by’ or ‘No charge’.

Daily Inventory						
Serial No.	Description	Used	Charge Code	Daily Cost	Cumulative Cost	
12005	ALS II Shaker	1	Full Charge	0.00	0.00	
12025	ALS II Shaker	1	Charge Code	0.00	0.00	
12173	ALS II Shaker	1	Full Charge	0.00	0.00	
21056	Mud Cleaner 8T4 w/o pump	1	Stand By	0.00	0.00	
50258	518 Centrifuge	1	No Charge	350.00	1390.00	
50138	518 Centrifuge	1	Full Charge	350.00	1390.00	
V120	100 HP Vacuum Unit	1	Full Charge	600.00	600.00	
HVD338	Horizontal Vacuum D-GASSER	1	Full Charge	0.00	0.00	
C0000	Vacuum Containers	4	Full Charge	80.00	80.00	
20KUC027	20K Choke	1	Full Charge	0.00	0.00	
58662	BEM 600 Shaker		Full Charge	0.00	0.00	

DAILY KEY DATA

1. “Used” quantity: used to calculate the **equipment rental cost** for today.

	Daily Cost	Cumulative Cost
Drilling Fluids	700.00	2780.00
Total	1380.00	3460.00

Drilling Fluids Only

Recalculate Cumulative Cost

8/21/2005

Daily Data Input – Drilling Fluids and Equipment Module

Tab #7 – Time Distribution

Time Distribution Last 24 hr	
Rig Up/Service	3
Drilling	16
Tripping	5
Non-Productive Time	
BOP Testing	24
BOP NU	
BOP Testing	
Cementing	
Condition Hole	
Condition Mud	
Coring	
Dir Survey	
Direction Work	
Fishing	
Lost Circ.	
Reaming	

Here you enter the amount of time (hrs) -for each rig activity- during the last 24 hours. Use the same “time distribution” data that’s reported for that day in the IADC sheet.

You can use the drop down menus to select even further categories.

DAILY KEY DATA

- Total Hours:** make sure the total time for each day adds up **24 hours**.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #7 – Time Distribution (continuation)

Time Distribution Last 24 hr	
Rig Up/Service	3
Drilling	16
Tripping	5
Non-Productive Time	
BOP Testing	
Cementing	
Total Hours	24

If a desired input is not on the drop-down list, new categories may be added here or in the General Setup tab (on the main ONE-TRAX screen).

Number	Description	Type
00001	Rig Up/Service	DF/CF
00002	Drilling	DF/CF
00003	Tripping	DF/CF
00004	Non-Productive Time	DF/CF
00005	BOP NU	DF
00006	BOP Testing	DF
00007	Cementing	DF
00008	Condition Hole	DF
00009	Condition Mud	DF
00010	Coring	DF
00011	Dir Survey	DF
00012	Direction Work	DF
00013	Fishing	DF
00014	Lost Circ.	DF/CF
00015	Reaming	DF
00016	Rig Repair	CF
00017	Running Casing	DF
00018	Testing	DF
00019	Wait on Cement	DF
00020	Wait on Weather	DF

This drop down menu is used to indicate if the specific category will be available for: the drilling fluids & equipment module (DF), the completion fluids module (CF), or both modules (DF/CF).

Daily Data Input – Drilling Fluids and Equipment Module

Tab #7 – Time Distribution (continuation)

Clicking on the 'Unscheduled Events' button will open the "Unscheduled Events" section of the program.

Daily Report [ABC Oil Company Gusher #2 12/14/1999]

5- Comments 6- Solids Equipment 7- Time Distribution
1- General 2- Pumps / Bits 3- Mud Properties

Time Distribution Last 24 hr

Rig Up/Service	3
Drilling	16
Tripping	5
Non-Productive Time	
BOP Testing	
Cementing	
Total Hours	24

Unscheduled Events

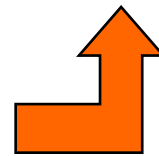
Time Distribution Setup

12/14/1999

Unscheduled Events

All list of Unscheduled Events

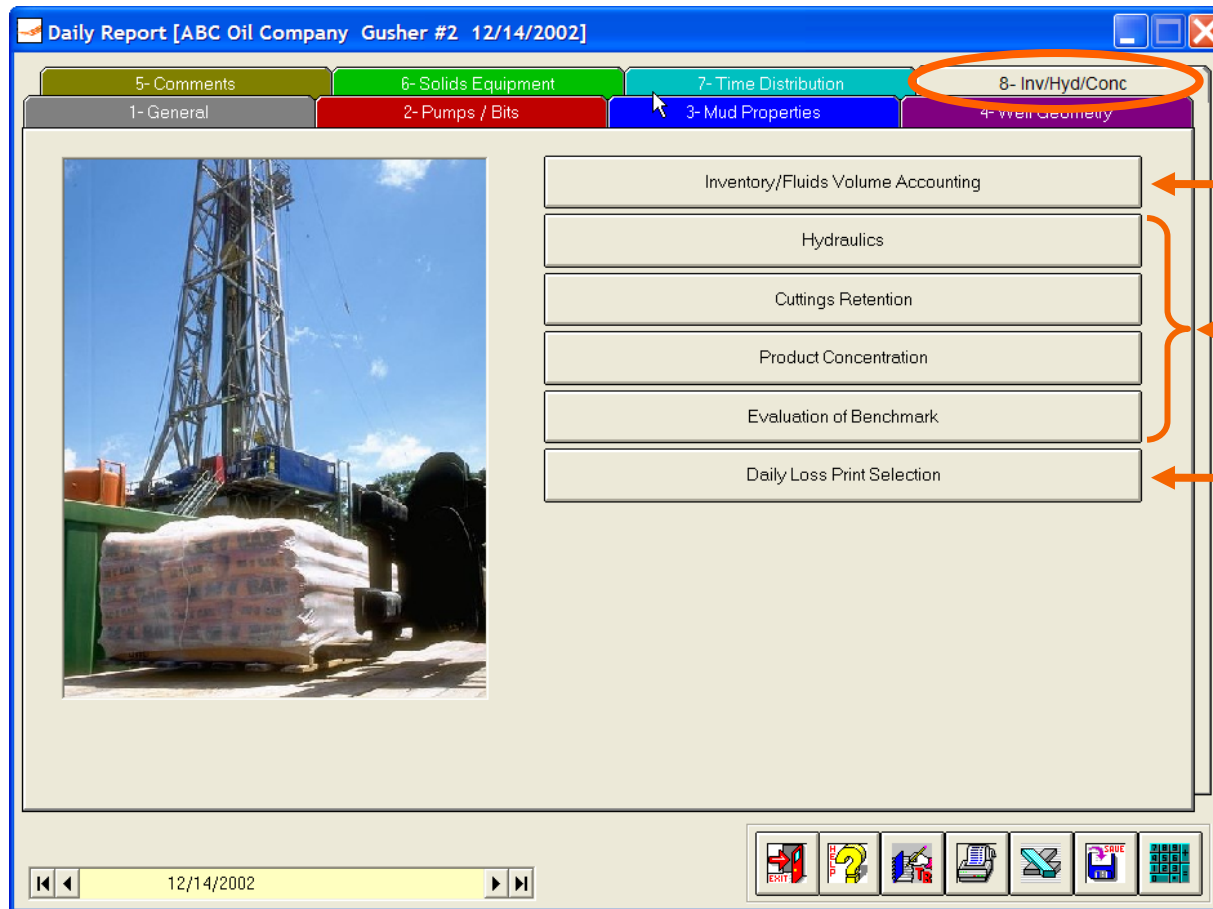
Type of Problem	Fluid Type	Description	Suspected Cause	Loss Description
Late Delivery	Spud Mud	Late delivery.	Port delay.	No lost time. Did NOT effect rig
Lost Circulation	KCL/Polymer	Partial lost returns.	High permeability formation.	Estimate 83 bbl seepage loss to
Lost Circulation	VERSACLEAN	Partial lost returns.	High permeability formation.	No lost time. Did NOT effect rig
Mechanical Failure	VERSACLEAN	Frame on ALS II #3	Metal fatigue.	Reduce flow rate and use remaining
Lost Circulation	Calcium	Lost CaCl2 brine to	Formation pressure less	Lost time spotting loss control pill.



Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – Inventory, Hydraulics and Concentrations

Tab 8 is made up of several independent sub-tabs, some of which are used only if required.



Data Entry section.

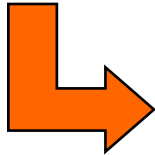
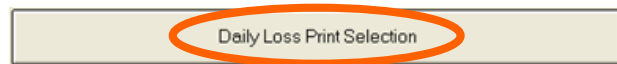
These four (4) buttons are used to **display** data (no data entry is done here).

Setup section.

Daily Data Input – Drilling Fluids and Equipment Module

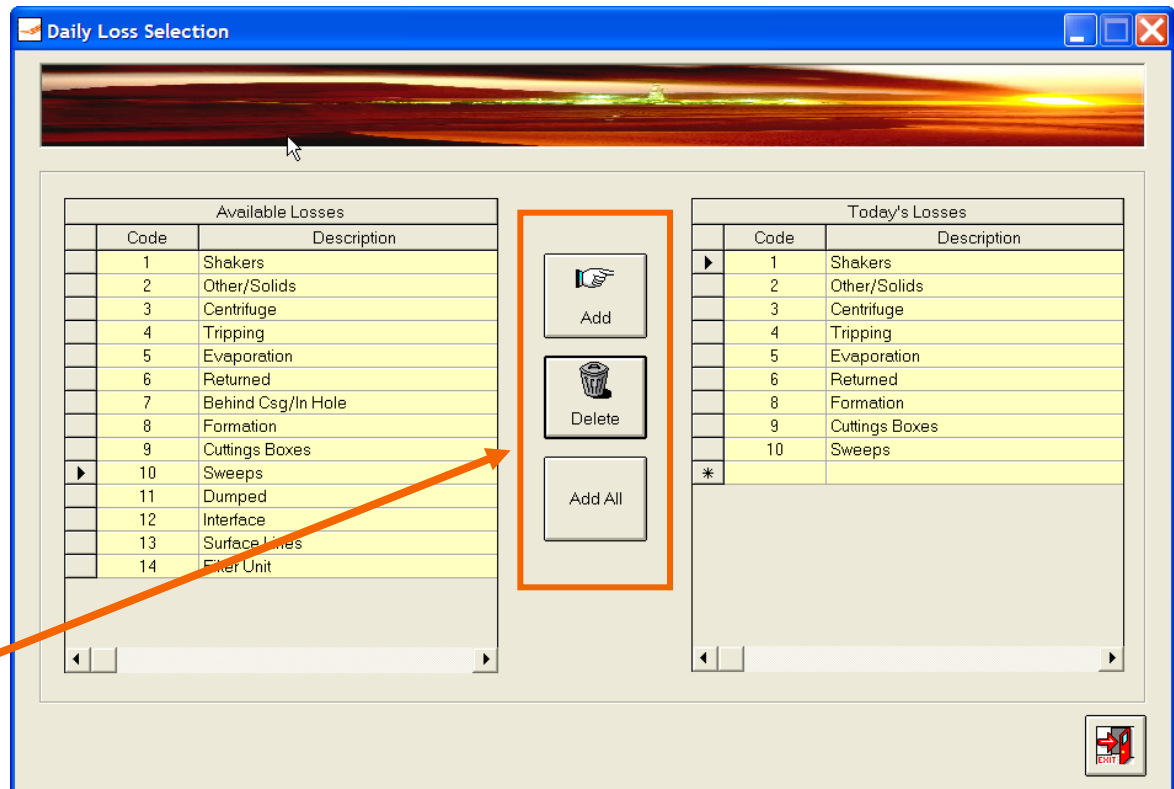
Tab #8 – ... / Daily Loss Print Selection

‘Today’s Losses’ may be modified so that relevant losses only show on the current report. These are also the categories that will be available for the “Volume Accounting” section and for your “Equipment Loss Type” categories (under “Equipment Performance”, tab #6).



As many as 20 loss categories are possible, BUT a maximum of 10 are allowed on the daily mud report.

Use these three (3) buttons to display (in the “Today’s Losses” list) the categories that you will need today.

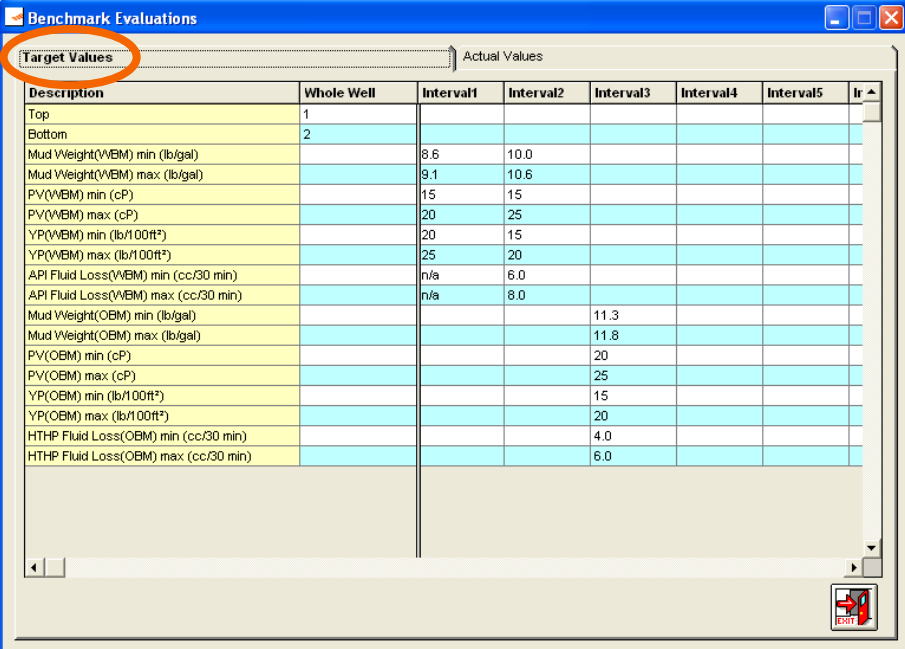


Daily Data Input – Drilling Fluids and Equipment Module

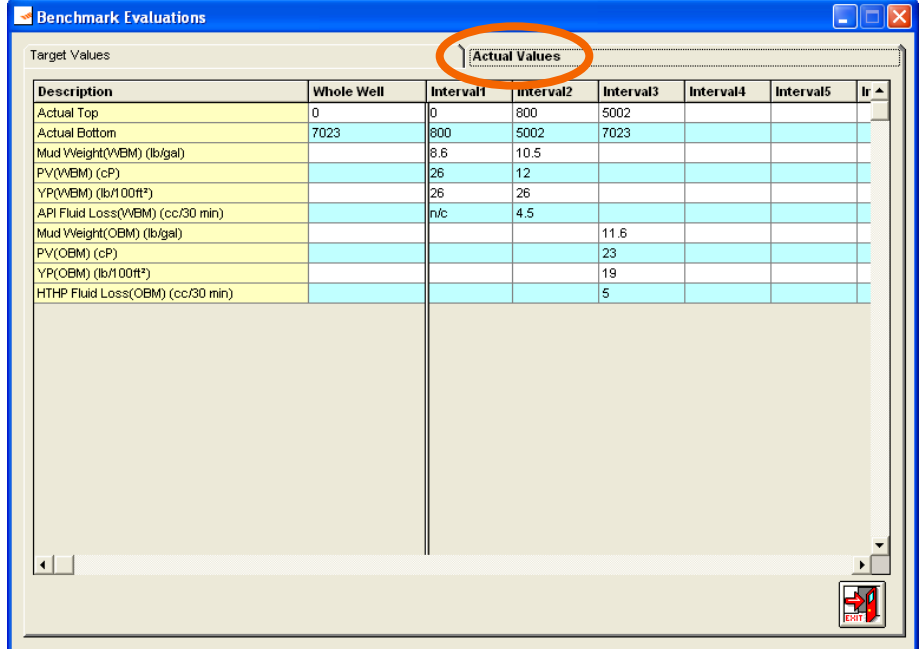
Tab #8 – ... / Benchmark Evaluations

Benchmark Evaluations

“Benchmark Evaluations” gives a visual picture of “What should be” (Target Values) vs. “What is (Actual Values)”.



Description	Whole Well	Interval1	Interval2	Interval3	Interval4	Interval5	Ir
Top	1						
Bottom	2						
Mud Weight(WBM) min (lb/gal)		8.6	10.0				
Mud Weight(WBM) max (lb/gal)		9.1	10.6				
PV(WBM) min (cP)		15	15				
PV(WBM) max (cP)		20	25				
YP(WBM) min (lb/100ft²)		20	15				
YP(WBM) max (lb/100ft²)		25	20				
API Fluid Loss(WBM) min (cc/30 min)		n/a	6.0				
API Fluid Loss(WBM) max (cc/30 min)		n/a	8.0				
Mud Weight(OBM) min (lb/gal)				11.3			
Mud Weight(OBM) max (lb/gal)				11.8			
PV(OBM) min (cP)				20			
PV(OBM) max (cP)				25			
YP(OBM) min (lb/100ft²)				15			
YP(OBM) max (lb/100ft²)				20			
HTHP Fluid Loss(OBM) min (cc/30 min)				4.0			
HTHP Fluid Loss(OBM) max (cc/30 min)				6.0			



Description	Whole Well	Interval1	Interval2	Interval3	Interval4	Interval5	Ir
Actual Top	0	0	800	5002			
Actual Bottom	7023	800	5002	7023			
Mud Weight(WBM) (lb/gal)		8.6	10.5				
PV(WBM) (cP)		26	12				
YP(WBM) (lb/100ft²)		26	26				
API Fluid Loss(WBM) (cc/30 min)		n/c	4.5				
Mud Weight(OBM) (lb/gal)				11.6			
PV(OBM) (cP)				23			
YP(OBM) (lb/100ft²)				19			
HTHP Fluid Loss(OBM) (cc/30 min)				5			

Target Values: where entered in the “Setup” section.

Actual Values: are calculated/extracted by ONE-TRAX

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Product Concentration

Product Concentration

Click on the '**Product Concentration**' button to view beginning and ending concentrations of any of the pits.

Product Concentration (8/21/2005)

Product Concentration - lb/bbl

Product Name	Start Conc.	End Conc.
MI BAR BULK	119.928	143.460
MI GEL BULK		
CALCIUM CHLORIDE		24.258
CAUSTIC SODA		
DUO-VIS		
GUAR GUM		
LIME	5.996	9.129
VERSAMUL	6.996	5.434
VERSACOAT		5.478
VERSAGEL	4.997	5.782
NUT PLUG		
POLYPAC UL		
POTASSIUM CHLORIDE		
SAFE-CARB 30		
SAFE-CARB 45		
SALT PVD		
SODAASH		

Pit Name: Active #1

Fluid Volume: 570

Options: 8.5 X 11 Letter, A-4 Paper

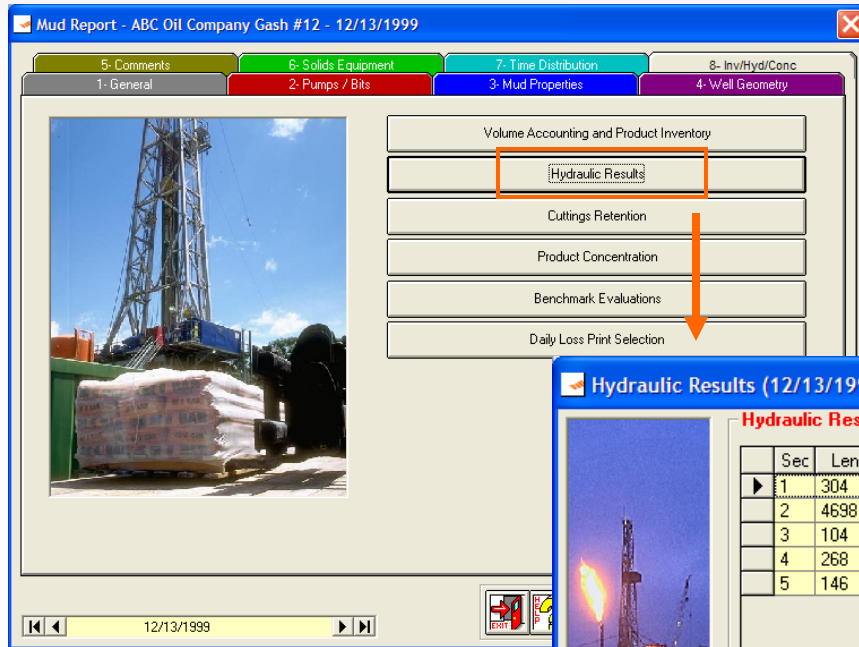
Report Type: Active, Pits

Use the drop down menu option to select the pit of interest.

These values are influenced by the daily transactions entered in the "Volume Accounting" section.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Hydraulic Results



When using the “API 5th edition” option (and provided that the additional required info was entered), ECD values are more accurate and three new parameters are now displayed: “AnnT”, “Ann PV”, Ann YP”.

Hydraulic Results (12/13/1999) - API 5th Edition



Hydraulic Results

	Sec	Length	Hole Dia	Pipe OD	Pipe ID	DS Loss	Ann Loss	MD	ECD	AnnT	AnnPV	AnnYP
▶ 1	304	19.500	5.000	4.276	20	2	304	11.65	104	23	23	
2	4698	12.515	5.000	4.276	974	58	5002	11.78	106	28	22	
3	104	12.553	5.000	4.276	94	3	5106	11.78	105	28	22	
4	268	12.553	8.500	3.000	195	8	5374	11.81	101	29	23	
5	146	12.553	9.500	3.000	115	7	5520	11.85	95	32	24	

DS Loss, psi

1462

Ann Loss, psi

79

Bit Loss, psi

1435

Total Loss, psi

2976

☐ Use API 4th Edition

☒ Use API 5th Edition







New Parameters

AnnT: annular temperature

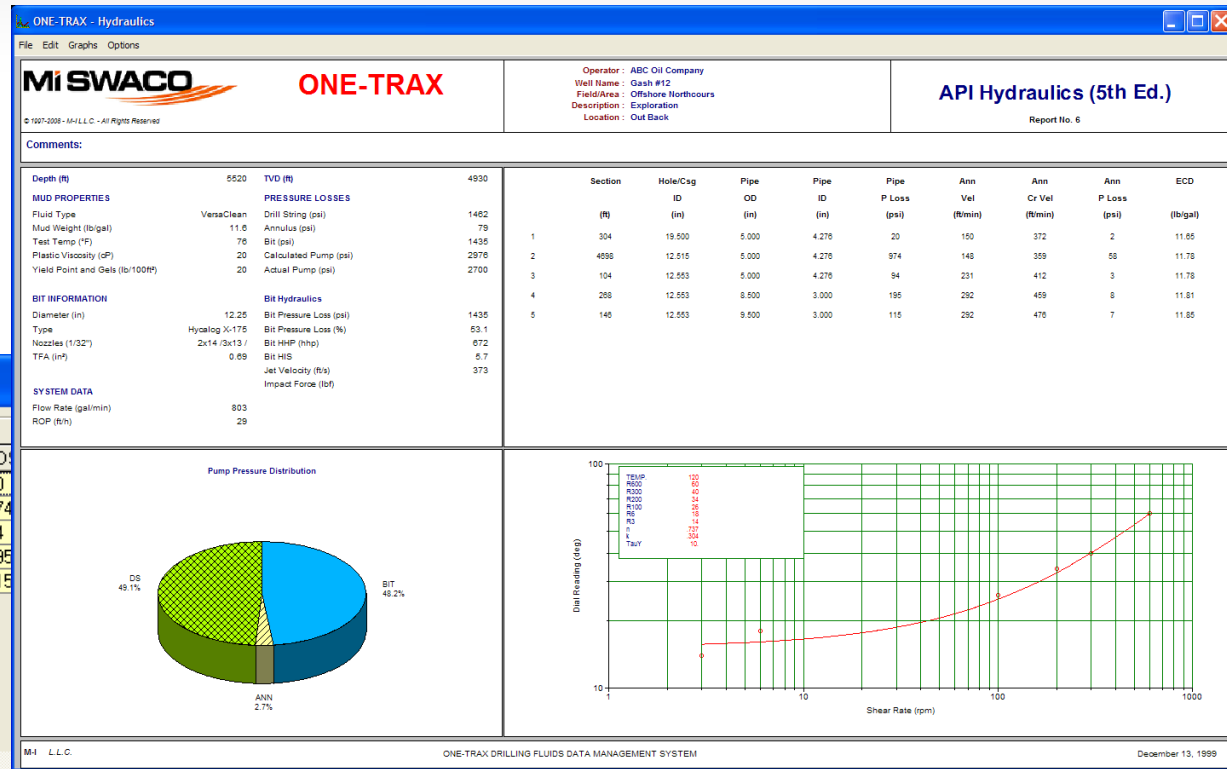
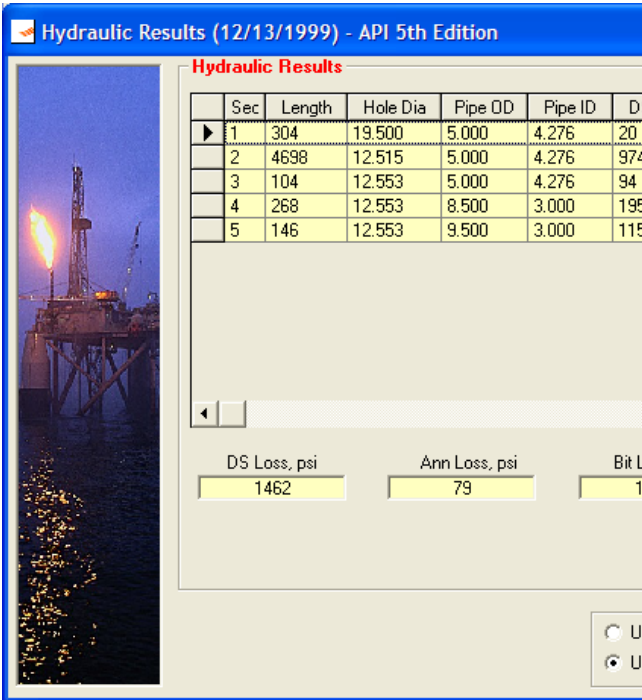
AnnPV: annular plastic viscosity

AnnYP: annular yield point

Tab #8 – ... / Hydraulic Results

Hydraulic Results

A basic hydraulics analysis is available by clicking on the **'Hydraulic Results'** button.



Back to: **Table of Content**



Pre-requisites for accurate Volume Accounting

Pre-requisite #1: “Depth” and “Bit Depth” values must be correct.

Make sure you don't miss the key data.

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

5- Comments
1- General

6- Solids Equipment
2- Pumps / Bits

7- Time Distribution
3- Mud Properties

8- Inv/Hyd/Conc
4- Well Geometry

Date	12/14/1999	Depth (ft)	7023	Fluid Type	VersaClean
Spud Date	12/7/1999	TVD (ft)	6486	Lithology	
Report	7	Bit Depth (ft)	7023		
Present Activity	Drig 12-1/4" hole			Lithology Setup	Well Survey

Operator Representative	John Able				
Contractor Representative	Peter Smart				
M-I SWACO Representative 1	Ian Scott	Rig Telephone Number	567-890-1234		
M-I SWACO Representative 2	David Jones	Warehouse Telephone	678-901-2345		
Telephones	123-456-7890 345-678-9012				
Pagers/FAX Numbers	234-567-8901 456-789-0123				



	DF Chem /Personnel	IFE/SC Engineer	Drilling Total	DF Equip. /Screens	Other Cost (DWM/CF)	Total
Daily Cost	48430.00	1450.00	49880.00	1774.00	7565.06	59219.06
Cumulative Cost	345735.50	5825.00	351560.50	6622.00	10464.29	368646.79

12/14/1999

EXIT ? HELP PRINT SAVE

DAILY KEY DATA

1. **Depth:** determines your well geometry, affecting: Downhole Vol., Tot. Vol., Depth Drilled, Cuttings Drilled & Cuttings Discharged.
2. **Bit Depth:** sets your drill string total depth, affecting Fluid Volumes, Bottom-up & Drill Pipe length.

Pre-requisite #2: “Bit Size” and “Washout” values **must** be correct.

Make sure you don't miss the key data.

Mud Report - ABC Oil Company Gash #12 - 12/14/1999

5- Comments 6- Solids Equipment 7- Time Distribution 8- Inv/Hyd/Conc
1- General 2- Pumps / Bits 3- Mud Properties 4- Well Geometry

Pump Data

Make and Model	Liner Diameter	Stroke Length	Rod Diam Duplex Only	Eff%	Pump Rate	Pump On Report	Riser Pump
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6.5	12		97	80	☒	☐
EMSCO F-1000	6	12		97		☐	☐
						☐	☐

Nozzles

Size	Quantity	TFA
14	x 2	0.69
13	x 3	
	x	

Extra ECD Input

Surface Code (1-5)	
Surface Pressure (psi)	
Ref. Flow Rate (gal/min)	
On/Off Bottom (psi)	

Bit and Drilling Data

Bit Description	Hycalog X-175	Bit Information
Bit Number	3	Bit Size/Hole Diameter 12.25
% Washout	5	Washout Hole Size 12.553

Rotary rpm 150 Pump Flow Rate 803
Rotating Hours 0 Pump Pressure 3000
Weight on Bit 24000 dP MWD
ROP 30 dP Motor
Riser Pump Flow Rate Motor RPM

12/14/1999

EXIT ? [Icons] SAVE [Icons]

DAILY KEY DATA

1. **% Washout:** is the % increase in volume of the open-hole section.

Note: **THIS IS NOT THE SAME VALUE GIVEN BY THE LOGGIN COMPANY.**

2. **Bit Size / Hole Diameter:** it determines the open hole geometry (together with the **Depth**), affecting: Downhole Vol., Tot. Vol., Depth Drilled, Cuttings Drilled & Cuttings Discharged.

Pre-requisite #3: “Drill String” and “Daily Casing/Volume” data **must** be complete and accurate.

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
DP	6609	5	4.276	6.375	3	
DC	268	8.5	3	9.5	3	
DC	146	9.5	3			

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number: 3
Print Order:

Pilot Hole

Hole Size: 0
Depth: 0

Make sure the data on this section is correct and complete.

The Well Geometry information **MUST** be completed properly before any Volume Accounting information is entered.

If a pilot hole was *previously* drilled, enter that data in the “pilot hole” frame.

Pre-requisite #4: properly estimate your solids control equipment fluid losses and assign to a loss category.

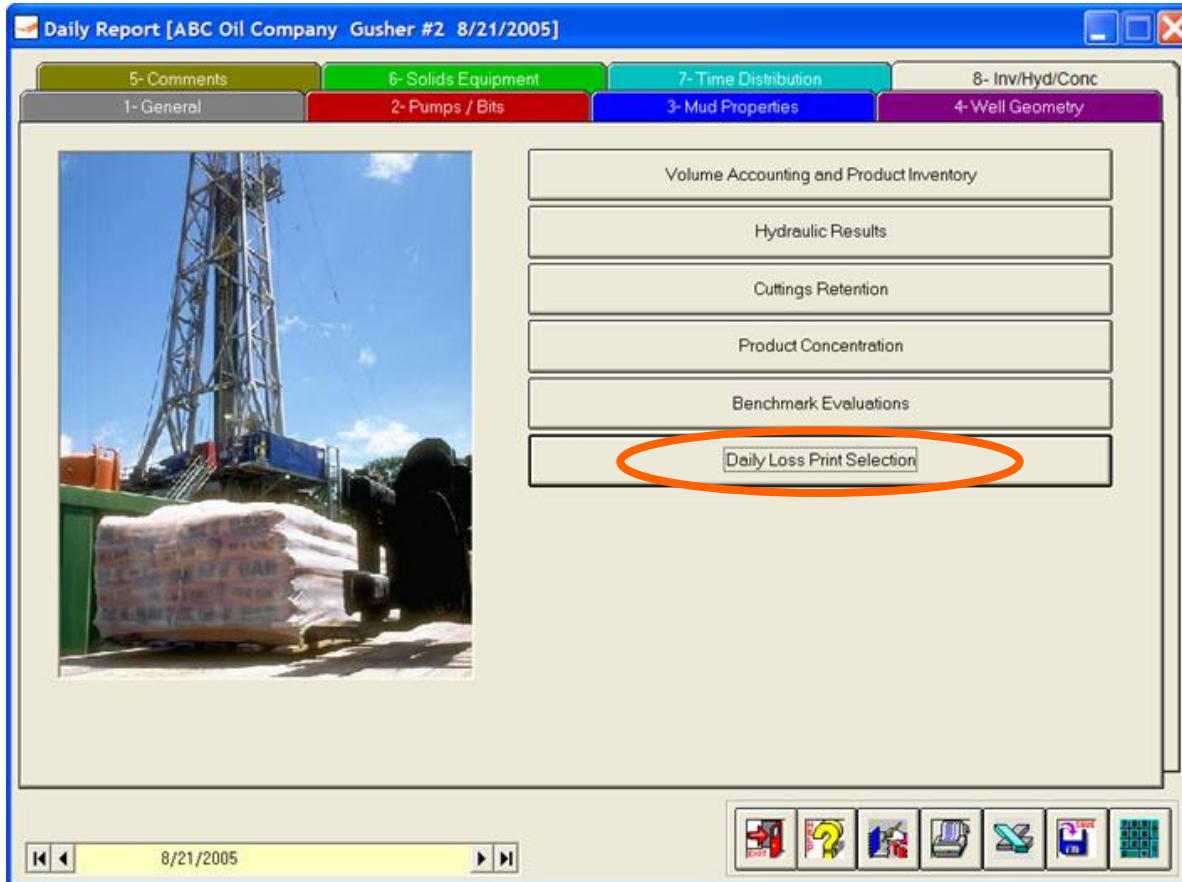
Provided that you enter all the key data for each equipment type, ONE-TRAX will calculate the “Vol. Mud On Solids Discharged”: the amount of fluid discharged on your wet cuttings.

Equipment Properties	Unit	#1 - Primary
Size / Specifications		120/200
Hours turned on		24
Cum. Hrs turned on		51.0
Mud On Cuttings	bbl/bbl	1.5
Loss Type		Shakers
Vol Discharged	bbl	143.8
Cum.Vol Discharged	bbl	2612.8
Vol Mud On Solids Discharged	bbl	86.3
Cum.Vol Mud On Solids Discharged	bbl	1378.3
% of Cuttings		25
Deck Angle	°	
G-Force	G's	
Vibration Angle	°	
Type of Motion		
Flow over Shaker	%	
Discharge Rate	lb/min	
Average Beach Length	in	

That vol. of fluid lost will be assigned to the loss category chosen in **this** drop-down menu.

“Loss” transactions have to be entered in Tab#8, matching the volumes calculated in this screen.

Pre-requisite #5: select the “Daily Loss” categories used today.



Select the proper Daily Loss categories for today following the instructions previously explained on this presentation.

After completing the
mentioned pre-requisites,
you can continue with the
Volume Accounting

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Volume Accounting and Product Inventory

Here is where all our fluid volume transactions & product additions are entered.

Pit Description	Capacity	Start Volume	Calculated End Vol	Fluid Weight	Temperature	Type	End Volume	Solids%
Pit #1	600		1300			Active	455	
Pit #2	600					Active		
Pit #3	600					Active		
Pit #4	400		350			Reserve	350	
Pit #5	400		250			Reserve	250	
Pit #6	400					Reserve		
Trip Tank	50							
Sand Trap	100							
Slug Pit	100							
BaseOil-1	500							
Boat 1	1000							
Boat 2	4000							

Calc Value	Actual Values	Not Accounted for
Reserve Pits	600	0
Premix Pits	0	0
Active Pits w/o Hole	455	
Mud Volume In Hole	845	
Active Pits w/ Hole	1300	0

Type	Volume
Base Oil	
Brine	

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes

Volume Accounting and Product Inventory [12/13/1999]

Pit Volumes | Volume Information | Inventory

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Solids%
Pit #1	600		1300			Active	455	
Pit #2	600					Active		
Pit #3	600					Active		
Pit #4	400		350			Reserve	350	
Pit #5	400		250			Reserve	250	
Pit #6	400					Reserve		
Trip Tank	50							
Sand Trap	100							
Slug Pit	100							
BaseOil-1	500							
Boat 1	1000							
Boat 2	1000							

Total Weight of Fluids in Pits (lb) 0

Sum of Fluid Volumes (End Value) - bbl

	Calc Value	Actual Values	Not Accounted for
Reserve Pits	600	600	0
Premix Pits	0	0	0
Active Pits w/o Hole		455	
Mud Volume In Hole		845	
Active Pits w/ Hole	1300	1300	0

Sum of Other Pit Volume

Type	Volume
Base Oil	
Brine	

Add Chemicals | Add Whole Mud | Transfer/Loss | View Trans
 Products Concentration | Reset | Recalc Conc | Update Inventory/Cost

12/13/1999 Ready

The “Pit Volumes” tab is where all volume data & transactions are recorded.

In addition products used to pits are entered in here.

The primary idea of this tab accounting is to balance the end-of-day actual volume in the pits & hole with the calculated amount (after entering the daily volume transactions and product usage).

At the end of each day **these values always** have to be equal to zero.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes (continuation)

End-of-day **actual** volumes

Volume Accounting and Product Inventory [12/13/1999]

Pit Volumes Volume Information Inventory

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Solids%
Pit #1	600		1300			Active	455	
Pit #2	600					Active		
Pit #3	600					Active		
Pit #4	400		350			Reserve	350	
Pit #5	400		250			Reserve	250	
Pit #6	400					Reserve		
Trip Tank	50							
Sand Trap	100							
Slug Pit	100							
BaseOil-1	500							
Boat 1	1000							
Boat 2	4000							

Total Weight of Fluids in Pits (lb) 0

Sum of Fluid Volumes (End Value) - bbl

	Calc Value	Actual Values	Not Accounted for
Reserve Pits	600	600	0
Premix Pits	0	0	0
Active Pits w/o Hole		455	
Mud Volume In Hole		845	
Active Pits w/ Hole	1300	1300	0

Sum of Other Pit Volume

Type	Volume
Base Oil	
Brine	

Add Chemicals Add Whole Mud Transfer/Loss View Trans

Products Concentration Reset Recalc Conc Update Inventory/Cost

12/13/1999 Ready

DAILY KEY DATA

- Pit Type:** use this column to indicate the category of each pit (e.g. Active, Reserve, Premix, Base Oil, Brine, Pill, Breaker, etc.) according to the "pit setup" section.
- End Volume:** manually enter in this column the **actual** final volume that you see in each pit at the end of the day.

These are the **REAL** fluid volumes on the pits so **DO NOT** change these values to balance your accounting.

Daily transactions used to determine the end-of-day **calculated** volumes.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes (continuation)

Use the “**Add Chemical**”, “**Add Whole Mud**”, “**Transfer/Loss**” buttons to enter volume and product addition transactions. These transactions will automatically update the products usage in your inventory.

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Solids%
Pit #1	600		1300			Active	455	
Pit #2	600					Active		
Pit #3	600					Active		
Pit #4	400		350			Reserve	350	
Pit #5	400		250			Reserve	250	
Pit #6	400					Reserve		
Trip Tank	50							
Sand Trap	100							
Slug Pit	100							
BaseOil-1	500							
Boat 1	1000							
Boat 2	4000							

Calc Value	Actual Values	Not Accounted for	
Reserve Pits	600	600	0
Premix Pits	0	0	0
Active Pits w/o Hole		455	
Mud Volume In Hole		845	
Active Pits w/ Hole	1300	1300	0

Type	Volume
Base Oil	
Brine	

Buttons: Add Chemicals, Add Whole Mud, Transfer/Loss, View Trans, Products Concentration, Reset, Recalc Conc, Update Inventory/Cost

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes / Add Chemicals

Add Chemicals

Inventory Detail

No.	Description	Code	Size	Unit	Packing	Quantity
2	CALCIUM CHLORIDE (94%)	M0005108	50	LB	BG	36
3	Calcium Chloride (11.6 ppg)	M0001210	1	BL	BK	0
4	CAUSTIC SODA	M0006659	50	LB	BG	0
5	DF-1 Base Oil	M0001715	1	BL	BK	25
6	DUOVIS	M0007915	50	LB	BG	0
7	GUAR GUM	M0006520	50	LB	BG	0
8	LIME	M0004111	50	LB	BG	0
9	MH BAR BULK	M0006011	1	TN	BK	3.6
10	MH GEL BULK	M0000321	1	TN	BK	0
11	VERSAMUL	M0002103	200	LT	DM	0
12	VERSACLEAN MUD (11.5 ppg)	M0002104	1	BL	BK	0
13	VERSACLEAN MUD (11.7 ppg)	M0002105	1	BL	BK	0
14	VERSACOAT	M0002106	200	LT	DM	0
15	VERSAGEL	M0002107	50	LB	BG	0
16	NUT PLUG	M0006401	50	LB	BG	0
17	POLYPAC UL	M0006211	50	LB	BG	0
18	Potassium Chloride Brine	M0006855	1	BL	BK	0
19	POTASSIUM CHLORIDE	M0006853	50	LB	BG	0
20	SAFECARB 30	M0000965	1	TN	BK	0
21	SALT (PVD)	M0000963	1	TN	BK	0
22	SODA ASH	M0004181	100	LB	BG	0
23	WATER-BASED MUD	M0000015	1	BL	BK	0
24	2 Micron ABS Filter Cartridge	P0012257	1		EA	0
25	Cartridge Filter Unit	P00122562	1		EA	0
26	MUD ENGINEERING SERVICE	R0088110	1		EA	0
27	SOLIDS CONTROL ENGINEER	R00881101	1		EA	0
28	IFE ENGINEERING SERVICE	R00881102	1		EA	0
29	SAFECARB 45	M0000960	50	LB	BG	0
30	SAFE-VIS E	M0001686	5	GA	CN	0
31	DIATOMACEOUS EARTH	M0007844	50	LB	BG	0
33	Calcium Chloride Spike	M0001210a	1	BL	BK	0
34	Delayed LTOPM 11.7 ppg	M0006000	1	BL	BK	0

Add Materials to Pit: Active-1

Add Base Fluid (bbl): 25

Add Water (bbl):

Volume of Chemicals (bbl): 8.12



1.- Select the pit you are adding products to.

2.- Enter the amount of Base Oil and or Water added to the pit.

3.- Enter the quantity of each product added to the pit.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes / Add Whole Mud

1.- When adding “whole mud” the **products concentration** is what’s entered in this table.

Add Whole Mud

Add Whole Mud

No.	Code	Description	Prod Conc
2	M0005108	CALCIUM CHLORIDE (94%)	34
3	M0001210	Calcium Chloride (11.6 ppg)	0
4	M0006659	CAUSTIC SODA	0
5	M0001715	DF-1 Base Oil	0
6	M0007915	DUOVIS	0
7	M0006520	GUAR GUM	0
8	M0004111	LIME	0
9	M0006011	MH BAR BULK	120
10	M0000321	MH GEL BULK	0
11	M0002103	VERSAMUL	6
12	M0002104	VERSACLEAN MUD (11.5 ppg)	0
13	M0002105	VERSACLEAN MUD (11.7 ppg)	0
14	M0002106	VERSACOAT	7
15	M0002107	VERSAGEL	8
16	M0006401	NUT PLUG	0
17	M0006211	POLYPAC UL	0
18	M0006855	Potassium Chloride Brine	0
19	M0006853	POTASSIUM CHLORIDE	0
20	M0000965	SAFECARB 30	0
21	M0000963	SALT (PvD)	0
22	M0004181	SODA ASH	0
23	M0000015	WATER-BASED MUD	0
24	P0012257	2 Micron ABS Filter Cartridge	0
25	P00122562	Cartridge Filter Unit	0
26	R0088110	MUD ENGINEERING SERVICE	0
27	R00881101	SOLIDS CONTROL ENGINEER	0
28	R00881102	IFE ENGINEERING SERVICE	0
29	M0000960	SAFECARB 45	0
30	M0001686	SAFE-VIS E	0
31	M0007844	DIATOMACEOUS EARTH	0

Ready

Add Whole Mud to Pit: Active-2

Product List No. for the Whole Mud: 12

Received from: Warehouse(New)

Volume (bbl): 230

Fluid Weight (lb/gal): 11.5



2.- Select the pit you are adding products to.

3.- Select the “whole mud” product number from the drop-down list.

4.- Specify if the whole mud brought from the warehouse or from another rig.

5.- Enter here the fluid volume added and the fluid weight.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes / “Transfer/Loss”

The screenshot shows the 'Transfer Fluids' dialog box. The 'Pit Description' dropdown is set to 'Active-2'. The 'To' dropdown is set to 'Reserve-3'. The 'Volume (bbl)' field contains the value '65'. At the bottom, the 'Between Pits' radio button is selected and highlighted with a black rectangle. There are also 'Returned' and 'Loss and Dumped' radio buttons, and two small icons at the bottom right.

There are three options on this section:

1.- Between Pits:

“Pit Description”: is used to designate which pit the volume will be moved from.

“To”: designate the receiving pit for the volume transfer.

The screenshot shows the 'Transfer Fluids' dialog box. The 'Pit Description' dropdown is set to 'Active-1'. The 'Return to' dropdown is set to 'Warehouse'. The 'Volume (bbl)' field contains the value '450'. At the bottom, the 'Returned' radio button is selected and highlighted with a black rectangle. There are also 'Between Pits' and 'Loss and Dumped' radio buttons, and two small icons at the bottom right.

2.- Returned:

“Pit Description”: same as above.

“Returned To”: designate the destination (rig, warehouse) of the volume returned.

The screenshot shows the 'Transfer Fluids' dialog box. The 'Pit Description' dropdown is set to 'Active-1'. The 'Loss Type' dropdown is set to 'Centrifuge'. The 'Volume (bbl)' field contains the value '14'. At the bottom, the 'Loss and Dumped' radio button is selected and highlighted with a black rectangle. There are also 'Between Pits' and 'Returned' radio buttons, and two small icons at the bottom right.

3.- Loss & Dumped.

“Pit Description”: same as above.

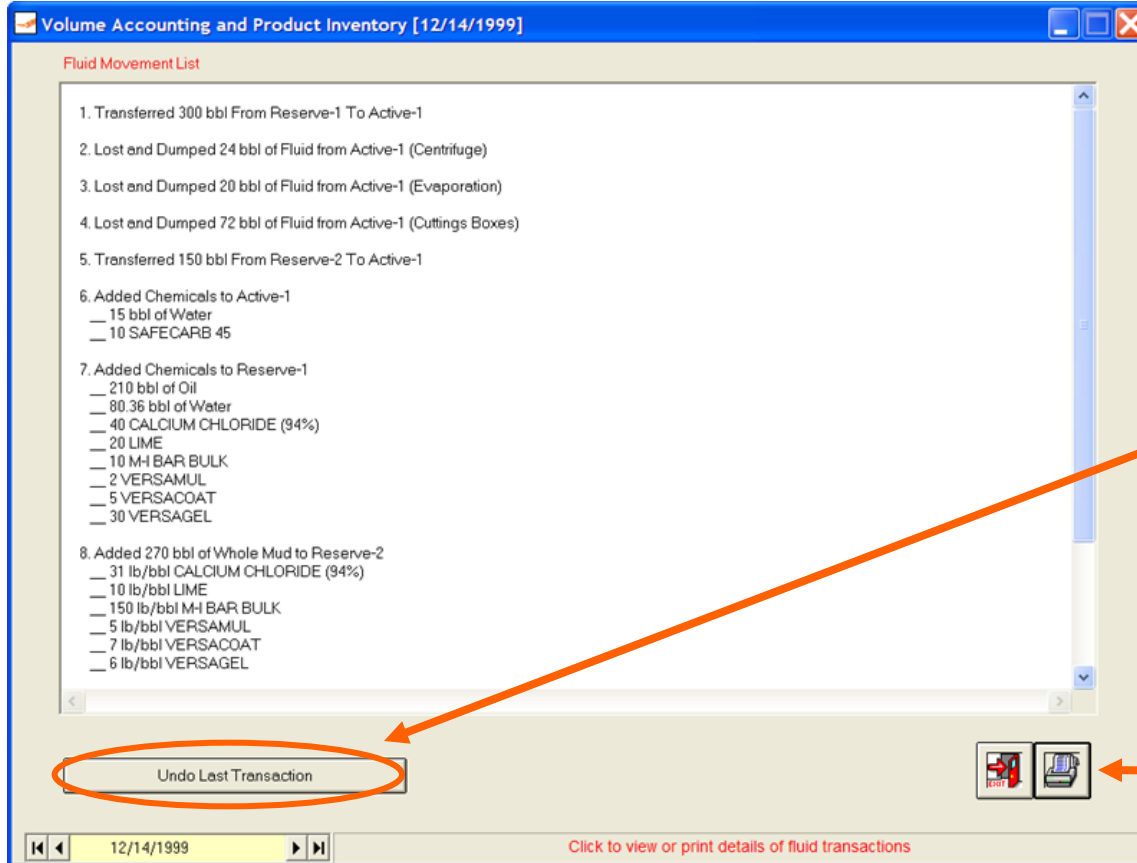
“Loss Type”: used to indicate the mud loss category.

In all three cases the “Volume” box is used to record the volume for the transaction.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes / View Transactions

ONE-TRAX keeps a volume accounting transaction log. This log can be viewed by clicking the “**View Trans**” button on the main volume accounting screen.



If you made a mistake entering a transaction, use this button to delete the last transaction done.

Print the transaction log.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Pit Volumes (special note balancing volumes)

These two values **DO NOT HAVE** to be equal

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Solids
Pit #1	600		1300			Active	455	
Pit #2	600		600			Active	600	
Pit #3	600		600			Active	600	
Pit #4	400		350			Reserve	350	

Reserve Pits	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Solids
Reserve Pits	600		600					
Premix Pits	0		0					

This is the “Active **System**” calculated end volume, which includes:

- Volume in the Hole
- Volume in the Surface Active Pits.

This is the “Active **Pit**” actual end volume, which includes:

- Volume in the Surface Active Pits.

As you can see the amount on the left includes “Volume in the Hole” and the value on the right doesn’t include it, so as long as you have some of your active system volume in the hole these two numbers **will not** agree.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Volume Information

Enter here how much volume (in the Annulus, Drill String & Below Bit) **IS NOT** occupied by your Active System's fluid.

Volume Accounting and Product Inventory [12/13/1999]

Pit Volumes **Volume Information** Inventory

Fluids in Hole - bbl

	Annulus	Total DS	Below Bit	Total
Total Hole Volume	751	94	0	845
Volume Not Fluids	0	0	0	0
Fluid Volume	751	94	0	845

Fluids Balance - bbl

	Active System	Reserve Pits	Premix Pits	Total
Start Volume	0	0	0	0
Oil Added	0	0	0	0
Water Added	0	0	0	0
Volume Chemicals Added	0	0	0	0
Chemical Not For Fluids	0	0	0	0
Total Volume Built	0	0	0	0
Received	1400	600	0	2000
Back Load	0	0	0	0
From Active to		0	0	
From Reserve to	0		0	
From Premix to	0	0		
Loss and Dumped	100	0	0	100
Final Fluid Volume	1300	600	0	1900

Total Loss Break Down

#	Description	Active Loss	Reserve Loss	Premix Loss	Sub Total	Discharge Volume
1	Shakers	78	0	0	78	
2	Other/Solids	0	0	0		
3	Centrifuge	0	0	0		
4	Tripping	0	0	0		
5	Evaporation	0	0	0		
6	Never Used	0	0	0		
8	Cuttings Boxes	22	0	0	22	
10	Returned	0	0	0		
					Total Sub-Surface Loss	22
					Total Surface Loss	78
					Total Loss	100

12/13/1999 Ready

This is a summary of all the volume information at report closing time.

This table displays the volume of fluid lost according to the data entered on the "Equipment Performance" section of tab #6.

This information should be used as reference when entering the "Transfer/Loss" transactions on the previous tab ("Pit Volumes").

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Inventory

In this screen the quantities on the blue or yellow cells come from other sections of the program.

- “Used for Fluid” values come from “Add Chemicals” and/or “Add Whole Mud”.
- “Received” and “Return” are updated through Delivery/Return Tickets entries (see next slides).

Volume Accounting and Product Inventory [12/13/1999]

Pit Volumes Volume Information **Inventory**

No.	Description	Code	Gross Price	Start	Received	Returned	Used for WMgt	Used for Fluid	Used Other Fluid Module	Adjustment
22	SODA ASH	M0004181	25.5	65			0			
23	WATER-BASED MUD	M0000015	52.5	1120			0			
24	2 Micron ABS Filter Cartridge	P0012257	30	0			0			
25	Cartridge Filter Unit	P00122562	120	0			0			
26	MUD ENGINEERING SERVICE	R0088110	450	0			0		2	
27	SOLIDS CONTROL ENGINEER	R00881101	400	0			0		1	
28	IFE ENGINEERING SERVICE	R00881102	525	0			0		2	

Cost Summary

	Daily (\$)	Cum. (\$)
Chemical	165000.00	291005.50
Mud Engineer	900.00	6300.00
S.C. Engineer	400.00	2800.00
IFE Engineer	1050.00	1575.00
Sub Total	167350.00	301680.50

Chemical Weight (lb) 1212095
Volume Chemicals for Fluids 2000.

Last Ticket No.

From Warehouse	G9962005
To Warehouse	ABC
From Other Well	
To Other Well	

Delivery/Return Tickets 

Add Chemicals Add Whole Mud Transfer/Loss View Trans
Products Concentration [Reset](#) [Recalc Conc](#) Update Inventory/Cost

12/13/1999 Ready

Only white cells can be edited in this screen.

- **Used Other Fluid Module:** this column is used to enter the personnel usage for each day (see example here).

- **Adjustment:** occasionally used when minor adjustments need to be made to the inventory.

View or Print the inventory report.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Inventory (continuation)

Volume Accounting and Product Inventory [12/13/1999]

Pit Volumes Volume Information Inventory

Daily Inventory									
No.	Description	DT_Qty	RT_Qty	On Order	Final	Daily Used	Daily Cost	Weight	Not Print
22	SODA ASH				65	0	0.00	600	☐
23	WATER-BASED MUD				1120	0	0.00	415.54	☐
24	2 Micron ABS Filter Cartridge				0	0	0.00		☒
25	Cartridge Filter Unit				0	0	0.00		☒
26	MUD ENGINEERING SERVICE				0	2	900.00		☐
27	SOLIDS CONTROL ENGINEER				0	1	400.00		☐
28	IFE ENGINEERING SERVICE				0	2	1050.00		☐

Cost Summary

	Daily (\$)	Cum. (\$)
Chemical	165000.00	291005.50
Mud Engineer	900.00	6300.00
S.C. Engineer	400.00	2800.00
IFE Engineer	1050.00	1575.00
Sub Total	167350.00	301680.50

Chemical Weight (lb) 1212095
Volume Chemicals for Fluids 2000.

Last Ticket No.

From Warehouse	G9962005
To Warehouse	ABC
From Other Well	
To Other Well	

Delivery/Return Tickets 

Add Chemicals Add Whole Mud Transfer/Loss View Trans
Products Concentration Reset Recalc Conc Update Inventory/Cost

12/13/1999 Ready

Scroll to the far right end and click on the 'Not Print' boxes for any inventory you do not want shown on a printed report.

Daily Data Input – Drilling Fluids and Equipment Module

Tab #8 – ... / Inventory / Delivery & Return Tickets

Delivery/Return Tickets

Ticket Transactions

Transfer Ticket Summary

Type	Ticket No.	Order By	Receive By	Warehouse Name	Warehouse
Receive from Warehouse	G9962003	John Small	John Small	Base Camp Warehouse	03569
Receive from Warehouse	G9962004	John Small	John Small	Cameron	00268
Receive from Warehouse	G9962005	John Small	John Small	Port Fouchon	00125

Click the "Add New" button, enter the information in the 'Transfer Ticket Summary' section, and fill in the 'Transfer Ticket detail' table.

Product Description	Size	Unit	Package	Price	Amount from Ticket	Actual Amount
VERSAGEL	50	LB	BG	71		
NUT PLUG	50	LB	BG	18.5	80	80
POLYPAC UL	50	LB	BG	123		
Potassium Chloride Brine	1	BL	BK	9.25	15	15
POTASSIUM CHLORIDE	50	LB	BG	8.5		
SAFECARB 30	1	TN	BK	265		
SALT (FVD)	1	TN	BK	156.5		
SODA ASH	100	LB	BG	25.5		
WATER-BASED MUD	1	BL	BK	52.5	80	80
2 Micron ABS Filter Cartridge	1		EA	30		
Cartridge Filter Unit	1		EA	120		
SAFECARB 45	50	LB	BG	7.5		
SAFE-VIS E	5	GA	CN	210		
DIATOMACEOUS EARTH	50	LB	BG	7.5		

- **Amount from ticket:** this is the quantity received according to the delivery ticket.

- **Actual Amount:** this is the actual quantity of product received on-site.

End of “Drilling Fluids and Equipment” module

ONE-TRAX[®] Wellsite

Reservoir Drill-in Fluids Module

Daily Data Input – Reservoir Drill-in Fluids Module

The screenshot shows a software window titled 'File Name: SAMPLE.MDB'. It contains a table with the following data:

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Below the table is a section titled 'Well Information' with several buttons: Well Header Information, Loss Setup, Well Casing Intervals (Cost), Active Products/Equipment/Screens, Well Survey, Benchmark Setup, Pit Information, Equipment Properties Setup, General Setup, Unscheduled Events, and Cost Overview.

Below this is a section titled 'ONE-TRAX Modules' with buttons for Drilling Fluids and Equipment, RDF Module (highlighted with a red box), Completion Fluids, and Waste Treatment and Disposal. There is also a 'Recap Reports' button and an 'Activate GRAFX' button with a small icon.

Advantages of the new “RDF” module

- Provides the ability to track displacements and drilling within the same module.
- Targets common RDF fluid properties, as part of the daily “mud” check (e.g. PPT, PSV, LSRV, etc).
- Records valuable information about the type of completion: cased-hole, open hole gravel pack, open hole pre-packed screen, etc.
- Documents key parameters with regards to RDF and/or completion related operations (e.g.: contact time during displacement, delay time, losses after breaker, type of breaker used, RDF fluid used, etc).
- Keeps track of RDF costs separately from the regular DF cost.

Daily Data Input – Reservoir Drill-in Fluids Module

Additional Setup

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
Cost Overview	

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports

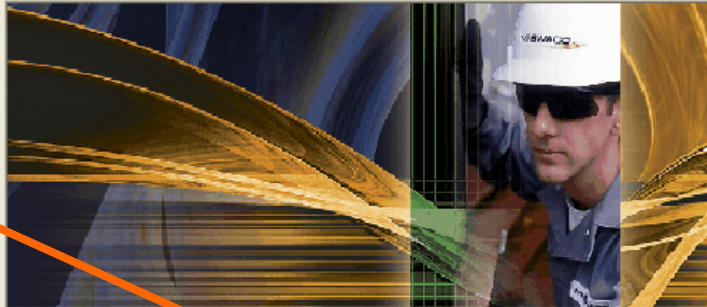
Activate GRAFX

EXIT

There are key data that needs to be entered in the “Well Completion Information” section.

Reservoir Drilling Fluids

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield



Daily RDF

New	Browse
Edit	Delete

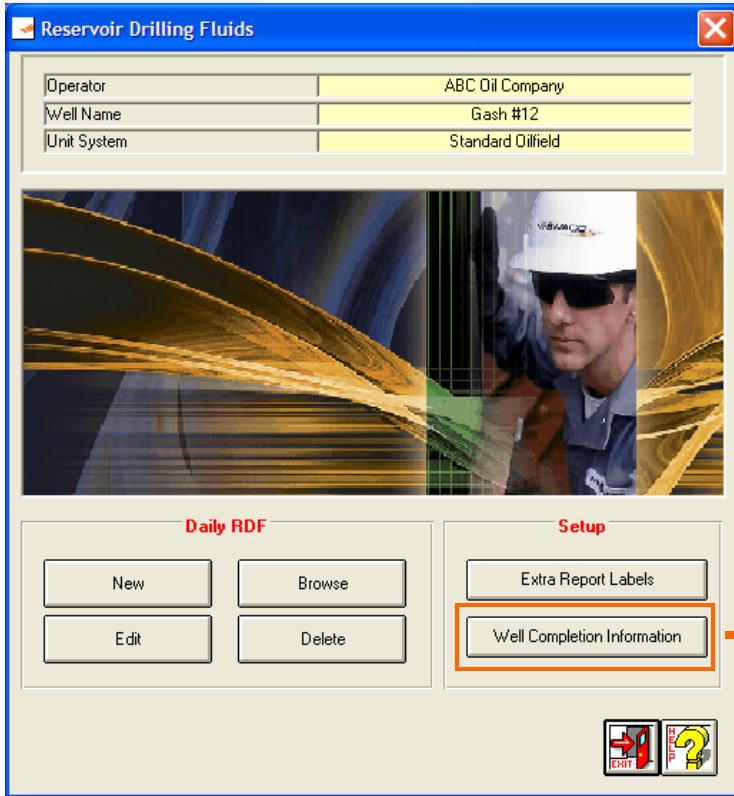
Setup

Extra Report Labels
Well Completion Information

EXIT ?

Daily Data Input – Reservoir Drill-in Fluids Module

Additional Setup



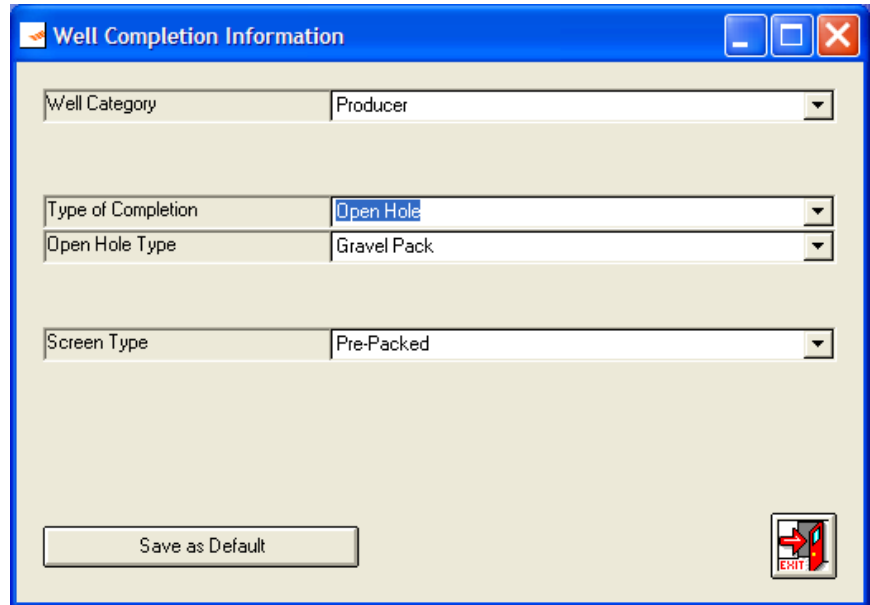
The "Reservoir Drilling Fluids" window features a blue title bar and a close button. It contains a table with the following data:

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

Below the table is a large image showing a worker in a hard hat and safety glasses. Underneath the image are two sections: "Daily RDF" with buttons for "New", "Browse", "Edit", and "Delete"; and "Setup" with buttons for "Extra Report Labels" and "Well Completion Information". The "Well Completion Information" button is highlighted with an orange border. At the bottom right are "EXIT" and "HELP" icons.

Use the drop down boxes to provide the key data required on this section.

It is extremely important to complete this section in order to build a useful database that can be queried on specific well completion criteria.



The "Well Completion Information" window has a blue title bar and standard window controls. It contains three dropdown menus:

- Well Category: Producer
- Type of Completion: Open Hole
- Open Hole Type: Gravel Pack

Below these is a "Screen Type" dropdown menu set to "Pre-Packed". At the bottom is a "Save as Default" button and an "EXIT" icon.

#1 – General

Tab #1 is for general information and current depth data.

The screenshot shows the 'RDF Fluids Report' window for 'ABC Oil Company' on '12/17/1999'. The 'General' tab is selected, showing various data entry fields and a cost summary table.

RDF Fluids Report [ABC Oil Company Gash #12 12/17/1999]			
6- Displacement 7- Time Distribution 8- Solids Equip 9- Inv/Vol Acct.			
1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information			
Date	12/17/1999	Depth (ft)	8002
Report	10	TVD	7516
Bit Depth	7490	PBTD - MD (ft)	
Operator Representative		John Able	
Contractor Representative		Peter Smart	
M-I SWACO Representative 1	Ian Scott	Rig Telephone Number	567-890-1234
M-I SWACO Representative 2	David Jones	Warehouse Telephone	678-901-2345
Telephones	123-456-7890	Pagers/FAX Numbers	234-567-8901
	345-678-9012		456-789-0123
Daily/Cum RDF Fluid Cost	0.00 /	0.00	
Daily/Cum RDF Engineering Cost	0.00 /	0.00	
Daily/Cum S.C. Engr Cost	0.00 /	0.00	
Daily/Cum IFE Engr Cost	0.00 /	0.00	
Daily/Cum RDF Screen Cost	0.00 /	0.00	
Daily/Cum RDF Equipment Cost	0.00 /	0.00	
Daily/Cum Total RDF Cost	0.00 /	0.00	
		Cumulative Total Well Cost	
		399730.78	

DAILY KEY DATA

Yarayara

1. **Depth & PBTD-MD:** are used to calculate the correct hole volumes.
2. **Bit Depth:** sets your drill string total depth (affecting Fluid Volumes, Bottom-up & Drill Pipe length) and can not be greater than the PBTD-MD.

#2 – Pumps & Bits

The rig pump data (liner diameter, stroke length, % efficiency & pump rate) is used to calculate the Pump Flow Rate, but this figure may be over-written if needed.

RDF Fluids Report [ABC Oil Company Gash #12 12/17/1999]

6- Displacement 7- Time Distribution 8- Solids Equip 9- Inv/Vol Acet

1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Pump Data

Make and Model	Liner Diameter	Stroke Length	Rod Diam Duplex Only	Eff%	Pump Rate	Pump On Report	Riser Pump
EMSCO F-1000	6.5	12		97	100	☒	☐
EMSCO F-1000	6.5	12		97		☒	☐
EMSCO F-1000	6	12		97		☐	☐
						☐	☐

Bit and Drilling Data

Bit Description	Bit Information
Bit Number	Bit Size/Hole Diameter 8.681
% Washout	Washout Hole Size 8.68
Rotary rpm	Pump Flow Rate 502
Rotating Hours 0	Pump Pressure 2000
Weight on Bit	Riser Pump Flow Rate
ROP	

Nozzles

Size	Quantity	TFA
	x	0
	x	
	x	

12/17/1999

DAILY KEY DATA

- Pump Rate:** this is used to calculate the total Pump Flow Rate, which affects the Total Circulating Time, Bottom-up and some indexes (i.e. pressure losses, annular velocity, etc.)
- Pressures:** used to calculate pressure losses, ECD, etc.
- % Washout:** is the % increase in volume of the open-hole section.
- Bit Size / Hole Diameter:** it determines the open hole geometry (together with the Depth), affecting: Downhole Vol., Tot. Vol., Depth Drilled, Cuttings Drilled & Cuttings Discharged.

#3 – Fluid Properties

RDF Fluids Report [ABC Oil Company Gash #12 12/17/1999]

6- Displacement 7- Time Distribution 8- Solids Equip 9- Inv/Vol Acct.

1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Fluid Properties		#1 - Primary	#2	#3	#4
Gels 10sec	lb/100ft ²	13			
Gels 10min	lb/100ft ²	14			
Gels 30min	lb/100ft ²	15			
LSRV 1min	lb/100ft ²	16			
LSRV 2min	lb/100ft ²	17			
LSRV 3min	lb/100ft ²	18			
PPT	psi				
API Fluid Loss	cc/30 min	20			
Modified HTHP	cc/30 min	8			
Cake API	1/32"	21			
Cake HTHP	1/32"	6			
Disk Mean Pore Diameter	microns	21			
Disk Mean Pore Diameter	microns				
pH		23			
Chlorides	mg/L	23			
Hardness	mg/L	26			

Extra Properties

12/17/1999

EXIT ? [Icons] SAVE [Icons]

The built-in fluid parameters on this tab target common reservoir drill-in fluids properties.

Extra properties can be added, just like in the "Drilling Fluids & Equipment" module.

#4 – Well Geometry

List drill string components from surface to TD.

RDF Fluids Report [ABC Oil Company Gash # 2 12/17/1999]

6- Displacement 7- Time Distribution 8- Solids Equip 9- Inv/Vol Acct.
1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
DP	7490.	5	4.276	6.375	3.75	

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number 4

Print Order

12/17/1999

EXIT ? F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12

DAILY KEY DATA

- Interval (Cost) Number:** is used to indicate where will today's cost be allocated. For example: if the number in the box is equal to "3", it means that ONE-TRAX will consider all the cost generated by today's activity (rental, personnel, mud mixing, etc.) as part of the cost that belongs to the 3rd interval.

This number is critical to get each 'project interval cost' correctly calculated in the Recap report.

#4 – Well Geometry / Daily Casing Volume

Daily Casing / Volume



Clicking the “Daily Casing / Volume” button on tab #4 will open the “Well Geometry” screen (below), which has a new category under “Type”: **Production Screen**.

Daily Intervals									
#	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size	Mesh	Gauge
1	F	20	19	800	800		28		
2	F	13.375	12.515	5002	4750		17.5		
3	F	9.625	8.681	8002	7516		12.25		
4	P			8003				122	244

The last interval depth should be equal to 8002.

Side-Track Setup Help

Hole Volume Summary (bbl)

Drill String	133
Annulus	456
Below Bit	23
Total Volume	612
Drilled Hole	
Drilled Depth (ft)	

Hole Volume Detail

Bit Size/Hole Diameter	8.681	Air Gap	84
Washout Hole Size	8.681	Water Depth	220
Depth	8002	Riser ID	19.5
TVD	7516	Riser OD	20.5
Bit Depth	7490	Riser Length	304
Pilot Hole Depth	0	Riserless Drilling	☐
Pilot Hole Size	0		

When the “Production Screen” is entered as part of the well geometry, the screen **mesh** and **gauge** must be provided.

#5 – Well Information

Supplying the information requested here populates the daily report and displacement worksheets.

The screenshot shows the 'RDF Fluids Report' window for 'ABC Oil Company Gash #12' dated '12/17/1999'. The '5- Well Information' tab is selected. The interface is divided into several sections:

- Down Hole:** A table with columns 'Tool' and 'Depth(ft)'.
- Perforations:** A table with columns 'From(ft)' and 'To(ft)'.
- Engineering Data:** A table with rows for 'B. Hole Temp (*F)', 'B. Hole Pressure (psi)', and 'Max Deviation'.
- Remarks:** A large text area for notes.
- Comments:** A button located below the 'Engineering Data' section, highlighted with an orange box.

The bottom of the window features a date field set to '12/17/1999' and a row of icons for navigation and file operations.

The “**Remarks**” box near the bottom of the screen is used to report daily activity, etc.

The “**Comments**” button opens four cells for data input.

#6 – Displacements

A displacement screen (similar to the one in the “Completions Fluids” module) is included here.

Displacement

Displacement 1 of 1

Add New Displacement Remove the Last Displacement

Type of Displacement Direct Forward

Fluid being Displaced - Type CALDRIL

Fluid being Displaced - Density (lb/gal) 11

RDF Fluid Name Sea Water

Displacing Fluid/Pill Type Breakers

Displacing Fluid/Pill Detail

Displacing Fluid/Pill Density (lb/gal) 11.5

Total Cost of Displacement Chemicals 5000

Pipe Rotation Rate 200

Mud Displacement Interface (bbl) 100

Brine Displacement Interface (bbl) 200

Volume Brine Interface Reclaimed (bbl) 50

Max Allowable Pump Pressure (psi) 1000

Max Pump Pressure Achieved (psi) 800

Pump Time for Displacement (Hours) 3

Down Time During Displacement (Hours) 4

Rig Cleanup Time - Total (Hours) 7

Rig Cleanup Time - On-line Only (Hours) 2

☐ Short-trip in Mud

☐ Short-trip in Seawater

☐ Short-trip in Brine

☐ Mud Observed on w/s After Displacement

☐ Mud Observed on Brushes After Displacement

☒ Brine Interface Reclaimed

Spacer/Pill/Breaker Fluids Data

Description	Fluid	Fluid Type	Weight (lb/gal)	Location	YP	LSRYP	Pump Rate (gal/min)	Volume (bbl)	Velocity (ft/min)
sfдав	Spacer #1	WBM	11	abva	22	33	44	10	55
	Solids Free Pill	WBM	11	abad	12	33	150	7	120

To Add>

Edit

EXIT

These two sections are only present in the RDF module.

#6 – Displacements / “Spacers/Pill/Breaker Fluids Data”

Clicking on the “Spacer/Pill/Breaker Fluids Data” edit button shown in the previous slide will open this window, where specific data for spacers, breakers, pills is entered.

The screenshot shows a software window titled "Displacement" with a blue title bar. Inside the window, there is a header section with a dropdown menu showing "Displacement 1 of 1", and two buttons: "Add New Displacement" and "Remove the Last Displacement". Below this is a section for "Type of Displacement" (set to "Direct Forward") and "Fluid being Displaced - Type" (set to "CALDRIL"). To the right of these are two input fields: "Max Allowable Pump Pressure (psi)" set to "1000" and "Max Pump Pressure Achieved (psi)" set to "800".

Below the header is a section titled "Spacer/Pill/Breaker Detail" in red text. This section contains a table of input fields:

Pit Name	SW Leg	pH	
Description	sfdav	% EDTA	
Fluid	Spacer #1	Acid #	
Fluid Type	WBM	Starch	
Weight (lb/gal)	11	PST	
YP	22		
LSRYP	33		
Pump Rate (gal/min)	44		
Volume (bbl)	10		
Velocity (ft/min)	55		
Contact Time (Hours)	66		
Delay Time (Hours)	77		
Losses after breaker (bbl)	88		
R600			
R300			
R200			
R100			
R6			
R3			

To the right of the "Spacer/Pill/Breaker Detail" section is a "Concentration Detail" section with a table:

Product Name	Concentration

At the bottom right of the window is a red "EXIT" button with a right-pointing arrow.

#7 – Time Distribution

RDF Fluids Report [ABC Oil Company Gash #12 12/17/1999]

6- Displacement 7- Time Distribution 8- Solids Equip 9- Inv/Vol Acct.

1- General 2- Pumps & Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Time Distribution Last 24 hr

Rig Up/Service
Drilling
Tripping
Non-Productive Time
Kill Well
Monitor Well
Pickup Tubing
Gravel
Circ. & Filter
Displ.
Total
Kill Well
Monitor Well
Pickup Tubing
Gravel
Circ. & Filter
Displace Well
Filter Fluid
Set Packer
Perforate
my time

Unscheduled Events
Time Distribution Setup

Here you enter the amount of time (hrs) -for each rig activity- during the last 24 hours. Use the same “time distribution” data that’s reported for that day in the IADC sheet.

You can use the drop down menus to select even further categories.

DAILY KEY DATA

1. **Total Hours:** make sure the total time for each day adds up **24 hours**.

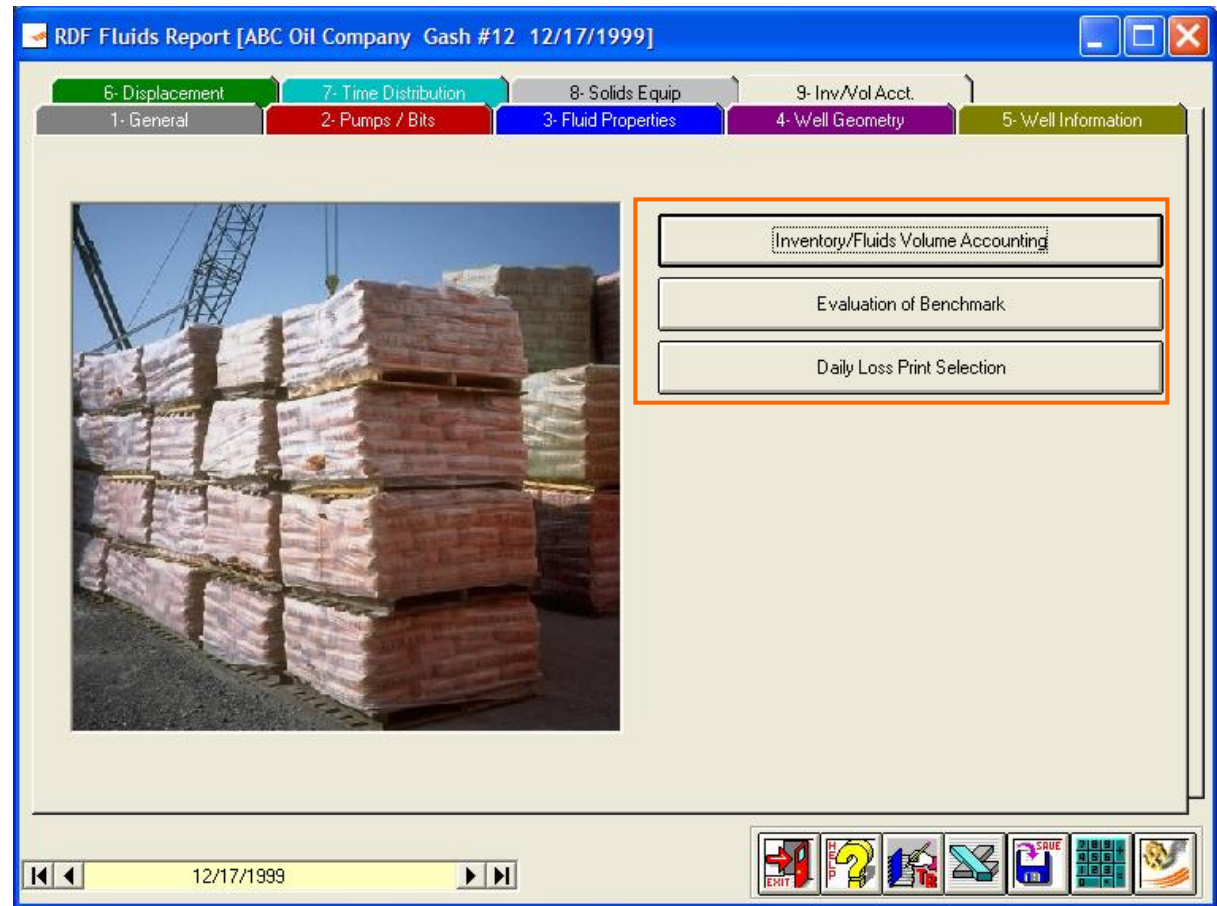
#8 – Solids Equipment

Fill in this section of the software just as it was indicated in the “Drilling Fluids and Equipment” module.

The screenshot shows the 'RDF Fluids Report' window for 'ABC Oil Company' and 'Gash #12' dated '12/17/1999'. The interface features a tabbed menu at the top with the following tabs: 1- General, 2- Pumps / Bits, 3- Fluid Properties, 4- Well Geometry, 5- Well Information, 6- Displacement, 7- Time Distribution, 8- Solids Equip (highlighted), and 9- Inv/Vol Acct. The main content area is divided into two sections. The left section contains a photograph of industrial equipment on a red platform. The right section, outlined in orange, lists six sub-modules: Shaker Screen Transactions, Shaker Screen Inventory, Equipment Details and Usage, SUS Compliance Testing, Cuttings Retention, and Equipment Solids Analysis. Below this, there is a section for 'IFE Remarks' and four summary/forward plan fields: Drilling Fluids Summary, Drilling Fluids Forward Plan, Solids-Control Summary, and Solids-Control Forward Plan. A 'DWM Extra Properties' button is located at the bottom right of the summary section. The bottom status bar shows the date '12/17/1999' and a series of navigation icons including EXIT, HELP, and various data entry tools.

#9 – Inventory & Volume Accounting

Use the same indications provided in the “Drilling Fluids and Equipment” module, to complete this section.



End of “Reservoir Drill-in Fluids” module

ONE-TRAX[®] Wellsite

Completion Fluids Module

Daily Data Input – Completion Fluids Module

Additional Setup

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield

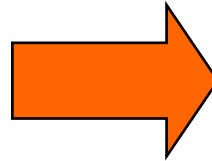
Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
Cost Overview	

ONE-TRAX Modules

Drilling Fluids and Equipment	RDF Module
Completion Fluids	Waste Treatment and Disposal

Recap Reports  Activate GRAFX 



These buttons work just like in the “Drilling Fluids and Equipment” module.

These buttons work just like in the “Reservoir Drill-in Fluids” module.

The “**Extra Report Labels**” section is explained in the following slide.

Completion Fluids

Operator	ABC Oil Company
Well Name	Gash #12
Unit System	Standard Oilfield



Daily Completion Fluids

New	Browse
Edit	Delete

Setup

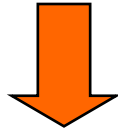
Extra Report Labels
Well Completion Information
Special Report
None

Daily Data Input – Completion Fluids Module

Additional Setup (continuation)

Extra Report Labels



Completion Fluids Setup

County, Parish or Offshore Area	Harris
State	Texas
Filtration Company 1	M-I / SWACO
DE Unit Size 1	1200
Filtration Company 2	SWACO / M-I
DE Unit Size 2	1200
Daily Spread Cost for Rig	5000
Average Trip Time	

Extra Properties Setup

Completing these cells will result in populating the appropriate blanks on the daily report and displacement worksheet.

Extra Properties Setup

#	Completion Fluids	Unit
	Description	
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		

Predefined Setting Custom Setting
Import Custom Setting Export Custom Setting

Clicking this button will open the 'Extra Properties Setup' screen, where you can enter up to 52 additional parameters.

#1 – General

Tab #1 is for general information and current depth data.

Daily Completion Fluids (12/18/2002)

1- General | 2- Pumps / Bits | 3- Fluid Properties | 4- Well Geometry | 5- Well Information | 6- Displacement | 7- Time Distribution | 8- Inv/Vol Acct | 9- Filtration | 10- Equipment

Date	12/18/2002	Depth	8002	PBTD - TVD	6860
Report	10	TVD	6880	Bit Depth	7962
		PBTD - MD	7962	Present Activity	Circ. & Filter

Operator Representative	John Able		
Contractor Representative	Peter Smart		
MH SWACO Representative 1	Ian Scott	Rig Telephone Number	123-456-7894
MH SWACO Representative 2	David Jones	Warehouse Telephone	123-456-7895
Telephones	123-456-7890		
	123-456-7892		
Pagers/FAX Numbers	123-456-7891		
	123-456-7893		
Daily Completion Fluids Cost	11135.00		
Cum. Completion Fluids Cost	40129.32		
Cumulative Cost	221866.82		

12/18/2002

DAILY KEY DATA

1. **Depth & PBTD-MD:** are used to calculate the correct hole volumes.
2. **Bit Depth:** sets your drill string total depth (affecting Fluid Volumes, Bottom-up & Drill Pipe length) and can not be greater than the PBTD-MD.

#2 – Pumps and Bits

The rig pump data (liner diameter, stroke length, % efficiency & pump rate) is used to calculate the Pump Flow Rate, but this figure may be over-written if needed.

Daily Completion Fluids (12/19/1999)

6- Displacement 7- Time Distribution 8- Inv/Vol Acct. 9- Filtration 10- Equipment
1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Pump Data

Make and Model	Liner Diameter	Stroke Length	Rod Diam Duplex Only	Eff%	Pump Rate	Pump On Report	Riser Pump
EMSCO F-1000	6.5	12		97	100	☒	☐
EMSCO F-1000	6.5	12		97		☒	☐
EMSCO F-1000	6	12		97		☐	☐
						☐	☐

Bit and Drilling Data

Nozzles

Size	Quantity	TFA
	x	0
	x	
	x	

Bit Description	
Bit Number	
% Washout	
Rotary rpm	
Rotating Hours	0
Weight on Bit	
ROP	
Bit Size/Hole Diameter	8.681
Washout Hole Size	8.68
Pump Flow Rate	502
Pump Pressure	2000
Riser Pump Flow Rate	

12/19/1999

DAILY KEY DATA

1. **Pump Rate:** this is used to calculate the total Pump Flow Rate, which affects the Total Circulating Time, Bottom-up and some indexes (i.e. pressure losses, annular velocity, etc.)
2. **Pressures:** used to calculate pressure losses, ECD, etc.

#3 – Fluid Properties

Fluid Properties		#1 - Primary	#2	#3	#4
Source		Pit			
Funnel Viscosity	s/qt				
Solids%		.01			
pH		8.5			
Temperature	°F	95			
Fluid Type		CaCl2			
SG		1.38			
Measured Density	lb/gal	11.495			
Corrected Density	lb/gal	11.575			
TCT	°F	40			
NTU's		30			
Iron Content		35			
SG of Fluid In Hole		1.02			
Temp of Fluid In Hole	°F	130			
Measured Density In Hole	lb/gal	8.497			
Corrected Density In Hole	lb/gal	8.667			

Extra Properties

The built-in fluid parameters on this tab target common completion fluids properties.

There is a place to enter the density of the fluid in the well bore as well as the pit.

DAILY KEY DATA

1. **SG**: both s.g. values (pit & hole) are used to calculate the measured densities (pit & hole).
2. **Temperature**: both temperature values (pit & hole) are used to calculate **corrected densities** (pit & hole), according to the measured density values.

“Extra Properties” are added just like in the “Drilling Fluids & Equipment” module.

Tab

In this section you will select the size and type of work string, tubing, etc.

Daily Completion Fluids (12/19/1999)

6- Displacement 7- Time Distribution 8- Inv/Vol Acct. 9- Filtration 10- Equipment
 1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	T.J. Len (in)
DP	8002	5	4.276	6.375	3.75	

Type Description

- DP Drill Pipe
- DC Drill Collar
- HWDP HEVI-WATE Drill Pipe
- MWD MWD Tool
- MudM Mud Motor
- T Tubing
- PT Production Tubing
- CT Coil Tubing
- CSG Casing

Interval (Cost)

Number 4

Print Order

12/19/1999

1. **Interval (Cost) Number:** is used to indicate where will today's cost be allocated. For example: if the number in the box is equal to "3", it means that ONE-TRAX will consider all the cost generated by today's activity (rental, personnel, mud mixing, etc.) as part of the cost that belongs to the 3rd interval.

This number is critical to get each 'project interval cost' correctly calculated in the Recap report.

Daily Data Input – Completion Fluids Module

#4 – Well Geometry (continuation)

Tab

Daily Completion Fluids (12/19/1999)

6- Displacement 7- Time Distribution 8- Inv/Vol Acct. 9- Filtration 10- Equipment
1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Drill String

Type	Length (ft)	Pipe O.D. (in)	Pipe I.D. (in)	Tool Jt O.D. (in)	Tool Jt I.D. (in)	TJ Len (in)
DP	8002	5	4.276	6.375	3.75	

Daily Casing / Volume

Well Casing Intervals (Cost)

Interval (Cost)

Number 4

Print Order

12/19/1999

EXIT ? ? ? ? ? ? ? ?

Enter the information required in these two sections the same way it was explained for the "Drilling Fluids & Equipment" module.

Daily Data Input – Completion Fluids Module

Tab

#5 – Well Information

Supplying the information requested here populates the daily report and displacement worksheets.

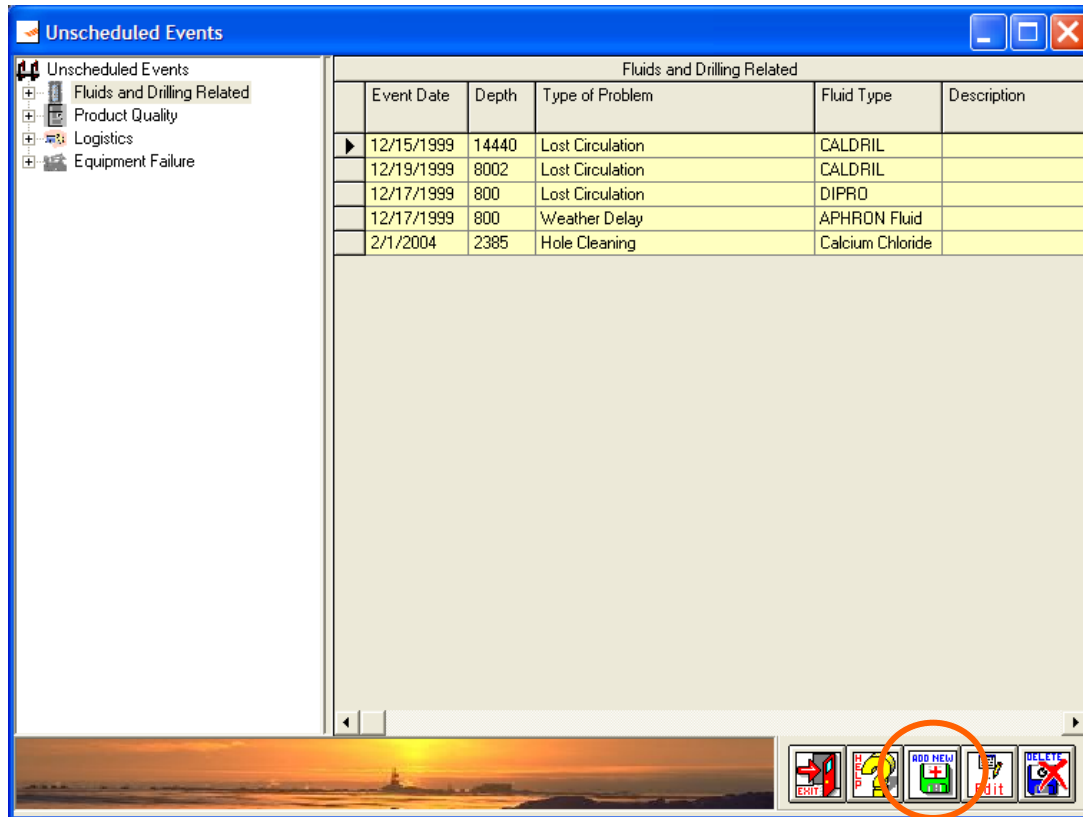
The “**Remarks**” box near the bottom of the screen is used to report daily activity, etc.

The “**Comments**” button opens four cells for data input.

Report in this section any issue that may have occurred today (see next slide for details) .

#5 – Well Information / Unscheduled Events

Clicking on the “**Unscheduled Events**” button opens the box below. This is the preview page of the events entered. A new event would be entered by clicking on the ‘Add New’ button.



#5 – Well Information / Unscheduled Events (cont.)

Event Details

Event Date	8/25/2005	Depth (#)	8002	Incident Category	Fluids and Drilling Related
Interval Number	4	Type of Problem	Lost Circulation	Type of Problem 2	Seepage
Fluid Type	Calcium Chloride				
Description	Lost CaCl ₂ brine to formation after perforating.				
Suspected Cause	Formation pressure less than predicted / porosity & permeability high.				
Loss Description	Lost time spotting loss control pill.				
Loss Time (Hr)	3				
Customer Assigned NPT (Hr)	0				
Fluid Density (lb/gal)	11.5				
Volume of Mud Lost (bbl)	247				

Back to: [Table of Content](#)

Here you can describe the Incident Category, Type of Problem, and complete description of the incident.

Daily Data Input – Completion Fluids Module

#6 – Displacement

The data entered here populates the “Displacement Worksheet”.

This data is usually a part of the displacement benchmarking process.

Displacement

Displacement 1 of 1

Add New Displacement Remove the Last Displacement

Type of Displacement	Indirect Forward	Max Allowable Pump Pressure	3700
Fluid being Displaced - Type	VERSACLEAN	Max Pump Pressure Achieved	2780
Completion Fluid - Type	Calcium Chloride	Pump Time for Displacement	5
		Down Time During Displacement	0
Displacement Chemical Supply Company	MISWACO	Rig Cleanup Time - Total	6
Fluid being Displaced - Density (lb/gal)	11.7	Rig Cleanup Time - On-line Only	
Completion Fluid - Density (lb/gal)	11.6		
Total Cost of Displacement Chemicals	8200	NTUs - Targeted Final	30
Pipe Rotation Rate	10	NTUs - At Start of Filtration Cycle	65
Mud Displacement Interface (bbl)	18	NTUs - After Short-trip	
Brine Displacement Interface (bbl)	36	NTUs - Maximum Level	250
Volume Brine Interface Reclaimed (bbl)		NTUs - Achieved Final	30
Brine Interface Reclaimed	☐	% Solids - At Start of Filtration Cycle	0.01
Short-trip in Mud	☐	% Solids - Maximum	0.25
Short-trip in Seawater	☐	% Solids - Achieved Final	0.04
Short-trip in Brine	☐		
Mud Observed on w/s After Displacement	☐		
Mud Observed on Brushes After Displacement	☐		

Spacer Detail

NTU Data / Pump Rate vs Pressure

There is room for additional daily or displacement comments under the “**Comments**” button in tab #5.

Displacement NTU data can be entered by clicking on the “**NTU Data / Pump Rate vs. Pressure**” button.

#6 – Displacement / NTU Data & Pump Rate vs. Pressure

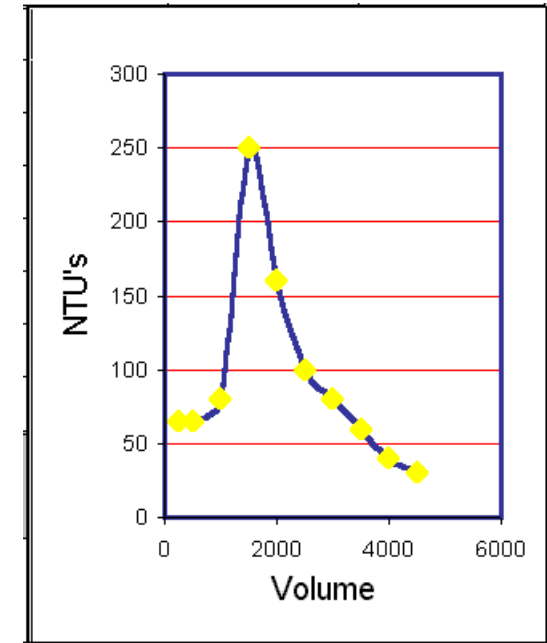
NTU Data / Pump Rate vs Pressure

Pumped Volume vs NTUs (Maximum 50 Rows)		
	Volume	NTUs
▶	250	65
	500	65
	1000	80
	1500	250
	2000	160
	2500	100
	3000	80
	3500	60
	4000	40
	4500	30
	5000	30
	5500	30
*		

Clicking the “**NTU Data & Pump Rate vs. Pressure**” button opens the window on the left. This is to report the amount of volume pumped vs. the NTU value.

As you enter the requested Volume and NTU data, a new line will appear for the next entry.

A graph is presented in the displacement worksheet that represents these values.



Daily Data Input – Completion Fluids Module

Tab

#6 – Displacement / Spacer Details

Spacer Detail

Clicking the “**Spacer Detail**” button will open this screen, which allows you to view individual spacers and their make-ups.

The screenshot shows the 'Displacement' software window. It has a title bar with standard window controls. Below the title bar is a header area with a background image of an oil rig at night. Below the header is a navigation bar with 'Displacement 1 of 1', 'Add New Displacement', and 'Remove the Last Displacement' buttons. The main area is divided into several sections: 'Type of Displacement' (Indirect Forward), 'Fluid being Displaced - Type' (VERSACLEAN), 'Completion Fluid - Type' (Calcium Chloride), 'Displacement Chemical Supply Company' (MISWACO), 'Fluid being Displaced - Density (lb/gal)' (11.7), 'Completion Fluid - Density (lb/gal)' (11.6), 'Total Cost of Displacement Chemicals' (8200), 'Pipe Rotation Rate' (10), 'Mud Displacement Interface (bbl)' (18), 'Brine Displacement Interface (bbl)' (36), 'Volume Brine Interface Reclaimed (bbl)', 'Brine Interface Reclaimed', 'Short-trip in Mud', 'Short-trip in Seawater', 'Short-trip in Brine', 'Mud Observed on w/s After Displacement', 'Mud Observed on Brushes After Displacement', 'Max Allowable Pump Pressure' (370), 'Max Pump Pressure Achieved' (278), 'Pump Time for Displacement' (5), 'Down Time During Displacement' (0), 'Rig Cleanup Time - Total' (6), 'Rig Cleanup Time - On-line Only', 'NTUs - Targeted Final' (30), 'NTUs - At Start of Filtration Cycle' (65), 'NTUs - After Short-trip', 'NTUs - Maximum Level' (250), 'NTUs - Achieved Final' (30), '% Solids - At Start of Filtration Cycle' (0.01), '% Solids - Maximum' (0.25), '% Solids - Achieved Final' (0.04), and a 'Spacer Detail' button circled in orange. Below the 'Spacer Detail' button is a button labeled 'NTU Data / Pump Rate vs Pressure'.

The screenshot shows the 'Spacer Detail' screen within the 'Displacement' software window. It has a title bar with standard window controls. Below the title bar is a header area with a background image of an oil rig at night. Below the header is a navigation bar with 'Displacement 1 of 1', 'Add New Displacement', and 'Remove the Last Displacement' buttons. The main area is divided into several sections: 'Spacer Description' (Spacer #3), 'Base Fluid' (Water), 'Location' (N/A), 'Spacer Volume (bbl)' (100), 'Products Concentration' table, and a 'Spacer Detail' button circled in orange. The 'Products Concentration' table has two columns: 'Product Name' and 'Concentration (lb/bbl)'. It contains two rows: 'CAUSTIC SODA' with a concentration of 2, and 'SAFE-VIS E' with a concentration of 2.082. Below the table is a large empty area for additional data entry.

[Back to: Table of Content](#)

MISWACO

Daily Data Input – Completion Fluids Module

Tab

#7 – Time Distribution

Time Distribution Last 24 hr

Activity	Hours
Rig Up/Service	
Drilling	
Tripping	6
Non-Productive Time	6
Displace Well	5
Circ. & Filter	7
Monitor Well	
Monitor Well	
Filter	
Total	24
Pickup Tubing	
Gravel	
Circ. & Filter	
Displace Well	
Filter Fluid	
Set Pecker	
Perforate	
my time	

Unscheduled Events

Time Distribution Setup

Here you enter the amount of time (hrs) -for each rig activity- during the last 24 hours. Use the same “time distribution” data that’s reported for that day in the IADC sheet.

You can use the drop down menus to select even further categories.

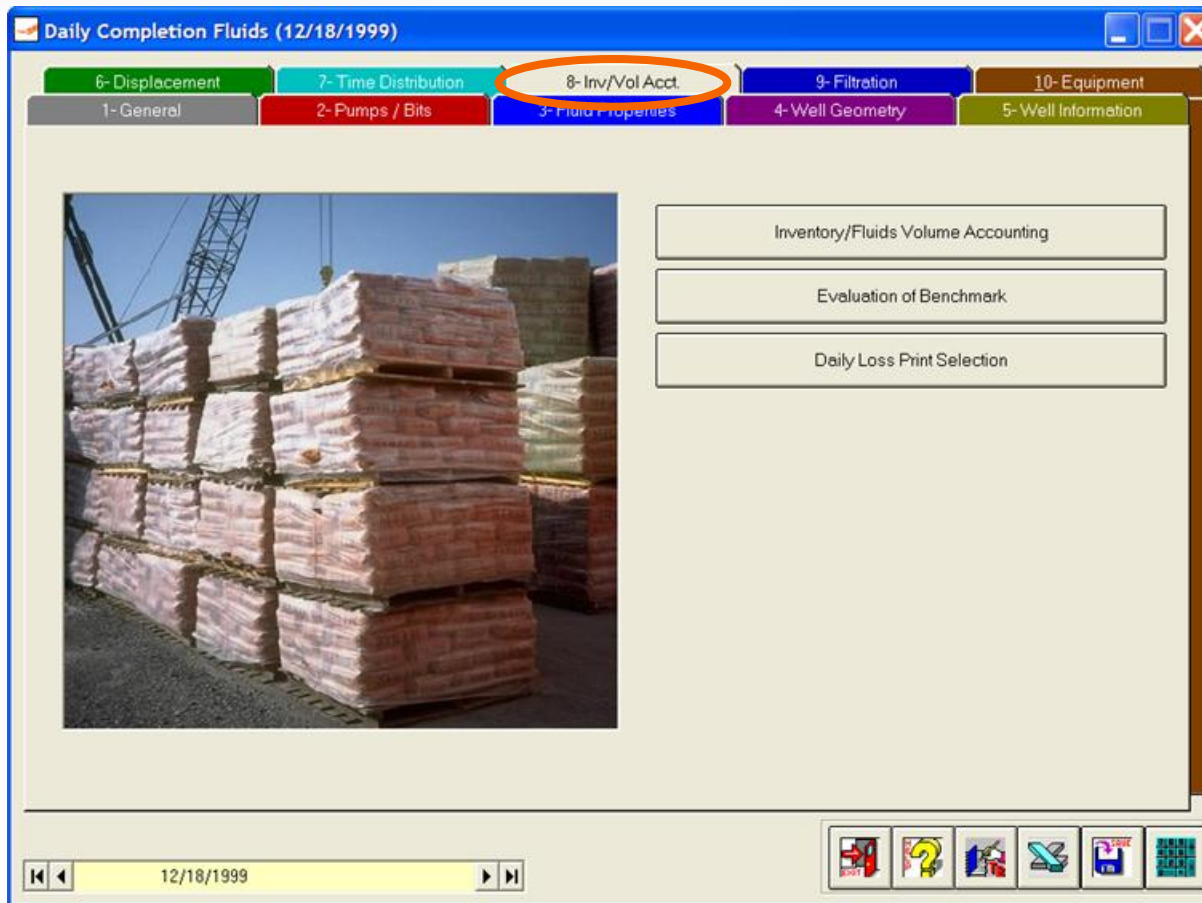
DAILY KEY DATA

1. **Total Hours:** make sure the total time for each day adds up **24 hours**.

Daily Data Input – Completion Fluids Module

#8 – Inventory & Volume Accounting

Tab



This is one of the most important sections of the ONE-TRAX reporting package.

It is essential that you understand how the transaction based Volume Accounting process works.

#8 – Inv. & Vol. Acct. / Daily Loss Print Selection

Daily Loss Print Selection

Clicking on this button opens the screen below

Available Losses	
Code	Description
1	Shakers
2	Other/Solids
3	Centrifuge
4	Tripping
5	Evaporation
6	Never Used
7	Lost to Formation
8	Cuttings Boxes
9	Sweeps
10	Returned
11	Interface
12	Surface Lines
13	Filter Unit
14	Operator Error

Add

Delete

Add All

Today's Losses	
Code	Description
4	Tripping
5	Evaporation
8	Cuttings Boxes
11	Interface
12	Surface Lines
13	Filter Unit
14	Operator Error

The Daily Fluid Loss Categories may be edited in this section by clicking the “Daily Loss Print Selection”. The screen on the left will appear.

To add, select a category from the “Available Losses” and click the “Add” button. The new category will appear in the “Today’s Losses” List.

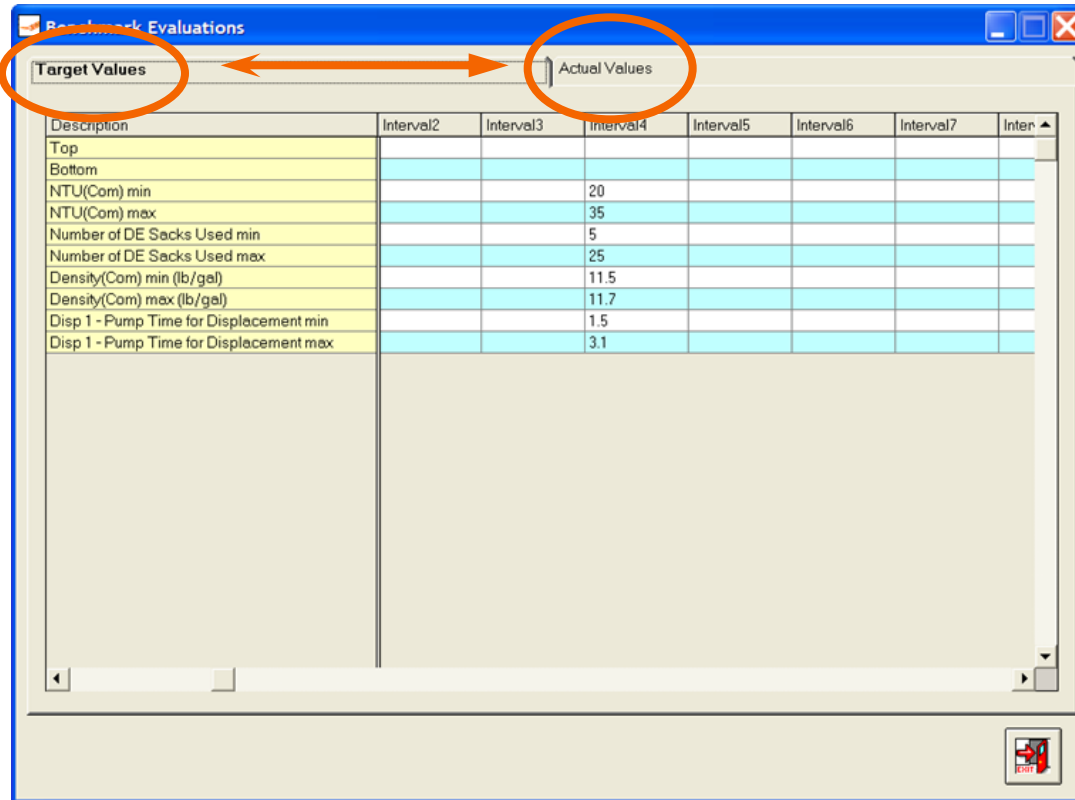
The categories shown in “Today’s Losses” will be the only ones that may be used for Volume Accounting purposes.

To delete, select a category from “Today’s Losses” and click the “Delete” button.

#8 – Inv. & Vol. Acct. / Evaluation Benchmark

Evaluation of Benchmark

Clicking on this button opens the screen below



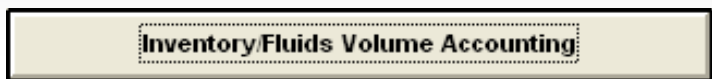
Description	Interval2	Interval3	Interval4	Interval5	Interval6	Interval7	Interval8
Top							
Bottom							
NTU(Com) min			20				
NTU(Com) max			35				
Number of DE Sacks Used min			5				
Number of DE Sacks Used max			25				
Density(Com) min (lb/gal)			11.5				
Density(Com) max (lb/gal)			11.7				
Disp 1 - Pump Time for Displacement min			1.5				
Disp 1 - Pump Time for Displacement max			3.1				

Alternating between these screens allows the user to visually compare the targeted benchmarks against the actual values.

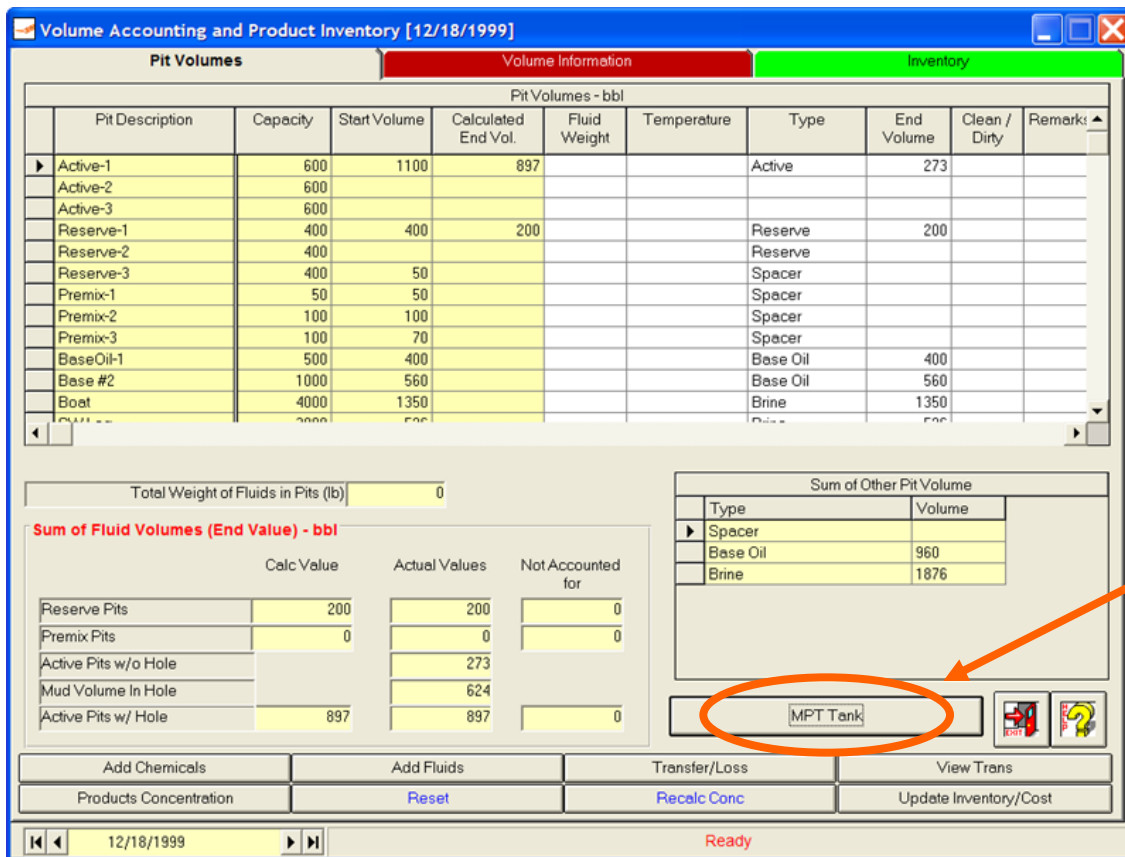
Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Pit Volumes



Clicking on this button opens the screen below

A screenshot of a software window titled "Volume Accounting and Product Inventory [12/18/1999]". The window has three tabs: "Pit Volumes", "Volume Information", and "Inventory". The "Pit Volumes" tab is active, showing a table of pit data. Below the table are sections for "Sum of Fluid Volumes (End Value) - bbl" and "Sum of Other Pit Volume". At the bottom, there are buttons for "Add Chemicals", "Add Fluids", "Transfer/Loss", and "View Trans", along with a date field and a "Ready" status.

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Clean / Dirty	Remarks
Active-1	600	1100	897			Active	273		
Active-2	600								
Active-3	600								
Reserve-1	400	400	200			Reserve	200		
Reserve-2	400					Reserve			
Reserve-3	400	50				Spacer			
Premix-1	50	50				Spacer			
Premix-2	100	100				Spacer			
Premix-3	100	70				Spacer			
BaseOil-1	500	400				Base Oil	400		
Base #2	1000	560				Base Oil	560		
Boat	4000	1350				Brine	1350		

Type	Volume
Spacer	
Base Oil	960
Brine	1876

	Calc Value	Actual Values	Not Accounted for
Reserve Pits	200	200	0
Premix Pits	0	0	0
Active Pits w/o Hole		273	
Mud Volume In Hole		624	
Active Pits w/ Hole	897	897	0

Buttons
Add Chemicals
Add Fluids
Transfer/Loss
View Trans
Products Concentration
Reset
Recalc Conc
Update Inventory/Cost

For Completion Fluid reporting purposes, there is a special area to track the "Spike Fluid" inventory that is stored in MPT tanks.

Click on the MPT button to open the screen that will allow you to enter the data for the MPT tanks.

#8 – Inv. & Vol. Acct. / Pit Volumes (cont.)



Clicking on this button opens the screen below.

Volume Accounting and Product Inventory [12/18/1999]

Pit Volumes Volume Information Inventory

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Clean / Dirty	Remarks
Active-1	600	1100	897			Active	273		
Active-2	600								
Active-3	600								
Reserve-1	400	400	200			Reserve	200		
Reserve-2	400					Reserve			
Reserve-3	400	50				Spacer			
Premix-1	50	50				Spacer			
Premix-2	100	100				Spacer			
Premix-3	100	70				Spacer			
BaseOil-1	500	400				Base Oil	400		
Base #2	1000	560				Base Oil	560		
Boat	4000	1350				Brine	1350		

Total Weight of Fluids in Pits (lb) 0

Sum of Fluid Volumes (End Value) - bbl

	Calc Value	Actual Values	Not Accounted for
Reserve Pits	200	200	0
Premix Pits	0	0	0
Active Pits w/o Hole		273	
Mud Volume In Hole		624	
Active Pits w/ Hole	897	897	0

MPT Tank Volume

Tank ID	Density	Volume
123	11.6	25
135	8.6	25
158	8.7	25
235	11.3	25
*		

Sum of Other Pit Volume

Add Chemicals Add Fluids Transfer/Loss View Trans

Products Concentration Reset Recalc Conc Update Inventory/Cost

12/18/1999 Ready

Enter the data for the fluid volumes stored in MPT tanks in this area. Adjust this table every time the inventory in the MPTs changes.

This data will populate the daily report.

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Pit Volumes (cont.)

Volume Accounting and Product Inventory [12/18/1999]

Pit Volumes | Volume Information | Inventory

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Clean / Dirty	Remarks
Active-1	600	1100	897	11.7		Active	273		
Active-2	600								
Active-3	600								
Reserve-1	400	400	200	11.5		Reserve	200		
Reserve-2	400					Reserve			
Reserve-3	400	50		7.1		Spacer	50		
Premix-1	50	50		8.6		Spacer	100		
Premix-2	100	100		11.3		Spacer	70		
Premix-3	100	70				Spacer			
BaseOil-1	500	400				Base Oil	400		
Base #2	1000	560				Base Oil	560		
Boat	4000	1350				Brine	1350		

Use the pull down menu in the "Type" column and select "Spacer".

Type	Volume
Spacer	220
Base Oil	960
Brine	1876

Sum of Other Pit Volume

Not Accounted for: 0

MPT Tank

12/18/1999

Ready

The Spacers are set up the same way the rest of the pits and tanks are.

The volumes that are entered in the "End Vol." column are added up and recorded as a total in the "Sum of Other Pit Volume".

The Spacers have no effect on the Active Fluid Inventory. This feature merely allows you to show the transaction involved with adding, then subtracting the spacers during a displacement.

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Pit Volumes (cont.)

Volume Accounting and Product Inventory [12/18/2002]

Pit Volumes Volume Information Inventory

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Clean / Dirty	Remarks
Active-1	600	558	356	11.6	105	Active	480	Dirty	Well Re
Active-2	600		460			Active	336	Clean	After Filt
Active-3	600								
Reserve-1	400	400	200	11.6	85	Reserve	200	Clean	
Reserve-2									
Reserve-3									
Pre-Mix-1									
Pre-Mix-2							60	Clean	
Pre-Mix-3							45	Clean	
Pre-Mix-4							120	Clean	
Base Oil									

Change Pit Type

Pit Description: Active-3

Old Pit Type:

New Pit Type:

Descrip

- Empty
- Active
- Reserve
- Premix
- Spacer

Other Pit Volume

Volume
225

Total Weight of Fluid:

Sum of Fluid Volumes (End):

Reserve Pits:

Premix Pits:

Active Pits w/o Hole:

Mud Volume In Hole: 542

Active Pits w/ Hole: 1358 1358 0

MPT Tank

Add Chemicals Add Fluids Transfer/Loss View Trans

Products Concentration Reset Recalc Conc Update Inventory/Cost

12/18/2002 Ready

Activate the "Pit Type" by clicking the drop-down button in the "Type" column.

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Pit Volumes (cont.)

Volume Accounting and Product Inventory [12/18/1999]

Pit Volumes | Volume Information | Inventory

Pit Description	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Clear / Dirty	Remarks
Active-1	897	11.7		Active	273		
Active-2							
Active-3							
Reserve-1	200	11.5		Reserve	200		
Reserve-2				Reserve			
Reserve-3		7.1		Spacer	50		
Premix-1		8.6		Spacer	100		MPT Tank #158
Premix-2		11.3		Spacer	70		MPT Tank #235
Premix-3				Spacer			
BaseOil-1				Base Oil	400		
Base #2				Base Oil	560		
Boat				Brine	1350		
...				...	...		

Total Weight of Fluids in Pits (lb) 315004

Sum of Fluid Volumes (End Value) - bbl

	Calc Value	Actual Values	Not Accounted for
Reserve Pits	200	200	0
Premix Pits	0	0	0
Active Pits w/o Hole		273	
Mud Volume In Hole		624	
Active Pits w/ Hole	897	897	0

Sum of Other Pit Volume

Type	Volume
Spacer	220
Base Oil	960
Brine	1876

MPT Tank

Add Chemicals | Add Fluids | Transfer/Loss | View Trans

Products Concentration | Reset | Recalc Conc | Update Inventory/Cost

12/18/1999 Ready

Enter the physical location of the spacers in the “**Remarks**” column. This data will populate the appropriate space in the displacement worksheet.

To find the “**Remarks**” column, scroll all the way across the screen using the arrows.

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Pit Volumes (cont.)

Volume Accounting and Product Inventory [12/18/2002]

Pit Volumes Volume Information Inventory

Pit Description	Capacity	Start Volume	Calculated End Vol.	Fluid Weight	Temperature	Type	End Volume	Clean / Dirty	Remarks
Active-1	600	558	356	11.6	105	Active	480	Dirty	Well Per
Active-2	600		460			Active	336	Clean	After Filtr
Active-3	600								
▶ Reserve-1	400	400	200	11.6	85	Reserve	200		
Reserve-2	400					Reserve			
Reserve-3	400					Reserve			
Pre-Mix-1	50	50		6.8	85	Spacer	6		
Pre-Mix-2	100	50		8.6	85	Spacer	45	Clean	
Pre-Mix-3	100	100		11.6	85	Spacer	120	Clean	
Pre-Mix-4	100	70				Spacer			
Base Oil	500								

Total Weight of Fluids in Pits (lb) 423150

Sum of Fluid Volumes (End Value) - bbl

	Calc Value	Actual Values	Not Accounted for
Reserve Pits	200	200	0
Premix Pits	0	0	0
Active Pits w/o Hole		816	
Mud Volume In Hole		542	
Active Pits w/ Hole	1358	1358	0

Sum of Other Pit Volume

Type	Volume
▶ Spacer	225

MPT Tank

Add Chemicals Add Fluids Transfer/Loss View Trans

Products Concentration Reset Recalc Conc Update Inventory/Cost

12/18/2002 Enter

Once you have designated the pit as Active, Reserve, or Spacer, you must give it a status of either “Clean” or “Dirty”.

Use the pull down menu under the “Clean / Dirty” column to make this designation.

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Volume Information

Enter here how much volume (in the Annulus, Drill String & Below Bit) **IS NOT** occupied by your Active System's fluid.

Volume Accounting and Product Inventory [12/19/1999]

Fluids in Hole - bbl

	Annulus	Total DS	Below Bit	Total
Total Hole Volume	482	142	0	624
Volume Not Fluids	0	0	0	0
Fluid Volume	482	142	0	624

Fluids Balance - bbl

	Active System	Reserve Pits	Premix Pits	Total
Start Volume	0	0	0	0
Oil Added	0	0	0	0
Water Added	0	0	0	0
Volume Chemicals Added	1100	0	0	1100
Chemical Not For Fluids	0	0	0	0
Total Volume Built	1100	0	0	1100
Received	0	0	0	0
Back Load	0	0	0	0
From Active to		0	0	0
From Reserve to	0		0	0
From Premix to	0	0		0
Loss and Dumped	0	0	0	0
Final Fluid Volume	1100	0	0	1100

Total Loss Break Down

#	Description	Active Loss	Reserve Loss	Premix Loss	Sub Total	Discharge Volume
1	Shakers	0	0	0		
2	Other/Solids	0	0	0		
3	Centrifuge	0	0	0		
4	Tripping	0	0	0		
5	Evaporation	0	0	0		
6	Never Used	0	0	0		
8	Cuttings Boxes	0	0	0		
10	Returned	0	0	0		
11	Interface	0	0	0		

Total Sub-Surface Loss: 0
Total Surface Loss: 0
Total Loss: 0

Buttons: Add Chemicals, Add Fluids, Transfer/Loss, View Trans, Products Concentration, Reset, Recalc Conc, Update Inventory/Cost

Ready

This is a summary of all the volume information at report closing time.

This table displays the volume of fluid lost according to the data entered on the "Equipment Performance" section of tab #6.

This information should be used as reference when entering the "Transfer/Loss" transactions on the previous tab ("Pit Volumes").

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Inventory

In this screen the quantities on the blue or yellow cells come from other sections of the program.

- “Used for Fluid” values come from “Add Chemicals” and/or “Add Whole Mud”.
- “Received” and “Return” are updated through Delivery/Return Tickets entries (see next slides).

Volume Accounting and Product Inventory [12/19/1999]

Pit Volumes Volume Information **Inventory**

No.	Description	Code	Gross Price	Start	Received	Returned	Used for Fluid	Used Other Fluid Module	Adj.
1	CALCIUM CHLORIDE (94%)	M0005108	12.5	100					
3	Calcium Chloride (11.6 ppg)	M0001210	22.6	0			400		
4	CAUSTIC SODA	M0006659	25.5	69			4		
5	DE-1 Base Oil	M0001715	85	285			50		
6	DUOVIS	M0007915	320	485					
7	GUAR GUM	M0006520	75	40					
8	LIME	M0004111	6.5	6					

Cost Summary

	Daily (\$)	Cum. (\$)		Daily (\$)	Cum. (\$)
Comp. Fluids	-9040.00	-9040.00	Filtra. All Inclusive	0.00	0.00
Spike Fluids	0.00	0.00	Filtration Products	0.00	0.00
Additive	8187.00	8187.00	Filtration Personnel	0.00	0.00
Engineering	900.00	900.00	Filtration Equipment	0.00	0.00
Rental Equipment	0.00	0.00	Filtration Other	0.00	0.00
Packer Fluid	0.00	0.00	Sub Total Filtration	0.00	0.00
Safety Equipment	0.00	0.00	Sub Total CF	47.00	47.00
Other	0.00	0.00	Sub Total	47.00	47.00

Chemical Weight (lb) 1183946
Volume Chemicals for Fluids -395.58

Last Ticket No.

From Warehouse	G9962005
To Warehouse	ABC
From Other Well	xxxx02
To Other Well	xxxx01

Delivery/Return Tickets

Add Chemicals Add Fluids Transfer/Loss View Trans
Products Concentration Reset Recalc Conc Update Inventory/Cost

12/19/1999 Ready

Only white cells can be edited in this screen.

- **Used Other Fluid Module:** this column is used to enter the personnel usage for each day (see example here).

- **Adjustment:** occasionally used when minor adjustments need to be made to the inventory.

View or Print the inventory report.

Daily Data Input – Completion Fluids Module

Tab

#8 – Inv. & Vol. Acct. / Inventory

Volume Accounting and Product Inventory [12/19/1999]

Pit Volumes Volume Information Inventory

No.	Adjustment	DT_Qty	RT_Qty	On Order	Final	Daily Used	Daily Cost	Weight	Not Print
					100	0	0.00	50.00	☐
3					400	-400	-9040.00	195.22	☐
4					65	4	102.00	3.50	☐
5					235	50	4250.00	61.63	☐
6					485	0	0.00	242.50	☐
7					40	0	0.00	20.00	☐
8					6	0	0.00	30.00	☐

Cost Summary

	Daily (\$)	Cum. (\$)		Daily (\$)	Cum. (\$)
Comp. Fluids	-9040.00	-9040.00	Filtra. All Inclusive	0.00	0.00
Spike Fluids	0.00	0.00	Filtration Products	0.00	0.00
Additive	8187.00	8187.00	Filtration Personnel	0.00	0.00
Engineering	900.00	900.00	Filtration Equipment	0.00	0.00
Rental Equipment	0.00	0.00	Filtration Other	0.00	0.00
Packer Fluid	0.00	0.00	Sub Total Filtration	0.00	0.00
Safety Equipment	0.00	0.00	Sub Total CF	47.00	47.00
Other	0.00	0.00	Sub Total	47.00	47.00

Chemical Weight (lb) 1183946
Volume Chemicals for Fluids -395.58

Last Ticket No.

From Warehouse	G9962005
To Warehouse	ABC
From Other Well	xxxx02
To Other Well	xxxx01

Delivery/Return Tickets

Add Chemicals Add Fluids Transfer/Loss View Trans
Products Concentration Reset Recalc Conc Update Inventory/Cost

12/19/1999 Ready

Scroll to the far right end and click on the 'Not Print' boxes for any inventory you do not want shown on a printed report.

Daily Data Input – Completion Fluids Module

Tab #8 – ... / Inventory / Delivery & Return Tickets

Delivery/Return Tickets

Ticket Transactions

Transfer Ticket Summary

Type	Ticket No.	Order By	Receive By	Warehouse Name	Warehouse
Receive from Warehouse	G9962003	John Small	John Small	Base Camp Wareh	03569
Receive from Warehouse	G9962004	John Small	John Small	Cameron	00268
Receive from Warehouse	G9962005	John Small	John Small	Port Fouchon	00125

Click the “Add New” button, enter the information in the ‘Transfer Ticket Summary’ section, and fill in the ‘Transfer Ticket detail’ table.

Product Description	Size	Unit	Package	Price	Amount from Ticket	Actual Amount
VERSAGEL	50	LB	BG	71		
NUT PLUG	50	LB	BG	18.5	80	80
POLYPAC UL	50	LB	BG	123		
Potassium Chloride Brine	1	BL	BK	9.25	15	15
POTASSIUM CHLORIDE	50	LB	BG	8.5		
SAFECARB 30	1	TN	BK	265		
SALT (FVD)	1	TN	BK	156.5		
SODA ASH	100	LB	BG	25.5		
WATER-BASED MUD	1	BL	BK	52.5	80	80
2 Micron ABS Filter Cartridge	1		EA	30		
Cartridge Filter Unit	1		EA	120		
SAFECARB 45	50	LB	BG	7.5		
SAFE-VIS E	5	GA	CN	210		
DIATOMACEOUS EARTH	50	LB	BG	7.5		

- **Amount from ticket:** this is the quantity received according to the delivery ticket.

- **Actual Amount:** this is the actual quantity of product received on-site.

#9 – Filtration

Daily Completion Fluids (12/18/2002)

6- Displacement 7- Time Distribution **8- Inv/Vol Acct.** **9- Filtration** 10- Equipment

1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Clarity And Data Profile

	Time of Sample	Well Return %Solids	TSS	Well Return NTU's	After Cartridges NTU's	Suction Pit NTU's	Volume Filtered (bbl)	Vol Circulated By Rig (bbl)
▶	12:00	0.4	2000	200	30	40	1100	1600
	15:00	0.2	500	180	20	30	1200	2000
	18:00	0.1	250	170	15	25	1350	1300
	21:00	0.1	250	170	15	25	1000	1100

DE Grade: Medium Cartridge Rating: 2 micron

Number of DE Sacks Used: 49 Time Required to Filter After Displacement: 8

DE Press Cycles to Clean Fluid: 4 Volume of Brine Filtered Getting to Spec.: 0

Average Volume Filtered Per Cycle: 1000 Downtime Due to Filtration Equipment: 0

Number of Filter Units Used: 1

Number of Cartridges Used: 80

Cycle Data

	Cycle No.	Start Time	End Time	Average Rate (gal/min)	D.E. Used	Fluid Lost (bbl)	Volume Filtered (bbl)
▶	1	12:00	15:00	502	10	12	
	2	15:00	18:00	502	8	10	
	3	18:00	21:00	502	6	8	
	4	22:00	23:30	500	5	3	

Filtration Technician: Tony the Tech

Filtration Assistant: Wishful Thinking

Comments:

12/18/2002

Data concerning the Filtration process is entered here.

Information entered will appear on the 'Filtration Report' portion of output spreadsheet.

Daily Data Input – Completion Fluids Module

Tab

#9 – Filtration (cont.)

Daily Completion Fluids (12/18/2002)

6- Displacement 7- Time Distribution 8- Inv/Vol Acct. 9- Filtration 10- Equipment

1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Clarity And Data Profile

	Time of Sample	Well Return %Solids	TSS	Well Return NTU's	After Cartridges NTU's	Suction Pit NTU's	Volume Filtered (bbl)	Vol Circulated By Rig (bbl)
▶	12:00	0.4	2000	200	30	40	1100	1600
	15:00	0.2	500	180	20	30	1200	2000
	18:00	0.1	250	170	15	25	1350	1300
	21:00	0.1	250	170	15	25	1000	1100

DE Grade: Medium

Cartridge Rating: 2 micron

Number of DE Sacks Used: 49

Time Required to Filter After Displacement: 8

DE Press Cycles to Clean Fluid: 4

Volume of Brine Filtered Getting to Spec.: 0

Average Volume Filtered Per Cycle: 1000

Downtime Due to Filtration Equipment: 0

Number of Filter Units Used: 1

Filtration Comments: Everything went pretty smoothly.

Filtration Technician: Tony the Tech

Filtration Assistant: Wishful Thinking

Cycle Data

12/18/2002

Clicking on this button toggles the box on the left between “Filtration Comments” and “Cycle Data”.

Daily Data Input – Completion Fluids Module

Tab

#9 – Filtration (cont.)

Daily Completion Fluids (12/18/2002)

6- Displacement 7- Time Distribution 8- Inv/Vol Acct 9- Filtration 10- Equipment

1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Clarity And Data Profile

	Time of Sample	Well Return %Solids	TSS	Well Return NTU's	After Cartridges NTU's	Suction Pit NTU's	Volume Filtered (bbl)	Vol Circulated By Rig (bbl)
▶	12:00	0.4	2000	200	30	40	1100	1600
	15:00	0.2	500	180	20	30	1200	2000
	18:00	0.1	250	170	15	25	1350	1300
	21:00	0.1	250	170	15	25	1000	1100

DE Grade: Medium Cartridge Rating: 2 micron

Number of DE Sacks Used: 49 Time Required to Filter After Displacement: 8

DE Press Cycles to Clean Fluid: 4 Volume of Brine Filtered Getting to Spec.: 0

Average Volume Filtered Per Cycle: 1000 Downtime Due to Filtration Equipment: 0

Number of Filter Units Used: 1

Number of Cartridges Used: 60

Cycle Data

	Cycle No.	Start Time	End Time	Average Rate (gal/min)	F.E. Used	Fluid Lost (bbl)	Volume Filtered (bbl)
▶	1	12:00	15:00	502	10	12	
	2	15:00	18:00	502	8	10	
	3	18:00	21:00	502	6	8	
	4	22:00	23:30	500	5	3	

Filtration Technician: Tony the Tech

Filtration Assistant: Wishful Thinking

Comments

12/18/2002

Clicking on this button toggles the box on the left between “Filtration Comments” and “Cycle Data”.

Daily Data Input – Completion Fluids Module

Tab

#9 – Filtration (cont.)

Daily Completion Fluids (12/18/2002)

6- Displacement 7- Time Distribution 8- Inv/Vol Acct. 9- Filtration 10- Equipment

1- General 2- Pumps / Bits 3- Fluid Properties 4- Well Geometry 5- Well Information

Clarity And Data Profile

	Time of Sample	Well Return %Solids	TSS	Well Return NTU's	After Cartridges NTU's	Suction Pit NTU's	Volume Filtered (bbl)	Vol Circulated By Rig (bbl)
▶	12:00	0.4	2000	200	30	40	1100	1600
	15:00	0.2	500	180	20	30	1200	2000
	18:00	0.1	250	170	15	25	1350	1300
	21:00	0.1	250	170	15	25	1000	1100

DE Grade: Medium

Cartridge Rating: 2 micron

Number of DE Sacks Used: 49

Time Required to Filter After Displacement: 8

DE Press Cycles to Clean Fluid: 4

Volume of Brine Filtered Getting to Spec.: 0

Average Volume Filtered Per Cycle: 1000

Downtime Due to Filtration Equipment: 0

Number of Filter Units Used: 1

Number of Cartridges Used: 80

Cycle Data

	Cycle No.	Start Time	End Time	Average Rate (gal/min)	D.E. Used	Fluid Lost (bbl)	Volume Filtered (bbl)
▶	1	12:00	15:00	502	10	12	
	2	15:00	18:00	502	8	10	
	3	18:00	21:00	502	6	8	
	4	22:00	23:30	500	5	3	

Filtration Technician: Tony the Tech

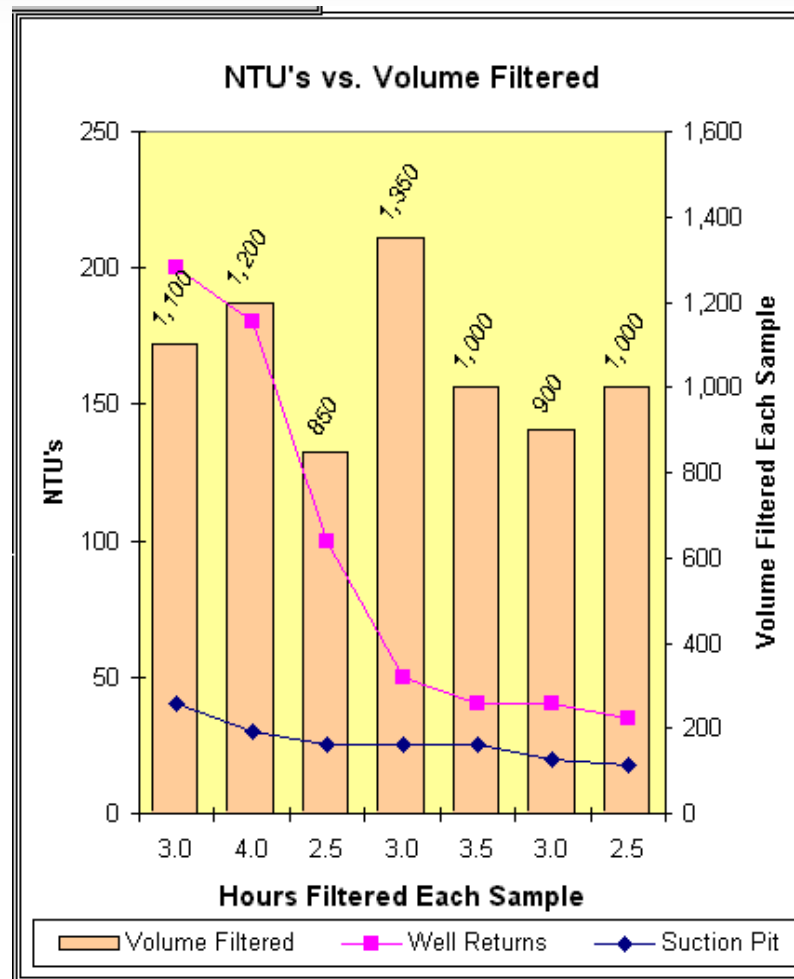
Filtration Assistant: Wishful Thinking

Comments:

12/18/2002

Some of the information in this section is responsible for populating a graph in the Filtration Daily Report.

#9 – Filtration (cont.)



This graph shows the relationship between NTU's from the well bore, NTU's in the suction pit, barrels pumped, and time.

Daily Data Input – Completion Fluids Module

Tab

#10 – Equipment



Clicking on this tab will allow the monitoring of equipment specific to Completion Fluids

Daily Data Input – Completion Fluids Module

Tab

#10 – Equipment

Equipment Details and Usage [8/24/2005]

Equipment Performance | Equipment Inventory | Equipment Usage Description

Equipment Type

- 1 - Centrifuge
- 4 - Mud Cleaner
- 5 - Shale Shaker
- 6 - Others
- 8 - Downhole Casing Cleaning Tool
- 23652 - 10-3/4" 'PUP' Scraper/Brush
- 9 - Filtration Equipment
- 33256 - 1150 Filter Unit
- 11 - Vacuum System
- 13 - Cuttings Container

33256 - 1150 Filter Unit (Completion Fluids Only)

Equipment Properties	Unit	#1 - Primary
Size / Specifications		1150
Hours		13
Cum. Hours		13
Filter Area	#2	
Screen mesh		2 micron

Full List
Recalculate Cumulative Values
Show Check #2

8/24/2005

Equipment Details and Usage [8/24/2005]

Equipment Performance | Equipment Inventory | Equipment Usage Description

Daily Inventory (Completion Fluids Only)

Serial No.	Description	Used	Charge Code	Daily Cost	Cumulative Cost
23652	10-3/4" 'PUP' Scraper/Brush		Full Charge	0.00	0.00
33256	1150 Filter Unit		Full Charge	0.00	0.00

* Filtration Equipment Cost is not yet implemented. Please enter Filtration Equipment Cost from product inventory.

	Daily Cost	Cumulative Cost
Completion Fluids	0.00	0.00
Total	0.00	4840.00

Full List
Recalculate Cumulative Cost

8/24/2005

[Back to: Table of Content](#)

MI SWACO

End of “Completion Fluids” module

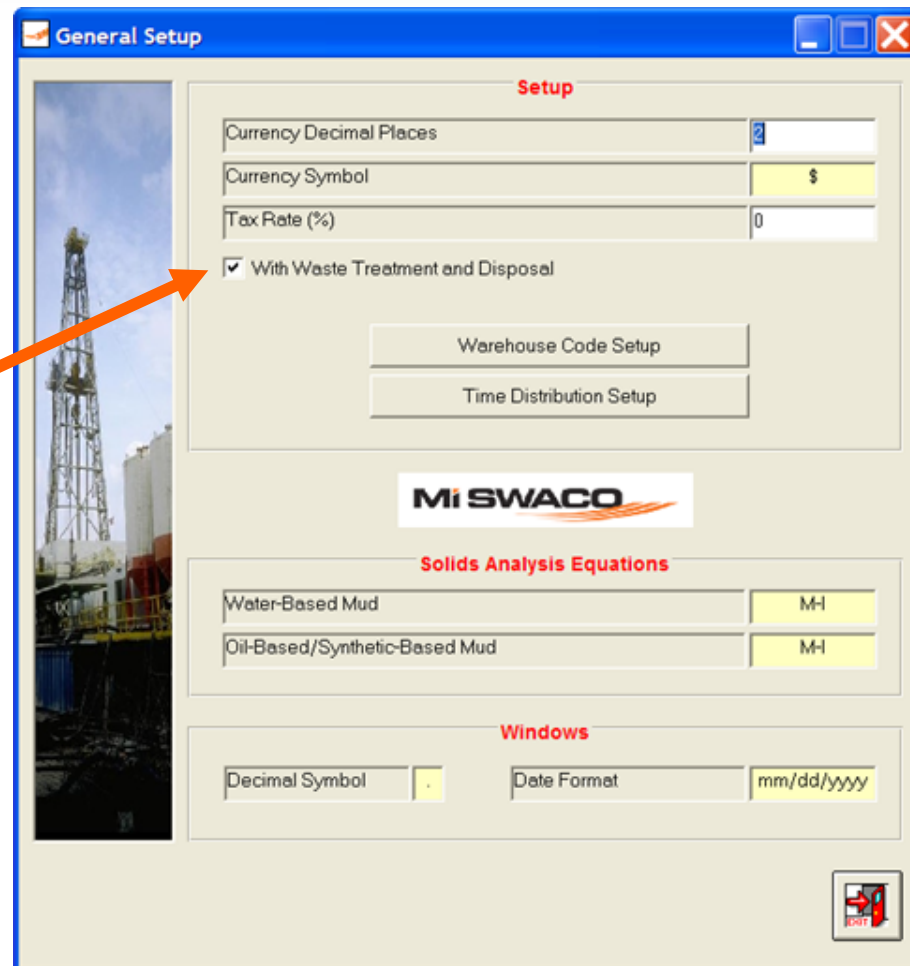
ONE-TRAX® Wellsite

Waste Treatment and Disposal Module

Daily Data Input – Waste Treatment and Disposal Module

Activating the Module

***Click Here To Select
Waste Treatment
and Disposal Option***



General Setup

Setup

Currency Decimal Places: 2

Currency Symbol: \$

Tax Rate (%): 0

☒ With Waste Treatment and Disposal

Warehouse Code Setup

Time Distribution Setup

Mi SWACO

Solids Analysis Equations

Water-Based Mud: MH

Oil-Based/Synthetic-Based Mud: MH

Windows

Decimal Symbol: .

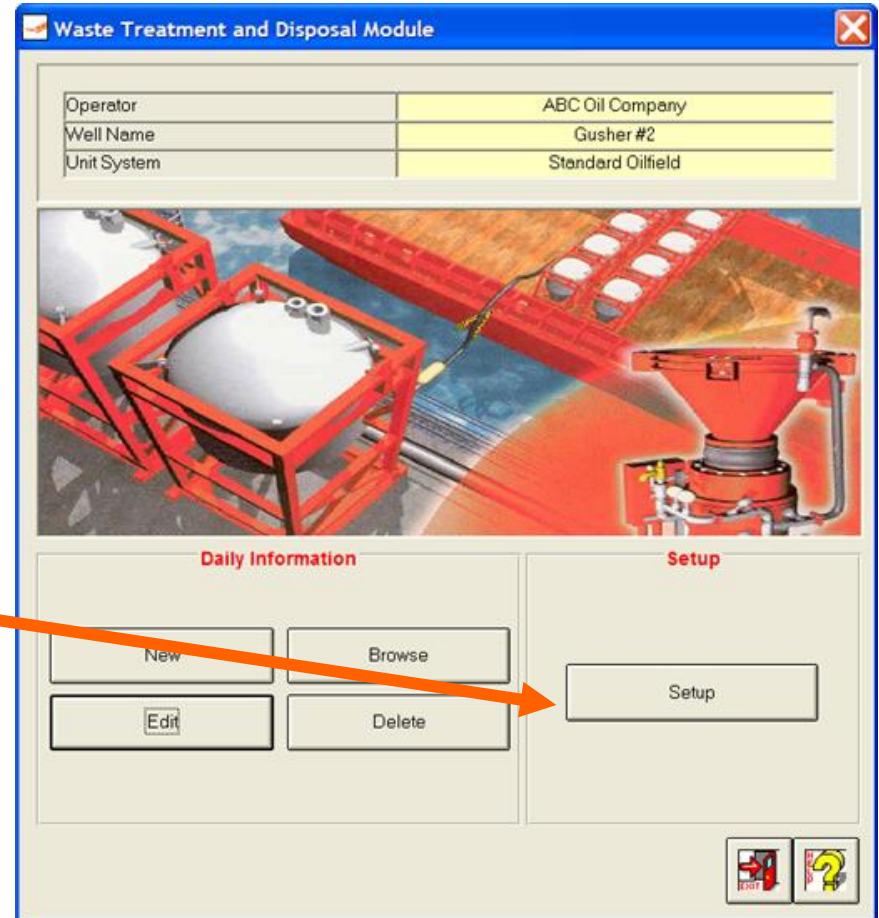
Date Format: mm/dd/yyyy

EXIT

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup

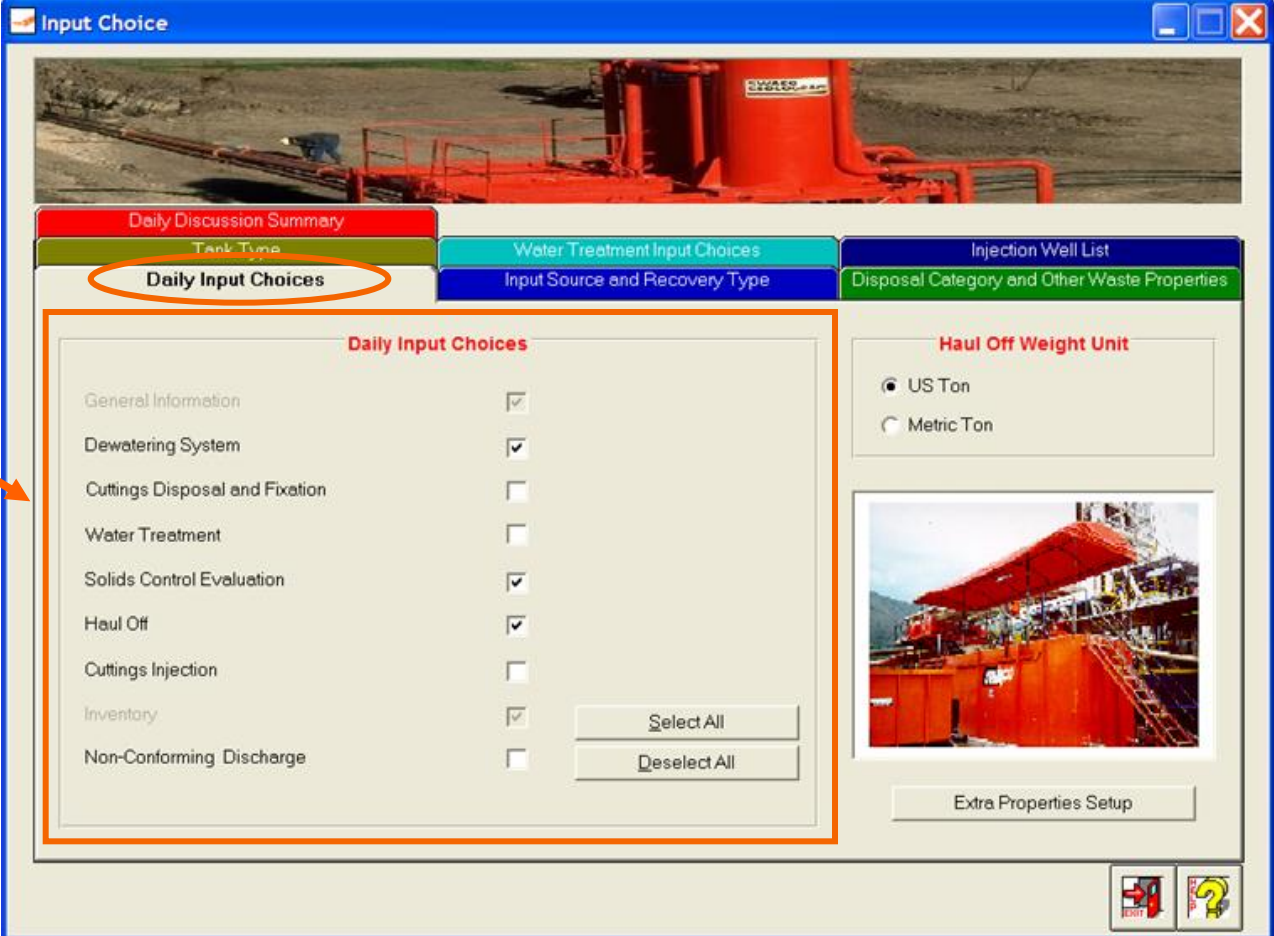
Click Here To Set up the Waste Treatment and Disposal Module



Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Daily Input Choices

Select here the “Waste Treatment & Disposal” services provided for this well/project.



The screenshot shows the 'Input Choice' software window. The 'Daily Input Choices' section is highlighted with an orange border and an orange arrow pointing to it from the text on the left. The section contains a list of services with checkboxes:

Daily Input Choices	
General Information	☒
Dewatering System	☒
Cuttings Disposal and Fixation	☐
Water Treatment	☐
Solids Control Evaluation	☒
Haul Off	☒
Cuttings Injection	☐
Inventory	☒
Non-Conforming Discharge	☐

Below the list are two buttons: 'Select All' and 'Deselect All'. To the right of the 'Daily Input Choices' section is the 'Haul Off Weight Unit' section, which has two radio buttons: 'US Ton' (selected) and 'Metric Ton'. Below this is a small image of an oil rig and an 'Extra Properties Setup' button. The top of the window has a blue title bar with the text 'Input Choice' and standard window controls. Below the title bar is a banner image of an oil rig. The main menu has four tabs: 'Daily Discussion Summary' (red), 'Tank Type' (green), 'Water Treatment Input Choices' (cyan), and 'Injection Well List' (blue). The 'Daily Input Choices' tab is currently selected and highlighted with an orange oval.

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Input Source

Input Source identifies all of the sources of waste that will have to be treated on location.

- Drilling Fluid.
- Drilling Fluid with Cuttings.
- Oil Waste.
- Wash Water.
- Camp Waste etc.

Be as specific as possible when you start out.

You can always add a source later in the well.

You cannot delete a source once you have used that source without going back and changing every day that used that source.

The screenshot shows the 'Input Choice' software window. At the top is a photograph of an industrial site. Below the photo are four tabs: 'Daily Discussion Summary', 'Tank Type', 'Input Source and Recovery Type' (which is selected and circled in orange), and 'Injection Well List'. The 'Input Source and Recovery Type' tab contains two tables. The first table, 'Other Input Source', has columns for 'Code' and 'Description'. The second table, 'Recovery Type', also has columns for 'Code' and 'Description'. Both tables have a list of items with checkboxes and a '*' button at the bottom.

Other Input Source	
Code	Description
1	D.F. Active System
2	Cuttings w/Mud Active System
3	Wash Water
4	Drain Water
5	Perimeter Water
*	

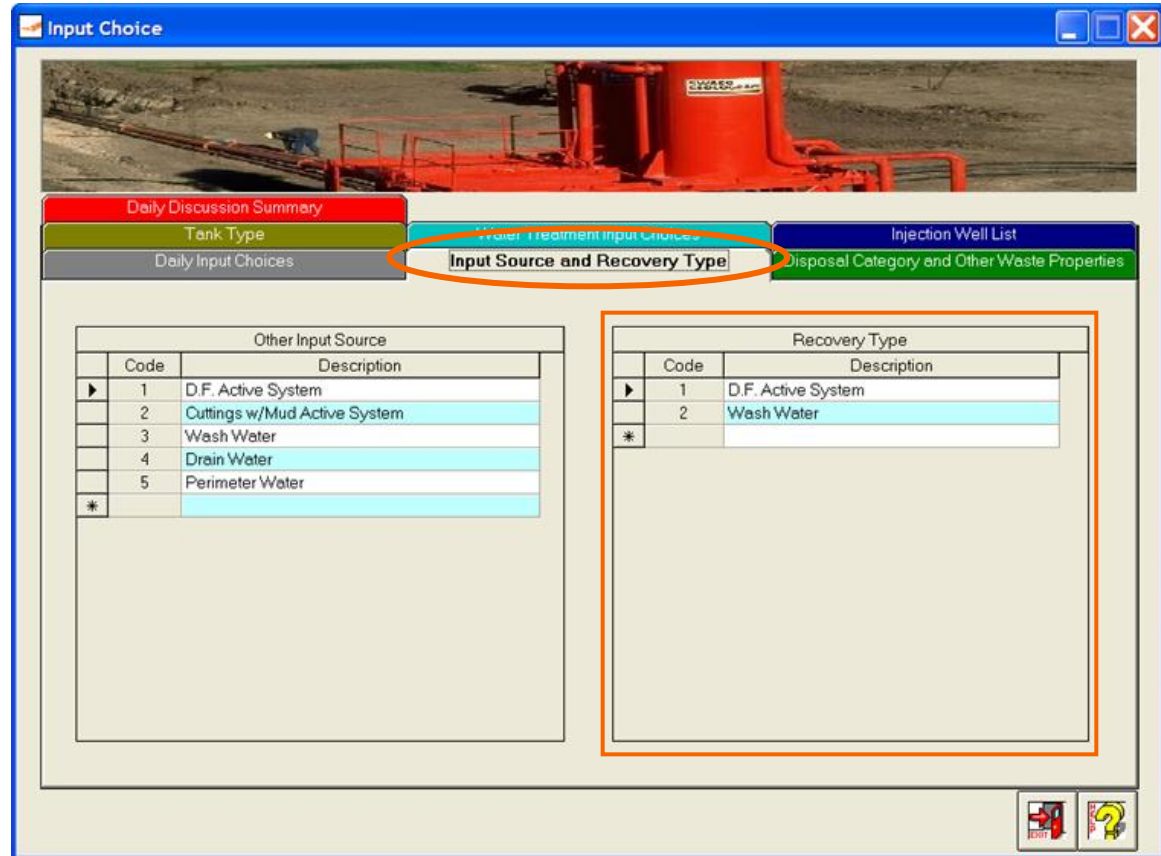
Recovery Type	
Code	Description
1	D.F. Active System
2	Wash Water
*	

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Recovery Type

Designate the waste streams you may recover fluid from:

- Active Drilling Fluid.
- Premix - Drilling Fluid or Dewatering.
- Wash Water.
- Injection Water.



The screenshot shows the 'Input Choice' software interface. The 'Waste Treatment Input Choices' button in the navigation pane is highlighted with an orange oval. The main content area displays two tables: 'Other Input Source' and 'Recovery Type'. The 'Recovery Type' table is highlighted with an orange border.

Other Input Source	
Code	Description
1	D.F. Active System
2	Cuttings w/Mud Active System
3	Wash Water
4	Drain Water
5	Perimeter Water
*	

Recovery Type	
Code	Description
1	D.F. Active System
2	Wash Water
*	

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Disposal Category

- Disposal Category allows you to designate the method for disposing the drilling wastes.

- Disposal.
- Land Farming.
- Bioremediation.
- Fixation.
- Irrigation.
- Injection.

The screenshot shows the 'Input Choice' software window. At the top is a photograph of a red industrial machine. Below it are several tabs: 'Daily Discussion Summary', 'Tank Type', 'Daily Input Choices', 'Water Treatment Input Choices', 'Input Source and Recovery Type', and 'Disposal Category and Other Waste Property'. The 'Disposal Category and Other Waste Property' tab is selected and highlighted with an orange oval. Below the tabs are two tables. The left table is titled 'Disposal Category' and has two columns: 'Code' and 'Description'. It contains two rows: '1 Disposal' and '2 RSL Remediation'. The right table is titled 'Waste Properties' and has two columns: '#' and 'Description'. It contains ten rows, with the first row being '1 Oil & Grease'. An orange arrow points from the list of disposal methods on the left towards the 'Disposal Category' table.

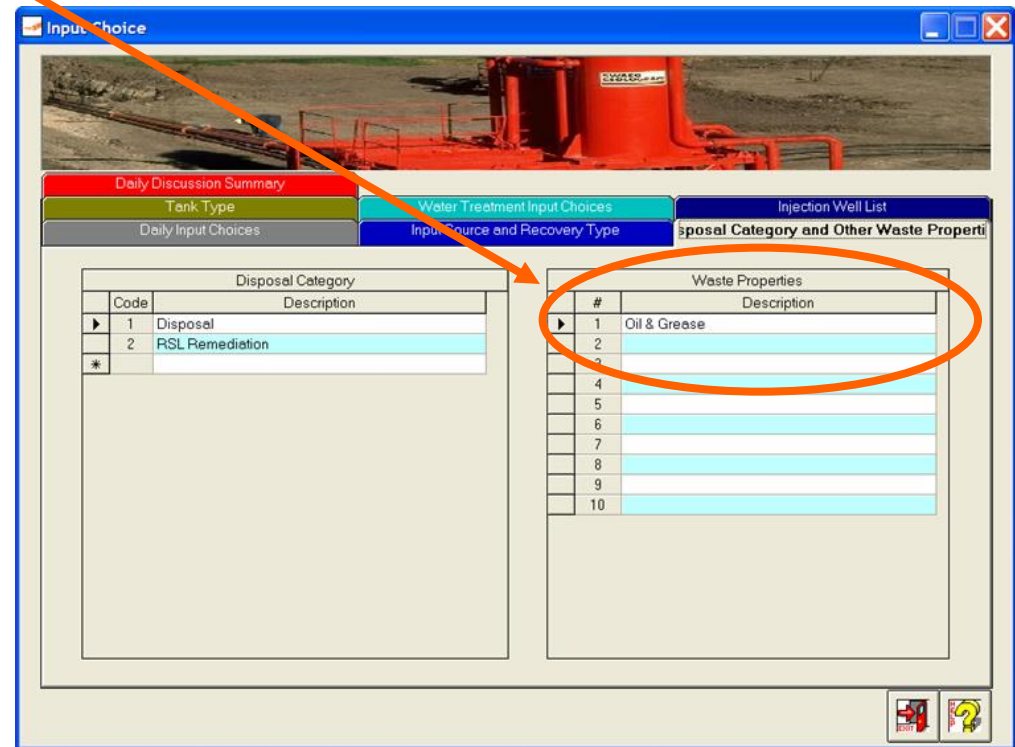
Code	Description
1	Disposal
2	RSL Remediation

#	Description
1	Oil & Grease
2	
3	
4	
5	
6	
7	
8	
9	
10	

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Waste Properties

- These are special properties -of the handled waste- that you may need to track, or that determine the cost for disposal.
 - % Oil.
 - Chlorides.
 - Solids.
 - pH.



The screenshot shows the 'Input Choice' software window. At the top is a photograph of an industrial site. Below it are several tabs: 'Daily Discussion Summary', 'Tank Type', 'Water Treatment Input Choices', 'Injection Well List', 'Daily Input Choices', 'Input Source and Recovery Type', and 'Disposal Category and Other Waste Properties'. The 'Disposal Category and Other Waste Properties' tab is active. It contains two tables. The left table is 'Disposal Category' with columns 'Code' and 'Description'. It has two rows: '1 Disposal' and '2 RSL Remediation'. The right table is 'Waste Properties' with columns '#', 'Description', and 'Value'. It has 10 rows, numbered 1 to 10. The first row is '1 Oil & Grease'. An orange arrow points from the text '% Oil' in the list to the first row of the 'Waste Properties' table. The 'Waste Properties' table is circled in orange.

Disposal Category	
Code	Description
1	Disposal
2	RSL Remediation

Waste Properties		
#	Description	Value
1	Oil & Grease	
2		
3		
4		
5		
6		
7		
8		
9		
10		

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Tank Type

- Allows you to set up the different storage and treatment facilities that you will have on location.
- A maximum of 30 different Tanks can be set up.



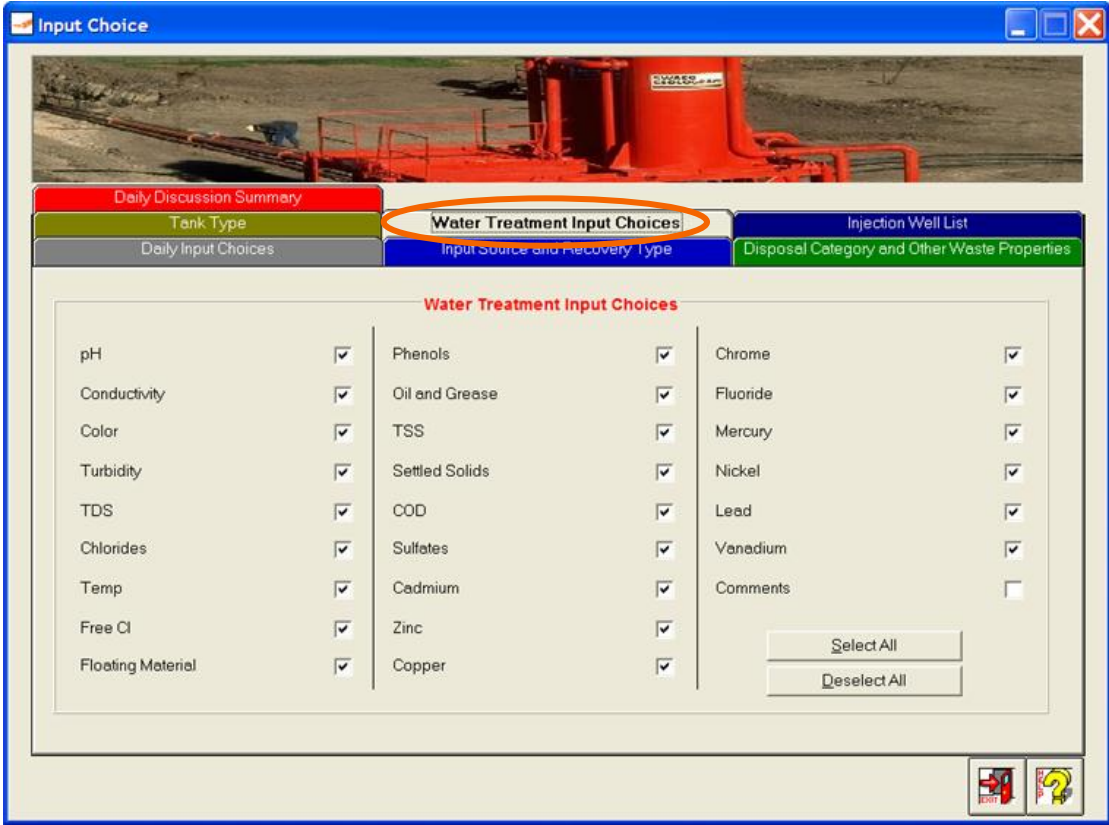
Number	Description	Capacity	Category	Type	Calc Conc?
1	Cuttings Container #1	25	H.O.	P	N
2	Cuttings Container #2	25	H.O.	P	N
3	Cuttings Container #3	25	H.O.	P	N
4	Cuttings Container #4	25	H.O.	S	N
5	Cuttings Container #5	25	H.O.	S	N
6	Cuttings Container #6	25	H.O.	S	N
7	Cuttings Shale Tank	200	H.O.	S	N
8	Dewatering Tank	250	D.W.	P	Y
9					
10					
11					
12					
13					
14					
15					

- Category: designates waste treatment service the tank will be used for (i.e. Dewatering, Water Treatment, Solids Processing & Fixation, Cuttings Injection, Haul Off).
- Type: to indicate if the tank is just for storage (no chemical addition required) or for processing (allowing chemical additions to it).
- Cal. Conc.: select “Y” to make the program calculate the tank’s chemical concentrations; “N” if you don’t require to do that.

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Water Treatment Input Choices

Indicate here the various properties and chemical compositions of the waste water to be tracked, depending on regulatory requirements or company mandates.



The screenshot shows a software window titled "Input Choice" with a blue title bar. Inside the window, there is a header image of an industrial facility. Below the image, there are four tabs: "Daily Discussion Summary" (red), "Tank Type" (green), "Water Treatment Input Choices" (blue, circled in orange), and "Injection Well List" (dark blue). Below the tabs, there are three sub-tabs: "Daily Input Choices" (grey), "Input Source and Recovery Type" (blue), and "Disposal Category and Other Waste Properties" (green). The main content area is titled "Water Treatment Input Choices" and contains a list of properties with checkboxes. The properties are arranged in three columns. The first column includes pH, Conductivity, Color, Turbidity, TDS, Chlorides, Temp, Free Cl, and Floating Material. The second column includes Phenols, Oil and Grease, TSS, Settled Solids, COD, Sulfates, Cadmium, Zinc, and Copper. The third column includes Chrome, Fluoride, Mercury, Nickel, Lead, Vanadium, and Comments. All checkboxes are checked except for "Comments". At the bottom right of the list, there are two buttons: "Select All" and "Deselect All".

Water Treatment Input Choices					
pH	☒	Phenols	☒	Chrome	☒
Conductivity	☒	Oil and Grease	☒	Fluoride	☒
Color	☒	TSS	☒	Mercury	☒
Turbidity	☒	Settled Solids	☒	Nickel	☒
TDS	☒	COD	☒	Lead	☒
Chlorides	☒	Sulfates	☒	Vanadium	☒
Temp	☒	Cadmium	☒	Comments	☐
Free Cl	☒	Zinc	☒		
Floating Material	☒	Copper	☒		

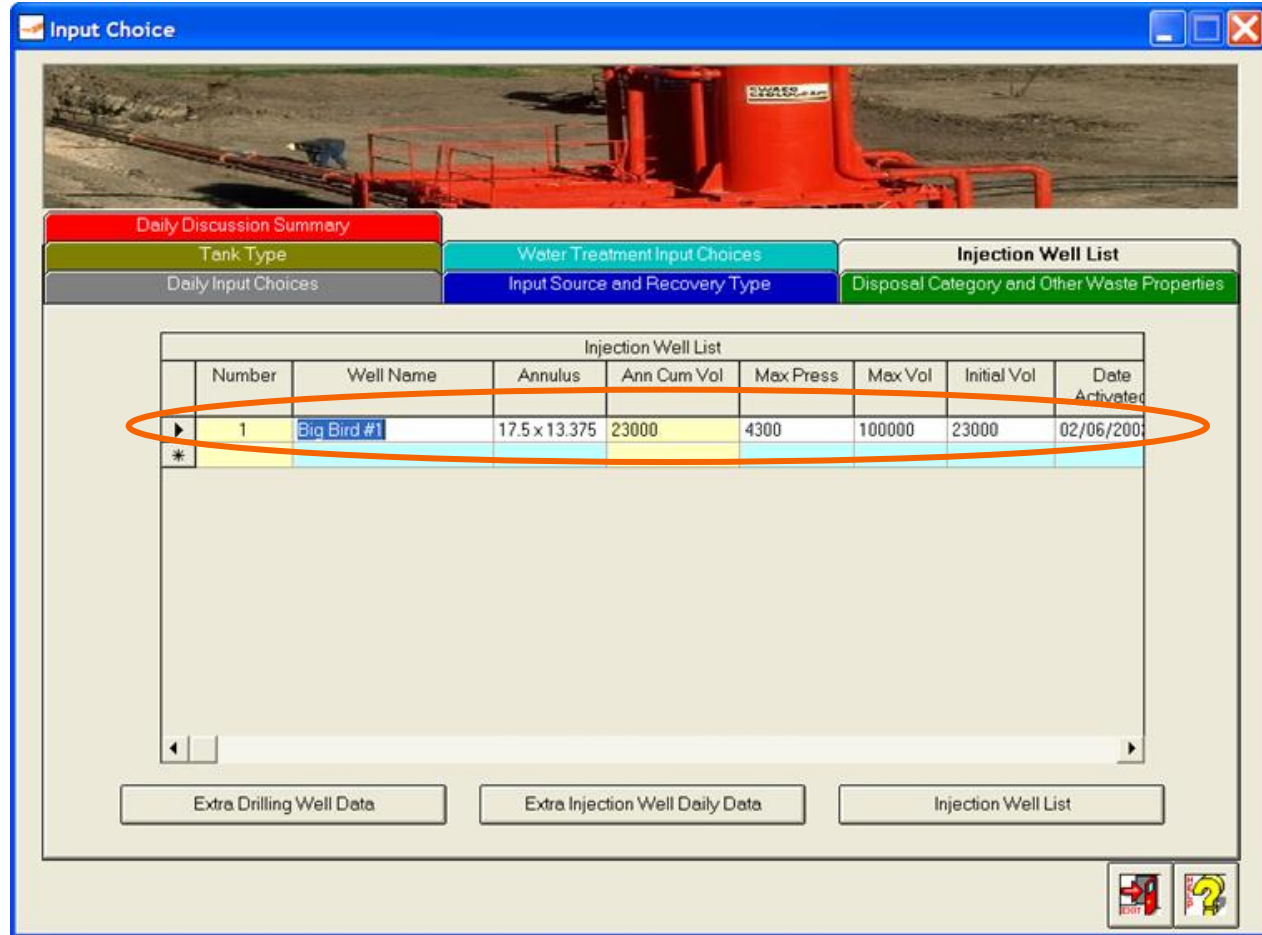
Select All
Deselect All

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Injection Well List

Provide here the injection well data:

- Maximum Pressure.
- Maximum Volume.
- Capacity.
- Well Configuration.
- Etc.



The screenshot shows the 'Input Choice' software window. At the top is a photograph of an industrial site. Below it are several tabs: 'Daily Discussion Summary' (red), 'Tank Type' (green), 'Daily Input Choices' (grey), 'Water Treatment Input Choices' (cyan), 'Input Source and Recovery Type' (blue), and 'Injection Well List' (white). The 'Injection Well List' tab is active, showing a table with the following data:

	Number	Well Name	Annulus	Ann Cum Vol	Max Press	Max Vol	Initial Vol	Date Activated
▶	1	Big Bird #1	17.5 x 13.375	23000	4300	100000	23000	02/06/2001
*								

An orange oval highlights the first row of the table. Below the table are three buttons: 'Extra Drilling Well Data', 'Extra Injection Well Daily Data', and 'Injection Well List'. At the bottom right are icons for a printer and a help document.

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Injection Well List

The data entered here is used to calculate Haul-off cost (transportation and disposal).

	Code	Transportation Company	Price	Type
▶	1	Tosia	8	5
	2	FAR SCAPA	10	5
*				

	Code	Disposal Site	Price	Type
▶	1	RSL Lowesoft	175	3
*				

1) Transportation:

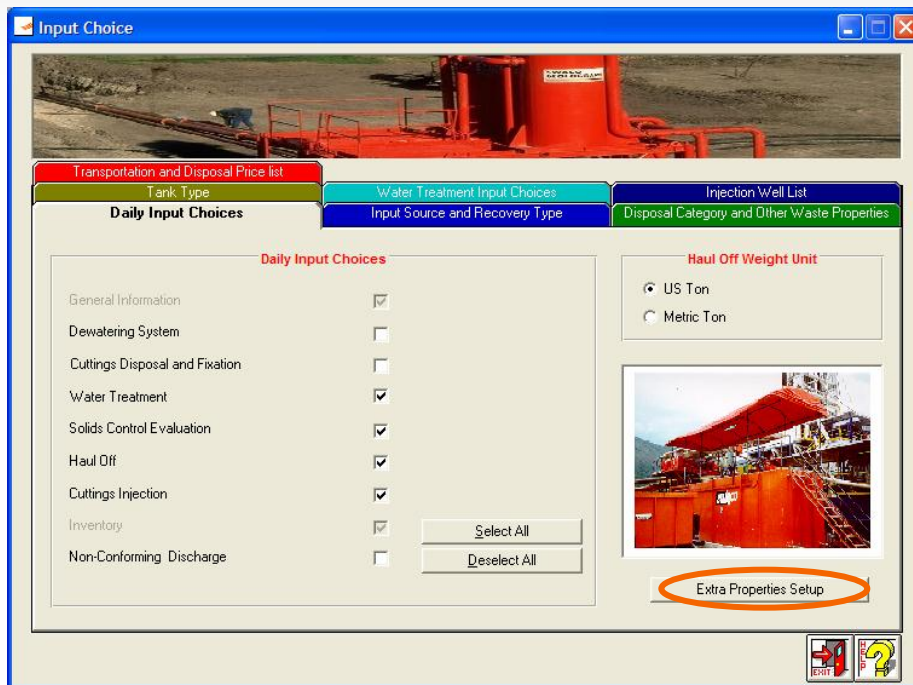
- Trucking Company.
- Cost.
- Cost Structure:
 - Flat.
 - Time Base.
 - Volume.
 - Metric Tonne.
 - Ton.

2) Disposal Cost:

- Company.
- Cost.
- Cost Structure:
 - Volume.
 - Metric Tonne.
 - Ton.

Daily Data Input – Waste Treatment and Disposal Module

Additional Setup – Extra Properties Setup



Input Choice

Transportation and Disposal Price list

Tank Type

Injection Well List

Daily Input Choices

Water Treatment Input Choices

Input Source and Recovery Type

Disposal Category and Other Waste Properties

Daily Input Choices

General Information ☒

Dewatering System ☐

Cuttings Disposal and Fixation ☐

Water Treatment ☒

Solids Control Evaluation ☒

Haul Off ☒

Cuttings Injection ☒

Inventory ☒

Non-Conforming Discharge ☐

Select All

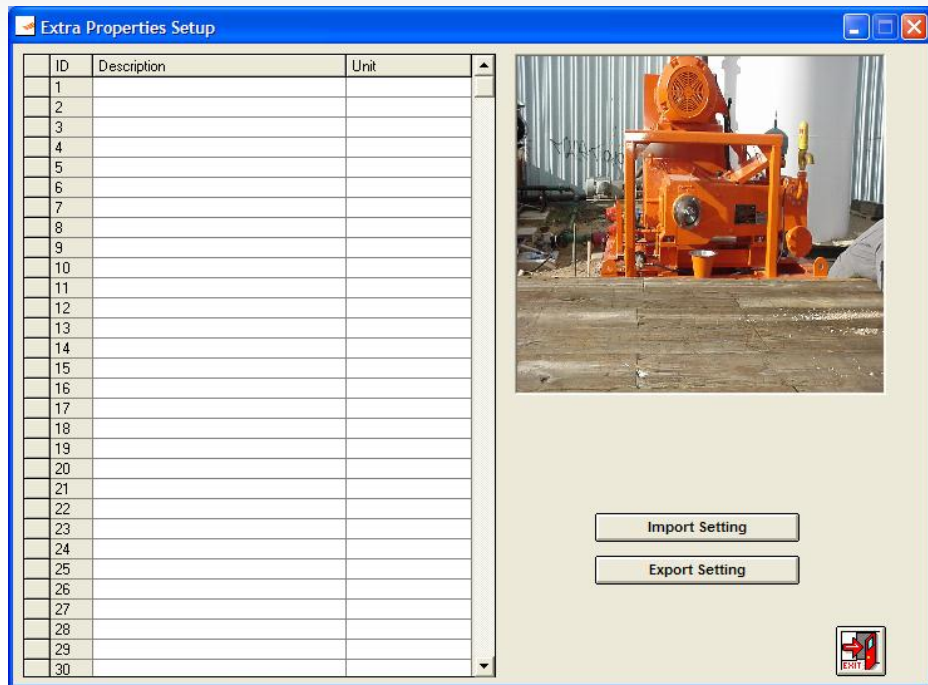
Deselect All

Haul Off Weight Unit

☒ US Ton

☐ Metric Ton

Extra Properties Setup

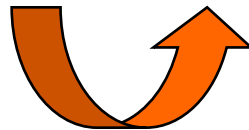


Extra Properties Setup

ID	Description	Unit
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		
25		
26		
27		
28		
29		
30		

Import Setting

Export Setting



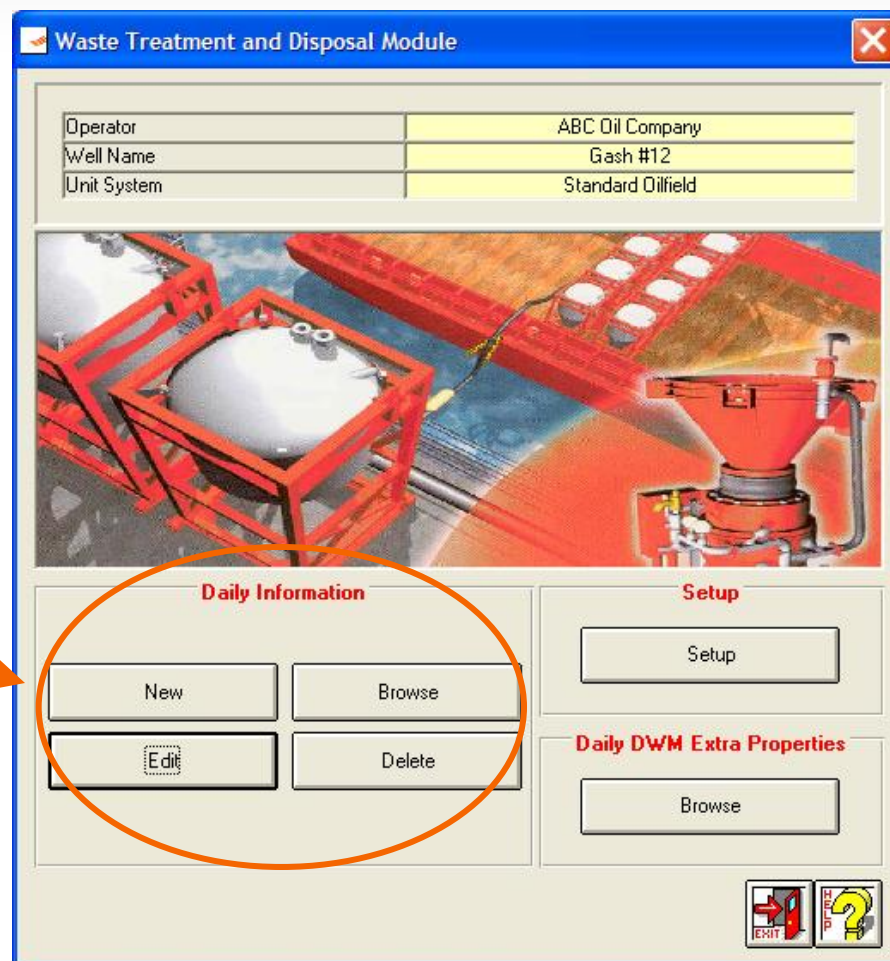
The “**Extra Properties Setup**” allows for 50 extra “wildcard” parameters per project. This provides a parallel capability to the fluids extra properties.

Daily Data Input – Waste Treatment and Disposal Module

Open New Daily Report, Edit, Browse or Delete.

Use the following buttons to either :

- Create a **NEW** day
- **BROWSE** through an existing day
- **EDIT** a previous day, or
- **DELETE** an entire day



Daily Data Input – Waste Treatment and Disposal Module

"General" tab

The screenshot shows the 'General' tab of the ONE-TRAX Report software. The window title is 'ONE-TRAX Report [ABC Oil Company Gash #12 12/13/1999]'. The interface includes a menu bar with 'Cuttings Injection', 'Inventory', 'Non-Conforming Discharge', 'General', 'Transactions', 'Cuttings Disposal and Fixation', 'Water Treatment', 'Solids Control Evaluation', and 'Haul Off'. The 'General' tab is active, displaying a form with various data entry fields. Some fields are highlighted in yellow, indicating data pulled from the 'Drilling Fluids and Equipment' module. The form is organized into several sections: a top data entry section, a personnel list, and three summary/forward plan sections.

ONE-TRAX Report [ABC Oil Company Gash #12 12/13/1999]					
General Transactions Cuttings Disposal and Fixation Water Treatment Solids Control Evaluation Haul Off					
Date	12/13/1999	Mud Type	VersaClean	Depth	5520
Operator	ABC Oil Company	Hole Diameter	12.25	Circ Rate	803
Report No.	6	Daily Rainfall	2.25	Recovered Mud	15
Mud Weight	11.6	Cumulative	2.25	Cumulative	15
Field Supervisor: James Bell		Waste Treatment Summary Started hauling off and injecting. Using vacuum transport system.			
Personnel 1: James McGregor		Waste Treatment Forward Plan			
Personnel 2: Bill SWACO					
Personnel 3: Hank SWECO					
Personnel 4: Yeah, right!					
Drilling Fluid Summary		Drilling Fluid Forward Plan			
Dewatering, Solids Control and Water Treatment Report Comments Started hauling off and injecting. Using vacuum transport system.					
Lithology: sand		Unscheduled Events			

At the bottom of the window, there is a status bar showing the date '12/13/1999' and a row of icons for navigation and actions.

This screen will automatically pull some of the data (those in yellow boxes) from the “Drilling Fluids and Equipment” module.

Insert comments as required by the project.

Waste Management Personnel and Daily Rainfall data is entered here, too.

Daily Data Input – Waste Treatment and Disposal Module

“Transactions” tab

- This tab handles over 75% of all of the data for the “Waste Treatment and Disposal” module, including:
 - Source of Waste (where is the waste coming from).
 - Waste management service used: disposal, haul-off, remediation.
 - Chemical additions for treatment.
 - Waste information: “Volume”, “Density” and “Weight”.
- Note:**
- The program only uses two to calculate the third.
 - Other Waste Information - Time, pH and Solids.
 - Volume transfers.
 - Ect.

Daily Data Input – Waste Treatment and Disposal Module

"Transactions" tab (cont.)

Action Buttons:

- **Input Source**: is used to declare that there is a volume amount subject to Waste Management.
- **Transfer**: used to move fluid from treatment tanks to other pits.
- **Disposal**: used when a waste volume is disposed or transferred out of the rig location.
- **Recovery**: click here if the treatment is intended for recovering valuable fluid.
- **Add Chemical**: used when treatment involves adding chemicals to the waste (e.g. adding polymers on dewatering jobs).

Tank No.	Description	Capacity (bbl)	Category	Type	Volume (bbl)
1	Cuttings Container #1	25	H.O.	P	
2	Cuttings Container #2	25	H.O.	P	
3	Cuttings Container #3	25	H.O.	P	
4	Cuttings Container #4	25	H.O.	S	
5	Cuttings Container #5	25	H.O.	S	
6	Cuttings Container #6	25	H.O.	S	
7	Cuttings Shale Tank	200	H.O.	S	148
8	Dewatering Tank	250	D.W.	P	10
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

Beginning	101
In	160
Out	103
Ending	158

Drilling Fluids Surface Losses 271

Print Grid View Trans Recalculate Reset View Conc Injection Wells
Input Sources Transfer Disposal Recovery Add Chemicals Fluids Activity Sum

Daily Data Input – Waste Treatment and Disposal Module

"Transactions" tab (cont.)

Sample of Haul-Off Disposal Transaction

[illegible]

Daily Data Input – Waste Treatment and Disposal Module

"Transactions" tab (cont.)

Waste Management Transactions

Summary of Processing and Storage Tank Summary of Transactions

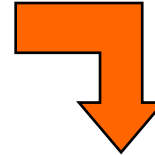
Tank No.	Description	Capacity (bbl)	Category	Type	Volume (bbl)
1	Cuttings Container #1	25	H.O.	P	
2	Cuttings Container #2	25	H.O.	P	
3	Cuttings Container #3	25	H.O.	P	
4	Cuttings Container #4	25	H.O.	S	
5	Cuttings Container #5	25	H.O.	S	
6	Cuttings Container #6	25	H.O.	S	
7	Cuttings Shale Tank	200	H.O.	S	148
8	Dewatering Tank	250	D.W.	S	
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

Waste Treatment and Disposal System Volume (

Beginning	101	Drilling Fluids Surface Losses
In	160	
Out	103	
Ending	158	

Print Grid **View Trans.** Recalculate Reset View

Input Sources Transfer Disposal Recovery Add Ch



Waste Management Transactions

Transactions

Trans No.	Description
1	Input 150 bbl from Cuttings w/Mud Active System to Cuttings Shale Tank
2	Transferred 16 bbl from Cuttings Shale Tank to Cuttings Container #1
3	Disposed 16 bbl (RSL Remediation) from Cuttings Container #1
4	Transferred 16 bbl from Cuttings Shale Tank to Cuttings Container #2
5	Disposed 16 bbl (RSL Remediation) from Cuttings Container #2
6	Transferred 16 bbl from Cuttings Shale Tank to Cuttings Container #3
7	Disposed 16 bbl (RSL Remediation) from Cuttings Container #3
8	Transferred 15 bbl from Cuttings Shale Tank to Cuttings Container #4
9	Disposed 15 bbl (RSL Remediation) from Cuttings Container #4
10	Transferred 20 bbl from Cuttings Shale Tank to Cuttings Container #5
11	Disposed 20 bbl (RSL Remediation) from Cuttings Container #5
12	Transferred 20 bbl from Cuttings Shale Tank to Cuttings Container #6
13	Disposed 20 bbl (RSL Remediation) from Cuttings Container #6
14	Added Chemicals to Dewatering Tank: - Added 8 bbl Water - Added 3 (50 LB BG) Caustic Soda - Added 2 (25 KG BG) Lime - Added 0.5 (1 TN BK) Salt - PVD
15	Input 50 bbl from Cuttings w/Mud Active System to Cuttings Container #2
16	Disposed 35 bbl (RSL Remediation) from Cuttings Container #2

Waste Treatment and Disposal System Volume (

Beginning	101	Drilling Fluids Surface Losses	271
In	210		
Out	138		
Ending	173		

Print Grid Back Trans. Properties Undo Last Trans.

Click Here to View Transactions

Back to: Table of Content

Daily Data Input – Waste Treatment and Disposal Module

"Transactions" tab (cont.)

- **View Transactions** - allows you to look at individual transactions. You can delete the last transaction. If an error occurs early in the transactions you must delete transactions one at a time until you delete the error. Then all of the transactions must be input again.
- **Reset** - Deletes all Transactions and any Inputs for the day and restores to previous days data.

Daily Data Input – Waste Treatment and Disposal Module

"Transactions" tab (cont.)

Summary Of Transactions

Waste Management Transactions

Summary of Processing and Storage Tank | **Summary of Transactions**

Date Closed	Dewatering		W. T.		Cut. Disp./Fix.		Cutting Inj.		Hauloff		Rec
	P	S	P	S	P	S	P	S	P	S	
From Dewatering P	—										
From Dewatering S		—									
From W. T. P			—								
From W. T. S				—							
From Cut. Disp./Fix. P					—						
From Cut. Disp./Fix. S						—					
From Cutting Inj. P							—				
From Cutting Inj. S								—			
From Hauloff P									—		
From Hauloff S									48		
From Input Sources									50	150	
From Chemical	10										
Total Transfer In	10								98	150	
Total Transfer Out									83	103	
Daily Balance(In-Out)	10								15	47	

Waste Treatment and Disposal System Volume (

Beginning	101
In	210
Out	138
Ending	173

Drilling Fluids Surface Losses 271

Print Grid View Trans. Recalculate Reset View Conc. Injection Wells

Input Sources Transfer Disposal Recovery Add Chemicals Fluids Activity Sum

Daily Data Input – Waste Treatment and Disposal Module

"Transactions" tab (cont.)

Remaining Buttons

- **Fluids Activities** - Fluid Movements.
- **Print Grid** - Prints movements of fluids and wastes.
- **Recalculate** - Costs and concentrations.
- **View Concentrations** - Shows concentrations of dewatering, water treatment or cuttings treatment.
- **Injection wells** - Injection well data.

Daily Data Input – Waste Treatment and Disposal Module

“Dewatering”, “Cuttings Disposal & Fixation” and “Water Treatment”

- Dewatering:
 - Data is filled in from Transactions
- Cutting Disposal and Fixation:
 - Some volume data is required to show water and oil base cuttings and sludge along with % oil and % water.
- Water Treatment:
 - Designate up to five (5) sampling point and input the water properties designated in the setup. Pit volumes are automatically transferred from Transactions.

Daily Data Input – Waste Treatment and Disposal Module

“Solids Control”, “Haul-Off” and “Cuttings Injection”

- Solids Control:
 - Evaluates the various pieces of equipment. Data can be input or will transfer from Drilling Fluids module.
- Haul Off:
 - Provides a table of all the loads moved that day along with cost information.
- Cuttings Injection:
 - Inputs are required for slurry properties along with unit running times and downtimes.

Daily Data Input – Waste Treatment and Disposal Module

"Inventory" tab

This section allows you to charge:

- Waste Management Personnel
- Waste Treatment Chemicals
- Rental Equipment

You can also create delivery or return tickets for:

- Shaker Screens
- Waste Treatment Products

The screenshot shows the 'Daily Inventory' window for the date 12/15/2002. It features a tabbed interface with the following tabs: 4 - Additional Equipment, 5 - Shaker Screens, 1 - Chemical (selected), 2 - Dewater Equipment, and 3 - Solids Control. The '1 - Chemical' tab displays a table of chemical inventory data.

No.	Product Name	Prod. Code	Unit	In Stock	Used for Mud	Cum. Used For Mud	Used for Dewat	Cum. Used for Dewat	Used for W Tr
4	Calcium Chloride (94%)	M0004017	LB	0	0	40	0	0	0
5	Calcium Chloride Brine (11.5%)	M0000934	BL	0	0	0	0	0	0
6	Caustic Soda	M0006659	LB	66	0	11	0	3	0
7	Duovis	M0007978	LB	285	0	75	0	0	0
8	Guar Gum	M0006781	LB	40	0	0	0	0	0
9	Lime	M0006806	KG	18	0	20	12	14	0
10	Versamul	M0005319	GA	18	0	2	0	0	0
11	Versaclean Mud - 11.5 ppg	M0002631	BL	430	0	2270	0	0	0
12	Versaclean Mud - 11.7 ppg	M0002624	BL	0	0	0	0	0	0
13	Versacoat	M0006339	GA	11	0	5	0	0	0
14	Versagel	M0002679	LB	50	0	30	0	0	0
15	Nut Plug	M0006403	LB	80	0	0	0	0	0

Below the table, there is a 'Full List' section with the following data:

Full List	
Daily Cost	2428
Cumulative Cost	315126.25

There is a 'Delivery/Return Tickets' button on the right. Below the cost summary, there is a red text label: 'Material Consumption Remarks: (on Solids Report)'. At the bottom, there are four buttons: 'View Full List', 'DWMgt. Chemicals', 'DWMgt. Personnel', and 'Recalculate'. On the far right, there are three small icons: a red 'X', a blue 'U', and a yellow question mark.

Daily Data Input – Waste Treatment and Disposal Module

"Inventory" tab (cont.)

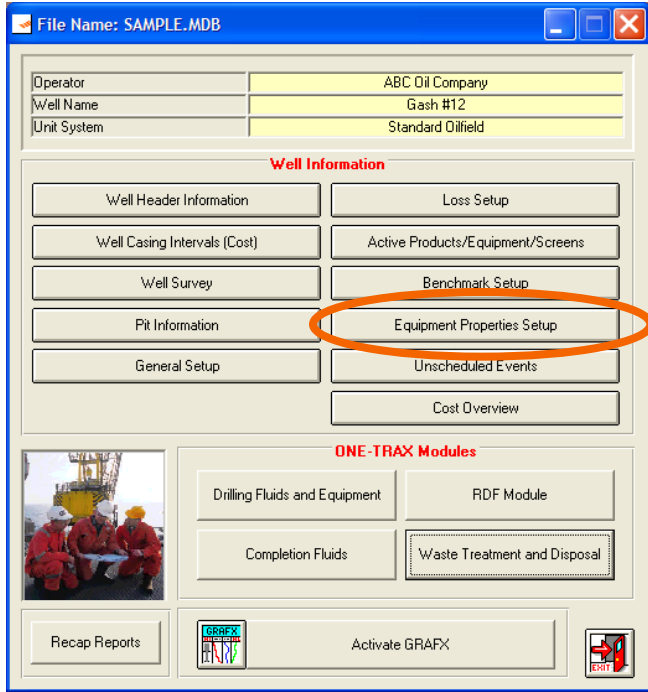
- Note: Some of this data will transfer from the Drilling Fluid Module if it was input and pulled over. If not you will have to input the data here.
- Calculates costs for Waste Treatment equipment and personnel.

Be careful not to duplicate materials received or materials used from drilling fluid inputs.

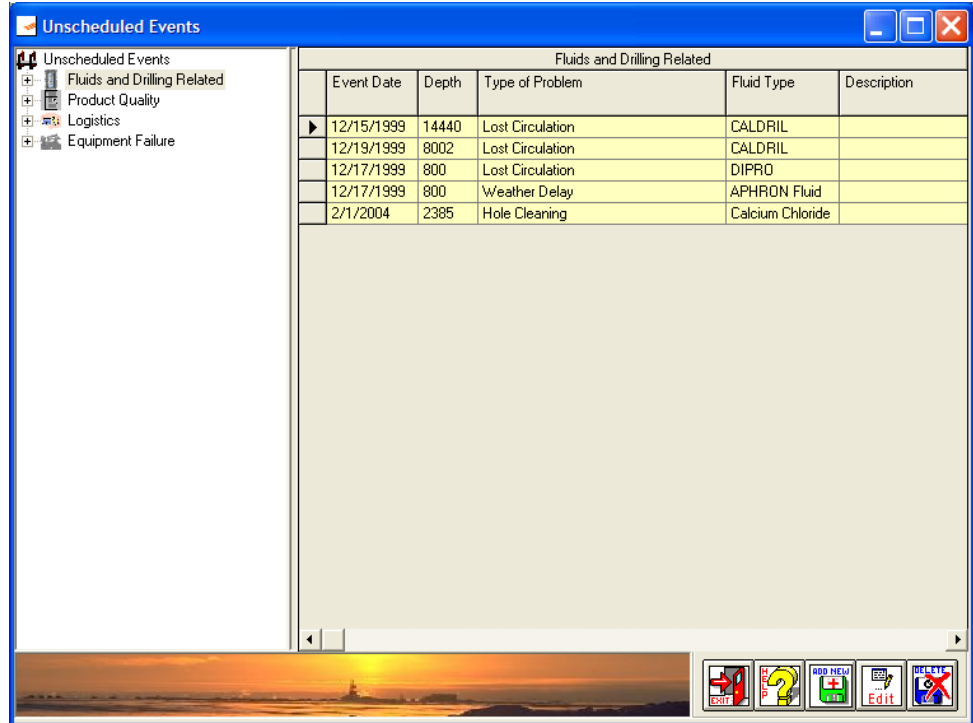
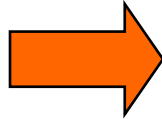
End of “Waste Treatment and Disposal” module

Unscheduled Events – All Modules

The “Unscheduled Events” section in ONE-TRAX can be accessed from several screens in each module.



The screenshot shows the ONE-TRAX main menu. At the top, it says "File Name: SAMPLE.MDB". Below this are fields for "Operator" (ABC Oil Company), "Well Name" (Gash #12), and "Unit System" (Standard Oilfield). A section titled "Well Information" contains several buttons: "Well Header Information", "Loss Setup", "Well Casing Intervals (Cost)", "Active Products/Equipment/Screens", "Well Survey", "Benchmark Setup", "Pit Information", "Equipment Properties Setup" (circled in orange), "General Setup", "Unscheduled Events", and "Cost Overview". At the bottom, there's a section for "ONE-TRAX Modules" with buttons for "Drilling Fluids and Equipment", "RDF Module", "Completion Fluids", and "Waste Treatment and Disposal". There are also buttons for "Recap Reports" and "Activate GRAFX".



The screenshot shows the "Unscheduled Events" window. It has a left sidebar with a tree view containing "Unscheduled Events", "Fluids and Drilling Related", "Product Quality", "Logistics", and "Equipment Failure". The main area displays a table titled "Fluids and Drilling Related" with the following data:

Event Date	Depth	Type of Problem	Fluid Type	Description
12/15/1999	14440	Lost Circulation	CALDRIL	
12/19/1999	8002	Lost Circulation	CALDRIL	
12/17/1999	800	Lost Circulation	DIPRO	
12/17/1999	800	Weather Delay	APHRON Fluid	
2/1/2004	2385	Hole Cleaning	Calcium Chloride	

At the bottom of the window, there's a status bar with a landscape image and several icons: "EXIT", "NEW", "ADD NEW", "Edit", and "DELETE".

Detailed information about the use for this section is presented in the following slides

Unscheduled Events – All Modules

- It's very important to use the “**Unscheduled Events**” capabilities in ONE-TRAX Wellsite to document problems that occur during a project. The Unscheduled Events fall into four major categories :
 - Drilling and Fluids Related (includes problems such as lost circulation & stuck pipe)
 - Equipment Related
 - Logistics Related (includes late deliveries)
 - Product Quality Related
- All events that fall into the above categories should be recorded regardless of fault. This is not simply a way to record problems caused by M-I SWACO.
- This information is not shown on the daily report (unless a custom mud template is designed to display).
- This information is transferred from ONE-TRAX Wellsite to an online system named the Incident Manager when the wellfile is uploaded. Online, additional information is added, lessons learned are documented and archived for future use.

Unscheduled Events – All Modules

Data Entry

File Name: SAMPLE.MDB

Operator	ABC Oil Company
Well Name	Gusher #2
Unit System	Standard Oilfield

Well Information

Well Header Information	Loss Setup
Well Casing Intervals (Cost)	Active Products/Equipment/Screens
Well Survey	Benchmark Setup
Pit Information	Equipment Properties Setup
General Setup	Unscheduled Events
	Cost Overview

ONE-TRAX Modules

	Drilling Fluids and Equipment
	Completion Fluids
	Waste Treatment and Disposal

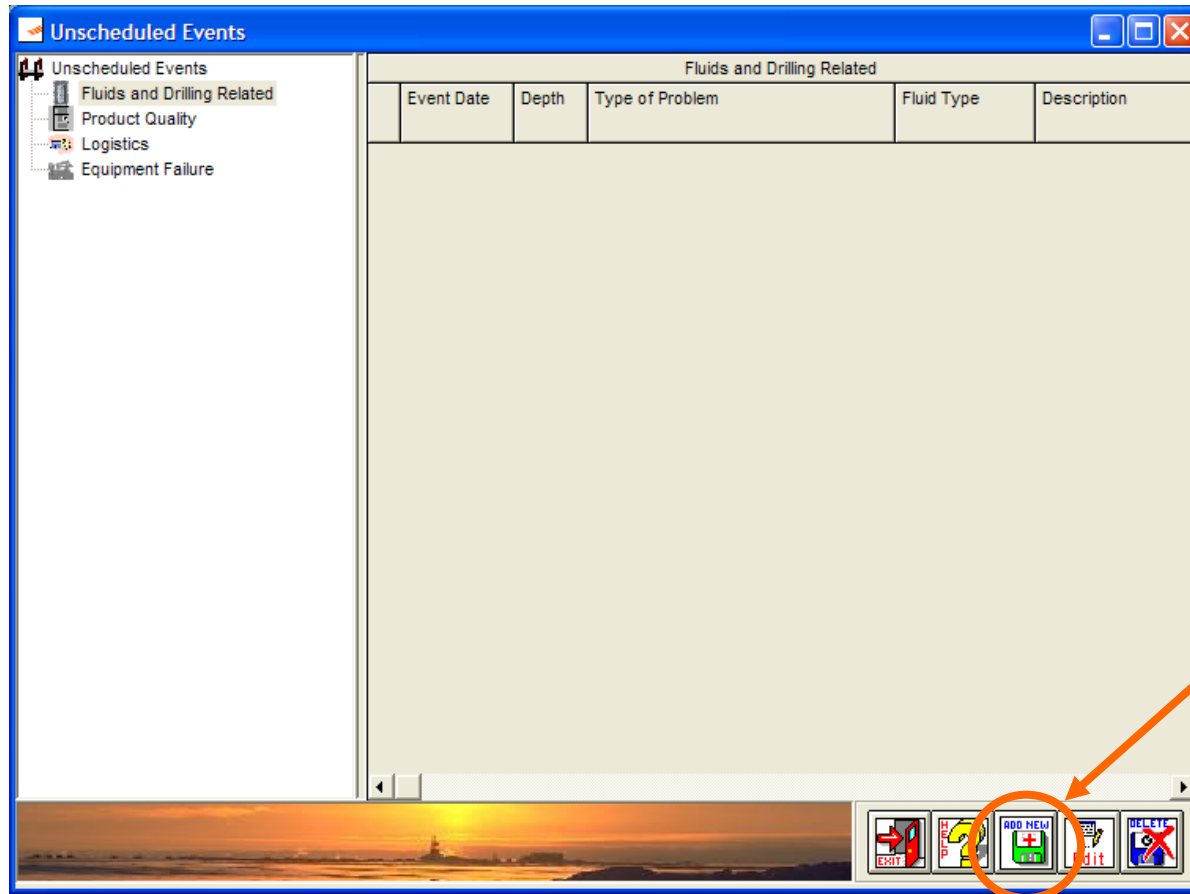
Recap Reports

 Activate GRAFX 

The Unscheduled Event data entry is accessed by clicking this button in ONE-TRAX Wellsite

Unscheduled Events – All Modules

Data Entry – Getting started



Click the “Add New” button to start the data entry for a new event.

Unscheduled Events – All Modules

Data Entry (cont.)

The screenshot shows the 'Event Details' window with the following fields:

Event Date	2/29/2008	Depth (ft)		Incident Category	
Interval Number	1			Type of Problem	
Fluid Type				Type of Problem 2	
Description					
Suspected Cause					
Loss Description					
Loss Time (Hr)	0	Product Code			
Customer Assigned NPT (Hr)	0	Product Description			
Fluid Density (lb/gal)	0	Transportation Type			
Equipment Description		Transportation Company			
Equipment Asset Number		M-I SWACO Operated?			
Rig Power	0	Is this Product Specific?			

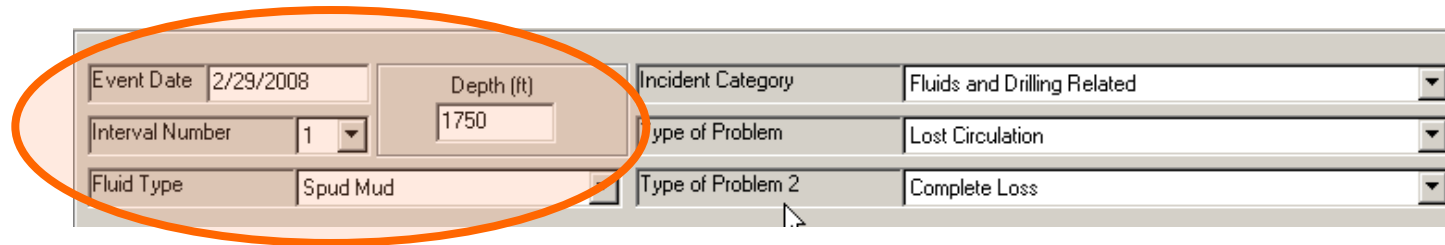
An orange oval highlights the 'Incident Category', 'Type of Problem', and 'Type of Problem 2' dropdown menus. An orange arrow points from the bottom of the screen up to the oval.

The next step is to select the “**Incident Category**”, “**Type of Problem**” and “**Type of Problem 2**”. The input fields at the bottom of the screen will change based on the category selections.

Unscheduled Events – All Modules

Data Entry (cont.) – Common Fields

Common entries to all Incidents



Event Date	2/29/2008	Depth (ft)	1750	Incident Category	Fluids and Drilling Related
Interval Number	1			Type of Problem	Lost Circulation
Fluid Type	Spud Mud			Type of Problem 2	Complete Loss

- Event Date : Enter the date when the problem first occurred. Note - the unscheduled event reporting is independent of the daily report creation. If the event lasts for more than one day it only needs to be entered once.
- Interval number – Select the current interval from the pull-down list.
- Depth – Enter the measured depth at the point when the event started.
- Fluid type – Select the fluid type from the pull-down list.

Unscheduled Events – All Modules

Data Entry (cont.) - Common Fields

Description	Partial losses starting at 1631', to complete losses at 3248'. Losses cured by spotting assorted LCM pills at 35 ppb and allowing 1/2 hour to heal. Water with Lime was only used to drill from 3248' to 3392', where circulation was again re-established. (2 cmt top jobs required to cmt the csg).
Suspected Cause	Fractured nature of the Teyarat limestone.
Loss Description	Partial to complete losses. (Volume lost below, is up to when casing was started to be run in the hole).
Loss Time (Hr)	6.5
Customer Assigned NPT (Hr)	

- **Description** : Provide a complete description of the problem.
- **Suspected Cause** : Provide details on the suspected cause if this information is known.
- **Loss Description** : Provide details on the loss.
- **Loss Time (Hr)** : Record the amount of time lost because of the problem. This is not NPT that has been assigned to a specific service company. This is simply the elapsed time from when the problem started until the planned activity resumed.
- **Customer Assigned NPT (Hr)** : This should only be recorded if the customer has a system for assigning NPT to contractors – and – has assigned NPT to M-I SWACO.

Unscheduled Events – All Modules

Data Entry (cont.) – Fluids and Drilling Related incidents

Suspected Cause	
Loss Description	
Loss Time (Hr)	0
Customer Assigned NPT (Hr)	0
Fluid Density (lb/gal)	0
Volume of Fluid Lost (bbl)	0

- **Fluid Density** : enter the density of the fluid
- **Volume of Fluid Lost** : for lost circulation incidents, enter the total volume of fluid lost.

Unscheduled Events – All Modules

Data Entry (cont.) – Equipment Failure incidents

The screenshot shows a data entry form with the following fields:

- Loss Time (Hr): 0
- Customer Assigned NPT (Hr): 0
- Equipment Description: (pull down list)
- Equipment Asset Number: (text field)
- Rig Power: 0
- Is M-I SWACO responsible for maintenance?: (pull down list)

- **Equipment Description** : the pull down list is populated by the active equipment list. Data should be entered directly into this field only if the equipment is not found in the pull down list. Note – make sure that equipment in the active list is described adequately (it should include the manufacturer name and model).
- **Equipment Asset Number** : if known enter this information.
- **Rig Power** : select from pull down list.
- **Is M-I SWACO responsible for maintenance?** : select from pull down list. It is a very important to complete this field.

Unscheduled Events – All Modules

Data Entry (cont.) – Logistics incidents

Loss Description	
Loss Time (Hr)	0
Customer Assigned NPT (Hr)	0
Product Code	
Product Description	
Transportation Type	
Transportation Company	
M-I SWACO Operated?	
Is this Product Specific?	

- **Product Code** : if the incident relates to a specific product then select the product code from the pull down list. This data comes from the active product list.
- **Product Description** : additional descriptive information about the product can be entered here.
- **Transportation Type** : select from pull down list (ship or truck)
- **Transportation Company** : enter the name of the transportation company
- **M-I SWACO Operated?** : “Yes”, if M-I SWACO operates the transport vessel; otherwise enter “No”.
- **Product Specific?** : “Yes”, if specific to a certain product; otherwise enter “No”.

Unscheduled Events – All Modules

Data Entry (cont.) – Product Quality incidents

The screenshot shows a data entry form with the following fields:

- Loss Description: A large text area with a mouse cursor pointing to it.
- Loss Time (Hr): A numeric input field with the value 0.
- Customer Assigned NPT (Hr): A numeric input field with the value 0.
- Product Code: A pull-down menu.
- Product Description: A text input field.

The Product Code and Product Description fields are highlighted with an orange border.

- **Product Code** : select the product code from the pull down list. This data comes from the active product list.
- **Product Description** : additional descriptive information about the product can be entered here.

Unscheduled Events – All Modules

Unscheduled Events Reports

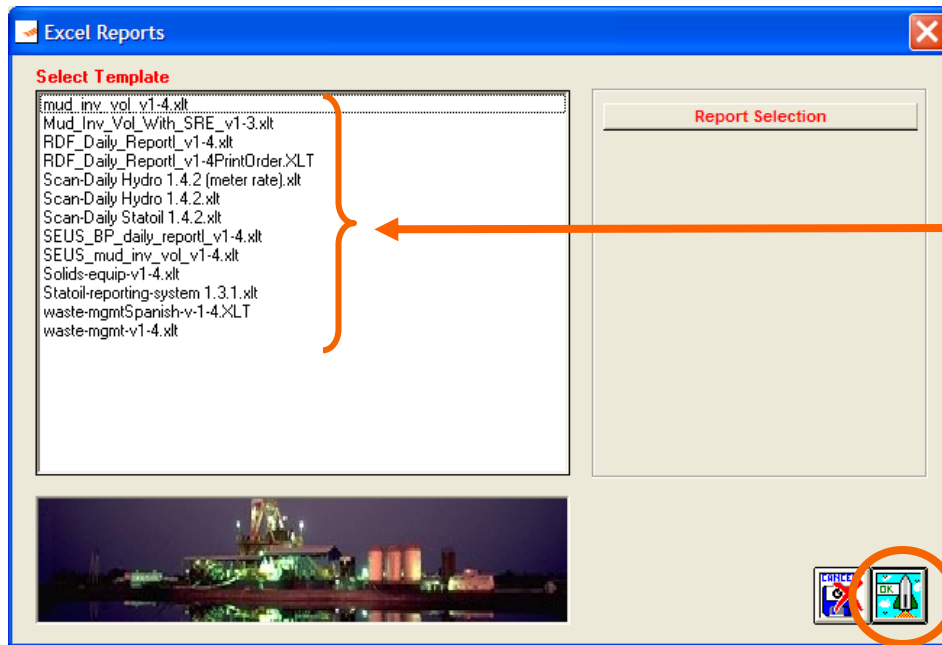
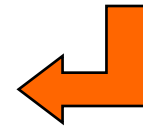
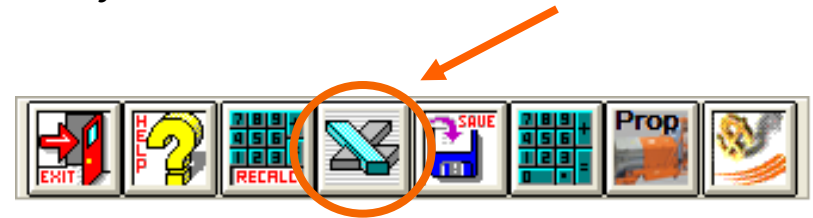
- By design, the “Unscheduled Events” information does not appear on the daily report templates included with ONE-TRAX. However, it is possible to design a custom report template and include this information.
- All of the “Unscheduled Events” data can be exported to a MS-Word format recap report. To generate this report, go to the Excel-based recap screen and then click the “Unscheduled Events” button.
- This data is sent to the both the daily and recap reporting systems. Custom Excel reports can be developed by linking to “ifedata.xls” for daily reports – or – “recapdata.xls” for recap reports.

End of “Unscheduled Events”

Daily Data Reporting – All Modules

M-I SWACO Daily Reports

The toolbar on the right is common to all modules in ONE-TRAX. Most daily reports are Excel based, so to select these types of reports you need to click in **this** button.



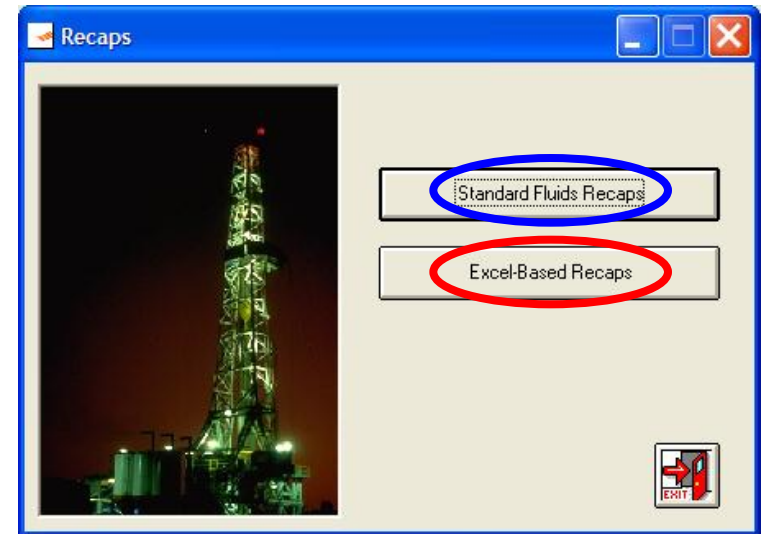
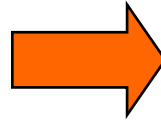
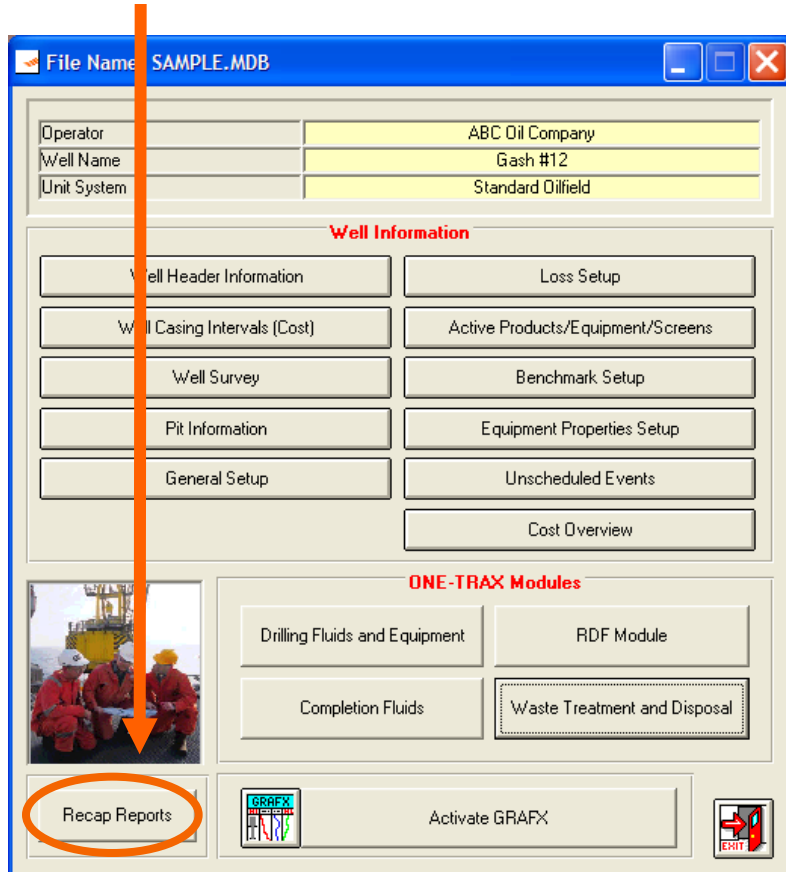
Once you have press the “Excel Reports” button, you can select the report to use from the list that will appear on your screen.

Finally, click the **OK** button.

Well Data Reporting – All Modules

M-I SWACO Recap Reports

To generating a well Recap go to the ONE-TRAX screen displayed below and click the “**Recap Reports**” button.



From the screen above you'll have access to “**Standard Recaps**” (that do not allow changes) and “**Excel Based Recaps**” (that can be adapted to the project/client specific requirements).

End of “Reports”

Special Cases: Side-Track

Side Track setup - Step #1

Normally, ONE-TRAX calculated the hole drilled by using the current depth minus the previous day's depth; however, this method does not work for the first day of a side-track. ONE-TRAX correctly calculates the drilled 'hole volume' and 'length' when a side-track occurs as long as the steps described below are followed. Note: only one sidetrack per day can be entered.

Important – it is very important to follow these steps (when a side-track happens) because several operators use: length drilled, volume of hole drilled, SCE and dilution volumes, as part of the key performance indicators to evaluate M-I SWACO service. This process ensures these values are correctly calculated in the event of a side-track.

The ONE-TRAX requirements -for a side-track- is a simple 3 steps process:

Step #1 – Enter the 'Plug Back Depth' the day before the side-track:

In ONE-TRAX, go to the date just previous to the side-track kick-off day and enter the Plug Back (or abandoned hole) depth, in the "Depth" column of the last interval in the "Daily/Casing Volume" section in tab #4. **Do not change the depth in tab #1; otherwise you will affect your volume accounting.**

As an example you can see (figure 1) a well where the Plug Back (or abandoned hole) depth is at 20,171'.

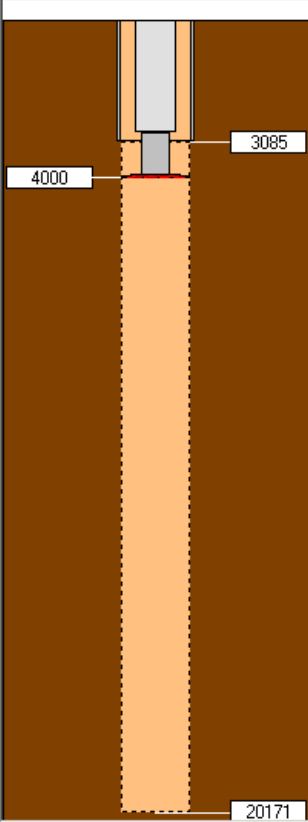
Note: This depth must appear only the day prior to the side-track.

Special Cases: Side-Track

Side Track setup - Step #1 (cont.)

Figure 1 - Plug Back Interval Depth – 20,171'

Well Geometry (1/18/2005)



Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size
1	F	16	15	114	114		
2	F	9.625	8.921	3085	2360	0	12.25
3	O			20171	484		8.5

Side-Track Setup Help

ADD NEW SAVE DELETE

Hole Volume Summary		Bit Size/Hole Diameter	8.5	Air Gap	
Drill String	64	Washout Hole Size	9.691	Water Depth	
Annulus	239	Depth	20171	Riser ID	
Below Bit	1475	TVD	4848	Riser OD	0
Total Volume	1775	Bit Depth	4000	Riser Length	
Drilled Hole		Pilot Hole Depth	0	Riserless Drilling	☐
		Pilot Hole Size	0		

Hole Volume Detail

EXIT

Special Cases: Side-Track

Side Track setup - Step #2

Step #2 – Enter the side-track data, on the kick-off day:

In the side-track kick-off day go to the “Daily/Casing Volume” section -in tab #4- and add a new interval (type “S”, for “Side-track”), which depth should be equal to the TD value (entered in tab #1). The kick-off depth for the side-track must be entered in the interval right above “S”. **Note:** editing this value will not affect your volume accounting.

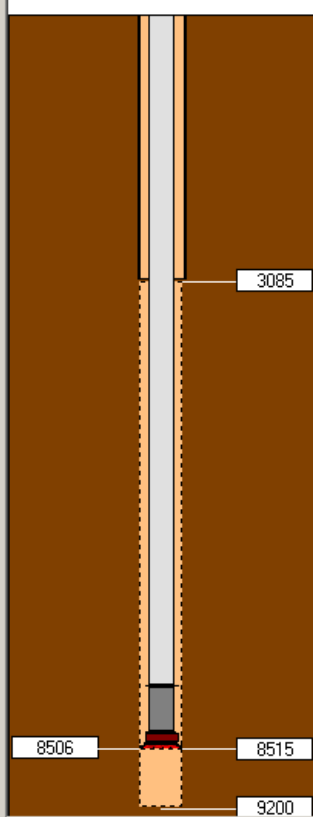
Following the same example, ‘figure 2’ shows the side-track kick-off depth of 8,515’ entered in open-hole interval, right above the side-track. Also, the added side-track interval (“S”) shows the final depth for the day: 9,200’.

Special Cases: Side-Track

Side Track setup - Step #2 (cont.)

Figure 2 - Kick-Off Depth 8,515' - New Side-Track Depth 9,200'

Well Geometry (1/19/2005)



Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size
1	F	16	15	114	114		
2	F	9.625	8.921	3085	2360	0	12.25
3	O			8515	5037		8.5
4	S			9200			8.5

Side-Track Setup Help

ADD NEW SAVE DELETE

Hole Volume Summary

Drill String	144
Annulus	409
Below Bit	63
Total Volume	616
Drilled Hole	62

Hole Volume Detail

Bit Size/Hole Diameter	8.5	Air Gap	
Washout Hole Size	9.69	Water Depth	
Depth	9200	Riser ID	
TVD	5037	Riser OD	0
Bit Depth	8506	Riser Length	
Pilot Hole Depth	0	Riserless Drilling	☐
Pilot Hole Size	0		

EXIT

Special Cases: Side-Track

Side Track setup - Step #3

Step #3 – Delete the side-track interval, the day after the side-track kick-off:

The day after the side-track is kicked-off go to the “Daily/Casing Volume” section -in tab #4- and delete the side-track interval (“S”) from the table. The last row in the ‘Daily Intervals’ table should be an open-hole interval (“O”) with a depth equal to the TD entered in tab #1 (just like it’s normally done in ONE-TRAX).

Note: as you can see the “S” interval is only used the day of the side-track kick-off.

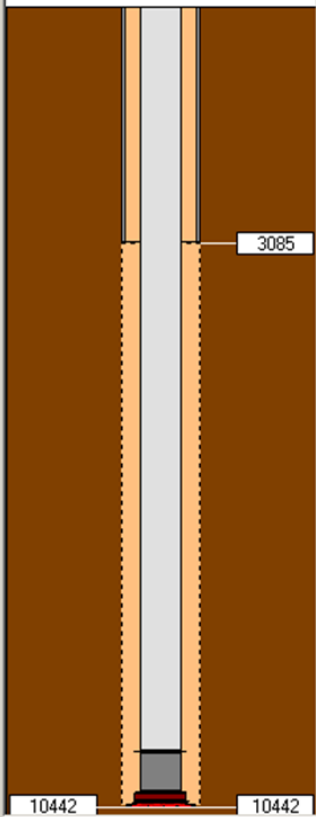
Following the example used before, ‘figure 3’ shows the open-hole section with a final depth equal to that day’s TD (same value entered in tab #1).

Special Cases: Side-Track

Side Track setup - Step #3 (cont.)

Figure 3 - New Open Hole Depth 10,442'

Well Geometry (1/20/2005)



Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size
1	F	16	15	114	114		
2	F	9.625	8.921	3085	2360	0	12.25
3	O			10442	5037		8.5

Side-Track Setup Help

Hole Volume Summary

Drill String	178
Annulus	652
Below Bit	
Total Volume	831
Drilled Hole	113

Hole Volume Detail

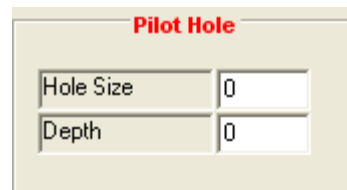
Bit Size/Hole Diameter	8.5	Air Gap	
Washout Hole Size	9.69	Water Depth	
Depth	10442	Riser ID	
TVD	5994	Riser OD	0
Bit Depth	10442	Riser Length	
Pilot Hole Depth	0	Riserless Drilling	☐
Pilot Hole Size	0		

EXIT

Special Cases: Pilot hole

When you are opening an existing hole, there are a few things to keep in mind when entering the data into ONE-TRAX: the well geometry has to be entered in such a way to allow ONE-TRAX to accurately calculate not only the hole volumes, but also the volume of drilled cuttings. Having the geometry and volumes data correct means your hydraulics calculations, bottoms-up, hole volume, etc. will be correct too. Having the volume of drilled cuttings correctly calculated also means that the SCE volume (mud and dry cuttings) discharged calculated in tab #6 will be correct. Without having both of these items calculated correctly, balancing your volumes will be difficult and inaccurate. Following are the instructions for getting the right data in the right places of ONE-TRAX, along with some screen captures to illustrate what is required in the software.

When you drill the pilot hole, there is not special data requirements for ONE-TRAX because this is just a “regular” hole of a certain size. The specific data requirements come into play when the larger bit starts ‘opening’ a hole that’s been already drilled. When that happens the volume of cuttings on surface are a fraction of the normal drilling scenario and also a fraction of the entire hole volume is actually circulating, so ONE-TRAX requires additional information to correctly calculates these parameters. This scenario is addressed using the “Pilot hole” box in tab #4.



Pilot Hole	
Hole Size	0
Depth	0

Special Cases: Pilot hole

Data Entry – Example: before start opening the hole

The current situation:

- 13-3/8" casing at 920 meters.
- 8-1/2" pilot hole drilled to 1710 meters
- Now: tripping in the hole

Tab-4 Well Geometry / Daily Casing & Volume

The screenshot displays the 'Tab-4 Well Geometry / Daily Casing & Volume' window. On the left is a wellbore diagram with depth markers at 112, 796, 920.11, and 1710 meters. The main area contains a 'Daily Intervals' table and a 'Hole Volume Summary' section.

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size
3	F	13.375	12.415	920.11	920.11	0	17.5
4	O			1710	1710		8.5

Hole Volume Summary (bbl)	
Drill String	39
Annulus	372
Below Bit	439
Total Volume	850
Drilled Hole	
Drilled Depth (m)	

Hole Volume Detail	
Bit Size/Hole Diameter	12.25
Washout Hole Size	12.25
Depth	1710
TVD	1710
Bit Depth	796
Pilot Hole Depth	0
Pilot Hole Size	0

Note: Hole geometry accurately reflects current geometry including the 8-1/2" pilot hole.

Tab-1 General

1- General		2- Pumps / Bits	
Date	7/29/2006	Depth (m)	1710
Spud Date	7/9/2006	TVD (m)	1710
Report	22	Bit Depth (m)	796
Present Activity	RIH 12.25" BHA		

Note: Depth is TOTAL depth of the 8-1/2" section.

Tab-2 Pumps & Bits

Bit and Drilling Data			
Bit Description	Hyc DSX65195-C3	Bit Information	
Bit Number	7	Bit Size/Hole Diameter	12.25
% Washout		Washout Hole Size	12.25

Note: Bit size is the 12-1/4"

Tab-4 Well Geometry

The screenshot shows the 'Tab-4 Well Geometry' window with the 'Pilot Hole' section highlighted. It contains two input fields: 'Hole Size' and 'Depth', both currently set to 0.

Note: NO PILOT HOLE INFO IS REQUIRED

Special Cases: Pilot hole

Data Entry – Example: first day opening the hole

The current situation:

- 13-3/8" casing at 920 meters.
- 8-1/2" pilot hole drilled to 1710 meters
- Now: opening the 8-1/2" hole to 12-1/4"
- 12-1/4" hole now at 1358 meters

Tab-4 Well Geometry / Daily Casing & Volume

The interface displays a wellbore diagram on the left with depth markers at 112, 920.11, 1358, and 1710 meters. The main area contains a 'Daily Intervals' table and a 'Hole Volume Summary (bbl)' table.

Interval Number	Type	Casing OD	Casing ID	Depth	TVD	Top (liners only)	Hole Size
3	F	13.375	12.415	920.11	920.11		17.5
4	O			1358			12.25

Hole Volume Summary (bbl)			
Drill String	72	Bit Size/Hole Diameter	12.25
Annulus	598	Washout Hole Size	12.25
Below Bit	81	Depth	1358
Total Volume	751	TVD	1358
Drilled Hole	109	Bit Depth	1358
Drilled Depth (m)	438	Pilot Hole Depth	1710
		Pilot Hole Size	8.5

Notes

- Open hole depth is depth of 12-1/4" bit – this would be adjusted daily as drilling progresses.
- Daily Intervals hole geometry does NOT include the 8-1/2" pilot hole.
- Drilled hole volume is calculated for use in solids removal calculations.
- Volume below bit calculated.
- Pilot hole information accurately displayed.

Tab-1 General

1-General		2-Pumps / Bits	
Date	7/30/2006	Depth (m)	1358
Spud Date	7/9/2006	TVD (m)	1358
Report	23	Bit Depth (m)	1358
Present Activity	Opening 8-1/2" hole		

Note: ALL Depths are now the 12-1/4" bit depth.

Tab-2 Pumps & Bits

Bit and Drilling Data			
Bit Description	Hyc DSX65195-C3		Bit Information
Bit Number	7	Bit Size/Hole Diameter	12.25
% Washout		Washout Hole Size	12.25

Note: Bit size is the 12-1/4"

Tab-4 Well Geometry

Pilot Hole	
Hole Size	8.5
Depth	1710

Note: 8-1/2" Pilot hole information is entered.

Special Cases: Solids Analysis in ONE-TRAX

Solids analysis considerations if using CaCO_3

WBM – Enter the SG of the barite for your weight material. You will have to estimate the pounds per barrel of Calcium Carbonate in the system and enter it as the “Chemical Concentration” in the solids analysis section. This will allow separation of %HGS, %LGS, and %DS (remember %LGS includes additives, it is NOT the same as %DS). If you do not enter the Calcium Carbonate as a “Chemical”, it will be reported as a drill solid.

OBM – Because there are no MBT or Chemical Concentration inputs in the OBM equations, you are limited to %HGS and %LGS results. The general recommendation is to input the SG your weight material, but be prepared to explain the “high” %LGS includes the Calcium Carbonate. The alternative is to adjust your weight material SG based on the ratio of the barite & calcium carbonate – this would get very messy and would probably change from day-to-day.

Special Cases: Solids Analysis in ONE-TRAX

Solids analysis considerations if using CaCO_3

If there are NO high gravity solids the engineers should adjust the total percent solids from the retort to get the HGS as close to zero as possible without getting negative numbers. If using OBM whatever adjustment is made to total solids should be subtracted from or added to the oil percentage.

A couple of suggestions before running your retort:

- 1.- If not already running a 50 cc retort you should be.
- 2.- You should also run a sample of pure base fluid (water or oil as the case may be) to see if you get 100% recovery. If not you should check:
 - a. - The final retort temperature, some allow adjustments. These can be CAREFULLY adjusted to see if the recovery percentage is improved. If there is no improvement, reduce the temperature to the initial setting.

Special Cases: Solids Analysis in ONE-TRAX

Solids analysis considerations if using CaCO_3

- b.- There may also be a discrepancy in the actual volumes of the "50 cc" retort cup and the "50 cc" graduated cylinder. Maybe the cup is worn (more than 50 cc) or corroded (less than 50 cc). If you have a 50 cc pipette use it to verify the volumes of cup and cylinder are equal to 50 cc.
- 3.- If the fluid is allowed to cool very much they need to reweigh it for the calculations to come out right – especially if using OBM!

END